The Fabius Function and Rvachev's Up-Function

Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics

Human-readable synthesis of the formal development

Vladimir Reshetnikov's `ProveIt` repository

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Abstract

The Fabius function is the unique smooth distribution function that is constant outside $[0, 1]$, symmetric about $1/2$, and satisfies the self-similar differential equation $F'(x) = 2F(2x)$ on the left half of the unit interval. Folding it about the midpoint gives Rvachev's compactly supported even function up, one of the canonical examples of a C^∞ function with compact support that is given by an exact refinement equation. The same object admits several sharply different descriptions: a contraction fixed point, a probability law of an infinite weighted sum of independent uniform random variables, an infinite convolution, an infinite product of sinc functions, a Thue–Morse translate sum, and a family of exact rational recurrences.

This article develops these descriptions and proves their equivalence. The elementary theory is carried further than is usual: the exact shape of F and up, including strict unimodality, the unique inflection point, and the fact that 2 is the *least* Lipschitz constant; the sharp uniform derivative bound $2^{\binom{m+1}{2}}$, which is attained; two effective flatness estimates with explicit constants; and the exact real-analytic locus of each of the three functions, endpoints included.

Particular emphasis is placed on two subjects for which the formal development goes substantially beyond the usual short accounts. First, every dyadic value is evaluated by an explicit finite Thue–Morse/moment formula, and also by a fast terminating recursion that removes one set bit of the numerator at each step. The mechanism behind the first of these is a half-base q-calculus, developed here in full: the Gaussian binomial row at base $1/2$ is the leading-coefficient functional for Lagrange interpolation through the geometric nodes $-1, -2, \ldots, -2^n$, and the Thue–Morse signs supply a second, independent triangular cancellation inside each dyadic block. Second, the small-argument asymptotic is derived by an exact negative-Laplace decomposition and a quantitative saddle-point argument. The correct main term contains a genuine nonconstant periodic fluctuation in the lower Lambert coordinate. Its Fourier coefficients are explicit Gamma–zeta values, and the fluctuation is *unique*: no other continuous periodic function completes the expansion. Omitting it, as in a frequently reproduced elementary formula, destroys the claimed inverse-logarithmic error bound. The first saddle correction and the full all-orders expansion are also described, together with a numerical check against exact rational values.

The presentation follows the theorem structure of the Lean formalization under `Analysis/FabiusFunction`, while translating the machine-oriented decomposition into conventional mathematical arguments. Source-level endpoint and normalization defects are corrected rather than repeated. In particular, a centered finite-spline approximation is given with an explicit uniform error, and this approximation is turned into a primitive-recursive dyadic algorithm proving computability of the bounded Fabius function in the sense of Grzegorczyk. Binary reduction also yields an absolutely convergent signed-global series; finite half-base q-binomial identities give arbitrary-complex translated presentations of its summands, and a bounded-variation Toeplitz argument proves the corresponding discrete-limit formula.

Contents

1 Historical note and scope

The probability distribution now attached to the name of Fabius is older than Fabius's two-page note. It occurs in the 1935 work of Jessen and Wintner on infinite convolutions of independent random variables [1], was rediscovered several times, and entered a different mathematical culture through the compactly supported "finite function" of V. A. Rvachev. Fabius used it in 1966 as a particularly economical probabilistic construction of a nowhere analytic C^∞ function [2]. Independently, and for entirely different reasons, the Rvachevs introduced the same object in 1971 as the canonical compactly supported solution of a functional–differential equation [3], and V. A. Rvachev later developed it into the theory of *atomic functions* used in approximation theory and numerical analysis [4, 5, 6]. Rvachev's normalization emphasizes the even compactly supported density, whereas the Fabius normalization emphasizes its cumulative distribution function. Arias de Reyna's 1982 paper gave a systematic treatment of the compactly supported function, including its Fourier product, probability interpretation, partitions of unity, derivatives, and dyadic values; an English translation appeared in 2017 [7]. His later arithmetic paper organized the exact rational sequences and denominator questions [8]. Haugland's note [9] gives a practical algorithm for evaluating F at dyadic rationals, closely related to the bit-recursion of Section 9.

The present article is not merely a paraphrase of those papers. It follows the proof dependency graph of the Lean development in the `ProveIt` repository [21]. That development first constructs the bounded distribution function, then derives the Rvachev and signed-global forms, and finally builds the arithmetic and asymptotic layers on top. This order has two advantages. It prevents identities valid for the signed extension from being silently applied to the bounded distribution, and it makes the exact dyadic algorithms independent of any numerical approximation.

The formalization also settles several questions that are only implicit, conjectural, or incorrectly normalized in the source material. Most importantly, the sharp endpoint asymptotic contains a nonconstant periodic function of an exact lower-Lambert phase. The periodic term is not a cosmetic improvement: every nonzero Fourier mode is explicit and nonzero, and an elementary formula without it fails at the error scale usually claimed. The development further proves a full inverse-Lambert expansion, establishes an exact bit-recursive evaluator for every dyadic rational, and supplies an absolutely and uniformly convergent Fourier–Legendre expansion of up.

Arias de Reyna's arithmetic paper [8] enumerates the independent occurrences of this function in chronological order, and two of them deserve mention beside Jessen–Wintner, Fabius, and the Rvachevs. In 1981 G. Kh. Kirov and G. A. Totkov arrived at it in a study of the distribution of the zeros of the derivatives of the function they wrote $\lambda(x)$ (*Godishnik Vissh. Uchebn. Zaved. Prilozhna Mat.* **17** (1981), no. 4, 157–165, in Bulgarian with English and Russian summaries); those zeros are the subject that reappears here as the terminating dyadic Taylor series (99). In 1985 R. Schnabl gave a further independent construction, "Über eine C^∞-Funktion," in *Zahlentheoretische Analysis*, Lecture Notes in Mathematics 1114, Springer, Berlin (1985), 134–142. Neither paper is used in the proofs below, which is why their data are given here rather than in the bibliography; they are recorded because a reader tracing the function through the literature will meet both names.

Our convention is fixed throughout:

- F is the bounded cumulative distribution function, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$;
- up is the even compactly supported density on $[-1, 1]$;
- $\mathcal{F}$ is the signed global extension satisfying $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on all of $\mathbb{R}$.

All logarithms are natural logarithms. We write $L = \log 2$ in the asymptotic sections, $\mathrm{wt}_2(n)$ for the sum of the binary digits of n, and $\operatorname{sinc} z = \sin z/z$ with the removable value 1 at the origin.

1.1 A guide to the main results

A reader who wants the shape of the theory before its details can follow a short path through what follows.

Everything rests on the bookkeeping of Section 2. Three functions carry three different global laws: the bounded distribution function F, pinned by (1) and (2); the even compactly supported bump up with its refinement equation (28); and the signed extension $\mathcal{F}$ of (11), the only one of the three that satisfies $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on all of $\mathbb{R}$. Nearly every apparent contradiction between published formulas for "the Fabius function" is a disagreement about which of the three is meant. That section should be read first, and returned to whenever a global identity is compared with a source.

Two independent constructions follow, and a reader in a hurry may take either. Section 3 builds F as the fixed point of a strict contraction; the same estimate that proves uniqueness also makes the construction a certified numerical scheme (515). Section 4 builds the same function as the law of a weighted sum of independent uniform variables, which explains positivity, monotonicity, symmetry, and smoothing at once, and which produces the finite atomic laws (46)–(48) and the centered finite spline of Section 4.4. That spline, with its explicit uniform error, is the one approximation used again later, in the computability theorem of Section 21.6.

The Fourier side is Section 5. Its central result is the entire sinc product (70); from the vanishing of its first factor at nonzero integers come Poisson summation (82) and the exact partitions of unity (83). Section 6 then converts the fold relation into a global scaling law and reads off every derivative in closed form (90). The consequence that the Taylor series terminates at every dyadic point, (99), is the hinge of the article: it yields nowhere analyticity on the one hand and the whole exact arithmetic on the other.

The exact arithmetic occupies Sections 7 to 9. The two moment recurrences (110) and (119) are elementary and triangular; the identification (136) of the half moments with the endpoint values $F(2^{-n})$ is what makes them useful. The two central algorithms should be read as a pair. The closed Thue–Morse formula (164) proves rationality, exhibits the cancellation, and certifies denominators; the bit-recursive Taylor evaluator built on (178) is that cancellation already carried out, makes one recursive call per set bit, and is what one actually runs. The arithmetic consequences – the exact valuation (156), the normalized odd integers (158), and the universal denominator bound (191) – all come from the same data.

Three sections explain why several visibly different finite expressions return the same rational number. The half-base q-binomial formula (205) in Section 9 presents the dyadic value as a Gaussian finite difference whose value is independent of the translation parameter Q; Section 11 iterates the Taylor block along the binary expansion of an arbitrary nonnegative real number, producing the telescope (278) and an absolutely convergent global series; and Section 12 shows that the same Gaussian row also converges as a genuine limit of finite shifted splines. A reader who only wants to evaluate the function may skip all three; a reader who wants to know why the published formulas agree should not.

The endpoint analysis is the technically deepest part and runs from Section 14 to Section 18. The exact decomposition (369) isolates a genuinely periodic function whose centered form Ψ of (370) has explicit Gamma–zeta Fourier coefficients (373). The sharp theorem (395) of Section 15 evaluates that fluctuation at the exact lower Lambert coordinate (390); Section 16 continues the expansion to every order; and Section 17 translates the result into elementary logarithms and proves that the frequently reproduced display without the periodic term is false at the error scale it claims, (470). The periodic term is the point of this whole sequence.

It is bounded and numerically minute, but it is nonconstant, and no choice of constant repairs a formula that omits it. Section 18 collects the coarse consequences, such as the quadratic logarithmic law (475), which are insensitive to it.

Two shorter analytic sections complete the picture: Section 19 gives the absolutely and uniformly convergent Fourier–Legendre expansion of up, with exact coefficients and least-squares optimality, and Section 20 collects the cosine series, scaled partitions, and the self-convolution law.

The last group is computational and critical. Section 21 decides which representation to use for which question, gives the precomputation recurrence, the operation count, the stable endpoint procedure, and a verification suite; Section 21.6 proves computability of F in the Grzegorczyk sense from the centered spline; Section 22 records where an attractive informal shortcut ceases to be valid and which printed statements have been corrected; and Section 23 summarizes what each viewpoint contributes. The appendices hold exact tables (Appendix A), the map to the formal development (Appendix B), the concordance with the source papers (Appendix C), and the notation crosswalk (Appendix D).

For a first reading of one hour: Section 2, then (90) and (99), then (164) and (178), then (395). Those five statements carry the article.

One object, four viewpoints. The bounded F is a cumulative distribution function, so its values are probabilities and its monotonicity is automatic. The bump up is the density of a centered version of the same random series, which is why it is even, positive on $(-1, 1)$, and C^∞. The transform $\widehat{\mathrm{up}}$ is an infinite product of sinc factors, which is why it is entire and rapidly decreasing on the real axis. The signed $\mathcal{F}$ is the Thue–Morse continuation of the same function, and it is the only one of the three that satisfies a pure dilation equation on all of $\mathbb{R}$. Every identity in this article is an identity about one of these four presentations of a single object; the work lies in keeping track of which.

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2 Three functions that must not be conflated

The literature often uses the same letter F for three related, but globally different, functions. That economy of notation is harmless in a local calculation and dangerous in a long proof. We therefore begin by separating them.

2.1 The bounded Fabius distribution function

Definition 2.1 (Bounded Fabius function)**.** A function $F : \mathbb{R} \to [0, 1]$ is called a *bounded Fabius function* if

$$F(x) = 0 \qquad (x \le 0), \tag{1}$$
$$F(x) = 1 \qquad (x \ge 1), \tag{2}$$
$$F(1-x) = 1 - F(x) \qquad (0 \le x \le 1), \tag{3}$$
$$F'(x) = 2F(2x) \qquad (0 \le x \le \tfrac{1}{2}), \tag{4}$$

and F is infinitely differentiable on $\mathbb{R}$.

The derivative equation on the right half follows from symmetry:

$$F'(x) = 2F(2-2x), \qquad \tfrac{1}{2} \le x \le 1. \tag{5}$$

In particular $F'(x) > 0$ for $0 < x < 1$, once positivity in the interior is known. Thus F is a genuine cumulative distribution function, not merely a smooth interpolation between 0 and 1.

Because F is constant on both exterior half-lines and is C^∞, every derivative of positive order vanishes at both endpoints:

$$F^{(m)}(0) = F^{(m)}(1) = 0 \qquad (m \geq 1). \tag{6}$$

This flatness is the source of both its extraordinary endpoint decay and its failure to be analytic there.

2.2 Rvachev's compactly supported bump

Definition 2.2 (Rvachev up-function)**.** Given F, define

$$\mathrm{up}(x) = \begin{cases} F(x+1), & x \leq 0, \\ F(1-x), & x > 0. \end{cases} \tag{7}$$

Equivalently,

$$\mathrm{up}(x) = F(1 - |x|). \tag{8}$$

Here the right side is interpreted using the globally bounded function F.

It follows immediately that up is even, nonnegative, supported on $[-1, 1]$, and $\mathrm{up}(0) = 1$. On the unit interval the two functions are simply translates:

$$F(x) = \mathrm{up}(x - 1), \qquad 0 \leq x \leq 1. \tag{9}$$

The graph of F is an infinitely flat sigmoid; the graph of up is its symmetric fold.

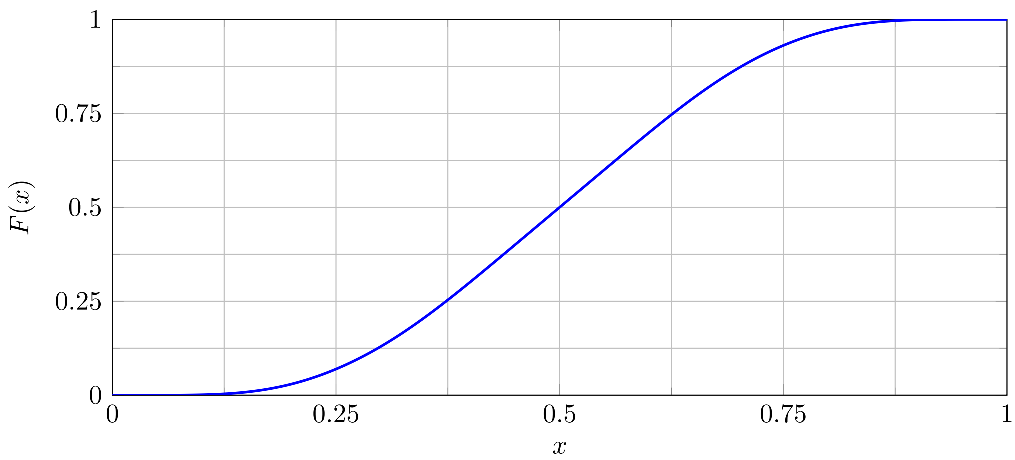

Figure 1: The bounded Fabius function on $[0, 1]$, drawn through exact dyadic values on a 2^{-8} grid. At this resolution the straight segments are visually indistinguishable from the smooth curve.

2.3 The signed global extension

Let $\mathrm{wt}_2(n)$ denote the sum of the binary digits of n, and put

$$\varepsilon_n = (-1)^{\mathrm{wt}_2(n)}. \tag{10}$$

The sequence (ε_n) is the $\{\pm 1\}$ form of the Thue–Morse sequence [14].

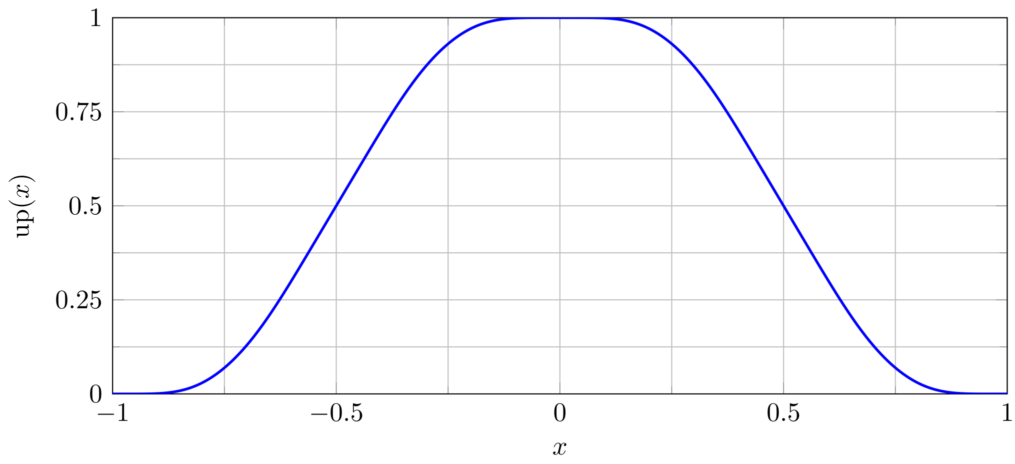

Figure 2: Rvachev's up-function on its support $[-1,1]$, obtained by folding the same exact dyadic data.

Definition 2.3 (Signed extension)**.** Define

$$\mathcal{F}(x) = \sum_{n\geq 0} \varepsilon_n \operatorname{up}(x - 2n - 1). \tag{11}$$

At a fixed x only finitely many summands can be nonzero, so this is a locally finite sum. On every block $[2b, 2b+2]$ only the b-th translate can be nonzero (and at the endpoints it too vanishes):

$$\mathcal{F}(x) = \varepsilon_b \operatorname{up}(x - 2b - 1), \qquad 2b \leq x \leq 2b + 2. \tag{12}$$

It vanishes for $x \leq 0$, and it agrees with the bounded F for every $x \leq 1$ (both functions vanish on $(-\infty, 0]$). It does *not* remain in $[0,1]$ to the right of that interval. This is the global object that satisfies the cleanest differential equation:

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \qquad (x \in \mathbb{R}). \tag{13}$$

Arias de Reyna's arithmetic paper uses this extension. Statements about its values outside $[0,1]$ must therefore not be transferred blindly to the bounded cumulative distribution.

Notation used in this article. F always means the bounded distribution function; up always means the compactly supported even bump; and $\mathcal{F}$ always means the signed global extension. We have $F = \mathcal{F}$ throughout $(-\infty, 1]$ (hence on $[0,1]$), but globally they are different functions.

2.4 Exact shape: positivity, bijectivity, and unimodality

The three defining equations (1)–(4) already determine the qualitative picture, but the statements that are actually used later are the sharp ones: not $F \geq 0$ but $F(x) > 0$ exactly for $x > 0$; not up ≤ 1 but $\operatorname{up}(x) = 1$ exactly at the origin. We record them here, in the form of equivalences, because the strict versions are what make the convexity and Lipschitz theorems below sharp.

Everything rests on a single formula that merges (4) and (5) into one identity valid on the whole line.

Lemma 2.4 (Unified derivative formula)**.** *For every real x,*

$$F'(x) = 2 \operatorname{up}(2x - 1). \tag{14}$$

Proof. By (8), $\mathrm{up}(2x-1) = F(1-|2x-1|)$. For $0 \le x \le \frac{1}{2}$ one has $|2x-1| = 1-2x$, so $\mathrm{up}(2x-1) = F(2x)$ and (14) is (4). For $\frac{1}{2} \le x \le 1$ one has $|2x-1| = 2x-1$, so $\mathrm{up}(2x-1) = F(2-2x)$ and (14) is (5). For $x < 0$ or $x > 1$ one has $|2x-1| > 1$, hence $1-|2x-1| < 0$ and $\mathrm{up}(2x-1) = 0$ by (1), while $F'(x) = 0$ because F is constant on each exterior half-line by (1) and (2). □

Since $\mathrm{up} \ge 0$, formula (14) shows at once that F is nondecreasing on all of $\mathbb{R}$. It is also convenient to note that the reflection symmetry (3), stated on the unit interval, in fact holds globally:

$$F(1-x) = 1 - F(x) \qquad (x \in \mathbb{R}). \tag{15}$$

Indeed, for $x < 0$ both sides equal 1 by (1) and (2), and for $x > 1$ both sides equal 0.

Theorem 2.5 (Positivity, strict monotonicity, bijectivity)**.** *The bounded Fabius function satisfies the following.*

1. $F(x) > 0$ *if and only if* $x > 0$.
2. $F(x) < 1$ *if and only if* $x < 1$.
3. $F'(x) > 0$ *for* $0 < x < 1$*, and* F *is strictly increasing on* $[0,1]$.
4. F *maps* $[0,1]$ *bijectively onto* $[0,1]$.

Proof. The heart of the matter is the following doubling step: if $0 \le x \le \frac{1}{2}$ and $F(x) = 0$, then $F(2x) = 0$. To see it, observe that $F \ge 0$ everywhere, so a zero of F is a global minimum of a function differentiable on all of $\mathbb{R}$; Fermat's rule gives $F'(x) = 0$, and (4) turns this into $2F(2x) = 0$.

Suppose now that $F(x) = 0$ for some $x > 0$. Since $2^n x \to \infty$, there is a smallest integer $n \ge 0$ with $2^n x \ge \frac{1}{2}$. For $0 \le m < n$ we have $2^m x < \frac{1}{2}$, so the doubling step applies at $2^m x$ and propagates the vanishing: from $F(x) = 0$ we obtain $F(2^m x) = 0$ for every $m \le n$, and in particular $F(2^n x) = 0$. But F is nondecreasing and $2^n x \ge \frac{1}{2}$, so $F(2^n x) \ge F(\frac{1}{2}) = \frac{1}{2}$, where the midpoint value comes from (3) at $x = \frac{1}{2}$. This contradiction proves that F has no zero on the positive half-line; combined with (1) it proves (i).

Assertion (ii) follows from (i) and (15): $F(x) < 1$ if and only if $F(1-x) > 0$, that is, if and only if $1 - x > 0$.

For (iii), let $0 < x < 1$. Then $2x - 1 \in (-1,1)$, so $1 - |2x-1| \in (0,1]$, and (i) gives $\mathrm{up}(2x-1) = F(1-|2x-1|) > 0$. Now (14) yields $F'(x) > 0$. A continuous function on $[0,1]$ whose derivative is positive throughout the interior is strictly increasing on $[0,1]$.

For (iv), F is continuous with $F(0) = 0$ and $F(1) = 1$, so the intermediate value theorem makes $F([0,1])$ contain $[0,1]$; since F takes values in $[0,1]$, the image is exactly $[0,1]$. Strict monotonicity from (iii) gives injectivity. □

The fold (8) now transfers the whole picture to Rvachev's bump, and it does so in strict form.

Theorem 2.6 (Strict unimodality of up)**.** *Rvachev's function satisfies the following.*

1. up *is strictly increasing on* $[-1,0]$ *and strictly decreasing on* $[0,1]$.
2. $\mathrm{up}(x) = 1$ *if and only if* $x = 0$*; the maximum value* 1 *is attained at the origin and nowhere else.*
3. $\mathrm{up}(x) = 0$ *if and only if* $x \notin (-1,1)$*. Thus* up *is nonzero exactly on* $(-1,1)$*, and* $\operatorname{supp}\mathrm{up} = [-1,1]$.
4. $\mathrm{up}'(x) > 0$ *for* $-1 < x < 0$ *and* $\mathrm{up}'(x) < 0$ *for* $0 < x < 1$.

Proof. By (8), $\mathrm{up}(x) = F(1-x)$ for $x \ge 0$. As x increases through $[0,1]$ the argument $1-x$ decreases through $[0,1]$, on which F is strictly increasing by theorem 2.5(iii); hence up is strictly decreasing on $[0,1]$, and evenness gives (i) on $[-1,0]$.

For (ii), $\mathrm{up}(x) = F(1-|x|)$ and theorem 2.5(ii) say that $\mathrm{up}(x) < 1$ unless $1-|x| \ge 1$, that is, unless $x = 0$; and $\mathrm{up}(0) = F(1) = 1$. For (iii), $\mathrm{up}(x) > 0$ if and only if $1-|x| > 0$ by theorem 2.5(i), and $\mathrm{up} \ge 0$, so the zero set is exactly the complement of $(-1,1)$; the closed support is the closure $[-1,1]$.

For (iv), the identity $\mathrm{up}(x) = F(1-x)$ holds on the whole open half-line $x > 0$, hence may be differentiated there: $\mathrm{up}'(x) = -F'(1-x)$, which is negative for $0 < x < 1$ by theorem 2.5(iii). Evenness gives $\mathrm{up}'(x) = -\mathrm{up}'(-x)$, so $\mathrm{up}' > 0$ on $(-1,0)$. □

Parts (i) and (iii) are the classical description of the bump [6, 7]; the strict forms, and the equivalences in theorem 2.5, are exactly what the formal development records.

2.5 Convexity and the unique inflection point

Formula (14) converts the first-order shape of up into the second-order shape of F. The affine substitution $x \mapsto 2x-1$ carries $(-\infty, \frac{1}{2}]$ onto $(-\infty, 0]$ and $[\frac{1}{2}, \infty)$ onto $[0,\infty)$, which are precisely the two halves on which up is monotone.

Theorem 2.7 (Convexity and the inflection point)**.** *The derivative F' is nondecreasing on $(-\infty, \frac{1}{2}]$ and nonincreasing on $[\frac{1}{2}, \infty)$, and it is strictly increasing on $[0, \frac{1}{2}]$ and strictly decreasing on $[\frac{1}{2}, 1]$. Consequently F is convex on $(-\infty, \frac{1}{2}]$ and concave on $[\frac{1}{2}, \infty)$, strictly convex on $[0, \frac{1}{2}]$ and strictly concave on $[\frac{1}{2}, 1]$. The midpoint $x = \frac{1}{2}$ is the unique inflection point, and*

$$F'(\tfrac{1}{2}) = 2 = \max_{x\in\mathbb{R}} F'(x), \tag{16}$$

the maximum being attained at $x = \frac{1}{2}$ and nowhere else.

Proof. By theorem 2.6, up is nondecreasing on $(-\infty, 0]$: it vanishes on $(-\infty, -1]$ and increases strictly on $[-1, 0]$. Symmetrically it is nonincreasing on $[0, \infty)$. Composing with $x \mapsto 2x - 1$ and using (14) shows that F' is nondecreasing on $(-\infty, \frac{1}{2}]$ and nonincreasing on $[\frac{1}{2}, \infty)$. If $0 \le x < y \le \frac{1}{2}$ then $-1 \le 2x-1 < 2y-1 \le 0$ and the increase of up on $[-1,0]$ is strict, so $F'(x) < F'(y)$; the second half is symmetric.

A differentiable function on an interval is convex precisely when its derivative is nondecreasing there, and strictly convex when the derivative is strictly increasing; the concave statements are the same assertions for $-F$. The intervals of strictness cannot be enlarged: F vanishes identically on $(-\infty, 0]$ and equals 1 identically on $[1, \infty)$, so it is affine, hence not strictly convex or concave, on either exterior half-line.

Since F is convex on the whole of $(-\infty, \frac{1}{2}]$ and concave on the whole of $[\frac{1}{2}, \infty)$, the sense of convexity can change only at the midpoint, and it does change there, because F is strictly convex immediately to the left of $\frac{1}{2}$ and strictly concave immediately to the right. So $\frac{1}{2}$ is the unique inflection point. Finally $F'(x) = 2\,\mathrm{up}(2x-1) \le 2\,\mathrm{up}(0) = 2$ by theorem 2.6(ii), with equality if and only if $2x - 1 = 0$. □

Differentiating (14) once more gives $F''(x) = 4\,\mathrm{up}'(2x-1)$, so $F''(\frac{1}{2}) = 4\,\mathrm{up}'(0) = 0$: the origin is an interior maximum of the differentiable function up. Together with $F(\frac{1}{2}) = \frac{1}{2}$ this locates the steepest point of the sigmoid exactly at its centre of symmetry, where the graph crosses its own midpoint with slope 2.

2.6 The optimal Lipschitz constant

The bound $F' \le 2$ of (16) is a Lipschitz estimate, and because the bound is attained the estimate cannot be improved.

Theorem 2.8 (Two is the least Lipschitz constant)**.** *Both F and* up *are 2-Lipschitz on $\mathbb{R}$:*

$$|F(x) - F(y)| \le 2|x-y|, \qquad |\operatorname{up}(x) - \operatorname{up}(y)| \le 2|x-y| \qquad (x, y \in \mathbb{R}). \tag{17}$$

Moreover 2 is the least admissible constant for each: if $K \ge 0$ is such that $|F(x) - F(y)| \le K|x-y|$ for all x, y, or such that $|\operatorname{up}(x) - \operatorname{up}(y)| \le K|x-y|$ for all x, y, then $K \ge 2$.

Proof. By (14) and $0 \le \operatorname{up} \le 1$ we have $0 \le F' \le 2$ on all of $\mathbb{R}$, so the mean value theorem gives the first estimate in (17). The map $x \mapsto 1 - |x|$ is 1-Lipschitz, so the second estimate follows from the first through the fold (8).

For optimality, let K be admissible for F. Dividing $|F(x) - F(\frac{1}{2})| \le K|x - \frac{1}{2}|$ by $|x - \frac{1}{2}|$ and letting $x \to \frac{1}{2}$ gives $K \ge |F'(\frac{1}{2})| = 2$ by (16). For up the same limit at the same point works: the proof of theorem 2.6(iv) gives $\operatorname{up}'(\frac{1}{2}) = -F'(\frac{1}{2}) = -2$, so $K \ge 2$. □

The two extremal points are the same point in the two pictures: $\frac{1}{2}$ is the centre of the sigmoid and the midpoint of the decreasing half of the bump, and the fold sends one to the other.

Corollary 2.9 (Linear majorants at the endpoints)**.** *For all admissible arguments,*

$$F(x) \le 2x \qquad (x \ge 0), \tag{18}$$
$$1 - F(x) \le 2(1-x) \qquad (x \le 1), \tag{19}$$
$$\operatorname{up}(x) \le 2(1 - |x|) \qquad (|x| \le 1). \tag{20}$$

Proof. Applying (17) with $y = 0$ and using $F(0) = 0$ gives (18). Replacing x by $1 - x$ there and using (15) gives (19). Substituting $1 - |x| \ge 0$ into (18) and using (8) gives (20). □

The content of (18) lies on $[0, \frac{1}{2}]$; for $x \ge \frac{1}{2}$ it is weaker than $F \le 1$. Its value is that it holds with no hypothesis at all on the size of x, so it can be used as a linear majorant in any estimate near the left endpoint, and (20) says that the bump vanishes at least linearly at both ends of its support. These majorants and the optimality of the constant 2 are contributions of the formal development; the plain 2-Lipschitz bound is used implicitly throughout [7, 6].

3 Existence and uniqueness by a contraction

The most economical construction of F is a Banach fixed-point argument. It has an additional advantage: the contraction estimate later gives a clean uniqueness theorem, without presupposing Fourier analysis or probability.

3.1 The admissible function space

Let $I = [0, 1]$ and let $C(I)$ carry the supremum norm. Define $\mathcal{A} \subset C(I)$ by

$$\mathcal{A} = \{f \in C(I) : 0 \le f \le 1,\ f(1-x) = 1 - f(x)\}. \tag{21}$$

The set $\mathcal{A}$ is closed in the Banach space $C(I)$, hence complete. It is nonempty; for example $f(x) = x$ belongs to it.

Extend $f \in \mathcal{A}$ to a continuous function $\widetilde{f}$ on $\mathbb{R}$ by clamping the argument to I:

$$\widetilde{f}(t) = \begin{cases} f(0), & t \le 0, \\ f(t), & 0 \le t \le 1, \\ f(1), & t \ge 1. \end{cases} \tag{22}$$

Symmetry gives

$$\int_0^1 f(t)\,\mathrm{d}t = \frac{1}{2}. \tag{23}$$

Indeed, replacing t by $1-t$ and adding the two integrals gives $2\int_0^1 f = 1$.

3.2 The integral transform

For $f \in \mathcal{A}$, define Tf by

$$(Tf)(x) = \begin{cases} \displaystyle\int_0^{2x} \widetilde{f}(t)\,\mathrm{d}t, & 0 \le x \le \frac{1}{2}, \\ 1 - \displaystyle\int_0^{2-2x} \widetilde{f}(t)\,\mathrm{d}t, & \frac{1}{2} < x \le 1. \end{cases} \tag{24}$$

The two definitions agree at $x = 1/2$ by (23). Positivity and the upper bound $f \le 1$ show that $0 \le Tf \le 1$, and a direct substitution shows $(Tf)(1-x) = 1 - (Tf)(x)$. Hence $T : \mathcal{A} \to \mathcal{A}$.

The central estimate is stronger than the naive integral bound.

Lemma 3.1 (Contraction estimate)**.** *For $f, g \in \mathcal{A}$,*

$$\|Tf - Tg\|_\infty \le \frac{1}{2}\|f - g\|_\infty. \tag{25}$$

Proof. Write $h = f - g$. Since both f and g have integral $1/2$, $\int_0^1 h = 0$. For $0 \le y \le 1$ we may therefore estimate the primitive in two ways:

$$\left|\int_0^y h(t)\,\mathrm{d}t\right| \le y\|h\|_\infty,$$
$$\left|\int_0^y h(t)\,\mathrm{d}t\right| = \left|\int_y^1 h(t)\,\mathrm{d}t\right| \le (1-y)\|h\|_\infty.$$

Consequently

$$\left|\int_0^y h(t)\,\mathrm{d}t\right| \le \min(y, 1-y)\|h\|_\infty \le \frac{1}{2}\|h\|_\infty.$$

Every value of $Tf - Tg$ is one of these primitives, possibly with a minus sign. Taking the supremum proves (25). □

Theorem 3.2 (Existence of the bounded Fabius function)**.** *The transform T has a unique fixed point $f_* \in \mathcal{A}$. Extending f_* by 0 on $(-\infty, 0]$ and by 1 on $[1, \infty)$ produces the bounded Fabius function F.*

Proof. The space $\mathcal{A}$ is complete and T is a strict contraction with constant $1/2$, so Banach's fixed-point theorem gives a unique fixed point. On $[0, 1/2]$ its fixed-point equation is

$$F(x) = \int_0^{2x} F(t)\,\mathrm{d}t. \tag{26}$$

Differentiating gives $F'(x) = 2F(2x)$. The second half of (24) supplies symmetry and the matching right-hand formula. The endpoint values are immediate. □

Starting from any admissible f_0, the iterates $f_{n+1} = Tf_n$ converge uniformly to F, with the explicit estimate

$$\|f_n - F\|_\infty \leq 2^{-n}\|f_0 - F\|_\infty. \tag{27}$$

This gives a conceptually simple approximation scheme, although exact dyadic evaluation will be much faster.

3.3 The differential equation for up and smoothness

Folding (4) and (5) yields the refinement-type differential equation

$$\mathrm{up}'(x) = 2\big(\mathrm{up}(2x+1) - \mathrm{up}(2x-1)\big) \qquad (x \in \mathbb{R}). \tag{28}$$

At $x = 0$ the two one-sided derivatives glue because $\mathrm{up}(1) = \mathrm{up}(-1) = 0$. Once up is continuously differentiable, (28) bootstraps regularity: if $\mathrm{up} \in C^r$, then the right side is C^r, so $\mathrm{up} \in C^{r+1}$. Induction gives $\mathrm{up} \in C^\infty$ and then $F \in C^\infty$.

Proposition 3.3 (Basic shape)**.** *The functions F and up have the following properties.*

1. *F is strictly increasing on $(0,1)$ and satisfies $0 < F(x) < 1$ there.*
2. *up is even, strictly positive on $(-1,1)$, and zero outside $[-1,1]$.*
3. *$\int_{-1}^{1} \mathrm{up}(x)\,\mathrm{d}x = 1$.*
4. *Every derivative of up vanishes at $x = \pm 1$.*

Proof. For $0 < x \leq 1/2$, (4) gives $F'(x) = 2F(2x)$. Positivity follows first near $1/2$ from $F(1/2) = 1/2$, and then propagates toward zero by repeatedly doubling the argument; symmetry handles the right half. This proves strict monotonicity and, by (8), positivity of up in the interior of its support. Finally,

$$\int_{-1}^{1} \mathrm{up}(x)\,\mathrm{d}x = 2\int_0^1 F(t)\,\mathrm{d}t = 1,$$

where the middle identity uses the fold and the last uses symmetry. Flatness at the support endpoints follows from smoothness and the fact that up is identically zero just outside the support. ☐

3.4 The original Rvachev characterization

At the level of solution spaces, the preceding construction is equivalent to the characterization historically associated with Rvachev. The fold alone cannot give a direct predicate iff for an arbitrary bounded function, because it does not inspect that function on $(1,\infty)$; the precise equivalence below retains a unique bounded Fabius witness.

Theorem 3.4 (Original characterization in the formal development)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be C^∞, strictly positive on $(-1,1)$, normalized by $\varphi(0) = 1$, and have topological support exactly $[-1,1]$. Let $k \in \mathbb{R}$. If*

$$\varphi'(x) = k\big(\varphi(2x+1) - \varphi(2x-1)\big) \qquad (x \in \mathbb{R}), \tag{29}$$

then $k = 2$ and $\varphi = \mathrm{up}$.

Proof. Continuity and the support hypothesis give $\varphi(\pm 1) = 0$, while strict interior positivity identifies its ordinary nonzero set as $(-1,1)$. Positivity of k need not be assumed: the mean-value theorem on $[-1,0]$ gives some $c \in (-1,0)$ for which $k\varphi(2c+1) = 1$, because the other translated term vanishes; hence $k > 0$. For $0 \le t \le 1$, differentiate $H(t) = \varphi(t) + \varphi(t-1)$. Equation (29) makes the two middle terms cancel and leaves

$$H'(t) = k\big(\varphi(2t+1) - \varphi(2t-3)\big) = 0,$$

because both arguments lie outside the open support. Since $H(0) = \varphi(0) + \varphi(-1) = 1$, we have $H \equiv 1$ on $[0,1]$. Integrating this identity shows noncircularly that

$$\int_{\mathbb{R}} \varphi = \int_0^1 \varphi(t)\,\mathrm{d}t + \int_0^1 \varphi(t-1)\,\mathrm{d}t = 1.$$

Now integrate (29) over $[-1,0]$. The left side is $\varphi(0) - \varphi(-1) = 1$; after the substitutions $y = 2x \pm 1$, the right side is $(k/2)\int_{-1}^{1} \varphi = k/2$. Hence $k = 2$. The Fourier argument in the next theorem, now with the mass and scale derived rather than assumed, gives $\varphi = \mathrm{up}$. □

The formal development also closes the converse direction. For a bounded Fabius solution G, write up_G for the fold (7) formed from G.

Corollary 3.5 (Equivalence of the two formal characterizations)**.** *For every $\varphi : \mathbb{R} \to \mathbb{R}$ and $k \in \mathbb{R}$, the original compact-support characterization holds for (φ, k) if and only if*

$$k = 2 \qquad \text{and} \qquad \text{there is a unique bounded Fabius solution } G \text{ with } \varphi = \mathrm{up}_G. \tag{30}$$

Proof. In the forward direction the theorem above forces $k = 2$ and $\varphi = \mathrm{up}$; the canonical bounded solution is therefore a witness, and uniqueness of bounded Fabius solutions makes it the only witness. Conversely, every bounded Fabius solution folds to a smooth function with topological support $[-1,1]$, strict positivity on $(-1,1)$, value one at the origin, and derivative law (29) with scale two. □

The missing tail can itself be described without any Fabius hypotheses.

Proposition 3.6 (Exact kernel of the fold)**.** *For arbitrary bounded candidates $G, H : \mathbb{R} \to [0,1]$,*

$$\mathrm{up}_G = \mathrm{up}_H \iff G(t) = H(t) \quad \text{for every } t \le 1. \tag{31}$$

Consequently, a bounded candidate G satisfies the Fabius characterization if and only if up_G satisfies the original characterization at scale two and $G(t) = 1$ for every $t > 1$.

Proof. If the folds agree and $t \le 1$, evaluate them at $t - 1 \le 0$; the left branch of (7) returns exactly $G(t)$ and $H(t)$. Conversely, each branch of the fold queries its candidate at either $x + 1 \le 1$ or $1 - x < 1$, so agreement through 1 makes the folds equal. For the final assertion, the forward implication is immediate. In the reverse direction, uniqueness of the original solution makes up_G canonical; (31) therefore identifies G with the canonical bounded solution on $(-\infty, 1]$, while the added hypothesis identifies their remaining values. Thus G is the canonical Fabius solution. The strict inequality is optimal: the fold still sees $G(1)$ at its input zero. □

The mass-normalized Fourier form is a useful corollary in which positivity and the value at the origin are replaced by a single integral hypothesis.

Theorem 3.7 (Integral-normalized Fourier characterization)**.** *There is a unique nonzero C^∞ function $\varphi : \mathbb{R} \to \mathbb{R}$ such that*

$$\operatorname{supp}\varphi = [-1,1], \qquad \int_{\mathbb{R}} \varphi(x)\,\mathrm{d}x = 1, \tag{32}$$

and

$$\varphi'(x) = k\big(\varphi(2x+1) - \varphi(2x-1)\big) \tag{33}$$

for some real scale k. Necessarily $k = 2$, and $\varphi = \mathrm{up}$.

Proof. Work throughout with the Fourier transform $\widehat{\varphi}$ of φ in the convention (69). Since φ is smooth with support in $[-1,1]$, the integral defining $\widehat{\varphi}$ converges for every complex z and $\widehat{\varphi}$ is entire, and integration by parts has no boundary terms, so $\widehat{\varphi'}(z) = 2\pi\mathrm{i}z\widehat{\varphi}(z)$.

The scale must be two. Substituting $y = 2x+1$ and $y = 2x-1$ gives, for every $z \in \mathbb{C}$,

$$\int_{\mathbb{R}} \varphi(2x+1)\mathrm{e}^{-2\pi\mathrm{i}xz}\,\mathrm{d}x = \tfrac{1}{2}\mathrm{e}^{\pi\mathrm{i}z}\widehat{\varphi}(z/2), \qquad \int_{\mathbb{R}} \varphi(2x-1)\mathrm{e}^{-2\pi\mathrm{i}xz}\,\mathrm{d}x = \tfrac{1}{2}\mathrm{e}^{-\pi\mathrm{i}z}\widehat{\varphi}(z/2). \tag{34}$$

Transforming (33) and using $\mathrm{e}^{\pi\mathrm{i}z} - \mathrm{e}^{-\pi\mathrm{i}z} = 2\mathrm{i}\sin(\pi z)$ therefore yields

$$2\pi\mathrm{i}z\,\widehat{\varphi}(z) = k\mathrm{i}\sin(\pi z)\,\widehat{\varphi}(z/2), \tag{35}$$

that is, after dividing by $2\pi\mathrm{i}z$ and filling in the removable value at the origin,

$$\widehat{\varphi}(z) = \frac{k}{2}\,\mathrm{sinc}(\pi z)\,\widehat{\varphi}(z/2). \tag{36}$$

Both sides of (36) are entire, so the identity holds at $z=0$ as well. There it reads $\widehat{\varphi}(0) = (k/2)\widehat{\varphi}(0)$. By the normalization (32), $\widehat{\varphi}(0) = \int_{\mathbb{R}}\varphi = 1 \neq 0$, and therefore $k=2$. Equation (36) becomes

$$\widehat{\varphi}(z) = \mathrm{sinc}(\pi z)\,\widehat{\varphi}(z/2). \tag{37}$$

Nondegeneracy at the origin. We claim that no entire solution of (37) other than the zero function can vanish at the origin. Suppose $\widehat{\varphi} \not\equiv 0$ and $\widehat{\varphi}(0) = 0$. Let

$$\widehat{\varphi}(z) = \sum_{r\geq 0} a_r z^r$$

be the Taylor expansion at the origin and let $m \geq 1$ be the least index with $a_m \neq 0$. Since $\mathrm{sinc}(\pi z) = 1 - \pi^2 z^2/6 + \mathcal{O}(z^4)$, in particular $\mathrm{sinc}(\pi z) = 1 + \mathcal{O}(z^2)$, the power series of $\mathrm{sinc}(\pi z)\widehat{\varphi}(z/2)$ begins at order m as well, and its coefficient of z^m receives no contribution from the correction terms of the sinc factor, because $\widehat{\varphi}(z/2)$ has no term below z^m. Comparing the coefficients of z^m on the two sides of (37) therefore gives

$$a_m = 2^{-m}a_m,$$

so $(1-2^{-m})a_m = 0$ and hence $a_m = 0$, contradicting the choice of m. Thus $\widehat{\varphi}(0) \neq 0$.

The product and the conclusion. Iterating (37) N times gives

$$\widehat{\varphi}(z) = \widehat{\varphi}(z/2^N)\prod_{n=0}^{N-1}\mathrm{sinc}\left(\frac{\pi z}{2^n}\right). \tag{38}$$

As $N \to \infty$ the first factor tends to $\widehat{\varphi}(0)$ by continuity, and the finite product tends to the entire function $\widehat{\mathrm{up}}$ locally uniformly by (70). Hence

$$\widehat{\varphi}(z) = \widehat{\varphi}(0)\,\widehat{\mathrm{up}}(z). \tag{39}$$

The normalization forces $\widehat{\varphi}(0) = 1$, so $\widehat{\varphi} = \widehat{\mathrm{up}}$. Both φ and up are continuous and integrable, so Fourier inversion (81) gives $\varphi = \mathrm{up}$.

Conversely up satisfies all the hypotheses: it is C^∞ with $\operatorname{supp}\mathrm{up} = [-1,1]$ by (7), it has integral 1, and it satisfies (33) with $k=2$ by (28). Existence and uniqueness are therefore both established, and necessarily $k=2$. □

Remark 3.8 (The two normalizations are one condition). Two normalizations circulate for the bump. The geometric one is $\mathrm{up}(0) = 1$, which is how (7) defines it out of $F(1) = 1$; the probabilistic one is $\int_{\mathbb{R}} \mathrm{up} = 1$, which makes up a probability density. For a solution of (33) with $k = 2$ and support in $[-1, 1]$ these are not merely proportional, they are the same condition. Integrating the equation over $[-1, 0]$ and using $\varphi(-1) = 0$ gives

$$\varphi(0) = 2\int_{-1}^{0} (\varphi(2x+1) - \varphi(2x-1))\,\mathrm{d}x = \int_{-1}^{1} \varphi(y)\,\mathrm{d}y - \int_{-3}^{-1} \varphi(y)\,\mathrm{d}y = \int_{\mathbb{R}} \varphi,$$

the last step because φ vanishes on $[-3, -1]$ except at the single point -1. This article uses the probabilistic normalization in every statement about $\widehat{\mathrm{up}}$, since $\widehat{\mathrm{up}}(0) = \int \mathrm{up}$ is what enters the transform argument at $z = 0$, and the geometric normalization whenever values of up are compared with values of F. The displayed identity says that no reconciliation is ever required. The mass normalization itself is `integral_rvachev_eq_one` in `AnalyticMoments.lean`.

4 The probability law behind the functions

The fixed-point construction is analytic. A probabilistic construction explains the same self-similarity almost without calculation and makes monotonicity and positivity transparent.

4.1 An infinite weighted sum of uniforms

Let $U_0, U_1, \ldots$ be independent random variables, each uniformly distributed on $[0, 1]$, and define

$$X = \sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}}. \tag{40}$$

The series converges absolutely for every outcome and satisfies $0 \le X \le 1$. Removing the first coordinate gives an independent copy X' of X, and

$$X = \frac{U_0 + X'}{2}. \tag{41}$$

Reflecting every coordinate, $U_n \mapsto 1 - U_n$, preserves the product measure and sends X to $1 - X$.

Let $H(x) = \mathbb{P}(X \le x)$ be the cumulative distribution function. Conditioning on U_0 in (41) gives

$$H(y) = \int_0^1 H(2y - u)\,\mathrm{d}u. \tag{42}$$

For $0 \le y \le 1/2$, the integrand vanishes when $u > 2y$, so

$$H(y) = \int_0^{2y} H(t)\,\mathrm{d}t. \tag{43}$$

Reflection gives $H(1-y) = 1 - H(y)$. Thus the restriction of H to $[0, 1]$ is a fixed point of the contraction T from section 3.

Theorem 4.1 (Global probabilistic representation). *For every $x, y \in \mathbb{R}$,*

$$F(x) = \mathbb{P}\left(\sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}} \le x\right), \tag{44}$$

$$\mathrm{up}(y) = \mathbb{P}\left(\sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}} \le 1 - |y|\right). \tag{45}$$

The second identity includes both tails: if $|y| \ge 1$, both sides vanish.

Formal proof. The product-space construction proves continuity and measurability of the uniformly convergent coordinate sum, identifies its pushforward distribution, establishes the head–tail self-similarity and the coordinate-reflection law, and invokes uniqueness of the bounded Fabius fixed point. The exact canonical specializations are `fabiusReal_eq_weightedSum_probability` and `rvachevUp_eq_weightedSum_probability_global` in `ProbabilityRepresentation.lean`. □

4.2 Finite atomic laws and weak convergence

There is also a completely discrete approximation theorem. For $k \geq 0$, let β_k be the probability measure assigning mass $1/2$ to each of $\pm 2^{-k-1}$. Starting from $\nu_0 = \delta_0$, define

$$\nu_{n+1} = \nu_n * \beta_{n+1}^{*(n+1)}. \tag{46}$$

Thus ν_n is a finite atomic probability measure. Its characteristic function is the finite weighted cosine product

$$\widehat{\nu}_n(t) = \prod_{k=0}^{n-1} \left(\cos \frac{t}{2^{k+2}} \right)^{k+1}. \tag{47}$$

The indexing in (46) is worth noticing: the new scale at stage $n+1$ is 2^{-n-2}, and it occurs with multiplicity $n+1$.

Let μ_{up} be the probability measure having density up. The finite laws converge weakly to μ_{up}:

$$\nu_n \Longrightarrow \mu_{\mathrm{up}}. \tag{48}$$

Equivalently, for every bounded continuous $g : \mathbb{R} \to \mathbb{R}$,

$$\lim_{n\to\infty} \int_{\mathbb{R}} g(x)\, \mathrm{d}\nu_n(x) = \int_{\mathbb{R}} g(x)\, \mathrm{up}(x)\, \mathrm{d}x. \tag{49}$$

The same test-function theorem is formalized for complex-valued, and more generally real-or-complex-like, bounded continuous functions.

Formal proof. The definitions are `centeredBernoulliMeasure` and `finiteConvolutionMeasure`. Their probability-measure instances and (47) are proved in `EarlyApproximants.lean`. Pointwise convergence of the characteristic functions, weak convergence (48), and the test-function formulations (49) are respectively `finiteConvolutionMeasure_charFun_tendsto`, `finiteConvolutionProbability_tendsto_fabius`, and `integral_finiteConvolutionMeasure_tendsto_fabius` in `WeakConvergence.lean`. □

4.3 Combinatorial step functions and the half-weight convention

A second finite model replaces each atom by a narrow interval. Define polynomials with nonnegative integer coefficients by

$$\mathcal{P}_0(X) = 1, \qquad \mathcal{P}_m(X) = \mathcal{P}_{m-1}(X^2)(1+X)^m \quad (m \geq 1). \tag{50}$$

They also have the finite product form

$$\mathcal{P}_m(X) = \prod_{j=1}^{m} (1 + X + \cdots + X^{2^j - 1}). \tag{51}$$

Writing $\mathcal{P}_m(X) = \sum_r a_{m,r} X^r$, the coefficient $a_{m,r}$ counts the tuples

$$(s_1, \ldots, s_m), \qquad 0 \leq s_j \leq 2^j - 1, \qquad s_1 + \cdots + s_m = r. \tag{52}$$

Moreover,

$$g_m := \deg \mathcal{P}_m = 2^{m+1} - m - 2, \qquad \mathcal{P}_m(1) = 2^{\binom{m+1}{2}}. \tag{53}$$

For $\alpha < \beta$, let $\mathbf{1}^*_{[\alpha,\beta]}$ be 1 in the interior, 1/2 at either endpoint, and 0 elsewhere. Put

$$I_{m,r} = \left[\frac{2r-1-g_m}{2^{m+1}}, \frac{2r+1-g_m}{2^{m+1}}\right] \tag{54}$$

and

$$\mathcal{W}_m(x) = \frac{2^m}{2^{\binom{m+1}{2}}} \sum_{r=0}^{g_m} a_{m,r}\, \mathbf{1}^*_{I_{m,r}}(x). \tag{55}$$

Theorem 4.2 (Corrected step approximation)**.** *For every $m \ge 0$ and $x \in \mathbb{R}$,*

$$0 \le \mathcal{W}_m(x) \le 1, \qquad \mathcal{W}_m(0) = 1, \qquad \int_{\mathbb{R}} \mathcal{W}_m(x)\,\mathrm{d}x = 1.$$

The function $\mathcal{W}_m$ is nondecreasing on $(-\infty, 0)$ and nonincreasing on $(0, \infty)$. For every real x,

$$\mathcal{W}_m(x) \longrightarrow \mathrm{up}(x). \tag{56}$$

The proof of pointwise convergence passes through a useful exact comparison. Set $h_m = 2^{-m-1}$, and use the finite atomic law ν_m from (46). Whenever $a \le b$,

$$\nu_m((a + h_m, b - h_m]) \;\le\; \int_a^b \mathcal{W}_m(x)\,\mathrm{d}x \;\le\; \nu_m((a - h_m, b + h_m]). \tag{57}$$

Consequently,

$$\lim_{m\to\infty} \int_a^b \mathcal{W}_m(x)\,\mathrm{d}x = \int_a^b \mathrm{up}(x)\,\mathrm{d}x. \tag{58}$$

Monotonicity on the two open half-lines, the exact value at the origin, and continuity of up then upgrade these interval limits to (56).

Formal proof. The polynomial recursion, product, degree, total coefficient mass, and bounded-partition interpretation are respectively proved by `approximationPolynomial_succ`, the definition `approximationPolynomial`, `approximationDegree_eq_sub`, `approximationPolynomial_eval_one`, and `approximationPolynomial_coeff_eq_card` in `EarlyApproximants.lean`. The finite atomic description is identified with `finiteConvolutionMeasure` by `polynomialMeasure_eq_finiteConvolutionMeasure` in `EarlyMeasureBridge.lean`.

For the analytic assertions, see `stepApproximant_nonneg`, `stepApproximant_le_one`, `stepApproximant_monotoneOn_Iio`, `stepApproximant_antitoneOn_Ioi`, `stepApproximant_apply_zero`, and `stepApproximant_tendsto_fabius` in `StepApproximationLimit.lean`. The normalization, the two sides of (57), and (58) are `integral_stepApproximant`, `finiteConvolutionMeasure_inner_le_intervalIntegral_stepApproximant`, `intervalIntegral_stepApproximant_le_finiteConvolutionMeasure_outer`, and `intervalIntegral_stepApproximant_tendsto_of_le` in `StepMeasureBridge.lean`. □

Warning 4.3 (Why the endpoint convention is explicit)*.* The half-weighted cell representative agrees almost everywhere with the ordinary interval indicator, so it has the same integral. Its point values are nevertheless part of the formal statement: adjacent cells share endpoints, and each contributes exactly half of its coefficient there. In particular, the corrected representative satisfies $\mathcal{W}_m(0) = 1$ at every level. These facts are formalized by `halfEndpointIntervalIndicator_ae_eq_indicator`, `intervalIntegral_halfEndpointIntervalIndicator_of_le`, and `stepApproximant_apply_zero`.

4.4 Centered finite splines and an explicit error bound

The preceding finite random sums also give a completely explicit polynomial approximation to F. Put

$$T_p = \sum_{i=0}^{p-1} \frac{U_i}{2^{i+1}}, \qquad C_p(x) = \mathbb{P}\left(T_p \le x - 2^{-p-1}\right), \tag{59}$$

where $p \ge 1$. The shift by half of the omitted tail is important: it centers the one-sided approximation and improves its error by a factor of two.

Let

$$N_p(x) = \max\left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor\right), \qquad \varepsilon_r = (-1)^{\mathrm{wt}_2(r)}, \tag{60}$$

and interpret a sum with $N_p(x) = 0$ as empty. Define

$$S_p(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} \varepsilon_r \left(r - 2^p x + \frac{1}{2}\right)^p. \tag{61}$$

Equivalently,

$$S_p(x) = \frac{1}{2^{\binom{p}{2}} p!} \sum_{r \ge 0} \varepsilon_r \left(2^p x - \frac{1}{2} - r\right)_+^p, \tag{62}$$

where only the terms $r < N_p(x)$ can be nonzero and $t_+ = \max(t, 0)$. The cutoff in (60) is a safe natural-number form of the inclusive upper limit

$$0 \le r \le \left\lfloor 2^p x - \frac{1}{2} \right\rfloor.$$

It handles the empty case without assigning a negative upper endpoint to a finite sum.

Theorem 4.4 (Centered spline as a finite distribution function)**.** *For $p \ge 1$ and $0 \le x \le 1$,*

$$S_p(x) = C_p(x). \tag{63}$$

Consequently $0 \le S_p(x) \le 1$ on the unit interval.

Proof. For $p = 1$, $T_1 = U_0/2$, and direct evaluation gives

$$C_1(x) = \max\left(0, \min\left(1, 2x - \frac{1}{2}\right)\right).$$

Formula (62) gives exactly the same broken-linear function: the term $r = 0$ starts at $x = 1/4$, and the term $r = 1$ subtracts the excess after $x = 3/4$.

For the induction step, split off the first uniform coordinate. The finite sums satisfy

$$T_{p+1} \overset{\mathrm{d}}{=} \frac{U_0 + T_p'}{2},$$

where T_p' is an independent copy of T_p. Conditioning on U_0 and accounting for the midpoint shifts gives the exact smoothing recurrence

$$C_{p+1}(x) = \int_0^1 C_p(2x - u)\, \mathrm{d}u. \tag{64}$$

On the spline side, integrate each truncated power in (62). The two binary children $2r$ and $2r + 1$ have opposite Thue–Morse signs, because $\varepsilon_{2r} = \varepsilon_r$ and $\varepsilon_{2r+1} = -\varepsilon_r$. After the change of scale this yields

$$S_{p+1}(x) = \int_0^1 S_p(2x - u)\, \mathrm{d}u. \tag{65}$$

The normalization is exactly the one required by

$$\sum_{r=0}^{2^p-1} \varepsilon_r r^p = (-1)^p 2^{\binom{p}{2}} p!,$$

while every lower power sum vanishes. Thus the two recurrences and their common base case prove (63). □

Theorem 4.5 (Quantitative centered approximation)**.** *For $p \geq 1$ and $0 \leq x \leq 1$,*

$$F(x - 2^{-p-1}) \leq S_p(x) \leq F(x + 2^{-p-1}), \tag{66}$$

and hence

$$|S_p(x) - F(x)| \leq 2^{-p}. \tag{67}$$

Proof. Write $X = T_p + R_p$, where the omitted tail satisfies

$$0 \leq R_p = \sum_{i=p}^{\infty} \frac{U_i}{2^{i+1}} \leq 2^{-p}.$$

If $X \leq x - 2^{-p-1}$, then certainly $T_p \leq x - 2^{-p-1}$. Conversely, if $T_p \leq x - 2^{-p-1}$, then $X \leq x + 2^{-p-1}$. Taking probabilities and using (44) and (63) proves (66).

The bounded Fabius function is globally 2-Lipschitz. Indeed, on the open unit interval its derivative is a rescaled value of up (or of the signed extension), whose absolute value is at most 1, and outside the unit interval F is constant. Thus

$$|F(x \pm 2^{-p-1}) - F(x)| \leq 2 \cdot 2^{-p-1} = 2^{-p}.$$

Combining these two inequalities with the sandwich gives (67). □

The bound is uniform on the closed interval and includes both endpoints. It is stronger than mere weak convergence of the finite distributions, and it is the quantitative input for the computability proof in section 21.6.

The same finite formula also has the correct signed global behavior. For $x \leq 0$ its safe cutoff is empty, so $S_p(x) = \mathcal{F}(x) = 0$. On the nonnegative half-line, binary block translation and midpoint reflection extend the local identification. Consequently, for every $p \geq 1$,

$$\sup_{x\in\mathbb{R}} |S_p(x) - \mathcal{F}(x)| \leq 2^{-p}, \qquad S_p \longrightarrow \mathcal{F} \quad \text{uniformly on } \mathbb{R}. \tag{68}$$

Indeed, every nonnegative two-unit block is a common signed copy of the first block, and on the second unit cell both functions use the same complement. Neither operation changes the absolute error in (67); the nonpositive half-line is exact. This global uniform form strengthens and packages the pointwise analytic input used by the complex-shift and discrete-limit results below.

5 Fourier transform, entire products, and Poisson summation

5.1 The sinc product

We use the Fourier convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x) \mathrm{e}^{-2\pi \mathrm{i} xz} \, \mathrm{d}x. \tag{69}$$

The transform refinement (37), specialized to up, determines its Fourier transform recursively.

Theorem 5.1 (Fourier product)**.** *For every $z \in \mathbb{C}$,*

$$\widehat{\mathrm{up}}(z) = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^n}\right). \tag{70}$$

The product converges locally uniformly and hence defines an entire function.

Proof. Iterating (37) with $\varphi = \mathrm{up}$ gives

$$\widehat{\mathrm{up}}(z) = \widehat{\mathrm{up}}(z/2^N) \prod_{n=0}^{N-1} \operatorname{sinc}\left(\frac{\pi z}{2^n}\right).$$

Continuity and the normalization $\widehat{\mathrm{up}}(0) = 1$ make the first factor tend to one. On compact subsets of $\mathbb{C}$,

$$\operatorname{sinc} w = 1 - \frac{w^2}{6} + \mathcal{O}(w^4),$$

so the sum of the deviations of the factors from 1 is dominated by $C_K \sum 4^{-n}$. The Weierstrass product theorem gives local uniform convergence to an entire function, and passage to the limit gives the formula for every complex z. □

The exact formal statement is `rvachevFourier_eq_product` in `FourierProduct.lean`. Separating the first factor gives the refinement equation already seen in (37):

$$\widehat{\mathrm{up}}(z) = \operatorname{sinc}(\pi z)\widehat{\mathrm{up}}(z/2). \tag{71}$$

Conversely, continuity at zero, normalization $\widehat{\mathrm{up}}(0) = 1$, and (71) force (70) by iteration. This is the core of the Fourier uniqueness proof.

5.2 Cosine and canonical products

The section title promises entire products in the plural, and there are three. The sinc product (70) is the analytic one; a cosine product exhibits the same function as a Bernoulli convolution; and a canonical Weierstrass product displays every zero with its exact multiplicity. The three are elementary transforms of one another, and the passage between them is a telescoping identity.

The mechanism is the half-angle identity for the unnormalized cardinal sine. For every $w \in \mathbb{C}$,

$$\operatorname{sinc} w = \cos\frac{w}{2} \operatorname{sinc}\frac{w}{2}, \tag{72}$$

which is $\sin w = 2\sin(w/2)\cos(w/2)$ divided by w, with the removable value at $w = 0$ filled in.

Lemma 5.2 (Telescoping the sinc product)**.** *For every $N \geq 1$ and every $z \in \mathbb{C}$,*

$$\prod_{n=0}^{N-1} \operatorname{sinc}\left(\frac{\pi z}{2^n}\right) = \left[\prod_{m=1}^{N} \left(\cos\frac{\pi z}{2^m}\right)^m\right] \operatorname{sinc}\left(\frac{\pi z}{2^N}\right)^N. \tag{73}$$

Proof. Induction on N. For $N = 1$ the assertion is (72) with $w = \pi z$. Assume (73) for N and multiply both sides by $\operatorname{sinc}(\pi z/2^N)$. The left side becomes the product through $n = N$, and the right side becomes

$$\left[\prod_{m=1}^{N} \left(\cos\frac{\pi z}{2^m}\right)^m\right] \operatorname{sinc}\left(\frac{\pi z}{2^N}\right)^{N+1}.$$

Now apply (72) with $w = \pi z/2^N$ to each of the $N+1$ identical factors. Each contributes one $\cos(\pi z/2^{N+1})$ and one $\operatorname{sinc}(\pi z/2^{N+1})$, so the expression equals

$$\left[\prod_{m=1}^{N+1}\left(\cos\frac{\pi z}{2^m}\right)^m\right]\operatorname{sinc}\left(\frac{\pi z}{2^{N+1}}\right)^{N+1},$$

which is (73) for $N+1$. □

Theorem 5.3 (Three equivalent entire products)**.** *For every $z \in \mathbb{C}$,*

$$\widehat{\mathrm{up}}(z) = \prod_{n=0}^{\infty}\operatorname{sinc}\left(\frac{\pi z}{2^n}\right) \tag{74}$$

$$= \prod_{m=1}^{\infty}\left(\cos\frac{\pi z}{2^m}\right)^m \tag{75}$$

$$= \prod_{m=1}^{\infty}\left(1-\frac{z^2}{m^2}\right)^{1+v_2(m)}. \tag{76}$$

All three converge absolutely and locally uniformly on $\mathbb{C}$. In the last product the symmetric pairing of the zeros at $\pm m$ is what makes exponential convergence factors unnecessary; because the convergence is absolute, the order of the factors is immaterial.

Proof. Formula (74) is (70).

For (75), fix a compact set $K \subset \mathbb{C}$ and let $R = \sup_{z\in K}|z|$. For $2^N > R$ the argument $\pi z/2^N$ lies in the disc of radius π, where sinc has no zero, and $\operatorname{sinc} w = 1 + \mathcal{O}(w^2)$ gives

$$\log\operatorname{sinc}\left(\frac{\pi z}{2^N}\right) = \mathcal{O}(R^2 4^{-N})$$

uniformly on K, with the principal branch. Hence $N\log\operatorname{sinc}(\pi z/2^N) = \mathcal{O}(NR^2 4^{-N}) \to 0$, so the trailing factor in (73) tends to 1 uniformly on K. Letting $N \to \infty$ in (73) therefore turns (74) into (75), and the same estimate shows that the cosine product converges locally uniformly.

For (76), insert Euler's sine product

$$\operatorname{sinc}(\pi w) = \prod_{r=1}^{\infty}\left(1-\frac{w^2}{r^2}\right) \tag{77}$$

into each factor of (74) with $w = z/2^n$:

$$\widehat{\mathrm{up}}(z) = \prod_{n=0}^{\infty}\prod_{r=1}^{\infty}\left(1-\frac{z^2}{(2^n r)^2}\right). \tag{78}$$

The double product converges absolutely and locally uniformly, since

$$\sum_{n\geq 0}\sum_{r\geq 1}\frac{|z|^2}{(2^n r)^2} = \frac{4}{3}\cdot\frac{\pi^2}{6}|z|^2 < \infty$$

uniformly on compact sets. Absolute convergence permits an arbitrary regrouping of the factors. Group them by the value $m = 2^n r$. For a fixed $m \geq 1$ the pairs (n,r) with $2^n r = m$ are exactly those with $0 \leq n \leq v_2(m)$ and $r = m/2^n$, so there are precisely $1 + v_2(m)$ of them, all contributing the same factor $1 - z^2/m^2$. This is (76). Absolute convergence of $\sum_m (1+v_2(m))|z|^2/m^2$, which holds because $v_2(m) \leq \log_2 m$, is the statement that no convergence-generating exponential factors are needed once the zeros at m and $-m$ are paired. □

Theorem 5.4 (Exact zero set and multiplicities). *The zero set of the entire function $\widehat{\mathrm{up}}$ is exactly $\mathbb{Z} \setminus \{0\}$, and for every nonzero integer m,*

$$\operatorname{ord}_{z=m} \widehat{\mathrm{up}} = 1 + v_2(|m|). \tag{79}$$

In particular $\widehat{\mathrm{up}}$ has no zeros off the real axis, and all of its zeros are real integers.

Proof. Work with (78), whose factors are $1 + a_{n,r}(z)$ with $a_{n,r}(z) = -z^2/(2^n r)^2$ and $\sum_{n,r} |a_{n,r}|$ locally uniformly convergent. For such a product the limit vanishes at a point exactly when one of the factors does, and the order of the zero is the sum of the orders of the vanishing factors. Indeed, on a disc D around a given point only finitely many factors have a zero in D, because $|a_{n,r}| < 1/2$ on D for all but finitely many pairs; splitting off those finitely many factors leaves a product that is nonvanishing on D.

The factor indexed by (n, r) vanishes precisely at $z = \pm 2^n r$, and each of those two zeros is simple. The union of the sets $\{\pm 2^n r : n \geq 0,\ r \geq 1\}$ is $\mathbb{Z} \setminus \{0\}$, which identifies the zero set. Fix a nonzero integer m. The pairs (n, r) for which $2^n r = |m|$ are those with $0 \leq n \leq v_2(|m|)$, so exactly $1 + v_2(|m|)$ factors vanish at $z = m$, each to first order. Summing gives (79).

The same count is visible directly in (74): the nth factor $\operatorname{sinc}(\pi z/2^n)$ vanishes exactly on $2^n\mathbb{Z} \setminus \{0\}$, simply, and 2^n divides m for exactly the indices $0 \leq n \leq v_2(|m|)$. □

Remark 5.5 (Where the Fourier side meets the arithmetic side). The exponent in (76) is the same 2-adic valuation that governs the arithmetic half of this article. There it appears as the exact valuation of the reciprocal-power values,

$$v_2\big(F(2^{-n})\big) = -\binom{n}{2} - 1 - v_2(n!),$$

which is (156), made explicit in binary by Legendre's formula (157). Here it appears as a zero order of an entire function. The two occurrences are not a coincidence: both are consequences of the single dilation $x \mapsto 2x$ in (4). On the arithmetic side the dilation is iterated against the binary expansion of the argument and produces powers of two in denominators; on the Fourier side the same dilation is iterated in (71) and stacks a new sinc zero on top of every frequency already divisible by a further power of two. This subsection is the one place in the article where the two sides are literally the same statement.

The zero structure also explains why up is a useful building block in approximation theory [6]. Since $\widehat{\mathrm{up}}(0) = 1$ and $\widehat{\mathrm{up}}$ vanishes at every nonzero integer, the integer translates of up reproduce constants exactly; that is the partition of unity (84). The zeros at odd integers are simple, so the integer grid alone reproduces nothing beyond constants. The extra multiplicity lives at the even integers, and it is recovered by widening the bump: the dilate $\mathrm{up}(\cdot/2^j)$ has transform $2^j\widehat{\mathrm{up}}(2^j\xi)$, whose zero at a nonzero integer m has order $1 + v_2(2^j m) \geq j + 1$ by (79). Poisson summation then makes all derivatives up to order j of the periodized transform vanish at every nonzero frequency, which is the Strang–Fix condition of order $j + 1$; consequently the integer translates of $\mathrm{up}(\cdot/2^j)$ reproduce every polynomial of degree at most j. Together with $\mathrm{up} \in C_c^\infty$ this is exactly the profile wanted of a smooth compactly supported element: arbitrary smoothness, exact support, and a reproduction order that can be raised at will by dilation.

5.3 Rapid decay and inversion

Since $\mathrm{up} \in C_c^\infty(\mathbb{R})$, repeated integration by parts gives, for every integer $N \geq 0$,

$$|\widehat{\mathrm{up}}(\xi)| \leq C_N(1 + |\xi|)^{-N}. \tag{80}$$

Thus the Fourier product is rapidly decreasing on the real axis despite being entire of finite exponential type. Fourier inversion is absolutely convergent:

$$\mathrm{up}(x) = \int_{\mathbb{R}} \widehat{\mathrm{up}}(\xi)\mathrm{e}^{2\pi \mathrm{i} x\xi}\,\mathrm{d}\xi. \tag{81}$$

The formal development records the equivalent polynomial-weight form of (80) as `rvachevFourier_real_rapidDecay`. Its stronger declaration `rvachevFourier_real_iteratedDeriv_rapidDecay` proves that for every $k, n \geq 0$ there is a positive constant $C_{k,n}$ such that

$$|\xi|^k \left\| \frac{\mathrm{d}^n}{\mathrm{d}\xi^n}\widehat{\mathrm{up}}(\xi)\right\| \leq C_{k,n} \qquad (\xi \in \mathbb{R}).$$

5.4 Poisson summation and partitions of unity

Rapid decay on both sides makes Poisson summation available without distributional qualifications.

Theorem 5.6 (Poisson summation)**.** *For $u > 0$ and $t \in \mathbb{R}$,*

$$\sum_{k\in\mathbb{Z}} \mathrm{up}(t + uk) = \frac{1}{u}\sum_{k\in\mathbb{Z}} \widehat{\mathrm{up}}(k/u)\mathrm{e}^{2\pi \mathrm{i} kt/u}. \tag{82}$$

Both series converge absolutely and locally uniformly.

Proof. Periodize the bump by $P_u(t) = \sum_{k\in\mathbb{Z}} \mathrm{up}(t + uk)$. Compact support makes this sum locally finite, so P_u is smooth and u-periodic. Unfolding one period shows that its nth Fourier coefficient is

$$\frac{1}{u}\int_0^u P_u(t)\mathrm{e}^{-2\pi \mathrm{i} nt/u}\,\mathrm{d}t = \frac{1}{u}\int_{\mathbb{R}} \mathrm{up}(x)\mathrm{e}^{-2\pi \mathrm{i} nx/u}\,\mathrm{d}x = \frac{1}{u}\widehat{\mathrm{up}}(n/u).$$

The rapid decay (80) makes the resulting Fourier series absolutely and locally uniformly convergent (indeed, uniformly with every derivative), and its Fourier expansion is exactly (82). □

For a positive integer m, every nonzero term $\widehat{\mathrm{up}}(mk)$ vanishes because the first sinc factor in (70) vanishes at nonzero integers. Taking $u = 1/m$ in (82) gives the exact partition of unity

$$\sum_{k\in\mathbb{Z}} \mathrm{up}\left(t + \frac{k}{m}\right) = m. \tag{83}$$

In particular,

$$\sum_{k\in\mathbb{Z}} \mathrm{up}(t + k) = 1. \tag{84}$$

This identity is one reason up is useful as a smooth compactly supported averaging kernel.

5.5 Moment expansion of the Fourier transform

Define the even moments

$$c_n = \int_{-1}^{1} x^{2n}\,\mathrm{up}(x)\,\mathrm{d}x. \tag{85}$$

Evenness eliminates the odd moments, and expansion of the exponential in (69) gives the entire series

$$\widehat{\mathrm{up}}(z) = \sum_{n=0}^{\infty} \frac{(-1)^n c_n}{(2n)!}(2\pi z)^{2n}. \tag{86}$$

Comparing this with the sinc refinement will produce the exact rational recurrence for c_n in section 7.

6 The global extension and exact derivative identities

The signed extension $\mathcal{F}$ is not merely a device for shortening notation. It turns the piecewise differential structure of F and up into a global scaling law, and from that law one can read off all higher derivatives.

6.1 Thue–Morse splitting

The binary weight satisfies

$$\mathrm{wt}_2(2n) = \mathrm{wt}_2(n), \qquad \mathrm{wt}_2(2n+1) = \mathrm{wt}_2(n) + 1. \tag{87}$$

Therefore

$$\varepsilon_{2n} = \varepsilon_n, \qquad \varepsilon_{2n+1} = -\varepsilon_n. \tag{88}$$

Differentiate the locally finite sum (11) term by term and use (28). The result is a sum over even and odd indices. Reindexing those two subseries with (88) makes them recombine into $2\mathcal{F}(2x)$.

Theorem 6.1 (Global differential equation)**.** *The signed extension is infinitely differentiable and*

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \tag{89}$$

for every real x.

Proof. Local finiteness permits differentiation of (11). By (28),

$$\begin{aligned}
\mathcal{F}'(x) &= 2\sum_{n\ge 0} \varepsilon_n \big[\mathrm{up}(2x-4n-1) - \mathrm{up}(2x-4n-3)\big] \\
&= 2\sum_{n\ge 0} \varepsilon_{2n}\,\mathrm{up}(2x-2(2n)-1) + 2\sum_{n\ge 0} \varepsilon_{2n+1}\,\mathrm{up}(2x-2(2n+1)-1) \\
&= 2\sum_{m\ge 0} \varepsilon_m\,\mathrm{up}(2x-2m-1) = 2\mathcal{F}(2x).
\end{aligned}$$

The same local-finiteness argument shows that the sum is C^∞ because each translate of up is. ☐

6.2 All iterated derivatives

Repeated differentiation of (89) gives a triangular power of two.

Theorem 6.2 (Derivative scaling)**.** *For every integer $m \ge 0$,*

$$\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}}\mathcal{F}(2^m x). \tag{90}$$

For $-1 \le x \le 1$,

$$\mathrm{up}^{(m)}(x) = 2^{\binom{m+1}{2}}\mathcal{F}(2^m x + 2^m). \tag{91}$$

Proof. The first assertion is immediate by induction. If it holds for m, then

$$\begin{aligned}
\mathcal{F}^{(m+1)}(x) &= 2^{\binom{m+1}{2}} 2^m \mathcal{F}'(2^m x) \\
&= 2^{\binom{m+1}{2}} 2^m \cdot 2\mathcal{F}(2^{m+1}x) \\
&= 2^{\binom{m+2}{2}}\mathcal{F}(2^{m+1}x).
\end{aligned}$$

For $-1 < x < 1$, the identity $\mathrm{up}(x) = \mathcal{F}(x+1)$ holds on a neighborhood of x, so it may be differentiated m times there. For fixed m, both resulting sides are continuous functions of x; letting x tend to either endpoint from inside proves the same identity at $x = \pm 1$. ☐

The formula is striking: the mth derivative is not an approximation or an asymptotic rescaling of the original function; it is an exact rescaling of the signed extension. The large factor $2^{m(m+1)/2}$ quantifies the rapid growth of high derivatives, while the argument $2^m x$ accounts for the increasingly fine oscillatory structure.

6.3 Sharp uniform bounds for all derivatives

Equation (90) converts any bound on $\mathcal{F}$ itself into a bound on all of its derivatives. The one ingredient it needs is that the signed extension is bounded by one in absolute value. That is not visible in (11), whose terms carry no uniform sign, but it is immediate from the block form (12).

Lemma 6.3 (The signed extension is bounded by one)**.** *For every real x one has $-1 \le \mathcal{F}(x) \le 1$.*

Proof. For $x \le 0$ the extension vanishes. For $x > 0$ let $b = \lfloor x/2 \rfloor$, a nonnegative integer with $2b \le x \le 2b+2$. Then (12) gives $\mathcal{F}(x) = \varepsilon_b \operatorname{up}(x - 2b - 1)$ with $\varepsilon_b = \pm 1$ by (10), and $0 \le \operatorname{up} \le 1$. $\square$

Theorem 6.4 (Sharp uniform derivative bounds)**.** *For every integer $m \ge 0$ and every real x,*

$$|\mathcal{F}^{(m)}(x)| \le 2^{\binom{m+1}{2}}, \qquad |\operatorname{up}^{(m)}(x)| \le 2^{\binom{m+1}{2}}, \qquad |F^{(m)}(x)| \le 2^{\binom{m+1}{2}}. \tag{92}$$

Each of the three constants is attained:

$$\mathcal{F}^{(m)}(2^{-m}) = F^{(m)}(2^{-m}) = \operatorname{up}^{(m)}(2^{-m} - 1) = 2^{\binom{m+1}{2}}. \tag{93}$$

Hence $2^{\binom{m+1}{2}}$ is not merely an upper bound but the exact supremum of $|\mathcal{F}^{(m)}|$, of $|\operatorname{up}^{(m)}|$, and of $|F^{(m)}|$ over $\mathbb{R}$, and each supremum is a maximum.

Proof. The bound for $\mathcal{F}$ is (90) combined with lemma 6.3. For up and $|x| \le 1$ the bound is (91) combined with the same lemma; for $|x| > 1$ the function up vanishes identically on a neighbourhood of x by theorem 2.6(iii), so every derivative of up vanishes at x.

For the bounded function, note that F and $\mathcal{F}$ agree on $(-\infty, 1]$: both vanish on $(-\infty, 0]$ and they agree on $[0, 1]$. Hence they have the same germ at every point $x < 1$, so $F^{(m)}(x) = \mathcal{F}^{(m)}(x)$ there and the bound holds for $x < 1$. For $x > 1$ the function F is constant near x, so $F^{(m)}(x) = 0$ when $m \ge 1$, while for $m = 0$ the bound reads $F \le 1$. The single remaining point $x = 1$ follows by letting $x \uparrow 1$, since $F^{(m)}$ is continuous and the bound is a closed condition.

For (93), take $x = 2^{-m}$ in (90): then $2^m x = 1$ and $\mathcal{F}(1) = 1$, so $\mathcal{F}^{(m)}(2^{-m}) = 2^{\binom{m+1}{2}}$. When $m \ge 1$ the point 2^{-m} lies in $(0, 1)$, where F and $\mathcal{F}$ have the same germ, so $F^{(m)}$ takes the same value there; when $m = 0$ the assertion is $F(1) = 1 = 2^0$. Finally take $x = 2^{-m} - 1 \in [-1, 0]$ in (91): then $2^m x + 2^m = 1$ and the same computation applies. $\square$

For $m = 1$ the theorem returns (16), with the extremal point $2^{-1} = \frac{1}{2}$; for $m = 2$ the sharp constant is $2^3 = 8$, for $m = 3$ it is $2^6 = 64$. Two features deserve emphasis. First, no estimate of the form $|F^{(m)}| \le C_m$ can have $C_m < 2^{\binom{m+1}{2}}$, so the triangular power of two in (90) is not an artefact of the induction. Second, the extremal points 2^{-m} march towards the origin as m grows: the derivatives become large exactly in the region where F itself is flattest. The next subsection quantifies that flatness, and it will do so at the very same points 2^{-m}.

6.4 Effective flatness at the origin

Equation (6) says that F vanishes to infinite order at the origin, so that $F(x) = o(x^n)$ as $x \to 0^+$ for every n. That statement carries no constant. Two elementary arguments supply one, and the second is strictly better than the first.

The qualitative theorem formalized in `FabiusFlatness.lean` is actually two-sided. For every $n \geq 0$,

$$\mathcal{F}(x) = o(|x|^n), \quad F(x) = o(|x|^n) \qquad (x \to 0),$$

and symmetry gives

$$1 - F(x) = o(|1-x|^n) \qquad (x \to 1).$$

Equivalently, the folded bump is flatter than every power at both endpoints: $\mathrm{up}(x) = o(|x-1|^n)$ as $x \to 1$ and $\mathrm{up}(x) = o(|x+1|^n)$ as $x \to -1$.

Lemma 6.5 (Self-improving estimate)**.** *For $0 \leq x \leq \frac{1}{2}$ one has $F(x) \leq 2x\,F(2x)$.*

Proof. At $x = 0$ both sides vanish. For $0 < x \leq \frac{1}{2}$, Lagrange's mean value theorem on $[0, x]$ provides a point $c \in (0, x)$ with $F(x) = F(x) - F(0) = x\,F'(c)$. Since $0 < c < x \leq \frac{1}{2}$, equation (4) gives $F'(c) = 2F(2c)$, and F is nondecreasing with $2c \leq 2x$, so $F'(c) \leq 2F(2x)$. □

Theorem 6.6 (Effective flatness, mean value form)**.** *Let $n \geq 0$. If $x \geq 0$ and $2^n x \leq 1$, then*

$$F(x) \leq 2^{\binom{n+1}{2}} x^n. \tag{94}$$

Proof. Induction on n. For $n = 0$ the claim is $F \leq 1$. Assume it for n, and let $x \geq 0$ satisfy $2^{n+1}x \leq 1$. Then $x \leq \frac{1}{2}$, because $2^{n+1} \geq 2$, so lemma 6.5 applies; and $2^n(2x) = 2^{n+1}x \leq 1$, so the inductive hypothesis applies at $2x$. Therefore

$$F(x) \leq 2x\,F(2x) \leq 2x \cdot 2^{\binom{n+1}{2}}(2x)^n = 2^{\binom{n+1}{2}+n+1} x^{n+1} = 2^{\binom{n+2}{2}} x^{n+1},$$

the last step being the Pascal identity $\binom{n+2}{2} = \binom{n+1}{2} + (n+1)$. □

Replacing the mean value theorem by the fundamental theorem of calculus recovers a factorial that the pointwise argument throws away. The reason is structural: the mean value theorem evaluates the integrand at a single unknown point and then bounds it by its largest value on the interval, whereas integrating the same bound keeps the whole profile.

Theorem 6.7 (Effective flatness, integral form)**.** *Let $n \geq 0$. If $x \geq 0$ and $2^n x \leq 1$, then*

$$F(x) \leq \frac{2^{\binom{n+1}{2}}}{n!}\, x^n. \tag{95}$$

Proof. Since $F(0) = 0$ and $F'(t) = 2F(2t)$ for $0 \leq t \leq \frac{1}{2}$ by (4),

$$F(x) = \int_0^x 2F(2t)\,\mathrm{d}t \qquad (0 \leq x \leq \tfrac{1}{2}). \tag{96}$$

Write $a_n = 2^{\binom{n+1}{2}}/n!$ and induct. For $n = 0$ the claim is $F \leq 1$. Assume $F(y) \leq a_n y^n$ whenever $y \geq 0$ and $2^n y \leq 1$, and let $x \geq 0$ satisfy $2^{n+1}x \leq 1$; then $x \leq \frac{1}{2}$. For $0 \leq t \leq x$ we have $2^n(2t) \leq 2^{n+1}x \leq 1$, so the inductive hypothesis gives $2F(2t) \leq 2a_n(2t)^n = 2^{n+1}a_n\,t^n$. Inserting this into (96) and integrating,

$$F(x) \leq 2^{n+1}a_n \int_0^x t^n\,\mathrm{d}t = \frac{2^{n+1}a_n}{n+1}\,x^{n+1} = \frac{2^{\binom{n+1}{2}+n+1}}{(n+1)!}\,x^{n+1} = a_{n+1}x^{n+1},$$

again by the Pascal identity. □

Both bounds are stated on the dyadic range $2^n x \le 1$, and the natural point to test them is the right endpoint $x = 2^{-n}$ of that range, which is also the point where the nth derivative attains its maximum in (93). There $\binom{n+1}{2} - n^2 = -\binom{n}{2}$, so (94) and (95) read

$$F(2^{-n}) \le 2^{-\binom{n}{2}} \qquad \text{and} \qquad F(2^{-n}) \le \frac{2^{-\binom{n}{2}}}{n!}, \tag{97}$$

so the entire gain of the integral argument over the mean value argument is the factorial. The second bound is now within an explicit factor of the truth. Indeed (136) gives the exact value $F(2^{-n}) = d_n/(n!\,2^{\binom{n}{2}})$ with d_n the half moment (115), so the right-hand side of (95) at $x = 2^{-n}$ overshoots the true value by exactly the factor $1/d_n$. This is consistent, and it also shows how little room is left: from (115) and $0 \le \mathrm{up} \le 1$, for $n \ge 1$,

$$d_n = n \int_0^1 t^{n-1}\,\mathrm{up}(t)\,\mathrm{d}t \le n \int_0^1 t^{n-1}\,\mathrm{d}t = 1,$$

and the inequality is strict because $\mathrm{up} < 1$ almost everywhere; separately $d_0 = 1$. At $n = 1$ the estimate reads $F(\frac{1}{2}) \le 1$ against the true value $F(\frac{1}{2}) = \frac{1}{2} = d_1$.

The fold (8) transports the estimate to both ends of the support of the bump: if $|x| \le 1$ and $2^n(1 - |x|) \le 1$, then

$$\mathrm{up}(x) \le \frac{2^{\binom{n+1}{2}}}{n!}(1 - |x|)^n. \tag{98}$$

For $n = 1$ this is the linear majorant (20) restricted to the two end zones. Both effective forms are contributions of the formal development; the mean value derivation is retained there alongside the sharper one because it uses nothing but the differential equation and monotonicity.

6.5 Polynomial behavior near dyadic points

Let $x_0 = a/2^n$ be a dyadic point in the interior of the unit interval. For $m > n$, the argument $2^m x_0$ in (90) is an even integer. The local translate formula (12) and flatness of up at its endpoints imply

$$F^{(m)}(x_0) = 0 \qquad (m > n). \tag{99}$$

Thus the Taylor series of F at a dyadic point terminates.

Proposition 6.8 (Terminating Taylor series)**.** *If $x_0 = a/2^n \in (0, 1)$ with a odd, then the formal Taylor series of F at x_0 is a polynomial of degree at most n. Nevertheless F does not agree with that polynomial on any neighborhood of x_0.*

Proof. Equation (99) proves termination. Suppose that F agreed with this degree-at-most-n polynomial on a nonempty open interval. That interval contains a reduced dyadic $r = a/2^m$ of arbitrarily large level, so choose $m > n$. Agreement on a neighborhood of r would make the mth derivative of F vanish there, whereas the exact calculation (100) below gives $|F^{(m)}(r)| = 2^{\binom{m+1}{2}} \ne 0$, a contradiction. □

6.5.1 The exact degree of the terminating polynomial

The proposition on terminating Taylor series bounds the degree of that polynomial by n. The bound is always attained, because the nth derivative at a level-n dyadic point with odd numerator can be computed exactly and is never zero.

Proposition 6.9 (Exact dyadic derivative)**.** *Let $x_0 = a/2^n \in (0,1)$ with a odd, say $a = 2b+1$; necessarily $n \geq 1$. Then*

$$F^{(n)}(x_0) = \varepsilon_b \, 2^{\binom{n+1}{2}} = -\varepsilon_a \, 2^{\binom{n+1}{2}}, \tag{100}$$

with ε as in (10)*. In particular $|F^{(n)}(x_0)| = 2^{n(n+1)/2} \neq 0$.*

Proof. The signed extension $\mathcal{F}$ agrees with F on $[0,1]$, hence all their derivatives agree at the interior point x_0. By the derivative scaling identity (90),

$$F^{(n)}(x_0) = \mathcal{F}^{(n)}(x_0) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n x_0) = 2^{\binom{n+1}{2}} \mathcal{F}(a).$$

The point $a = 2b+1$ lies in the block $[2b, 2b+2]$, so the single-block formula (12) applies and gives

$$\mathcal{F}(a) = \varepsilon_b \, \mathrm{up}(a - 2b - 1) = \varepsilon_b \, \mathrm{up}(0) = \varepsilon_b,$$

since $\mathrm{up}(0) = 1$. This is the first equality in (100). For the second, (87) gives $\mathrm{wt}_2(a) = \mathrm{wt}_2(2b+1) = \mathrm{wt}_2(b) + 1$, hence $\varepsilon_b = -\varepsilon_a$. Finally $\binom{n+1}{2} = n(n+1)/2$ and $|\varepsilon_b| = 1$. □

The computation isolates exactly what makes $m = n$ the last surviving order. In (90) the argument $2^m x_0 = 2^{m-n} a$ is an even integer for every $m > n$, and $\mathcal{F}$ vanishes at even integers; this is (99). For $m = n$ that argument is the odd integer a, and $\mathcal{F}$ takes the value ± 1 at every odd integer. The truncation of the Taylor series is therefore governed by a parity, and the last nonzero coefficient is forced.

Proposition 6.10 (Exact degree)**.** *Let $x_0 = a/2^n \in (0,1)$ with a odd, and let*

$$T_{x_0}(x) = \sum_{k=0}^{n} \frac{F^{(k)}(x_0)}{k!} (x - x_0)^k \tag{101}$$

be the Taylor polynomial of F at x_0. Then T_{x_0} is the whole formal Taylor series of F at x_0 and has degree exactly n, with leading coefficient $-\varepsilon_a 2^{\binom{n+1}{2}}/n!$.

Proof. By (99) every coefficient of order $m > n$ in the formal Taylor series of F at x_0 vanishes, so that series is the polynomial (101) and $\deg T_{x_0} \leq n$. Its coefficient of $(x-x_0)^n$ is $F^{(n)}(x_0)/n!$, which by (100) equals $-\varepsilon_a 2^{\binom{n+1}{2}}/n!$ and is nonzero. Hence $\deg T_{x_0} = n$. □

Remark 6.11. The same computation applies to Rvachev's bump on all of $(-1,1)$, and not merely on the half where up is a translate of F. Let $n \geq 0$ and let $y_0 = a/2^n - 1$ with a odd and $0 < a < 2^{n+1}$, so that $y_0 \in (-1,1)$. For $m \geq n$ the identity (91) gives $\mathrm{up}^{(m)}(y_0) = 2^{\binom{m+1}{2}} \mathcal{F}(2^{m-n} a)$, whose argument is an even integer when $m > n$ and the odd integer a when $m = n$. Hence $\mathrm{up}^{(m)}(y_0) = 0$ for $m > n$ while $\mathrm{up}^{(n)}(y_0) = -\varepsilon_a 2^{\binom{n+1}{2}} \neq 0$, so the Taylor polynomial of up at y_0 has degree exactly n. This sharpening is what makes the nowhere-analyticity corollary effective rather than merely soft: a convergent power series centered at a level-n odd dyadic point would force up to be a polynomial of degree n on a whole interval, while every neighborhood of that point contains odd dyadic points of arbitrarily large level m, at which the mth derivative is nonzero by (100).

Corollary 6.12 (Nowhere analytic)**.** *Neither F on $(0,1)$ nor* up *on $(-1,1)$ is real analytic at any point.*

Proof. Suppose first that F were analytic at some $x \in (0,1)$. It would then be analytic on a nonempty open interval I about x. Choose in I a reduced dyadic $q = a/2^n$. The Taylor series at q terminates at degree n, so analyticity would make F equal that polynomial on a smaller interval. That interval contains a reduced dyadic of level $m > n$, where (100) says that the mth derivative is nonzero, contradicting the polynomial identity. For up, use the shifted reduced dyadics $a/2^n - 1$ described above: they are dense in $(-1,1)$, have terminating Taylor series of exact degree n, and every neighborhood contains such points of arbitrarily larger level. The identical derivative contradiction applies. □

6.6 Elementary endpoint bounds

The derivative scaling (90) already contains a quantitative flatness statement at the left endpoint, and extracting it costs three lines. No transform, no saddle point, and no asymptotic analysis is involved.

Theorem 6.13 (A priori flatness bound)**.** *For every integer $n \geq 0$ and every x with $0 \leq x \leq 2^{-n}$,*

$$0 \leq F(x) \leq \frac{2^{\binom{n+1}{2}}}{n!}\, x^n. \tag{102}$$

Proof. The condition $0 \leq x \leq 2^{-n}$ is equivalent to $x \geq 0$ and $2^n x \leq 1$, so the upper bound is precisely (95). The lower bound is the nonnegativity of the bounded distribution function. Thus this restatement records the same estimate in the endpoint notation used in the Fourier sections, without duplicating its proof. □

Corollary 6.14 (Super-polynomial decay, elementarily)**.** *For every real $N > 0$,*

$$F(x) = o(x^N) \qquad (x \downarrow 0). \tag{103}$$

Proof. Let n be an integer with $n > N$. For $0 < x \leq 2^{-n}$, (102) gives

$$0 \leq \frac{F(x)}{x^N} \leq \frac{2^{\binom{n+1}{2}}}{n!}\, x^{n-N},$$

and the right side tends to 0 as $x \downarrow 0$ because $n - N > 0$. Taking $n = \lfloor N \rfloor + 1$ suffices. □

Remark 6.15 (Removing a circularity in the order of exposition)*.* The statement (103) also appears later as (477) in Corollary 18.2, where it is deduced from the quadratic logarithmic law (475) and hence from the entire Mellin and saddle-point apparatus of sections 14 and 15. That derivation is correct but expensive, and it makes a simple qualitative fact hostage to two thousand lines of asymptotic analysis. A reader who skips the asymptotics is then left without an independent proof that F is flatter than every power, even though the differential structure supplies one immediately. Theorem 6.13 and Corollary 6.14 supply exactly that proof, at this point of the article, from (90) and (6) alone. The later corollary is then a sharpening rather than the only source: the asymptotic machinery is what pins the decay between all powers and every scale $\mathrm{e}^{-c/x}$, which the elementary bound does not attempt.

Remark 6.16 (How lossy the bound is)*.* The bound can be compared with an exact value at the natural test point $x = 2^{-n}$, where the constraint of theorem 6.13 is tight. Substituting gives

$$\frac{2^{\binom{n+1}{2}}}{n!}\, 2^{-n^2} = \frac{2^{-\binom{n}{2}}}{n!}, \tag{104}$$

whereas (136) gives the exact value

$$F(2^{-n}) = \frac{d_n}{n!\, 2^{\binom{n}{2}}}.$$

The two differ by exactly the factor $1/d_n$. Since $d_n = n \int_0^1 t^{n-1} \operatorname{up}(t)\, \mathrm{d}t \leq 1$ by (115) and $\operatorname{up} \leq 1$, the bound always overshoots, and it overshoots by the reciprocal of the nth half moment and by nothing else. The bound therefore captures the whole of the doubly exponential factor $2^{-\binom{n}{2}}/n!$ and loses only the slowly decaying half moment.

6.7 The exact analytic locus

The corollary above can be sharpened to an exact description of the analytic locus of each of the three functions. Two points are worth isolating in advance. The endpoints are not neutral: both 0 and 1 belong to the non-analytic set of F, and both -1 and 1 belong to the non-analytic set of up, even though each function is analytic arbitrarily close to them on the outside. And the right endpoint of the unit interval must be treated separately, because the fold identity that handles every other point degenerates there.

Theorem 6.17 (Analytic locus of F)**.** *Let $x \in \mathbb{R}$. The bounded Fabius function is real analytic at x if and only if $x \notin [0, 1]$.*

Proof. If $x < 0$, then F vanishes identically on the neighbourhood $(-\infty, 0)$ of x by (1); if $x > 1$, then F equals 1 identically on the neighbourhood $(1, \infty)$ of x by (2). A function that is constant near a point is analytic at that point.

Conversely, let $x \in [0, 1]$. For $0 < x < 1$ this is the corollary above.

Suppose F were analytic at 0. Its Taylor coefficients at 0 are $F(0) = 0$ and $F^{(m)}(0)/m! = 0$ for $m \geq 1$ by (6), so the Taylor series at 0 is identically zero and analyticity would force F to vanish on a neighbourhood of the origin. That contradicts theorem 2.5(i), which gives $F(t) > 0$ for every $t > 0$.

Suppose F were analytic at 1. Its Taylor coefficients at 1 are $F(1) = 1$ and, again by (6), $F^{(m)}(1)/m! = 0$ for $m \geq 1$, so analyticity would force F to equal 1 on a neighbourhood of 1. That contradicts theorem 2.5(ii), which gives $F(t) < 1$ for every $t < 1$. □

Remark 6.18. The right endpoint really does need its own argument. The fold (8) gives $F(y) = \mathrm{up}(1 - y)$ for every $y \leq 1$, and for no $y > 1$: there $\mathrm{up}(1 - y) = F(2 - y)$, which is smaller than $F(y) = 1$. So the reflection is an identity of germs at every point $y < 1$, and at $y = 1$ it is only a one-sided identity; non-analyticity of up at the origin therefore cannot be transported to F at 1 along it. The flatness argument above sidesteps the difficulty. The formal development takes the other available route and transports analyticity at 1 to analyticity at 0 along the global symmetry (15), which, unlike the fold, is a genuine identity of germs at every point of the line.

Corollary 6.19 (Analytic locus of up)**.** *Rvachev's function is real analytic at x if and only if $|x| > 1$. In particular it is not analytic at either endpoint of its support.*

Proof. If $|x| > 1$, then up vanishes identically near x by theorem 2.6(iii). If $|x| < 1$, this is the corollary above. At $x = 1$, use that $\mathrm{up}(y) = F(1-y)$ for all $y > 0$, hence on a neighbourhood of 1; therefore $\mathrm{up}(1) = F(0) = 0$ and $\mathrm{up}^{(m)}(1) = (-1)^m F^{(m)}(0) = 0$ for $m \geq 1$ by (6). Analyticity at 1 would force up to vanish on a neighbourhood of 1, contradicting theorem 2.6(iii). Evenness gives the same conclusion at $x = -1$. □

The statement of the corollary on the closed interval $[-1, 1]$ is the unnumbered corollary following equation (24) of [8]; non-analyticity in the interior is Fabius's original point [2].

Theorem 6.20 (Formalized signed-extension scope)**.** *The signed extension $\mathcal{F}$ is not real analytic at any point of $[0, 2)$.*

Formal proof. This is `extendedFabius_not_analyticAt` in `NowhereAnalytic.lean`. The theorem is deliberately stated on the exact interval covered by that declaration; a proposed global analytic-locus classification is recorded in the research-frontier notebook rather than asserted here. □

6.8 The totalized inverse

The restriction $F : [0,1] \to [0,1]$ is a strictly increasing continuous bijection. Let

$$I(y) = F^{-1}(y) \qquad (0 \le y \le 1)$$

and extend it to the whole real line by clamping the argument to $[0,1]$. Thus $I(y) = 0$ for $y \le 0$ and $I(y) = 1$ for $y \ge 1$.

Theorem 6.21 (Inverse order and calculus)**.** *The totalized inverse $I : \mathbb{R} \to [0,1]$ is continuous and nondecreasing, and its restriction to $[0,1]$ is a strictly increasing bijection of that interval onto itself. For $x, y \in [0,1]$,*

$$F(I(y)) = y, \qquad I(F(x)) = x.$$

For $0 < y < 1$, the inverse belongs to C^∞ locally and

$$I'(y) = \frac{1}{F'(I(y))} = \frac{1}{2\ \mathrm{up}(2I(y) - 1)} > 0. \tag{105}$$

Formal proof. The order isomorphism and totalized inverse are `fabiusOrderIso` and `fabius Inv`. The range, inverse, order, continuity, and bijectivity statements are `fabiusInv_mem_Icc`, `fabiusReal_fabiusInv`, `fabiusInv_fabiusReal`, `strictMonoOn_fabiusInv`, `monotone_f abiusInv`, `continuous_fabiusInv`, and `bijOn_fabiusInv`. Interior smoothness and (105) are `fabiusInv_contDiffOn_Ioo`, `deriv_fabiusInv_eq_inv_two_mul_rvachevUp`, and `deriv_fabiusInv_pos`, all in `FabiusInverse.lean`. □

The inverse inherits exact arithmetic and symmetry from F:

$$I(1 - y) = 1 - I(y), \qquad I(1/2) = 1/2, \tag{106}$$

and, for $0 \le a \le 2^n$,

$$I\left(F\left(\frac{a}{2^n}\right)\right) = \frac{a}{2^n}. \tag{107}$$

These are `fabiusInv_one_sub`, `fabiusInv_half`, and `fabiusInv_fabiusDyadicUnit_of_le`.

The flatness of F becomes infinite endpoint steepness of I. The formalized quantitative form says that, for $y \in [0,1]$ and $2^n I(y) \le 1$,

$$y \le 2^{\binom{n+1}{2}} I(y)^n, \qquad y \le \frac{2^{\binom{n+1}{2}}}{n!} I(y)^n. \tag{108}$$

It also gives the two exact one-sided limits

$$\frac{I(y)}{y} \longrightarrow +\infty \quad (y \downarrow 0), \qquad \frac{1 - I(y)}{1 - y} \longrightarrow +\infty \quad (y \uparrow 1). \tag{109}$$

The counterparts are `le_two_pow_mul_fabiusInv_pow`, `le_two_pow_div_factorial_mul_f abiusInv_pow`, `tendsto_fabiusInv_div_atTop`, and `tendsto_one_sub_fabiusInv_div_on e_sub_atTop` in `FabiusInverse.lean`.

Theorem 6.22 (Analytic locus of the inverse)**.** *The totalized inverse I is real analytic at y if and only if $y \notin [0,1]$. In particular, it is nonanalytic at every point of the unit interval, although it is C^∞ and has no critical points in its interior.*

Formal proof. This is `fabiusInv_analyticAt_iff`; the interior no-critical-point statement is `deriv_fabiusInv_ne_zero`. Both are in `InverseNotElementary.lean`. □

6.9 A precise no-elementary-form theorem

Here *elementary* means the formal class `IsElementary`: the smallest class of total real functions containing constants and the identity, and closed under arithmetic, reciprocal, fixed real powers, exponential, logarithm, sine, cosine, arcsine, arctangent, and composition. Its analytic locus is dense. This makes nonanalyticity on an interval an obstruction rather than a heuristic absence of a familiar formula.

Theorem 6.23 (Non-elementarity)**.** *In this formal class, none of the bounded Fabius function, Rvachev's function, the signed extension, or the totalized inverse is elementary. More locally, no elementary function agrees with F or I throughout the open unit interval.*

Formal proof. The dense-analytic-locus theorem is `IsElementary.dense_analyticLocus` in `ElementaryFunction.lean`. The four global obstructions are `canonical_fabius_not_isElementary`, `rvachev_not_isElementary`, `extendedFabius_not_isElementary`, and `fabiusInv_not_isElementary`; the local forms are `canonical_fabius_not_isElementary_on_Ioo` and `canonical_fabiusInv_not_isElementary_on_Ioo`. See `NotElementary.lean` and `InverseNotElementary.lean`. □

The formal development also closes the elementary class under repeated continuous inverse branches. Even this enlarged class cannot agree with F or I on $(0,1)$; the exact statements are `canonical_fabius_not_isElementaryOrInverse_on_Ioo` and `canonical_fabiusInv_not_isElementaryOrInverse_on_Ioo` in `InverseNotElementary.lean`.

7 Moments and generating functions

Exact arithmetic enters through two rational moment sequences. One is attached to the even moments of up; the other records its one-sided moments and, at the same time, the values of F at reciprocal powers of two.

7.1 The even moments c_n

Recall

$$c_n = \int_{-1}^{1} x^{2n}\,\mathrm{up}(x)\,\mathrm{d}x, \qquad c_0 = 1.$$

Theorem 7.1 (Moment recurrence)**.** *For $n \geq 1$,*

$$(2n+1)(2^{2n}-1)c_n = \sum_{k=0}^{n-1} \binom{2n+1}{2k} c_k. \tag{110}$$

Consequently every c_n is rational and positive.

Proof. Insert the power series (86) into (71). Since

$$\operatorname{sinc}(\pi z) = \sum_{j\geq 0} \frac{(-1)^j(\pi z)^{2j}}{(2j+1)!}$$

and

$$\widehat{\mathrm{up}}(z/2) = \sum_{k\geq 0} \frac{(-1)^k c_k (\pi z)^{2k}}{(2k)!},$$

comparison of the coefficient of $(-1)^n(\pi z)^{2n}$ gives

$$\frac{2^{2n}c_n}{(2n)!} = \sum_{k=0}^{n} \frac{c_k}{(2k)!(2n-2k+1)!}.$$

Multiplying by $(2n+1)!$ yields

$$(2n+1)2^{2n}c_n = \sum_{k=0}^{n} \binom{2n+1}{2k} c_k.$$

The term $k=n$ is $(2n+1)c_n$; moving it to the left gives (110). Positivity follows either from the integral definition or by induction from the recurrence. □

The first values are

$$1, \quad \frac{1}{9}, \quad \frac{19}{675}, \quad \frac{583}{59535}, \quad \frac{132809}{32531625}, \ldots \tag{111}$$

A useful denominator clearing is

$$c_n = \frac{C_n}{(2n+1)!! \prod_{j=1}^{n}(2^{2j}-1)}, \tag{112}$$

where C_n is a positive integer. It can be defined without division by

$$C_0 = 1, \tag{113}$$

$$C_{n+1} = \sum_{k=0}^{n} C_k \binom{2n+3}{2k} \prod_{j=k+1}^{n} (2j+1) \prod_{j=k+1}^{n} (2^{2j}-1). \tag{114}$$

The formal development proves directly that every C_n is odd. Modulo 2, all factors in the two products are 1, and Lucas's theorem [13] shows that among the coefficients $\binom{2n+3}{2k}$ with $0 \le k \le n$ an odd number are odd. This closes the induction.

7.2 The half moments d_n

Define

$$d_0 = 1, \qquad d_n = n \int_0^1 t^{n-1}\, \mathrm{up}(t)\, \mathrm{d}t \quad (n \ge 1). \tag{115}$$

Lean's totalized analytic half-moment definition assigns the value 1 at $n=0$, so the exact and analytic sequences also admit a single all-index equality; the displayed integral itself remains restricted to $n \ge 1$. Introduce the entire exponential generating function

$$G(z) = 1 + z \int_0^1 \mathrm{up}(t) \mathrm{e}^{zt}\, \mathrm{d}t = \sum_{n=0}^{\infty} d_n \frac{z^n}{n!}. \tag{116}$$

Theorem 7.2 (Generating-function equation)**.** *For every $z \in \mathbb{C}$,*

$$G(2z) = \frac{\mathrm{e}^z - 1}{z} G(z), \tag{117}$$

where the quotient has its removable value 1 at $z = 0$.

Proof. One may derive the identity directly from (28) by integration by parts. An especially transparent route uses Fourier analysis. From evenness and the fold relation,

$$G(z) = \mathrm{e}^{z/2} \widehat{\mathrm{up}}\left(\frac{\mathrm{i}z}{4\pi}\right). \tag{118}$$

Applying (71) at $\mathrm{i}z/(2\pi)$ and simplifying $\mathrm{sinc}(\mathrm{i}z/2) = 2\sinh(z/2)/z$ gives

$$G(2z) = \mathrm{e}^z \frac{2\sinh(z/2)}{z} \widehat{\mathrm{up}}\left(\frac{\mathrm{i}z}{4\pi}\right) = \frac{\mathrm{e}^z - 1}{z} G(z).$$

□

Corollary 7.3 (Half-moment recurrence)**.** *For $n \geq 1$,*

$$(n+1)(2^n-1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k. \tag{119}$$

In particular $d_n \in \mathbb{Q}$.

Proof. Expand $(\mathrm{e}^z-1)/z = \sum_{j\geq 0} z^j/(j+1)!$ in (117). Comparison of the coefficient of $z^n/n!$ gives

$$2^n d_n = \frac{1}{n+1} \sum_{k=0}^{n} \binom{n+1}{k} d_k.$$

Isolating the $k=n$ term proves the recurrence. □

As for c_n, one may clear denominators without division:

$$d_n = \frac{D_n}{(n+1)! \prod_{j=1}^{n}(2^j-1)}, \tag{120}$$

where $D_n \in \mathbb{N}$. A division-free recurrence follows by substituting (120) into (119). The normalized parity theorem says that, for $n \geq 1$, the reduced numerator and denominator of $2d_n$ are both odd.

7.3 The probability–Laplace bridge

Let μ_X be the distribution of the weighted sum X from (40). Its survival function on the nonnegative half-line is exactly up:

$$\mu_X((t,\infty)) = \mathrm{up}(t) \qquad (t \geq 0). \tag{121}$$

Consequently, if g is continuously differentiable on $[0,1]$, then

$$\int_{[0,1]} g(x)\ \mathrm{d}\mu_X(x) = g(0) + \int_0^1 g'(t)\,\mathrm{up}(t)\,\mathrm{d}t. \tag{122}$$

This is a probability-measure form of integration by parts, proved without assuming a density for μ_X.

For $s \in \mathbb{R}$ and $k \in \mathbb{N}$, define the raw tilted moments

$$L_k(s) = \int_{[0,1]} \mathrm{e}^{-sx} x^k\ \mathrm{d}\mu_X(x). \tag{123}$$

They agree exactly with the analytic Fabius Laplace moments used later. At zero tilt, reflection invariance identifies both the ordinary and endpoint moments with the rational half-moment sequence:

$$L_n(0) = \int_{[0,1]} (1-x)^n\ \mathrm{d}\mu_X(x) = d_n. \tag{124}$$

Formal proof. Equation (121) is `weightedSumDistribution_real_Ioi_eq_rvachevUp_of_nonneg`, and (122) is `integral_unit_eq_zero_add_integral_deriv_mul_rvachevUp`. The tilted identification and the two zero-tilt identities are `unitLaplaceMoment_weightedSumDistribution_eq_fabiusLaplaceMoment`, `unitLaplaceMoment_weightedSumDistribution_zero_eq_halfMoment`, and `unitEndpointMoment_weightedSumDistribution_eq_halfMoment` in `ProbabilityLaplaceMoments.lean`. □

7.4 Relation between the two moment sequences

Expanding (118) in powers of z gives a finite binomial transform at each order.

Proposition 7.4. *For every $n \geq 0$,*

$$d_n = 2^{-n} \sum_{0 \leq k \leq n/2} \binom{n}{2k} c_k. \tag{125}$$

Proof. By (86),

$$\widehat{\mathrm{up}}\left(\frac{\mathrm{i}z}{4\pi}\right) = \sum_{k \geq 0} c_k \frac{(z/2)^{2k}}{(2k)!}.$$

Multiplying by $\mathrm{e}^{z/2}$ and extracting the coefficient of $z^n/n!$ gives exactly (125). □

7.4.1 Integer numerators and a divisibility bridge

The normalizations (112) and (120) produce two integer sequences, and the relation (125) between the moments becomes an exact divisibility statement between them. Throughout, empty products equal 1 and $(-1)!! = 1$.

Re-indexing (114) by $n \mapsto n-1$ puts the division-free recurrence for C_n in the form used below: $C_0 = 1$ and, for $n \geq 1$,

$$C_n = \sum_{k=0}^{n-1} C_k \binom{2n+1}{2k} \prod_{j=k+1}^{n-1} (2j+1) \prod_{j=k+1}^{n-1} (2^{2j}-1). \tag{126}$$

The primary text records only that a companion recurrence for D_n follows by substitution. It is the following.

Proposition 7.5 (Division-free recurrence for D_n). *Set $D_0 = 1$. For every $n \geq 1$,*

$$D_n = \sum_{k=0}^{n-1} D_k \binom{n+1}{k} \frac{n!}{(k+1)!} \prod_{j=k+1}^{n-1} (2^j - 1). \tag{127}$$

Every factor on the right is a positive integer, so $D_n \in \mathbb{N}$ for all n.

Proof. Substitute (120) into (119). The left side becomes

$$(n+1)(2^n-1)\frac{D_n}{(n+1)!\prod_{j=1}^{n}(2^j-1)} = \frac{D_n}{n!\prod_{j=1}^{n-1}(2^j-1)},$$

and the right side becomes $\sum_{k=0}^{n-1}\binom{n+1}{k} D_k \left[(k+1)!\prod_{j=1}^{k}(2^j-1)\right]^{-1}$. Multiplying through by $n!\prod_{j=1}^{n-1}(2^j-1)$ gives (127). Since $k \leq n-1$, the quotient $n!/(k+1)!$ is a product of consecutive integers and hence an integer, and the remaining product is an integer as well. Induction from $D_0 = 1$ proves integrality and positivity. □

The first values are

n	C_n	D_n
0	1	1
1	1	1
2	19	5
3	2915	84
4	2788989	4004
5	14754820185	494760
6	402830065455939	150120600
7	54259734183964303995	107969547840
8	34931036957548128175343565	179605731622464

Substituting (120) into (136) clears all denominators of the reciprocal-power values at once:

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}} D_n}{n!\,(n+1)!\prod_{j=1}^{n}(2^j-1)}. \tag{128}$$

This is the exact closed form behind (143): the entire dependence on n beyond elementary factors is carried by the single integer D_n.

The two integer sequences are linked through the odd half moments. The identity responsible is never displayed in the primary text, so we record it.

Lemma 7.6. *For every $k \geq 0$,*

$$d_{2k+1} = \frac{2k+1}{2} c_k. \tag{129}$$

Proof. By (115), $d_{2k+1} = (2k+1)\int_0^1 t^{2k}\,\mathrm{up}(t)\,\mathrm{d}t$. The integrand is even, so $\int_0^1 t^{2k}\,\mathrm{up}(t)\,\mathrm{d}t = \frac{1}{2}\int_{-1}^{1} t^{2k}\,\mathrm{up}(t)\,\mathrm{d}t = \frac{1}{2}c_k$ by (85). □

Theorem 7.7 (Bridge between the integer numerators)**.** *For every $n \geq 0$,*

$$\frac{D_{2n+1}}{2n+1} = 2^n (n+1)!\, C_n \prod_{k=0}^{n} (2^{2k+1}-1). \tag{130}$$

In particular $(2n+1)C_n$ divides D_{2n+1}.

Proof. By (120) and (129),

$$D_{2n+1} = d_{2n+1}(2n+2)! \prod_{j=1}^{2n+1}(2^j-1) = \frac{2n+1}{2}\, c_n\,(2n+2)! \prod_{j=1}^{2n+1}(2^j-1).$$

Insert (112) and divide by $2n+1$:

$$\frac{D_{2n+1}}{2n+1} = \frac{C_n}{2}\cdot\frac{(2n+2)!}{(2n+1)!!}\cdot\frac{\prod_{j=1}^{2n+1}(2^j-1)}{\prod_{j=1}^{n}(2^{2j}-1)}.$$

The last quotient retains exactly the factors with odd index, that is $\prod_{k=0}^{n}(2^{2k+1}-1)$, since the even indices $j = 2k$ with $1 \leq k \leq n$ are precisely those canceled by the denominator. For the factorial quotient, separate the even factors of $(2n+2)!$:

$$(2n+2)! = \big[2\cdot 4\cdots(2n+2)\big]\big[1\cdot 3\cdots(2n+1)\big] = 2^{n+1}(n+1)!\,(2n+1)!!,$$

so $(2n+2)!/(2n+1)!! = 2^{n+1}(n+1)!$. Combining the two evaluations gives

$$\frac{D_{2n+1}}{2n+1} = \frac{C_n}{2}\cdot 2^{n+1}(n+1)!\prod_{k=0}^{n}(2^{2k+1}-1),$$

which is (130). The right side is an integer multiple of C_n, so $D_{2n+1} = (2n+1)C_n \cdot 2^n(n+1)!\prod_{k=0}^{n}(2^{2k+1}-1)$ and the divisibility follows. □

Remark 7.8 (Independent numerical verification). Every numeric value printed in this subsubsection was recomputed independently in exact rational arithmetic: the moments were generated from (110) and (119), the numerators were obtained both by clearing denominators in (112) and (120) and by running the division-free recurrences (126) and (127), and the two computations were compared entry by entry. The closed form (128), the identity (129), the bridge (130), and the divisibility $(2n+1)C_n \mid D_{2n+1}$ were checked for all indices $n \leq 20$. For instance, at $n = 2$, $D_5/5 = 98952 = 2^2\cdot 3!\cdot C_2\cdot(2-1)(2^3-1)(2^5-1)$ with $C_2 = 19$, and $5\cdot 19 = 95$ divides $D_5 = 494760$.

7.5 Bernoulli-number recurrences

The two refinement equations can be inverted before coefficients are compared. Doing so replaces the binomial convolutions of (110) and (119) by recurrences whose coefficients are Bernoulli numbers.

Throughout this subsection B_k denotes the Bernoulli numbers in the convention

$$\frac{z}{e^z - 1} = \sum_{k\geq 0} B_k \frac{z^k}{k!}, \qquad B_1 = -\frac{1}{2}, \tag{131}$$

so that $B_0 = 1$, $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $B_6 = \frac{1}{42}$, and $B_{2m+1} = 0$ for $m \geq 1$. The sign of B_1 is not cosmetic here. It is what produces the isolated first-order term in (135); the opposite convention $B_1 = +\frac{1}{2}$ reverses that term's sign and destroys the recurrence.

Rewrite (71) and (117) by moving the refinement factor to the other side:

$$\frac{\pi z}{\sin \pi z} \widehat{\mathrm{up}}(z) = \widehat{\mathrm{up}}(z/2), \qquad \frac{z}{e^z - 1} G(2z) = G(z). \tag{132}$$

Both reciprocal factors have classical Bernoulli expansions. The second is (131) itself, and the first comes from

$$\frac{w}{\sin w} = \sum_{k\geq 0} (-1)^k (2 - 2^{2k}) B_{2k} \frac{w^{2k}}{(2k)!}, \qquad |w| < \pi, \tag{133}$$

evaluated at $w = \pi z$. The first two terms of (133) are $1 + w^2/6$, as they must be.

Theorem 7.9 (Bernoulli recurrences)**.** *For every $n \geq 1$,*

$$c_n = \frac{1}{2^{2n} - 1} \sum_{k=1}^{n} 2^{2n-2k} (2^{2k} - 2) \binom{2n}{2k} B_{2k}\, c_{n-k}, \tag{134}$$

and

$$d_n = \frac{n\, 2^{n-2}}{2^n - 1} d_{n-1} - \frac{1}{2^n - 1} \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n}{2k} 2^{n-2k} B_{2k}\, d_{n-2k}. \tag{135}$$

In (135) the sum is empty for $n = 1$.

Proof. For the even moments, put $w = \pi z$. By (86),

$$\widehat{\mathrm{up}}(z) = \sum_{n\geq 0} (-1)^n c_n \frac{(2w)^{2n}}{(2n)!}, \qquad \widehat{\mathrm{up}}(z/2) = \sum_{n\geq 0} (-1)^n c_n \frac{w^{2n}}{(2n)!}.$$

Multiply the first series by (133) and use the left identity in (132). Both sides are entire in w, so their coefficients agree. The coefficient of w^{2n} on the left is

$$\begin{aligned} &\sum_{k=0}^{n} (-1)^k (2 - 2^{2k}) B_{2k} \frac{1}{(2k)!} \cdot (-1)^{n-k} c_{n-k} \frac{2^{2n-2k}}{(2n-2k)!} \\ &\qquad = \frac{(-1)^n}{(2n)!} \sum_{k=0}^{n} \binom{2n}{2k} (2 - 2^{2k}) 2^{2n-2k} B_{2k} c_{n-k}, \end{aligned}$$

and on the right it is $(-1)^n c_n/(2n)!$. Canceling the common factor $(-1)^n/(2n)!$ gives

$$\sum_{k=0}^{n} \binom{2n}{2k} (2 - 2^{2k}) 2^{2n-2k} B_{2k} c_{n-k} = c_n.$$

The term $k = 0$ equals $2^{2n}c_n$, because $B_0 = 1$ and $2 - 2^0 = 1$. Moving it to the right gives $-(2^{2n} - 1)c_n$ there, and reversing the sign of the bracket $2 - 2^{2k}$ in the remaining terms yields (134).

For the half moments, expand $G(2z) = \sum_{n\ge 0} 2^n d_n z^n/n!$ from (116) and multiply by (131). The right identity in (132) and comparison of the coefficient of $z^n/n!$ give

$$\sum_{m=0}^{n} \binom{n}{m} B_m 2^{n-m} d_{n-m} = d_n.$$

The term $m = 0$ is $2^n d_n$. The term $m = 1$ is $nB_1 2^{n-1}d_{n-1} = -n2^{n-2}d_{n-1}$, by the convention (131). Every term with m odd and $m \ge 3$ vanishes, so the remaining terms are those with $m = 2k$ and $1 \le k \le \lfloor n/2 \rfloor$. Isolating d_n and dividing by $2^n - 1$ produces (135). □

The two forms are equivalent to (110) and (119), since each determines its sequence uniquely from $c_0 = d_0 = 1$. The binomial recurrences are better for exact computation, because they involve only integers and one division; the Bernoulli forms are better for asymptotic estimates, because the size of B_{2k} is known explicitly.

8 Exact values at reciprocal powers of two

The half moments are already endpoint values in disguise.

Theorem 8.1 (Reciprocal-power values)**.** *For every $n \ge 0$,*

$$F(2^{-n}) = \frac{d_n}{n!\, 2^{\binom{n}{2}}}. \tag{136}$$

Equivalently,

$$d_n = n!\, 2^{\binom{n}{2}} F(2^{-n}). \tag{137}$$

Proof. The case $n = 0$ is $F(1) = d_0 = 1$, so assume $n \ge 1$. The whole argument is a single chain of primitives supplied by (28). For $r \ge 0$ set

$$f_r(t) = 2^{\binom{r}{2}}\, \mathrm{up}\left(\frac{t}{2^r} - \frac{1}{2^r} - \frac{1}{2^{r-1}} - \cdots - \frac{1}{2}\right), \tag{138}$$

where the subtracted sum is empty for $r = 0$, so that $f_0 = \mathrm{up}$. Since $\sum_{j=1}^{r} 2^{-j} = 1 - 2^{-r}$, the shift collapses and

$$f_r(t) = 2^{\binom{r}{2}}\, \mathrm{up}((t+1)2^{-r} - 1). \tag{139}$$

The chain has two properties on $[-1, 1]$. First, $f_r(-1) = 2^{\binom{r}{2}}\, \mathrm{up}(-1) = 0$ for every r. Second,

$$f'_{r+1} = f_r \qquad \text{on } [-1, 1]. \tag{140}$$

To see (140), differentiate (139) at index $r + 1$ and write $x = (t + 1)2^{-(r+1)} - 1$, so that $2x + 1 = (t + 1)2^{-r} - 1$ and $2x - 1 = (t + 1)2^{-r} - 3$. For $t \in [-1, 1]$ and $r \ge 0$ we have $0 \le (t + 1)2^{-r} \le 2$, hence $2x - 1 \le -1$; since up is supported on $[-1, 1]$ and $\mathrm{up}(-1) = 0$, the second term of (28) vanishes and $\mathrm{up}'(x) = 2\,\mathrm{up}(2x + 1)$. Therefore

$$f'_{r+1}(t) = 2^{\binom{r+1}{2}} 2^{-(r+1)}\, \mathrm{up}'(x) = 2^{\binom{r+1}{2}-r}\, \mathrm{up}((t+1)2^{-r} - 1) = f_r(t),$$

because $\binom{r+1}{2} - r = \binom{r}{2}$. The exponent bookkeeping is exactly what makes the chain close.

Put $J_n = \int_{-1}^{0} t^{n-1}\,\mathrm{up}(t)\,\mathrm{d}t$. Substituting $t \mapsto -t$ and using evenness of up gives

$$\int_0^1 t^{n-1}\,\mathrm{up}(t)\,\mathrm{d}t = (-1)^{n-1} J_n. \tag{141}$$

Integrate J_n by parts $n-1$ times, replacing f_j by f'_{j+1} at each step. The boundary contribution at $t = -1$ vanishes at every step because $f_{j+1}(-1) = 0$, and the contribution at $t = 0$ vanishes because the surviving polynomial factor t^{n-1-j} still has positive degree for $0 \le j \le n-2$. Hence

$$J_n = (-1)^{n-1}(n-1)! \int_{-1}^{0} f_{n-1}(t)\,\mathrm{d}t = (-1)^{n-1}(n-1)!\big(f_n(0) - f_n(-1)\big) = (-1)^{n-1}(n-1)!\, f_n(0),$$

where the last integral was evaluated by (140) once more. By (139), evenness, and (8),

$$f_n(0) = 2^{\binom{n}{2}}\,\mathrm{up}(2^{-n} - 1) = 2^{\binom{n}{2}}\,\mathrm{up}(1 - 2^{-n}) = 2^{\binom{n}{2}} F(2^{-n}).$$

Combining this with (141) makes the two signs cancel and yields

$$\int_0^1 t^{n-1}\,\mathrm{up}(t)\,\mathrm{d}t = (n-1)!\, 2^{\binom{n}{2}} F(2^{-n}). \tag{142}$$

Multiplying by n and using (115) gives (137), and dividing gives (136). $\square$

The first values are therefore

n	$F(2^{-n})$
0	1
1	$\frac{1}{2}$
2	$\frac{5}{72}$
3	$\frac{1}{288}$
4	$\frac{143}{2073600}$
5	$\frac{19}{33177600}$
6	$\frac{1153}{561842749440}$
7	$\frac{583}{179789679820800}$

(143)

The denominators grow far faster than exponentially, in agreement with the quadratic leading term in the logarithmic asymptotic derived later.

8.1 A triangular recurrence and its nonrecursive solution

The values in (143) may be generated without first computing the moments. Write

$$V_n = F(2^{-n}). \tag{144}$$

Eliminating d_n between (119) and (136) gives, for $n \ge 1$,

$$V_n = \frac{2^{-\binom{n}{2}}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}}}{(n-k+1)!} V_k, \qquad V_0 = 1. \tag{145}$$

The restriction $n \ge 1$ is essential: at $n = 0$ the denominator $2^n - 1$ is zero and the sum is empty, whereas $V_0 = F(1) = 1$. Thus the initial value is part of the recurrence data, not a removable endpoint convention.

The point survives every attempt to legislate it away. In a formal development in which division by zero is totalized, so that $x/0$ is given the value 0, the instance $n = 0$ of (145) is not vacuous but false: its right side reduces to zero, both because the sum is empty and because the prefactor has a zero denominator, while $V_0 = 1$. No convention makes the recurrence assertable at $n = 0$, and the restriction is therefore part of the statement rather than an artifact of how it is written.

The triangular powers of two disappear after the normalization

$$A_n = 2^{\binom{n}{2}} V_n. \tag{146}$$

Indeed, $A_0 = 1$ and

$$A_n = \sum_{k=0}^{n-1} \frac{A_k}{(2^n - 1)(n - k + 1)!}. \tag{147}$$

This is a weighted-path recurrence on the directed graph with one edge $k \to n$ whenever $k < n$. Iterating it terminates, and the resulting paths can be indexed by ordered compositions.

The recurrence is triangular: once $A_0 = 1$ is fixed, each A_n is a linear combination of earlier values. Induction therefore gives uniqueness. In particular, the argument below is purely algebraic and applies to any sequence satisfying (147); no further analytic property of F is used.

An ordered composition of n is a tuple $\rho = (r_1, \ldots, r_m)$ of positive integers with sum n. Put

$$s_j = r_1 + \cdots + r_j, \qquad W(\rho) = \prod_{j=1}^{m} \frac{1}{(2^{s_j} - 1)(r_j + 1)!}. \tag{148}$$

For $n = 0$ there is one empty composition; its empty product is 1.

Theorem 8.2 (Finite composition formula)**.** *For every $n \geq 0$,*

$$\boxed{F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\rho \vDash n} \prod_{j=1}^{\mathrm{len}(\rho)} \frac{1}{(2^{s_j} - 1)(r_j + 1)!}} \tag{149}$$

where $\rho = (r_1, \ldots, r_m)$ and $s_j = r_1 + \cdots + r_j$. For $n = 0$ the unique empty composition makes the right side equal to 1.

Proof. Let $\Sigma_n = \sum_{\rho \vDash n} W(\rho)$. The empty-composition convention gives $\Sigma_0 = 1$. Fix $n \geq 1$ and split every composition of n immediately before its last block. If the preceding blocks sum to k, then $0 \leq k < n$, the last block is $n - k$, and the last factor of (148) is

$$\frac{1}{(2^n - 1)(n - k + 1)!}.$$

Conversely, appending the block $n - k$ to any composition of k gives exactly one composition of n with penultimate partial sum k. Therefore

$$\Sigma_n = \sum_{k=0}^{n-1} \frac{\Sigma_k}{(2^n - 1)(n - k + 1)!}.$$

Thus (Σ_n) and (A_n) have the same initial value and the same triangular recurrence. Induction gives $\Sigma_n = A_n$, and multiplying by $2^{-\binom{n}{2}}$ proves (149). $\square$

Every point 2^{-n} lies in $[0,1]$, where the bounded Fabius function and the signed global extension agree. Consequently the same formula holds with $F(2^{-n})$ replaced by $\mathcal{F}(2^{-n})$.

Equivalently, put $i_0 = 0$ and $i_j = s_j$. Compositions of length m correspond bijectively to chains

$$0 = i_0 < i_1 < \cdots < i_m = n, \qquad r_j = i_j - i_{j-1}.$$

Hence, for $n \geq 1$,

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{m=1}^{n} \sum_{0=i_0<i_1<\cdots<i_m=n} \prod_{j=1}^{m} \frac{1}{(2^{i_j}-1)(i_j - i_{j-1}+1)!}. \tag{150}$$

There are 2^{n-1} compositions of a positive integer n, so this is finite and nonrecursive, although exponential in size. For fixed m, the inner sum may instead be written over $1 \leq i_1 < \cdots < i_{m-1} \leq n-1$, with $i_0 = 0$ and $i_m = n$; when $m = 1$ there are no internal indices and there is one direct path $0 \to n$.

The common factor $2^{-\binom{n}{2}}$ has a direct path interpretation. Before normalization, the edge $k \to r$ has weight

$$U_{r,k} = \frac{2^{\binom{k}{2}-\binom{r}{2}}}{(2^r-1)(r-k+1)!}. \tag{151}$$

Along a chain $0 = i_0 < i_1 < \cdots < i_m = n$, the powers of two telescope:

$$\prod_{j=1}^{m} 2^{\binom{i_{j-1}}{2}-\binom{i_j}{2}} = 2^{\binom{i_0}{2}-\binom{i_m}{2}} = 2^{-\binom{n}{2}}. \tag{152}$$

The remaining factors are precisely the normalized chain weight in (150). Thus normalization extracts an endpoint factor common to every path, rather than merely shortening the recurrence.

Example 8.3 (The first nontrivial path sums)**.** For $n = 2$, the compositions are (2) and $(1,1)$, with weights

$$W(2) = \frac{1}{(2^2-1)3!} = \frac{1}{18}, \qquad W(1,1) = \frac{1}{(2^1-1)2!}\frac{1}{(2^2-1)2!} = \frac{1}{12}.$$

Hence $A_2 = 5/36$ and $F(2^{-2}) = 2^{-1}A_2 = 5/72$. For $n = 3$, the four ordered compositions $(3), (1,2), (2,1), (1,1,1)$ have weights

$$\frac{1}{168}, \qquad \frac{1}{84}, \qquad \frac{1}{252}, \qquad \frac{1}{168}.$$

The unequal weights of $(1,2)$ and $(2,1)$ record their different intermediate partial sums. Their total is $A_3 = 1/36$, giving $F(2^{-3}) = 2^{-3}A_3 = 1/288$.

For a precise matrix form, fix N and define a matrix indexed by $0, \ldots, N$ by

$$T^{[N]}_{i,j} = \begin{cases} \dfrac{1}{(2^i-1)(i-j+1)!}, & j < i, \\ 0, & j \geq i. \end{cases}$$

With $\mathbf{A} = (A_0, \ldots, A_N)^\mathsf{T}$ and $\mathbf{e}_0 = (1, 0, \ldots, 0)^\mathsf{T}$, the initial value and recurrence become

$$\mathbf{A} = \mathbf{e}_0 + T^{[N]}\mathbf{A}.$$

Since $T^{[N]}$ is strictly lower triangular, $(T^{[N]})^{N+1} = 0$ and

$$\mathbf{A} = (I - T^{[N]})^{-1}\mathbf{e}_0 = (I + T^{[N]} + \cdots + (T^{[N]})^N)\mathbf{e}_0.$$

The entry $(T^{[N]})^m_{n,0}$ is the sum of products of edge weights over all length-m increasing paths from 0 to n, so its nth component is exactly (150). This is a finite geometric identity for a nilpotent matrix, not an analytic convergence argument. Finally, if

$$D = \operatorname{diag}(2^{\binom{0}{2}}, 2^{\binom{1}{2}}, \dots, 2^{\binom{N}{2}}),$$

then the matrix of the original recurrence is $U = D^{-1}T^{[N]}D$. This diagonal similarity is the matrix form of (152).

Corollary 8.4 (Positivity)**.** *For every $n \ge 0$, $F(2^{-n}) > 0$; equivalently $A_n > 0$.*

Proof. For $n = 0$ the single empty composition contributes the empty product 1. Let $n \ge 1$. Every partial sum s_j of a composition of n satisfies $s_j \ge 1$, so $2^{s_j} - 1 \ge 1 > 0$, and every factorial $(r_j + 1)!$ is positive. Hence each weight (148) is a positive rational, the sum (149) has at least the one term $\rho = (n)$, and the factor $2^{-\binom{n}{2}}$ is positive. □

The argument uses no property of F beyond the recurrence: no integral representation, no probabilistic interpretation, no monotonicity, no continuity. Positivity of the reciprocal-power values is a statement about a triangular linear system with positive coefficients. Combined with monotonicity of F it recovers $F(x) > 0$ for $0 < x \le 1$, but the corollary itself is purely arithmetic, and it is the step that shows the numerators in (143) cannot vanish.

The worked example above stops at $n = 3$. The first case in which compositions of three distinct lengths occur is $n = 4$.

Example 8.5 (The eight compositions of 4)**.** There are $2^3 = 8$ ordered compositions of 4. Their partial sums and weights (148) are

ρ	$(s_1, \dots, s_m)$	$W(\rho)$
(4)	(4)	$\frac{1}{(2^4-1)5!} = \frac{1}{1800}$
$(1,3)$	$(1,4)$	$\frac{1}{(2^1-1)2!} \cdot \frac{1}{(2^4-1)4!} = \frac{1}{720}$
$(2,2)$	$(2,4)$	$\frac{1}{(2^2-1)3!} \cdot \frac{1}{(2^4-1)3!} = \frac{1}{1620}$
$(3,1)$	$(3,4)$	$\frac{1}{(2^3-1)4!} \cdot \frac{1}{(2^4-1)2!} = \frac{1}{5040}$
$(1,1,2)$	$(1,2,4)$	$\frac{1}{(2^1-1)2!} \cdot \frac{1}{(2^2-1)2!} \cdot \frac{1}{(2^4-1)3!} = \frac{1}{1080}$
$(1,2,1)$	$(1,3,4)$	$\frac{1}{(2^1-1)2!} \cdot \frac{1}{(2^3-1)3!} \cdot \frac{1}{(2^4-1)2!} = \frac{1}{2520}$
$(2,1,1)$	$(2,3,4)$	$\frac{1}{(2^2-1)3!} \cdot \frac{1}{(2^3-1)2!} \cdot \frac{1}{(2^4-1)2!} = \frac{1}{7560}$
$(1,1,1,1)$	$(1,2,3,4)$	$\frac{1}{(2^1-1)2!} \cdot \frac{1}{(2^2-1)2!} \cdot \frac{1}{(2^3-1)2!} \cdot \frac{1}{(2^4-1)2!} = \frac{1}{5040}$

Their total is

$$A_4 = \frac{1}{1800} + \frac{1}{720} + \frac{1}{1620} + \frac{1}{5040} + \frac{1}{1080} + \frac{1}{2520} + \frac{1}{7560} + \frac{1}{5040} = \frac{143}{32400}, \tag{153}$$

and since $\binom{4}{2} = 6$,

$$F(2^{-4}) = 2^{-6}A_4 = \frac{143}{2073600},$$

which is the entry for $n = 4$ in (143). Note that $(3,1)$ and $(1,1,1,1)$ happen to share the weight $1/5040$; the coincidence is numerical, and the two compositions have different lengths and different chains. As at $n = 3$, the reversed pairs $(1, 3)$ and $(3, 1)$, and $(1, 1, 2)$ and $(2, 1, 1)$, carry different weights, because the Mersenne factors depend on the partial sums and not on the multiset of block sizes.

Both the eight individual weights and the total (153) were recomputed in exact rational arithmetic directly from (148), with the assignment of each weight to its composition checked term by term; the composition sum was also verified against the recurrence (147) for $0 \le n \le 7$.

Finally, the three presentations given in this subsection are not three theorems but one.

Proposition 8.6 (One object, three presentations)**.** *Let $(V_n)_{n\ge0}$ be a sequence in $\mathbb{Q}$ with $V_0 = 1$, and put $A_n = 2^{\binom{n}{2}} V_n$. The following are equivalent.*

(i) *(V_n) satisfies the triangular recurrence (145) for every $n \ge 1$.*

(ii) *(V_n) is given by the composition sum (149) for every $n \ge 0$.*

(iii) *For every $N \ge 0$, the vector $\mathbf{A} = (A_0, \dots, A_N)^\mathsf{T}$ satisfies $\mathbf{A} = (I - T^{[N]})^{-1}\mathbf{e}_0 = (I + T^{[N]} + \cdots + (T^{[N]})^N)\mathbf{e}_0$.*

Each of the three determines (V_n) uniquely, and the common value is $V_n = F(2^{-n})$.

Proof. Statement (i) is equivalent to the normalized recurrence (147), since $A_n = 2^{\binom{n}{2}} V_n$ and the powers of two cancel. Splitting a composition before its last block turns the composition sum into (147), so (ii) implies (i); conversely the recurrence is triangular, hence has a unique solution with the given initial value, and the composition sum is one such solution, so (i) implies (ii). For (iii), the identity $\mathbf{A} = \mathbf{e}_0 + T^{[N]}\mathbf{A}$ is exactly (147) written for the indices $0, \dots, N$ together with $A_0 = 1$; since $T^{[N]}$ is strictly lower triangular, $(T^{[N]})^{N+1} = 0$ and $I - T^{[N]}$ is invertible with the displayed nilpotent geometric inverse. Expanding the matrix powers gives the chain sum (150), which the bijection between chains and compositions carries to (ii). The identification $V_n = F(2^{-n})$ is (136) together with (119). □

This settles what "closed form" means here, and what it does not mean. Formula (149) is closed in the sense that it is nonrecursive: each summand is determined by the blocks of a single composition and by n alone, with no reference to earlier values of the sequence, and the matrix presentation is a finite polynomial in a nilpotent matrix rather than a limit. It is not closed in the sense of being cheap: the sum has 2^{n-1} terms, so as an evaluation procedure it is exponentially worse than the $\mathcal{O}(n^2)$ rational operations of the recurrence itself. Eliminating recursion and reducing arithmetic cost are different goals, and only the first is achieved here. The value of the composition form is structural — it exhibits positivity termwise, it displays the telescoping of the powers of two along paths as in (152), and it makes the edge weights (151) visible — while the recurrence and the bit recursion remain the instruments of computation.

8.2 An exact two-adic valuation

8.2.1 An exact count of odd binomial coefficients

The parity statements used above and below rest on a single counting lemma, which Lucas's theorem makes exact rather than merely qualitative.

Lemma 8.7 (Odd entries in row $2n + 1$)**.** *Let $n \ge 1$. Among the coefficients $\binom{2n+1}{2k}$ with $0 \le k \le n$, exactly $2^{\mathrm{wt}_2(n)}$ are odd. Among all the coefficients $\binom{2n+1}{k}$ with $0 \le k \le 2n + 1$, exactly $2^{\mathrm{wt}_2(n)+1}$ are odd.*

Proof. Lucas's theorem modulo 2 [13] states that $\binom{N}{K}$ is odd exactly when every binary digit of K is at most the corresponding binary digit of N; equivalently, when the set of positions of the 1-bits of K is contained in that of N.

The binary expansion of $N = 2n+1$ is the expansion of n shifted up by one position, with a final digit 1 appended. Hence K satisfies the Lucas condition precisely when its lowest digit is arbitrary and its remaining digits, read as the binary expansion of $\lfloor K/2 \rfloor$, select a subset of the 1-bits of n. There are $2^{\mathrm{wt}_2(n)}$ such subsets and two choices of the lowest digit, giving $2^{\mathrm{wt}_2(n)+1}$ odd entries in the whole row. Restricting to even $K = 2k$ fixes the lowest digit to 0 and leaves $2^{\mathrm{wt}_2(n)}$ odd entries; as K runs over the even integers in $[0, 2n+1]$, the index k runs over $[0, n]$. $\square$

Corollary 8.8. *Let $n \geq 1$. The number of odd coefficients $\binom{2n+1}{2k}$ with $0 \leq k \leq n-1$ is $2^{\mathrm{wt}_2(n)} - 1$, and the number of odd coefficients $\binom{2n+1}{k}$ with $1 \leq k \leq 2n-1$ is $2^{\mathrm{wt}_2(n)+1} - 3$. Both numbers are odd.*

Proof. The omitted coefficient in the first count is $\binom{2n+1}{2n} = 2n+1$, which is odd. The omitted coefficients in the second count are $\binom{2n+1}{0} = 1$, $\binom{2n+1}{2n} = 2n+1$ and $\binom{2n+1}{2n+1} = 1$, all odd. Subtract from the two totals of Lemma 8.7. Since $n \geq 1$ forces $\mathrm{wt}_2(n) \geq 1$, both $2^{\mathrm{wt}_2(n)}$ and $2^{\mathrm{wt}_2(n)+1}$ are even, so $2^{\mathrm{wt}_2(n)} - 1$ and $2^{\mathrm{wt}_2(n)+1} - 3$ are odd. $\square$

Proposition 8.9 (Oddness of the even-moment numerators)**.** *Every C_n is odd.*

Proof. Induct on n, the case $C_0 = 1$ being immediate. Let $n \geq 1$ and assume $C_0, \ldots, C_{n-1}$ are odd. In (126) every factor other than the binomial coefficient is odd: the entries $2j+1$ are odd, each $2^{2j} - 1$ is odd, and each C_k is odd by hypothesis. Hence, modulo 2, the sum counts the odd coefficients $\binom{2n+1}{2k}$ with $0 \leq k \leq n-1$. By Corollary 8.8 that count is $2^{\mathrm{wt}_2(n)} - 1$, an odd number. Therefore C_n is odd. $\square$

We can now prove the parity statement for the half moments that was asserted after (120). Work in the local ring $\mathbb{Z}_{(2)}$ of rational numbers with odd denominator; its residue field is $\mathbb{Z}/2\mathbb{Z}$, and a rational number x lies in $\mathbb{Z}_{(2)}$ with $v_2(x) = 0$ exactly when its reduced numerator and denominator are both odd.

Theorem 8.10 (Normalized parity)**.** *For every $n \geq 1$,*

$$v_2(2d_n) = 0. \tag{154}$$

Equivalently, the reduced numerator and denominator of $2d_n$ are both odd, and $v_2(d_n) = -1$.

Proof. For an odd index $n = 2k+1$, combine (129) with (112):

$$2d_{2k+1} = (2k+1)c_k = \frac{(2k+1)\,C_k}{(2k+1)!!\,\prod_{j=1}^{k}(2^{2j}-1)}.$$

The numerator is odd by Proposition 8.9, and the denominator is a product of odd integers. Hence $v_2(2d_{2k+1}) = 0$. This covers every odd index, including $n = 1$.

For an even index, write (119) at $2n$ and multiply by 2:

$$(2n+1)(2^{2n}-1)\,2d_{2n} = \sum_{k=0}^{2n-1} \binom{2n+1}{k} 2d_k. \tag{155}$$

Proceed by strong induction on $n \geq 1$, assuming $v_2(2d_k) = 0$ for $1 \leq k \leq 2n-1$; the odd indices in that range are already settled unconditionally, and the even ones are covered by the induction hypothesis. Every term on the right of (155) then lies in $\mathbb{Z}_{(2)}$, so the right side does, and the prefactor $(2n+1)(2^{2n}-1)$ on the left is an odd integer and hence a unit of $\mathbb{Z}_{(2)}$. Solving for $2d_{2n}$ shows $2d_{2n} \in \mathbb{Z}_{(2)}$.

Reduce (155) modulo 2 in $\mathbb{Z}_{(2)}$. The prefactor is congruent to 1. The term $k = 0$ contributes $2d_0 = 2 \equiv 0$. Each term with $1 \le k \le 2n-1$ has $2d_k$ a unit, so it is congruent to $\binom{2n+1}{k}$. Therefore

$$2d_{2n} \equiv \#\{\, 1 \le k \le 2n-1 : \ \tbinom{2n+1}{k} \text{ odd} \,\} = 2^{\mathrm{wt}_2(n)+1} - 3 \equiv 1 \pmod 2,$$

using Corollary 8.8. Hence $v_2(2d_{2n}) = 0$, completing the induction. The final assertion is the definition of v_2 on $\mathbb{Q}^\times$ together with $v_2(d_n) = v_2(2d_n) - 1$. □

Theorem 8.10 is exactly the input required by the valuation theorem that follows, which is therefore now proved outright rather than conditionally.

The parity theorem for d_n immediately determines the power of two in the reduced denominator.

Theorem 8.11 (Exact 2-adic valuation)**.** *For $n \ge 1$,*

$$v_2(F(2^{-n})) = -\binom{n}{2} - 1 - v_2(n!). \tag{156}$$

Proof. The reduced numerator and denominator of $2d_n$ are odd, so $v_2(d_n) = -1$. Apply (136) and add valuations:

$$v_2(F(2^{-n})) = v_2(d_n) - v_2(n!) - \binom{n}{2}.$$

□

Legendre's formula

$$v_2(n!) = n - \mathrm{wt}_2(n) \tag{157}$$

then makes the binary dependence explicit.

8.3 A normalized odd integer

For $n \ge 1$ define

$$R_n = 2^{\binom{n-1}{2}} (2n)! F(2^{-n}) \prod_{j=1}^{\lfloor n/2 \rfloor} (2^{2j} - 1). \tag{158}$$

The exponent of two in (158) is *positive*; a negative sign that appears in an abstract of the source paper is a typographical error.

That these numbers are integers at all was not obvious. The question was raised by Reshetnikov [20] on the basis of numerical evidence, and answered affirmatively by Arias de Reyna [8], whose paper devotes a section to it.

Theorem 8.12 (Reshetnikov's integers)**.** *For every $n \ge 1$ the number R_n of* (158) *is an odd positive integer. It has the half-moment form*

$$R_n = 2d_n\,(2n-1)!! \prod_{m=1}^{\lfloor n/2 \rfloor} (2^{2m} - 1), \tag{159}$$

the odd-index closed form

$$R_{2n+1} = C_n \frac{(4n+1)!!}{(2n-1)!!} \qquad (n \ge 0), \tag{160}$$

and the even-index expansion

$$R_{2n} = \sum_{k=0}^{n} \frac{2\,C_k}{2^{2n}} \binom{2n}{2k} \frac{(4n-1)!!}{(2k+1)!!} \prod_{l=k+1}^{n} (2^{2l} - 1) \qquad (n \ge 1). \tag{161}$$

Empty products equal 1 *and* $(-1)!! = 1$.

Proof. The half-moment form. Substitute (136) into (158):

$$R_n = 2^{\binom{n-1}{2}-\binom{n}{2}} \frac{(2n)!}{n!} d_n \prod_{m=1}^{\lfloor n/2 \rfloor} (2^{2m} - 1).$$

Now $\binom{n}{2} - \binom{n-1}{2} = n - 1$, and separating the even factors of $(2n)!$ gives $(2n)! = 2^n n!\,(2n-1)!!$, hence $(2n)!/n! = 2^n(2n-1)!!$. The two powers of two combine into $2^{-(n-1)} \cdot 2^n = 2$, which is (159). Every factor after $2d_n$ in (159) is odd, so

$$v_2(R_n) = v_2(2d_n) = 0 \tag{162}$$

by Theorem 8.10.

Odd indices. Take $n \mapsto 2n+1$ in (159), note $\lfloor (2n+1)/2 \rfloor = n$, and use (129) and (112):

$$R_{2n+1} = (2n+1)c_n(4n+1)!! \prod_{m=1}^{n} (2^{2m} - 1) = (2n+1) \frac{C_n(4n+1)!!}{(2n+1)!!} = C_n \frac{(4n+1)!!}{(2n-1)!!},$$

since $(2n+1)!! = (2n+1)(2n-1)!!$. This is (160). The quotient $(4n+1)!!/(2n-1)!!$ is the product of the odd integers $2n+1, 2n+3, \ldots, 4n+1$, so R_{2n+1} is a product of odd integers and is therefore an odd integer.

Even indices. Take $n \mapsto 2n$ in (159), insert the binomial transform (125), and replace each c_k by (112):

$$\begin{aligned} R_{2n} &= 2 \cdot 2^{-2n} \sum_{k=0}^{n} \binom{2n}{2k} c_k\,(4n-1)!! \prod_{m=1}^{n} (2^{2m} - 1) \\ &= \sum_{k=0}^{n} \frac{2\,C_k}{2^{2n}} \binom{2n}{2k} \frac{(4n-1)!!}{(2k+1)!!} \prod_{l=k+1}^{n} (2^{2l} - 1), \end{aligned}$$

because the product $\prod_{m=1}^{k}(2^{2m} - 1)$ in the denominator of c_k cancels the initial segment of $\prod_{m=1}^{n}(2^{2m} - 1)$. This is (161). Since $k \le n \le 2n-1$ for $n \ge 1$, each quotient $(4n-1)!!/(2k+1)!!$ is a product of consecutive odd integers and hence an integer; the binomial coefficients, the C_k and the remaining products are integers as well. Consequently $2^{2n-1}R_{2n}$ is an integer, so the only prime that can occur in the denominator of R_{2n} is 2. By (162) that denominator is odd, so it equals 1 and R_{2n} is an integer, necessarily odd.

Positivity. Every factor in (158) is positive, and $F(2^{-n}) > 0$ because F is strictly increasing on $[0, 1]$. □

Remark 8.13 (Extended value list). The first nine values are

$$1,\ 5,\ 15,\ 1001,\ 5985,\ 2853675,\ 26261235,\ 72808620885,\ 915304354965, \tag{163}$$

extending the table of Appendix A by $R_8 = 72808620885$ and $R_9 = 915304354965$. All nine were recomputed independently in exact rational arithmetic from the definition (158), from the half-moment form (159), and, according to the parity of the index, from (160) or (161); the three routes agree, and every value is odd. The check was carried out for all $n \le 24$.

9 Exact evaluation at arbitrary dyadic rationals

The rationality theorem for dyadic arguments is much stronger than the statement "F happens to take rational values there." It comes with two complementary exact algorithms:

1. a closed finite formula involving Thue–Morse signs and the rational moments c_k;
2. a bit-recursive Taylor algorithm whose number of recursive calls is the number of set bits in the numerator.

The first is best for proofs; the second is best for computation.

9.1 The closed Thue–Morse formula

Theorem 9.1 (Exact dyadic formula)**.** *Let $n, a \in \mathbb{N}$ with $0 \le a \le 2^n$. Then*

$$\boxed{F\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} (-1)^{\mathrm{wt}_2(h)} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2a-2h-1)^{n-2k}.} \tag{164}$$

An empty outer sum has value 0.

Proof. Write $x = a/2^n$ and use $F(x) = \mathrm{up}(x-1)$. Since up and all its derivatives vanish at -1, Taylor's theorem with integral remainder gives

$$\mathrm{up}(x-1) = \frac{1}{n!} \int_{-1}^{x-1} (x-1-t)^n \, \mathrm{up}^{(n+1)}(t) \, \mathrm{d}t. \tag{165}$$

Insert the exact derivative identity (91):

$$\mathrm{up}^{(n+1)}(t) = 2^{\binom{n+2}{2}} \mathcal{F}(2^{n+1}(t+1)).$$

Put $s = 2^{n+1}(t+1)$. As t runs from -1 to $x-1$, s runs from 0 to $2a$. Split this interval into the blocks $[2h, 2h+2]$, $0 \le h < a$. On the hth block, (12) gives

$$\mathcal{F}(s) = (-1)^{\mathrm{wt}_2(h)} \, \mathrm{up}(s-2h-1).$$

The substitution $u = s-2h-1$ maps the block to $[-1,1]$. Gathering the powers of two gives

$$F(a/2^n) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} (-1)^{\mathrm{wt}_2(h)} \int_{-1}^{1} (2a-2h-1-u)^n \, \mathrm{up}(u) \, \mathrm{d}u. \tag{166}$$

Expand the nth power. Odd powers of u integrate to zero because up is even, while

$$\int_{-1}^{1} u^{2k} \, \mathrm{up}(u) \, \mathrm{d}u = c_k.$$

The surviving even terms are exactly the inner sum in (164). □

Every quantity on the right of (164) is rational, so dyadic rationality is immediate. More importantly, the formula lives entirely in $\mathbb{Q}$ and can be evaluated without roundoff.

Theorem 9.2 (Global dyadic formula)**.** *For all $n, a \in \mathbb{N}$ the signed global extension satisfies*

$$\boxed{\mathcal{F}\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2a-2h-1)^{n-2k}.} \tag{167}$$

No upper bound on a is imposed. An empty outer sum has value 0.

Proof. By (90) the mth derivative of $\mathcal{F}$ is $2^{\binom{m+1}{2}} \mathcal{F}(2^m x)$, and $\mathcal{F}$ vanishes on $(-\infty, 0]$. Hence $\mathcal{F}^{(m)}(0) = 0$ for every $m \ge 0$, and Taylor's theorem with integral remainder at the base point 0, applied through order n, gives for $x \ge 0$

$$\mathcal{F}(x) = \frac{1}{n!} \int_0^x (x-t)^n \mathcal{F}^{(n+1)}(t) \, \mathrm{d}t. \tag{168}$$

Put $x = a/2^n$ and insert $\mathcal{F}^{(n+1)}(t) = 2^{\binom{n+2}{2}}\mathcal{F}(2^{n+1}t)$. The substitution $u = 2^{n+1}t$ carries $[0, x]$ onto $[0, 2a]$ and produces $(x-t)^n = 2^{-n(n+1)}(2a-u)^n$ together with $\mathrm{d}t = 2^{-(n+1)}\,\mathrm{d}u$. Since

$$\binom{n+2}{2} - n(n+1) - (n+1) = -\binom{n+1}{2},$$

this is exactly

$$\mathcal{F}\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!}\int_0^{2a}(2a-u)^n\mathcal{F}(u)\,\mathrm{d}u. \tag{169}$$

Split $[0, 2a]$ into the a blocks $[2h, 2h+2]$ with $0 \le h < a$. On the hth block (12) gives $\mathcal{F}(u) = \varepsilon_h\,\mathrm{up}(u - 2h - 1)$, and the substitution $v = u - 2h - 1$ maps that block onto $[-1, 1]$:

$$\int_{2h}^{2h+2}(2a-u)^n\mathcal{F}(u)\,\mathrm{d}u = \varepsilon_h\int_{-1}^{1}(2a-2h-1-v)^n\,\mathrm{up}(v)\,\mathrm{d}v.$$

Expand the nth power by the binomial theorem. The function up is even, so every odd moment vanishes, while $\int_{-1}^{1} v^{2k}\,\mathrm{up}(v)\,\mathrm{d}v = c_k$ and the surviving sign $(-1)^{2k}$ is $+1$. Summing over h gives (167). □

Two special cases are the ones used in practice.

Corollary 9.3 (Bounded case)**.** *If $0 \le a \le 2^n$ then $\mathcal{F}(a/2^n) = F(a/2^n)$, so (167) reduces to (164).*

Proof. The two functions agree on $[0, 1]$ and $a/2^n \in [0, 1]$. □

Corollary 9.4 (Bump case)**.** *Let $n \in \mathbb{N}$ and $q \in \mathbb{Z}$. If $|q| < 2^n$, then*

$$\mathrm{up}\left(\frac{q}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!}\sum_{h=0}^{q+2^n-1}\varepsilon_h\sum_{k=0}^{\lfloor n/2\rfloor}\binom{n}{2k}c_k\left(2q + 2^{n+1} - 2h - 1\right)^{n-2k}, \tag{170}$$

that is, (167) evaluated at $a = q + 2^n$. If $|q| \ge 2^n$, then $\mathrm{up}(q/2^n) = 0$.

Proof. Taking $b = 0$ in (12) gives $\mathcal{F}(x) = \mathrm{up}(x-1)$ for $0 \le x \le 2$. With $x = 1 + q/2^n = (q+2^n)/2^n$ and $|q| < 2^n$ we have $0 < x < 2$, so $\mathrm{up}(q/2^n) = \mathcal{F}((q+2^n)/2^n)$, and (167) with $a = q + 2^n$ applies. The last statement is the support condition $\operatorname{supp}\mathrm{up} = [-1, 1]$. □

Corollary 9.4 is the global form invoked in the proof of the uniform denominator multiplier theorem below. The point is that it genuinely requires Theorem 9.2 and not (164): for $0 < q < 2^n$ the numerator $a = q + 2^n$ exceeds 2^n, so the bounded statement does not apply. At the endpoints $|q| = 2^n$ the two clauses agree, both giving 0; for $q = -2^n$ the outer sum is empty and for $q = 2^n$ the value is $\mathcal{F}(2) = 0$.

Remark 9.5 (Why the sum is finite). Finiteness has two independent sources, and neither involves a limit. Horizontally, only the a complete translated bumps indexed by $0 \le h < a$ meet the integration range in (169); the locally finite sum (11) therefore contributes finitely many terms. Vertically, expanding $(2a - 2h - 1 - v)^n$ produces only powers v^j with $j \le n$, and all odd j integrate to zero, leaving the $\lfloor n/2\rfloor + 1$ even moments $c_0, \ldots, c_{\lfloor n/2\rfloor}$. Every one of these is an explicit rational number produced by (110), so (167) is an exact algorithm over $\mathbb{Q}$ with no roundoff and no truncation. It is not, however, an efficient one: the outer range has length a, whereas the bit recursion of Algorithm 9.12 needs only $\mathrm{wt}_2(a)$ steps.

Example 9.6 (The closed formula evaluated)**.** The formula is stated, globalized, used to bound denominators, and proved finite above without ever being evaluated. Three small cases, using $c_0 = 1$ and $c_1 = 1/9$.

For $n = 2$, $a = 1$ the outer sum has the single term $h = 0$ with $\varepsilon_0 = +1$ and $2a - 2h - 1 = 1$, so

$$F\left(\frac{1}{4}\right) = \frac{2^{-3}}{2!}\left[\binom{2}{0}c_0 \cdot 1^2 + \binom{2}{2}c_1 \cdot 1^0\right] = \frac{1}{16}\left(1 + \frac{1}{9}\right) = \frac{5}{72}.$$

For $n = 3$, $a = 1$ the same single term gives

$$F\left(\frac{1}{8}\right) = \frac{2^{-6}}{3!}\left[\binom{3}{0}c_0 \cdot 1^3 + \binom{3}{2}c_1 \cdot 1^1\right] = \frac{1}{384}\left(1 + \frac{3}{9}\right) = \frac{1}{288}.$$

For $n = 3$, $a = 3$ the Thue–Morse signs finally do some work: $h = 0, 1, 2$ carry $\varepsilon_h = +1, -1, -1$ and $2a - 2h - 1 = 5, 3, 1$, so

$$\begin{aligned} F\left(\frac{3}{8}\right) &= \frac{2^{-6}}{3!}\left[\left(5^3 + 3 \cdot \frac{5}{9}\right) - \left(3^3 + 3 \cdot \frac{3}{9}\right) - \left(1^3 + 3 \cdot \frac{1}{9}\right)\right] \\ &= \frac{1}{384}\left[\left(125 + \frac{5}{3}\right) - 28 - \left(1 + \frac{1}{3}\right)\right] = \frac{1}{384} \cdot \frac{292}{3} = \frac{73}{288}, \end{aligned}$$

in agreement with the level-4 table below. The third case is the smallest one in which the signs are not all positive, and it shows the characteristic feature of the formula: the individual bracketed terms are much larger than the answer, and the cancellation among them is what produces a value in $[0, 1]$.

9.2 Thue–Morse cancellation

The mechanism that makes the dyadic formulas simplify is the Prouhet cancellation carried by the Thue–Morse signs.

Lemma 9.7 (Affine power cancellation)**.** *If $d < b$, then for all $X, Y \in \mathbb{Q}$,*

$$\sum_{h=0}^{2^b-1} (-1)^{\mathrm{wt}_2(h)}(X + Yh)^d = 0. \tag{171}$$

Proof. It is enough to prove the claim for h^j, $0 \le j \le d$, and then expand the affine power. Let

$$S_{b,j} = \sum_{h=0}^{2^b-1} (-1)^{\mathrm{wt}_2(h)} h^j.$$

Split the sum for $S_{b+1,j}$ at 2^b and use $\varepsilon_{2^b+h} = -\varepsilon_h$:

$$\begin{aligned} S_{b+1,j} &= \sum_{h<2^b} \varepsilon_h\left(h^j - (2^b + h)^j\right) \\ &= -\sum_{r=0}^{j-1} \binom{j}{r} 2^{b(j-r)} S_{b,r}. \end{aligned}$$

If $j < b + 1$, every index r on the right satisfies $r < b$, so the induction hypothesis makes all terms zero. The base case is immediate. ∎

A compact generating-function version is

$$\sum_{h<2^b} (-1)^{\mathrm{wt}_2(h)} \mathrm{e}^{ht} = \prod_{j=0}^{b-1} (1 - \mathrm{e}^{2^j t}), \tag{172}$$

whose right side has a zero of order b at $t = 0$. Differentiating fewer than b times at zero gives the same cancellation.

This lemma explains why apparently large translated sums in (164) collapse to short Taylor blocks when the numerator crosses a binary boundary.

The cancellation lemma states that the first b moments of the signed block vanish. The first surviving moment has an equally clean closed value, and that value is the constant that governs every leading coefficient in the q-binomial analysis of Section 9.9.

Proposition 9.8 (The first surviving Thue–Morse moment). *For every $k \geq 0$,*

$$\boxed{\sum_{r=0}^{2^k-1} \varepsilon_r\, r^k = (-1)^k 2^{\binom{k}{2}} k!.} \tag{173}$$

Proof. By (172),

$$\sum_{r<2^k} \varepsilon_r \mathrm{e}^{rt} = \prod_{j=0}^{k-1} (1 - \mathrm{e}^{2^j t}). \tag{174}$$

Each factor is an entire function with a simple zero at $t = 0$, and

$$1 - \mathrm{e}^{2^j t} = -2^j t + \mathcal{O}(t^2) \qquad (t \to 0),$$

so the product has a zero of order exactly k at the origin, with

$$\prod_{j=0}^{k-1} (1 - \mathrm{e}^{2^j t}) = \Big(\prod_{j=0}^{k-1} (-2^j)\Big) t^k + \mathcal{O}(t^{k+1}) = (-1)^k 2^{\binom{k}{2}} t^k + \mathcal{O}(t^{k+1}),$$

because $\sum_{j=0}^{k-1} j = \binom{k}{2}$. The left side of (174) is the exponential generating function

$$\sum_{d\geq 0} \Big(\sum_{r<2^k} \varepsilon_r r^d\Big) \frac{t^d}{d!}.$$

Comparing coefficients of t^d for $d < k$ recovers (171); comparing coefficients of t^k gives

$$\frac{1}{k!} \sum_{r<2^k} \varepsilon_r r^k = (-1)^k 2^{\binom{k}{2}}.$$

The case $k = 0$ is the empty product, both sides being 1. □

Corollary 9.9 (Translation invariance). *For every $k \geq 0$ and every Q in a field of characteristic zero,*

$$\sum_{r=0}^{2^k-1} \varepsilon_r\, (r + Q)^k = (-1)^k 2^{\binom{k}{2}} k!. \tag{175}$$

Proof. Expand $(r + Q)^k = \sum_{d=0}^{k} \binom{k}{d} Q^{k-d} r^d$ and interchange the two finite sums. For $d < k$ the inner sum $\sum_{r<2^k} \varepsilon_r r^d$ vanishes by (171), so only $d = k$ survives, with coefficient 1. Apply (173). □

Thus the value is unchanged if r is replaced by $r+Q$ for any constant Q, precisely because all the lower moments vanish: a shift can only alter the coefficients of degrees below the top, and those coefficients are already zero. The centered choice $Q=-2^k$ gives

$$\sum_{r=0}^{2^k-1} \varepsilon_r\,(r-2^k)^k = (-1)^k 2^{\binom{k}{2}} k!, \tag{176}$$

which is exactly the constant term of the block factorization (208) established later: dividing the centered block sum $C_k(X)$ by X^k leaves the value $(-1)^k 2^{\binom{k}{2}}$ at $X=0$, because the remaining exponential factors there all tend to 1. This also shows that the order of vanishing in (206) is exactly k and not more, which is what makes the removal of the forced zero legitimate.

Remark 9.10 (Numerical check). The Thue–Morse signs for $0\le r\le 7$ are $+,-,-,+,-,+,+,-$. Reading off the first three cases of (173):

$$\begin{aligned}
k=1:&\quad 0-1=-1, && (-1)^1 2^0 1! = -1;\\
k=2:&\quad 0-1-4+9=4, && (-1)^2 2^1 2! = 4;\\
k=3:&\quad 0-1-8+27-64+125+216-343=-48, && (-1)^3 2^3 3! = -48.
\end{aligned}$$

These were recomputed in exact integer arithmetic, together with $k=4,\dots,7$, whose values are 1536, −122 880, 23 592 960, and −10 569 646 080, in agreement with (173). The shift invariance (175) was checked in exact rational arithmetic for $k\le 5$ at $Q=-2^k$, $Q=1/2$, and $Q=7$, and the vanishing of all lower moments for $k\le 6$.

9.3 A Taylor block that removes the leading bit

Suppose $0<x<1$, and let r be the unique integer such that

$$2^{-r}\le x<2^{-r+1}. \tag{177}$$

Put $y=x-2^{-r}$. Then $0\le y<2^{-r}$.

Theorem 9.11 (Exact Taylor-block recursion)**.** *With the notation above,*

$$F(x) = -F(y) + \sum_{j=0}^{r} \frac{2^{\binom{j+1}{2}} F(2^{j-r})}{j!}\, y^j. \tag{178}$$

Proof. Apply Taylor's theorem at the point 2^{-r} through degree r:

$$F(2^{-r}+y) = \sum_{j=0}^{r} \frac{F^{(j)}(2^{-r})}{j!} y^j + R_r(y). \tag{179}$$

The derivative scaling formula gives

$$F^{(j)}(2^{-r}) = 2^{\binom{j+1}{2}} F(2^{j-r}), \qquad 0\le j\le r. \tag{180}$$

It remains to identify the integral remainder. The scaling equation carries the interval $[2^{-r},2^{-r}+y]$ into the adjacent signed translate of the interval $[0,y]$; on that translate the Thue–Morse sign is negative. Reversing the substitutions in the derivative identity shows

$$R_r(y) = -\frac{1}{r!}\int_0^y (y-t)^r F^{(r+1)}(t)\,\mathrm{d}t = -F(y),$$

where the last equality is Taylor's integral formula at the flat point 0. Substitution in (179) proves (178). ☐

Using (136), the coefficients may be written entirely in terms of the half moments:

$$\frac{2^{\binom{j+1}{2}} F(2^{j-r})}{j!} = \frac{2^{\binom{j+1}{2}-\binom{r-j}{2}} d_{r-j}}{j!(r-j)!}. \tag{181}$$

This is the form that appears in the original arithmetic recurrence.

9.4 The bit-recursive evaluator

Let $x = a/2^e$ with $0 < a < 2^e$. Write

$$b = \lfloor \log_2 a \rfloor, \qquad r = e - b, \qquad q = a - 2^b. \tag{182}$$

Then 2^{-r} is the leading binary term of x and

$$y = \frac{q}{2^e}. \tag{183}$$

Applying (178) reduces $F(a/2^e)$ to $F(q/2^e)$; the highest set bit of the numerator has disappeared.

Algorithm 9.12 (Exact dyadic evaluator)**.** Precompute $V_j = F(2^{-j})$ for $0 \le j \le e$ from (136). Define $\mathrm{Eval}(e, a)$ recursively by

$$\mathrm{Eval}(e,a) = \begin{cases} 0, & a \le 0, \\ 1, & a \ge 2^e, \\ T_r(q/2^e) - \mathrm{Eval}(e,q), & 0 < a < 2^e, \end{cases}$$

where b, r, q are given by (182) and

$$T_r(z) = \sum_{j=0}^{r} \frac{2^{\binom{j+1}{2}} V_{r-j}}{j!} z^j. \tag{184}$$

Evaluate T_r by Horner's rule.

A literal pseudocode version is:

```
Eval(e, a, V):
    if a <= 0:      return 0
    if a >= 2^e:    return 1

    b = floor_log2(a)
    r = e - b
    q = a - 2^b
    y = q / 2^e

    p = V[0]                        // highest Taylor coefficient, j = r
    for m = 1 .. r:
        j = r - m
        p = V[m] + (2^(j+1) * y / (j+1)) * p

    return p - Eval(e, q, V)
```

The displayed loop is one convenient Horner ordering; any exact evaluation of (184) is equivalent.

Theorem 9.13 (Correctness and termination). *For every $e \in \mathbb{N}$ and $a \in \mathbb{Z}$,*

$$\mathrm{Eval}(e, a) = F(a/2^e) \tag{185}$$

under the bounded convention $F = 0$ to the left and $F = 1$ to the right. The recursion makes exactly $\mathrm{wt}_2(a)$ *nontrivial calls when $0 \le a < 2^e$.*

Proof. The boundary clauses agree with (1) and (2). In the interior, (178) is precisely the recursive clause. Since $0 \le q < a$, strong induction proves correctness and termination. Replacing a by q clears its highest set bit and leaves all lower bits unchanged, so the number of nontrivial steps is the popcount $\mathrm{wt}_2(a)$. □

Precomputing the reciprocal-power table costs $O(e^2)$ rational operations using the triangular recurrence for d_n. A single value then costs $O(e\ \mathrm{wt}_2(a))$ arithmetic operations in a straightforward implementation. This is much smaller than the a outer summands in (164) when a is large.

Example 9.14 (The value at 5/16). Here $a = 5 = 2^2 + 1$, $e = 4$, so $r = 2$, $q = 1$, and $y = 1/16$. Formula (178) gives

$$\begin{aligned} F(5/16) &= F(1/4) + 2F(1/2)y + \frac{2^3 F(1)}{2} y^2 - F(1/16) \\ &= \frac{5}{72} + \frac{1}{16} + \frac{1}{64} - \frac{143}{2073600} \\ &= \boxed{\frac{305857}{2073600}}. \end{aligned}$$

Only two recursive calls are needed, corresponding to the two set bits of 5.

9.5 Reduction of an arbitrary dyadic argument

Algorithm 9.12 evaluates the bounded function F on $[0, 1]$. The signed global extension is defined on all of $\mathbb{R}$ and is unbounded there, so a total evaluator for $\mathcal{F}$ needs one preliminary step: fold the argument back into the unit interval and record the sign that the fold costs. The fold is triangular in the sense that it is a reduction modulo 2 followed by a reflection.

Theorem 9.15 (Triangular reduction). *Let $n \in \mathbb{N}$ and put $s = 2^n$. Let $a \in \mathbb{N}$ and write*

$$a = 2sB + R, \qquad B = \left\lfloor \frac{a}{2s} \right\rfloor, \qquad 0 \le R < 2s, \tag{186}$$

and set

$$C = \begin{cases} R, & R \le s, \\ 2s - R, & R > s. \end{cases} \tag{187}$$

Then $0 \le C \le s$ and

$$\boxed{\mathcal{F}\left(\frac{a}{2^n}\right) = \varepsilon_B\, F\left(\frac{C}{2^n}\right).} \tag{188}$$

For $a \in \mathbb{Z}$ with $a < 0$ one has $\mathcal{F}(a/2^n) = 0$.

Proof. Write $x = a/2^n = 2B + R/s$. Since $0 \le R < 2s$ we have $2B \le x < 2B + 2$, so x lies in the Bth block and (12) gives

$$\mathcal{F}(x) = \varepsilon_B\, \mathrm{up}(x - 2B - 1) = \varepsilon_B\, \mathrm{up}\left(\frac{R}{s} - 1\right).$$

Put $t = R/s - 1$, so that $-1 \le t < 1$. The fold relation (7) says $\mathrm{up}(t) = F(t+1)$ for $t \le 0$ and $\mathrm{up}(t) = F(1-t)$ for $t > 0$; on $[-1,1]$ both clauses read $\mathrm{up}(t) = F(1-|t|)$. If $R \le s$ then $|t| = 1 - R/s$ and $F(1-|t|) = F(R/s)$. If $R > s$ then $|t| = R/s - 1$ and $F(1-|t|) = F(2-R/s) = F((2s-R)/2^n)$. In both cases the argument is $C/2^n$ with C as in (187), and $0 \le C \le s$ is immediate from $0 \le R < 2s$. The final statement is the vanishing of $\mathcal{F}$ on the negative half-line. ☐

Combining (188) with Algorithm 9.12 gives a total exact evaluator for $\mathcal{F}$ at every dyadic rational: reduce with (186)–(187), compute the block sign $\varepsilon_B = (-1)^{\mathrm{wt}_2(B)}$ from the binary digits of $B = \lfloor a/2^{n+1} \rfloor$, and call $\mathrm{Eval}(n, C)$. The cost of the reduction is one shift, one mask, and one popcount.

The reduction is compatible with unreduced presentations. Replacing (n,a) by $(n+1, 2a)$ doubles s and R, leaves B unchanged, and doubles C; the right side of (188) is therefore unchanged, as it must be.

Remark 9.16 (Recognizing dyadic arguments exactly). A reduced rational p/q with $q > 0$ is dyadic exactly when q is a power of two, and in that case $n = \log_2 q$ and $a = p$ are read off directly. An exact rational interface to F, up, and $\mathcal{F}$ therefore tests a single integer condition on the reduced denominator, returns "not dyadic" when it fails, and otherwise invokes the signed integer evaluator above. No floating-point recognition of powers of two is involved anywhere, so the interface is exact on every input and cannot be defeated by rounding in the recognition step.

9.6 A complete level-4 table

The exact values at denominator 16 are shown below. The second half follows from $F(1-x) = 1 - F(x)$.

x	$F(x)$	x	$F(x)$
0	0	1/2	1/2
1/16	143/2073600	9/16	1295857/2073600
1/8	1/288	5/8	215/288
3/16	46657/2073600	11/16	1767743/2073600
1/4	5/72	3/4	67/72
5/16	305857/2073600	13/16	2026943/2073600
3/8	73/288	7/8	287/288
7/16	777743/2073600	15/16	2073457/2073600

9.7 The odd numerators at level 5

One tier further is worth recording, because every numerator at level 4 has at most two set bits, so the level-4 table never exercises a three-step bit recursion. At level 5 the values with odd numerator all share the denominator 33 177 600, and the even numerators reduce to earlier levels. Writing $F(a/32) = N_a/33\,177\,600$:

a	N_a	a	N_a
1	19	9	3 470 381
3	25 219	11	6 555 581
5	334 781	13	10 393 219
7	1 396 781	15	14 515 219

Each fraction is already in lowest terms, and the reflection $N_{32-a} = 33\,177\,600 - N_a$ supplies the upper half. The numerator $a = 7$, with three set bits, is the first entry that costs the bit recursion three reduction steps; $a = 15$ costs four. These make better regression tests than the level-4 values for exactly that reason.

9.8 Denominator bounds

Let

$$M_m = \prod_{j=1}^{m} (2^{2j} - 1), \qquad (2m + 1)!! = 1 \cdot 3 \cdots (2m + 1). \tag{189}$$

Theorem 9.17 (Uniform denominator multiplier)**.** *Let $n \geq 1$ and $a \in \mathbb{Z}$ with $|a| < 2^n$. Then*

$$\mathrm{up}(a/2^n)\; n!\, 2^{\binom{n+1}{2}} (2\lfloor n/2 \rfloor + 1)!!\; M_{\lfloor n/2 \rfloor} \tag{190}$$

is an integer.

Proof. Apply the global dyadic form of (164). Replace each c_k by (112). For $k \leq \lfloor n/2 \rfloor$, both the odd double factorial in the denominator of c_k and its even-Mersenne product divide the two common products in (190). The remaining factors are integers. Thus every summand becomes integral after multiplication by the displayed common factor. □

Let Δ_n be the least common multiple of the reduced denominators of the values $\mathrm{up}(a/2^n)$ with odd a in the support. The theorem gives the explicit divisibility bound

$$\Delta_n \mid n!\, 2^{\binom{n+1}{2}} (2\lfloor n/2 \rfloor + 1)!! M_{\lfloor n/2 \rfloor}. \tag{191}$$

A complementary lower statement is available at no cost, and it is what makes the normalization of the conjecture below meaningful.

Proposition 9.18 (The reciprocal power is one of the samples)**.** *For every $n \geq 1$ the reduced denominator of $F(2^{-n})$ divides Δ_n. Consequently*

$$v_2(\Delta_n) \geq \binom{n}{2} + 1 + v_2(n!). \tag{192}$$

Proof. By the fold (8), $F(2^{-n}) = \mathrm{up}(1 - 2^{-n}) = \mathrm{up}\big((2^n - 1)/2^n\big)$, and the numerator $2^n - 1$ is odd. So $F(2^{-n})$ is itself one of the samples whose denominators Δ_n is the least common multiple of, and its reduced denominator therefore divides Δ_n. The valuation statement follows from (156), which gives $v_2\big(F(2^{-n})\big) = -\binom{n}{2} - 1 - v_2(n!)$, so the reduced denominator contains exactly that power of two. □

Together, (191) and (192) trap Δ_n between an explicit multiple and an explicit power of two. In particular the power of two in Δ_n is pinned from below, which is what the normalizer of the conjecture below relies on: dividing Δ_n by $2^{\binom{n}{2}}$ is only informative because that much of a power of two is known to be present.

The exact closed form of Δ_n is subtler. The source paper records a conjectural factorization; the formal development correctly retains it as a conjecture rather than a theorem.

9.8.1 Numerical values of the level denominators

The divisibility bound (191) is easy to prove and far from sharp. It is worth seeing the actual denominators. For $0 \le n \le 9$,

$$\begin{aligned}\Delta_0, \dots, \Delta_9 = 1, \quad 2, \quad 72, \quad 288, \quad 2\,073\,600, \quad 33\,177\,600,\\ 2\,809\,213\,747\,200, \quad 179\,789\,679\,820\,800,\\ 704\,200\,217\,922\,109\,440\,000,\\ 180\,275\,255\,788\,060\,016\,640\,000.\end{aligned} \tag{193}$$

The next two levels continue

$$\begin{aligned}\Delta_{10} &= 6\,231\,974\,256\,792\,696\,936\,191\,754\,240\,000,\\ \Delta_{11} &= 6\,381\,541\,638\,955\,721\,662\,660\,356\,341\,760\,000.\end{aligned} \tag{194}$$

Write B_n for the proved multiplier of (191),

$$B_n = n!\, 2^{\binom{n+1}{2}} (2\lfloor n/2 \rfloor + 1)!!\, M_{\lfloor n/2 \rfloor}, \tag{195}$$

with M_m as in (189). The quotients B_n/Δ_n are

n	B_n/Δ_n	n	B_n/Δ_n
0	1	6	160
1	1	7	2 240
2	2	8	2 688
3	12	9	48 384
4	8	10	161 280
5	80	11	3 548 160

The bound is exact only at $n = 0, 1$. Its slack is not bounded: it already exceeds 3.5×10^6 at $n = 11$ and continues to grow. The bound is therefore useful as a divisibility certificate and as a source of the exact power of two through (156), but it does not identify Δ_n.

Remark 9.19 (Recomputation). Every value in (193) and (194) was recomputed independently in exact rational arithmetic for this article, not copied from the source paper. The reciprocal-power table $V_j = F(2^{-j})$ was generated from the triangular recurrence (145) and cross-checked against the composition formula (149); every level-n value $\mathrm{up}(a/2^n)$ with odd a was then produced by the bit recursion (178), reduced to lowest terms, and the least common multiple of the resulting denominators taken. As a control, the recomputed level-4 values reproduce the table of (143) and the complete level-4 table above. All ten published values agree with the recomputation, and the divisibility (191) was verified numerically for $0 \le n \le 11$.

9.8.2 The denominator conjecture

The list (193) has visible structure: the terms come in equal consecutive pairs after a power of two is removed. The right normalization is

$$\mathfrak{a}_n = 2^{-\binom{n}{2}} \Delta_n, \tag{196}$$

which is not to be confused with the composition normalization A_n of Section 8.1. Its first values are

$$\begin{aligned}\mathfrak{a}_0, \dots, \mathfrak{a}_{11} = 1, \quad 2, \quad 36, \quad 36, \quad 32\,400, \quad 32\,400,\\ 85\,730\,400, \quad 85\,730\,400,\\ 2\,623\,350\,240\,000, \quad 2\,623\,350\,240\,000,\\ 177\,123\,361\,504\,320\,000, \quad 177\,123\,361\,504\,320\,000.\end{aligned} \tag{197}$$

Conjecture 9.20 (Arias de Reyna denominator pattern)**.** *With $\mathfrak{a}_n$ as in (196):*

(i) $\mathfrak{a}_{2n} = \mathfrak{a}_{2n+1}$ *for every* $n \geq 1$*;*

(ii) $2(2n-1)!$ *divides* $K_n := \mathfrak{a}_{2n-1}$ *for every* $n \geq 1$*;*

(iii) *the quotient*

$$H_n = \frac{K_n}{2(2n-1)!} \tag{198}$$

is an odd integer for every $n \geq 1$.

Equivalently, if the conjecture holds then for every $n \geq 1$

$$\Delta_{2n-1} = 2^{1+\binom{2n-1}{2}}(2n-1)!\,H_n, \qquad \Delta_{2n} = 2^{1+\binom{2n}{2}}(2n+1)!\,H_{n+1}, \tag{199}$$

so that the entire sequence (Δ_n) would be determined by the single odd sequence (H_n) together with explicit powers of two and factorials. The first values are

$$H_1, \dots, H_6 = 1, \quad 3, \quad 135, \quad 8\,505, \quad 3\,614\,625, \quad 2\,218\,656\,825. \tag{200}$$

The restriction $n \geq 1$ in part (i) is essential and not a convention: $\mathfrak{a}_0 = 1$ while $\mathfrak{a}_1 = 2$, so the pairing fails at the bottom of the range.

The conjecture is due to Arias de Reyna [8]. The proved bound (191) and the exact 2-adic valuation (156) constrain it — the latter pins down the power of two in each reduced denominator, which is why the normalization (196) is the natural one — but neither settles the odd part, and no proof of (i), (ii), or (iii) is known. The formal development records the statement as a conjecture and derives nothing from it.

Remark 9.21 (Verification)*.* The values (197) and (200) were computed for this article from the recomputed denominators of Remark 9.19, in exact integer arithmetic. Part (i) was verified for $1 \leq n \leq 5$, that is, for the five pairs $(\mathfrak{a}_2, \mathfrak{a}_3)$ through $(\mathfrak{a}_{10}, \mathfrak{a}_{11})$. Part (ii) was verified for $1 \leq n \leq 6$, the largest case being $2 \cdot 11! = 79\,833\,600$ dividing $K_6 = \mathfrak{a}_{11} = 177\,123\,361\,504\,320\,000$. Every H_n in (200) is an odd integer, as (iii) requires, and the two presentations in (199) were checked against (193) for $1 \leq n \leq 5$.

9.9 A q-binomial form, double cancellation, and scalar translation

The formal development derives a second exact dyadic formula organized by Gaussian binomial coefficients. Put

$$(a;q)_n = \prod_{j=0}^{n-1}(1 - aq^j), \tag{201}$$

and write $\left[{n \atop k}\right]_q$ for the Gaussian or q-binomial coefficient [15, 16, 17, 18]. The base $q = 1/2$ is forced by the dilation grid $1, 2, 4, \dots$: the corresponding Pochhammer product is the node polynomial for reciprocal dyadic scales.

The finite half-base q-binomial theorem is

$$(z;1/2)_n = \sum_{k=0}^{n}(-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} z^k. \tag{202}$$

To see its interpolation content, put

$$\mathfrak{m}_n = \prod_{\ell=1}^{n}(2^\ell - 1), \qquad x_k = -2^k, \qquad w_{n,k} = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}.$$

Then

$$\sum_{k=0}^{n} w_{n,k} x_k^d = \begin{cases} 0, & d < n, \\ \mathfrak{m}_n, & d = n, \end{cases} \qquad \frac{w_{n,k}}{\mathfrak{m}_n} = \frac{1}{\prod_{\ell \neq k}(x_k - x_\ell)}. \tag{203}$$

Thus the normalized Gaussian row is exactly the leading-coefficient functional for the Lagrange interpolant through $-1, -2, \ldots, -2^n$. The normalization also satisfies

$$(1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n. \tag{204}$$

Likewise $\binom{n}{k}_{1/2} = 2^{-k(n-k)} \binom{n}{k}_2$, so the half-base coefficients are power-of-two normalizations of the integer Gaussian coefficients that count k-planes in $\mathbb{F}_2^n$.

Theorem 9.22 (Global q-binomial dyadic formula)**.** *For $m, n \in \mathbb{N}$ and every scalar translation $Q \in \mathbb{C}$,*

$$\mathcal{F}(m/2^n) = \frac{1}{2^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\binom{n}{k}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \times \sum_{r=0}^{m2^k-1} (-1)^{\mathrm{wt}_2(r)} (r - m2^k + Q)^{n+k}. \tag{205}$$

Here the real value on the left is embedded in $\mathbb{C}$. The right side is independent of Q, so the identity also holds for real, rational, and Gaussian-rational shifts. If $0 \le m \le 2^n$, then $\mathcal{F}(m/2^n) = F(m/2^n)$, giving the bounded formula.

The shape (205) was first found experimentally, with the particular shift $Q = 1/2$, and posted as a conjecture in 2019 [19]. The formal development proves it for all $m, n \in \mathbb{N}$, for every complex Q, and for every unreduced representation $m/2^n$ of a dyadic rational.

Proof architecture. There are two nested finite differences. Inside a block of length 2^k, the generating polynomial factors as

$$\sum_{r=0}^{2^k-1} (-1)^{\mathrm{wt}_2(r)} z^r = \prod_{j=0}^{k-1} (1 - z^{2^j}). \tag{206}$$

It has a zero of order k at $z = 1$, so the inner power sum annihilates every polynomial of degree below k. After this forced zero is removed, its degree-n coefficient is a polynomial of degree at most n in the geometric scale node -2^k.

Here is the coefficient calculation explicitly. Normalize the half moments by

$$a_j = \frac{d_j}{j!}, \qquad \mathscr{A}(X) = \sum_{j \ge 0} a_j X^j, \qquad E(X) = \frac{\mathrm{e}^X - 1}{X}.$$

The half-moment recurrence is equivalent, coefficient by coefficient, to

$$\mathscr{A}(2X) = E(X)\mathscr{A}(X). \tag{207}$$

For the centered complete block set

$$C_k(X) = \sum_{r < 2^k} (-1)^{\mathrm{wt}_2(r)} \mathrm{e}^{(r-2^k)X}.$$

Binary factorization gives

$$\frac{C_k(X)}{X^k} = (-1)^k 2^{\binom{k}{2}} \mathrm{e}^{-X} \prod_{j=0}^{k-1} E(-2^j X). \tag{208}$$

Iterating (207) along the negative dyadic orbit gives the same product:

$$\mathscr{A}(-2^k X) = \left(\prod_{j<k} E(-2^j X)\right) \mathscr{A}(-X).$$

Consequently the coefficient of X^n after the forced X^k is removed is, up to the displayed sign and power of two, the value at $z = -2^k$ of

$$[X^n]\,\mathscr{R}(z;X), \qquad \mathscr{R}(z;X) = \frac{\mathrm{e}^{-X}\mathscr{A}(zX)}{\mathscr{A}(-X)}.$$

For fixed n this is a polynomial in z of degree at most n, and its leading coefficient is a_n.

Across the scales, the weights in (203) form the divided-difference stencil on the nodes $-1, -2, \dots, -2^n$. The finite q-binomial theorem says that this stencil annihilates degrees below n and extracts the leading coefficient in degree n. It therefore returns $\mathfrak{m}_n a_n$. Using (204) and $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$ gives the one-block formula. For a general numerator, the finite prefix series

$$B_m(X) = \sum_{h=0}^{m-1} (-1)^{\mathrm{wt}_2(h)} \mathrm{e}^{(m-1-h)X}$$

enters because writing $r = h2^k + s$ factors the long block as $B_m(-2^k X) C_k(X)$. The same divided difference then extracts the coefficient $[X^n](B_m(X)\mathscr{A}(X))$, which is exactly the signed dyadic value $2^{\binom{n}{2}} \mathcal{F}(m/2^n)$.

For translation invariance, package the entire numerator as a polynomial $N_{m,n}(X)$ over $\mathbb{Q}$. Multiplying every block exponential by e^{QX} cannot change its degree-n coefficient, because the aggregated coefficients of all lower degrees have vanished. Thus $N_{m,n}$ is a constant polynomial. Evaluation in any characteristic-zero field proves the rational, real, and complex statements algebraically; no density or continuity argument is involved. Decomposing the long prefix into m blocks of length 2^k supplies the arbitrary numerator m. □

The endpoint cases are literal finite identities. When $m = 0$, every inner range is empty, so both sides are zero. When $n = 0$, the formula is the scale-zero signed block identity. A nonreduced presentation $m/2^n = 2m/2^{n+1}$ changes every finite sum but not its value, because both are identified with the same global function value.

Binary complementation gives a useful raw-coordinate version. Reflection $r \mapsto 2^k - 1 - r$ changes the Thue–Morse sign by $(-1)^k$. Once the exponent $n + k$ is included, the remaining sign is independent of k and is absorbed by $(-2)^{n^2}$. Consequently, for every $n \in \mathbb{N}$ and $Q \in \mathbb{C}$,

$$\boxed{\mathcal{F}(2^{-n}) = \frac{1}{(-2)^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\mathrm{wt}_2(r)} (r+Q)^{n+k}.} \tag{209}$$

Again the left side is embedded in $\mathbb{C}$, and the nested expression is independent of Q.

The q-binomial formula is not competitive with the bit recursion as an evaluator. Its value is structural: the inner difference is adapted to binary blocks, the outer Gaussian difference is adapted to geometric scales, and the same identity survives arbitrary complex scalar translation.

10 The half-base q-calculus behind the dyadic formula

Section 9.9 stated (205) and compressed its mechanism into a single proof architecture. That compression hides the fact that every ingredient of the formula is forced, and that each forcing is an elementary identity. This section supplies the missing mathematics. It develops the finite q-calculus at base $q = 1/2$ from the definitions, proves the geometric interpolation identity (203) that section 9.9 merely asserted, derives the Thue–Morse exponential block factorization behind (206) and (208), and turns the coefficient extraction into a theorem with a complete proof. Translation invariance, arbitrary dyadic numerators, worked numerical examples, and the practical consequences of the formula's catastrophic cancellation follow. Nothing here is analytic: every step before the final identification with a Fabius value is a finite identity in $\mathbb{Q}$ or in a ring of formal power series over $\mathbb{Q}$.

Throughout, $\varepsilon_r = (-1)^{\mathrm{wt}_2(r)}$ is the Thue–Morse sign (10), $\mathfrak{m}_n = \prod_{\ell=1}^{n}(2^\ell - 1)$ is the odd Mersenne product, d_n are the half moments (115), and

$$a_n = \frac{d_n}{n!}, \qquad \mathscr{A}(X) = \sum_{n\geq 0} a_n X^n, \qquad E(X) = \frac{\mathrm{e}^X - 1}{X} = \sum_{j\geq 0} \frac{X^j}{(j+1)!}, \tag{210}$$

as in section 9.9. In this normalization the generating-function equation (117) is exactly the formal refinement identity (207), namely $\mathscr{A}(2X) = E(X)\mathscr{A}(X)$, the half-moment recurrence (119) reads

$$(2^n - 1)a_n = \sum_{j=0}^{n-1} \frac{a_j}{(n-j+1)!} \qquad (n \geq 1), \qquad a_0 = 1, \tag{211}$$

and the reciprocal-power theorem (136) reads $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$. The first values are $a_0 = 1$, $a_1 = 1/2$, $a_2 = 5/36$, $a_3 = 1/36$, $a_4 = 143/32400$, $a_5 = 19/32400$, $a_6 = 1153/17146080$.

One analytic fact is worth recording before the finite theory begins, because it shows that the geometric grid of section 10.1 is already visible in the moment series itself, and not only in the interpolation stencil built later. By (210) and (116) the two normalizations agree, $\mathscr{A}(z) = G(z)$, so $\mathscr{A}$ is an entire function and not merely a formal series.

Proposition 10.1 (Entire dyadic product for the moment series)**.** *For every $z \in \mathbb{C}$,*

$$\mathscr{A}(z) = \prod_{j=1}^{\infty} E\left(\frac{z}{2^j}\right) = \prod_{j=1}^{\infty} \frac{\mathrm{e}^{z/2^j} - 1}{z/2^j}, \tag{212}$$

each factor being given its removable value 1 at $z = 0$. The product converges absolutely and uniformly on every disk $|z| \leq R$, so its limit is entire.

Proof. Each factor is entire, since $E(w) = \sum_{i\geq 0} w^i/(i+1)!$. For $|w| \leq R$,

$$|E(w) - 1| \leq \sum_{i\geq 1} \frac{|w|^i}{(i+1)!} \leq |w| \sum_{i\geq 1} \frac{|w|^{i-1}}{(i-1)!} = |w|\mathrm{e}^{|w|},$$

using $(i+1)! \geq (i-1)!$. Taking $w = z/2^j$ with $|z| \leq R$ gives $|E(z/2^j) - 1| \leq R\mathrm{e}^R 2^{-j}$, and $\sum_{j\geq 1} R\mathrm{e}^R 2^{-j} = R\mathrm{e}^R < \infty$. A product $\prod_j (1 + u_j)$ of holomorphic factors whose deviations u_j are uniformly summable on a compact set converges absolutely and uniformly there, and its limit is holomorphic; since R is arbitrary, the limit is entire.

It remains to identify the limit. Putting $z/2$ for z in (117) gives $G(z) = E(z/2)\,G(z/2)$, and iterating J times,

$$G(z) = \left(\prod_{j=1}^{J} E\left(\frac{z}{2^j}\right)\right) G\left(\frac{z}{2^J}\right).$$

The function G is continuous with $G(0) = d_0 = 1$, so $G(z/2^J) \to 1$ uniformly on $|z| \leq R$ as $J \to \infty$. Letting $J \to \infty$ in the display and using $\mathscr{A} = G$ gives (212). □

The identity (362) is the restriction of (212) to the negative real axis, where it is used as a Laplace transform; the statement above is the entire version, valid on all of $\mathbb{C}$, and it costs no extra work. Note also what the product says structurally. Its scales are $1/2, 1/4, 1/8, \ldots$, a geometric progression of ratio $1/2$, and the factor attached to each scale is one and the same function E. Thus $\mathscr{A}$ is a multiplicative object over a base-$1/2$ geometric grid before any Gaussian binomial coefficient has been written down. The half-base q-calculus that follows does not impose that grid on the problem; it reads off a grid the refinement equation (207) has already fixed. The zeros of (212) are transparent as well, and they confirm the identification. The factor $E(z/2^j)$ vanishes exactly at the points $z = 2^{j+1}\pi i\ell$ with $\ell \in \mathbb{Z} \setminus \{0\}$, so $\mathscr{A}$ vanishes precisely on $4\pi i(\mathbb{Z} \setminus \{0\})$, and the number of factors responsible for the zero at $4\pi i m$ is the number of indices $j \geq 1$ with 2^{j-1} dividing m, namely $1 + v_2(|m|)$. That is exactly the multiplicity supplied by the canonical product (76) through (118), so the dyadic product and the sinc product are two factorizations of one entire function, grouped by scale in the first case and by zero in the second.

10.1 Why a geometric calculus should be expected

Ordinary finite differences are adapted to arithmetic progressions. The nth forward difference at equally spaced nodes $0, 1, \ldots, n$ annihilates every polynomial of degree below n and returns $n!$ times the leading coefficient in degree n; its weights are ordinary binomial coefficients, and the identity that certifies the annihilation is the ordinary binomial theorem applied to $(1-z)^n$ at $z = 1$. The whole calculus is a shadow of one symmetry: the translation $x \mapsto x + 1$.

The Fabius equation has no such symmetry. Its defining relation (4) and its global form (90) do not translate the argument; they *dilate* it. Differentiating once replaces x by $2x$, so differentiating n times replaces x by $2^n x$ and accumulates the triangular factor $2^{\binom{n+1}{2}}$. Consequently the natural node set of the problem is not arithmetic but geometric,

$$1,\ 2,\ 4,\ \ldots,\ 2^n, \qquad \text{or, after the sign change dictated by (207),} \qquad -1,\ -2,\ -4,\ \ldots,\ -2^n.$$

Any linear functional that isolates one coefficient of a quantity depending on the scale index k must therefore be a divided difference on *these* nodes, not on $0, 1, \ldots, n$.

The finite-difference calculus of a geometric grid with ratio λ is precisely the finite q-calculus with reciprocal base $q = \lambda^{-1}$. Its node polynomial is the finite q-Pochhammer product (201), since $\prod_{j=0}^{n-1}(1 - z\lambda^{-j}) = (z; \lambda^{-1})_n$ vanishes exactly at $z = 1, \lambda, \ldots, \lambda^{n-1}$; its expansion coefficients are the Gaussian binomial coefficients at that base, by the finite q-binomial theorem; and the endpoint value of the node polynomial at the first node that is not a root supplies the normalization. For the Fabius function $\lambda = 2$, so $q = 1/2$. The base is not a choice of notation and cannot be deformed: it is the reciprocal of the dilation ratio in the refinement equation. The rest of this section makes each of these three roles explicit.

10.2 Finite q-Pochhammer symbols at base one half

Recall (201). For a base q, a parameter a, and $n \in \mathbb{N}$,

$$(a; q)_n = \prod_{j=0}^{n-1}(1 - aq^j), \qquad (a; q)_0 = 1, \tag{213}$$

and, when $|q| < 1$, the infinite counterpart is $(a; q)_\infty = \prod_{j \geq 0}(1 - aq^j)$. For several parameters one writes $(a_1, \ldots, a_s; q)_n = \prod_{i=1}^{s}(a_i; q)_n$. The name "shifted factorial" can mislead: the ordinary Pochhammer symbol $(a)_n = a(a+1)\cdots(a+n-1)$ shifts its argument additively, whereas (213) shifts it multiplicatively. Both objects are the basic coefficient packages of their respective hypergeometric theories [15, 16, 17, 18].

Three elementary rules are used repeatedly. The *step rule*

$$(a;q)_{n+1} = (a;q)_n\,(1-aq^n) \tag{214}$$

is the analogue of $(n+1)! = n!\,(n+1)$. The *splitting rule*

$$(a;q)_{m+n} = (a;q)_m\,(aq^m;q)_n \tag{215}$$

follows by breaking the product after m factors and reindexing. The *inversion rule* extracts $-aq^{-j}$ from each factor of a reciprocal base:

$$(a;q^{-1})_n = (-a)^n q^{-\binom{n}{2}}(a^{-1};q)_n \qquad (aq \neq 0). \tag{216}$$

Indeed $1 - aq^{-j} = -aq^{-j}(1-a^{-1}q^j)$, and the accumulated power of q is $q^{-(0+1+\cdots+(n-1))} = q^{-\binom{n}{2}}$.

The zeros of (213) matter more than anything else here. If $a = q^{-d}$ with $0 \le d < n$, the factor with index $j = d$ is $1 - q^{-d}q^d = 0$, so

$$(q^{-d};q)_n = 0 \qquad (0 \le d < n). \tag{217}$$

At $q = 1/2$ this says $(2^d;1/2)_n = 0$ for $d < n$: up to its nonzero leading coefficient, $(z;1/2)_n$ is the monic polynomial whose root set is exactly the geometric grid $1, 2, 4, \ldots, 2^{n-1}$ generated by dyadic dilation. At the first node that is *not* a root,

$$(2^n;1/2)_n = \prod_{j=0}^{n-1}(1-2^{n-j}) = \prod_{i=1}^{n}(1-2^i) = (-1)^n \mathfrak{m}_n. \tag{218}$$

The classical objects are recovered by a limit. For $q \neq 1$ put

$$[n]_q = \frac{1-q^n}{1-q} = 1 + q + \cdots + q^{n-1}, \qquad [n]_q! = \prod_{j=1}^{n}[j]_q, \tag{219}$$

so that $(q;q)_n = (1-q)^n[n]_q!$, since $(q;q)_n = \prod_{j=1}^{n}(1-q^j)$. As $q \to 1$ one has $[n]_q \to n$ and $[n]_q! \to n!$. More precisely, for fixed real α,

$$\lim_{q \to 1^-} \frac{(q^\alpha;q)_n}{(1-q)^n} = \alpha(\alpha+1)\cdots(\alpha+n-1), \tag{220}$$

because each quotient $(1-q^{\alpha+j})/(1-q)$ tends to $\alpha+j$. This is the sense in which q-Pochhammer symbols deform ordinary rising factorials. The Fabius identities use none of this limiting theory: they live at the fixed nonclassical base $q = 1/2$. The limit is recorded only because it explains the terminology.

Specialize now to $a = q = 1/2$, which is the product occurring in (205):

$$(1/2;1/2)_n = \prod_{j=0}^{n-1}\left(1 - \tfrac{1}{2}\cdot 2^{-j}\right) = \prod_{j=1}^{n}\left(1 - 2^{-j}\right) > 0. \tag{221}$$

Clearing one power of two from each factor gives (204), $(1/2;1/2)_n = 2^{-\binom{n+1}{2}}\mathfrak{m}_n$, and therefore the arithmetic form of the global denominator of (205),

$$2^{n^2}(1/2;1/2)_n = 2^{\binom{n}{2}}\mathfrak{m}_n. \tag{222}$$

The apparently mixed normalizer is a pure power of two times an odd integer. The first values of (222) are $1, 1, 6, 168, 20160, 9999360, 20158709760$ for $n = 0, \ldots, 6$.

The finite products (221) decrease and converge to a positive constant,

$$(1/2;1/2)_\infty = \prod_{j\ge 1}\left(1-2^{-j}\right) = 0.288788095086602421278899721 92\ldots, \tag{223}$$

because $\sum_j 2^{-j} < \infty$. Only the finite products occur in the exact formulas, but the nonzero limit is what makes the two-sided bound $0.2887\ldots \le (1/2;1/2)_n \le 1$ available uniformly in n; it will be used in section 10.13 and again in section 12. It also shows that in (222) the entire growth is carried by the explicit factor 2^{n^2} and not by a vanishing q-product.

10.3 Gaussian binomial coefficients on the dyadic grid

Definition 10.2 (Gaussian coefficient)**.** For $0 \le k \le n$ and $q \ne 1$, the Gaussian or q-binomial coefficient is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_k\,(q;q)_{n-k}} = \frac{[n]_q!}{[k]_q!\,[n-k]_q!}, \tag{224}$$

and $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_q = 0$ when $k > n$ or $k < 0$.

Although (224) is written as a quotient, $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_q$ is a polynomial in q with nonnegative integer coefficients and constant term 1. The quotient form makes the symmetry

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q \tag{225}$$

immediate, and letting $q \to 1$ gives $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_q \to \binom{n}{k}$. Two q-Pascal recurrences hold:

$$\begin{bmatrix} n+1 \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{\,n+1-k}\begin{bmatrix} n \\ k-1 \end{bmatrix}_q, \tag{226}$$

$$\begin{bmatrix} n+1 \\ k \end{bmatrix}_q = q^k\begin{bmatrix} n \\ k \end{bmatrix}_q + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q. \tag{227}$$

Each follows from (224) by writing $(q;q)_{n+1} = (q;q)_n(1-q^{n+1})$ and splitting $1-q^{n+1}$ as $(1-q^{n+1-k}) + q^{\,n+1-k}(1-q^k)$ or as $q^k(1-q^{n+1-k}) + (1-q^k)$; the two forms are exchanged by (225). Exactly as the ordinary Pascal rule proves the binomial theorem, either recurrence proves the following.

Theorem 10.3 (Finite q-binomial theorem)**.** *For every $n \in \mathbb{N}$ and every base q,*

$$(z;q)_n = \sum_{k=0}^{n} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k. \tag{228}$$

Proof. Both sides equal 1 when $n = 0$. Assume (228) at n. By the step rule (214),

$$(z;q)_{n+1} = (1-zq^n)\sum_{k=0}^{n} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k.$$

The coefficient of z^k on the right is

$$(-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q - q^n\,(-1)^{k-1} q^{\binom{k-1}{2}} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = (-1)^k q^{\binom{k}{2}} \left(\begin{bmatrix} n \\ k \end{bmatrix}_q + q^{\,n+1-k}\begin{bmatrix} n \\ k-1 \end{bmatrix}_q\right),$$

where the exponent bookkeeping uses $\binom{k}{2} = \binom{k-1}{2} + k - 1$, so that $q^n q^{\binom{k-1}{2}} = q^{\binom{k}{2}} q^{\,n+1-k}$. By (226) the bracket is $\left[\begin{smallmatrix} n+1 \\ k \end{smallmatrix}\right]_q$, which is the asserted coefficient at $n+1$. ☐

At $q = 1$ the theorem degenerates to $(1-z)^n = \sum_k (-1)^k \binom{n}{k} z^k$, and at $q = 1/2$ it is (202). The triangular exponent $\binom{k}{2}$ is not decoration: it is the cumulative exponent $0+1+\cdots+(k-1)$ produced by choosing the second term from k successive factors of $\prod_{j<n}(1-zq^j)$.

Combining (202) with the vanishing (217) and the endpoint (218) gives the two nodal identities on which everything below rests:

$$\sum_{k=0}^{n} (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} (2^d)^k = 0 \qquad (0 \le d < n), \tag{229}$$

$$\sum_{k=0}^{n} (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} (2^n)^k = (-1)^n \mathfrak{m}_n. \tag{230}$$

With the weights and nodes of (203),

$$w_{n,k} = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}, \qquad x_k = -2^k, \tag{231}$$

one has $x_k^d = (-1)^d (2^d)^k$, so the extra sign turns (230) into $+\mathfrak{m}_n$ and reproduces the first half of (203):

$$\sum_{k=0}^{n} w_{n,k} x_k^d = \begin{cases} 0, & 0 \le d < n, \\ \mathfrak{m}_n, & d = n. \end{cases} \tag{232}$$

It remains to record the arithmetic of the coefficients themselves. The value $q = 1/2$ is not a prime power, so the subspace interpretation does not apply directly; but when q is a prime power, $\begin{bmatrix} n \\ k \end{bmatrix}_q$ counts the k-dimensional subspaces of $\mathbb{F}_q^n$, which is the source of the adjective "Gaussian". Applying the inversion rule (216) to each of the three products in (224) gives the reciprocal-base identity $\begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}} = q^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_q$ and hence the relation already quoted in section 9.9,

$$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = 2^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_2. \tag{233}$$

Because $\begin{bmatrix} n \\ k \end{bmatrix}_2$ is an integer with odd value (its polynomial has constant term 1, so $\begin{bmatrix} n \\ k \end{bmatrix}_2 \equiv 1 \pmod 2$), (233) determines the 2-adic valuation exactly:

$$v_2\left(\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} \right) = -k(n-k). \tag{234}$$

Equivalently, in terms of the odd Mersenne products,

$$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = \frac{\mathfrak{m}_n}{\mathfrak{m}_k\, \mathfrak{m}_{n-k}\, 2^{k(n-k)}}. \tag{235}$$

Thus every apparent nonintegrality in the Gaussian row is a known power of two, and the odd part is finite-field combinatorics. This exact denominator control is one reason the q-notation is arithmetically useful and not merely compact.

Table 1 shows the first rows. The Mersenne numerators $2^j - 1$ visible there are the same factors that occur in $(1/2;1/2)_n$ through (204) and in the final denominator through (222). The corresponding rows of the signed weights (231) are

$$\begin{aligned}
n = 0 &: \quad 1, \\
n = 1 &: \quad 1,\ -1, \\
n = 2 &: \quad 1,\ -\tfrac{3}{2},\ \tfrac{1}{2}, \\
n = 3 &: \quad 1,\ -\tfrac{7}{4},\ \tfrac{7}{8},\ -\tfrac{1}{8}, \\
n = 4 &: \quad 1,\ -\tfrac{15}{8},\ \tfrac{35}{32},\ -\tfrac{15}{64},\ \tfrac{1}{64}.
\end{aligned}$$

Table 1: The first half-base Gaussian rows $\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$ and, for comparison, the integer rows $\begin{bmatrix} n \\ k \end{bmatrix}_2$ that count k-planes in $\mathbb{F}_2^n$. The two are related by (233).

$n\backslash k$	0	1	2	3	4	5	6
0	1						
1	1	1					
2	1	3/2	1				
3	1	7/4	7/4	1			
4	1	15/8	35/16	15/8	1		
5	1	31/16	155/64	155/64	31/16	1	
6	1	63/32	651/256	1395/512	651/256	63/32	1
4	1	15	35	15	1		
5	1	31	155	155	31	1	
6	1	63	651	1395	651	63	1

They are exactly the coefficients of the node polynomial $(z;1/2)_n$, read as weights on the nodes -2^k.

10.4 The barycentric reading of the Gaussian row

Equation (232) says that the normalized functional

$$\mathcal{J}_n[f] = \frac{1}{\mathfrak{m}_n} \sum_{k=0}^{n} w_{n,k} f(x_k), \qquad x_k = -2^k, \tag{236}$$

annihilates every polynomial of degree below n and returns the leading coefficient of every polynomial of degree at most n: if $f(x) = c_0 + c_1 x + \cdots + c_n x^n$, then $\mathcal{J}_n[f] = c_n$. That is the defining property of the nth divided difference on the nodes $x_0, \ldots, x_n$. The second half of (203) makes the identification with Lagrange interpolation exact, and it is proved by splitting one product.

Proposition 10.4 (Barycentric form of the half-base weights)**.** *For $x_k = -2^k$ and $0 \le k \le n$,*

$$\frac{w_{n,k}}{\mathfrak{m}_n} = \frac{1}{\prod\limits_{\substack{0 \le j \le n \\ j \ne k}} (x_k - x_j)}. \tag{237}$$

Proof. Split the product at $j = k$. For $j < k$,

$$x_k - x_j = -2^k + 2^j = -2^j\,(2^{\,k-j} - 1),$$

so those k factors contribute

$$(-1)^k\, 2^{\,0+1+\cdots+(k-1)} \prod_{j=0}^{k-1} (2^{\,k-j} - 1) = (-1)^k 2^{\binom{k}{2}} \mathfrak{m}_k.$$

For $j > k$,

$$x_k - x_j = 2^j - 2^k = 2^k\,(2^{\,j-k} - 1),$$

so those $n-k$ factors contribute $2^{k(n-k)}\mathfrak{m}_{n-k}$. Multiplying,

$$\prod_{j \ne k} (x_k - x_j) = (-1)^k 2^{\binom{k}{2} + k(n-k)} \mathfrak{m}_k \mathfrak{m}_{n-k}. \tag{238}$$

On the other side, substituting (235) into (231) gives

$$\frac{w_{n,k}}{\mathfrak{m}_n} = \frac{(-1)^k}{2^{\binom{k}{2}+k(n-k)}\mathfrak{m}_k\mathfrak{m}_{n-k}},$$

which is the reciprocal of (238) because $(-1)^k$ is its own reciprocal. □

The right-hand side of (237) is the classical coefficient of $f(x_k)$ in the leading coefficient of the Lagrange interpolant through $x_0, \ldots, x_n$. Hence the Gaussian row in (205) is a table of barycentric weights for a geometric grid, and (232) is not a coincidence but the standard vanishing of an nth divided difference on lower-degree data. This is exactly the content that (203) asserted without justification.

Example 10.5 (The $n = 2$ stencil)**.** The nodes are $-1, -2, -4$ and $\mathfrak{m}_2 = 3$. From (231),

$$(w_{2,0}, w_{2,1}, w_{2,2}) = \left(1, -\tfrac{3}{2}, \tfrac{1}{2}\right),$$

so

$$\mathcal{J}_2[f] = \tfrac{1}{3}f(-1) - \tfrac{1}{2}f(-2) + \tfrac{1}{6}f(-4).$$

Direct verification confirms all three moment conditions: $\frac{1}{3} - \frac{1}{2} + \frac{1}{6} = 0$; $-\frac{1}{3} + 1 - \frac{4}{6} = 0$; $\frac{1}{3} - 2 + \frac{16}{6} = 1$. The three coefficients are also the reciprocals of the node-difference products $(-1+2)(-1+4) = 3$, $(-2+1)(-2+4) = -2$, and $(-4+1)(-4+2) = 6$, as proposition 10.4 requires.

10.5 The Thue–Morse exponential block

The second cancellation is combinatorial rather than interpolatory. Its source is that $\varepsilon_r = (-1)^{\mathrm{wt}_2(r)}$ is a multiplicative character of the binary digit expansion.

Lemma 10.6 (Block multiplicativity and reflection)**.** *Let $k \in \mathbb{N}$ and $0 \le s < 2^k$. For every $h \in \mathbb{N}$,*

$$\varepsilon_{h2^k+s} = \varepsilon_h \varepsilon_s, \tag{239}$$

and for every $0 \le r < 2^k$,

$$\varepsilon_{2^k-1-r} = (-1)^k \varepsilon_r. \tag{240}$$

Proof. In $h2^k + s$ the binary digits of h occupy positions $\ge k$ and those of s occupy positions $< k$, so no carries occur and $\mathrm{wt}_2(h2^k + s) = \mathrm{wt}_2(h) + \mathrm{wt}_2(s)$; exponentiating -1 gives (239). For (240), the map $r \mapsto 2^k - 1 - r$ complements each of the k low binary digits, so $\mathrm{wt}_2(2^k - 1 - r) = k - \mathrm{wt}_2(r)$. □

Proposition 10.7 (Digitwise factorization)**.** *For every $k \in \mathbb{N}$ and every z,*

$$\sum_{r=0}^{2^k-1} \varepsilon_r z^r = \prod_{j=0}^{k-1} (1 - z^{2^j}), \tag{241}$$

which is (206).

Proof. Every $0 \le r < 2^k$ has a unique expansion $r = \sum_{j=0}^{k-1} b_j 2^j$ with $b_j \in \{0, 1\}$, and $\mathrm{wt}_2(r) = \sum_j b_j$. Therefore

$$\sum_{r=0}^{2^k-1} \varepsilon_r z^r = \sum_{b_0,\ldots,b_{k-1}\in\{0,1\}} \prod_{j=0}^{k-1} (-1)^{b_j} z^{b_j 2^j} = \prod_{j=0}^{k-1} \sum_{b=0}^{1} (-1)^b z^{b2^j} = \prod_{j=0}^{k-1} (1 - z^{2^j}),$$

the middle step being the distributive law for a product of k two-term sums. □

Now introduce the translated centered power sums. For $m, k, d \in \mathbb{N}$ and a scalar Q, put

$$\sigma_{m;k,d}(Q) = \sum_{r=0}^{m2^k-1} \varepsilon_r \left(r - m2^k + Q\right)^d, \qquad \sigma_{k,d}(Q) = \sigma_{1;k,d}(Q), \tag{242}$$

the range being empty, hence the sum zero, when $m = 0$. These are exactly the inner sums of (205). Their exponential generating functions are the objects to work with. Set, as in section 9.9,

$$C_k(X) = \sum_{r=0}^{2^k-1} \varepsilon_r \mathrm{e}^{(r-2^k)X} = \sum_{d\geq 0} \sigma_{k,d}(0)\frac{X^d}{d!}, \tag{243}$$

and observe the *translated-block one-liner*: multiplying every summand of (243) by e^{QX} replaces $r - 2^k$ by $r - 2^k + Q$ in each exponent, so

$$\sum_{d\geq 0} \sigma_{k,d}(Q)\frac{X^d}{d!} = \mathrm{e}^{QX} C_k(X). \tag{244}$$

That single identity is the entire source of translation invariance in section 10.8.

Theorem 10.8 (Centered block factorization)**.** *Let $\mathscr{E}_k(X) = \mathrm{e}^{-X}\prod_{j=0}^{k-1} E(-2^j X)$, a formal power series over $\mathbb{Q}$ with $\mathscr{E}_k(0) = 1$. Then*

$$C_k(X) = (-1)^k 2^{\binom{k}{2}} X^k \mathscr{E}_k(X), \tag{245}$$

so that C_k has a zero of exact order k at $X = 0$. Consequently

$$\sigma_{k,d}(Q) = 0 \quad (d < k), \qquad \sigma_{k,k}(Q) = (-1)^k 2^{\binom{k}{2}} k! \tag{246}$$

for every Q, and, writing $\mathscr{H}_k(X) = C_k(X)/X^k$,

$$\mathscr{H}_k(X) = \sum_{n\geq 0} \frac{\sigma_{k,n+k}(0)}{(n+k)!} X^n = (-1)^k 2^{\binom{k}{2}} \mathscr{E}_k(X), \tag{247}$$

which is (208).

Proof. Apply (241) with $z = \mathrm{e}^X$ and multiply by $\mathrm{e}^{-2^k X}$:

$$C_k(X) = \mathrm{e}^{-2^k X} \prod_{j=0}^{k-1} (1 - \mathrm{e}^{2^j X}). \tag{248}$$

Each factor is $1 - \mathrm{e}^u = -u\, E(u)$ with $u = 2^j X$, so

$$\prod_{j=0}^{k-1} (1 - \mathrm{e}^{2^j X}) = (-1)^k \left(\prod_{j=0}^{k-1} 2^j\right) X^k \prod_{j=0}^{k-1} E(2^j X) = (-1)^k 2^{\binom{k}{2}} X^k \prod_{j=0}^{k-1} E(2^j X).$$

The reflection identity $E(-u) = \mathrm{e}^{-u}E(u)$, valid because both sides equal $(1 - \mathrm{e}^{-u})/u$, converts the positive-argument product into a negative-argument one: using $\sum_{j<k} 2^j = 2^k - 1$,

$$\prod_{j=0}^{k-1} E(-2^j X) = \mathrm{e}^{-(2^k-1)X} \prod_{j=0}^{k-1} E(2^j X), \qquad \text{i.e.} \qquad \mathscr{E}_k(X) = \mathrm{e}^{-2^k X} \prod_{j=0}^{k-1} E(2^j X). \tag{249}$$

Substituting this into the display above gives (245). Since $\mathscr{E}_k(0) = 1$, the order of the zero is exactly k, and comparing coefficients in (243) yields the first part of (246) at $Q = 0$; by (244) multiplication by e^{QX} cannot lower the order, so the vanishing holds for every Q, and the coefficient of X^k in $\mathrm{e}^{QX} C_k(X)$ equals that of $C_k(X)$, which gives the stated value of $\sigma_{k,k}(Q)$. Dividing (245) by X^k gives (247). ☐

Identity (246) is the classical Prouhet cancellation for dyadic Thue–Morse blocks [14]: a complete block of length 2^k annihilates all power moments of degree below k, in any translate of the coordinate. The factor $2^{\binom{k}{2}}$ in the first surviving moment is the product $\prod_{j<k} 2^j$ contributed by the k dyadic scales that occur in (241).

Writing $\beta_{k,n} = [X^n]\mathscr{E}_k(X)$ for the coefficients of the refinement factor, (247) says

$$\beta_{k,n} = (-1)^k 2^{-\binom{k}{2}} \frac{\sigma_{k,n+k}(0)}{(n+k)!}, \tag{250}$$

and multiplying by the interpolation weight $w_{n,k}$ of (231) makes the two signs $(-1)^k$ cancel and the two triangular powers of two combine:

$$w_{n,k}\, \beta_{k,n} = \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!}\, \sigma_{k,n+k}(0). \tag{251}$$

The right-hand side is precisely one summand of the outer sum in (205).

Why the formula contains $4^{\binom{k}{2}}$ and not $2^{\binom{k}{2}}$. Two independent copies of $2^{-\binom{k}{2}}$ are needed, and their product is $4^{-\binom{k}{2}}$.

The first copy comes from the Thue–Morse side. The block C_k has a forced zero of order k whose leading coefficient is $(-1)^k 2^{\binom{k}{2}}$ by (245). To read a degree-n coefficient of the refinement factor $\mathscr{E}_k$, one must first divide that leading constant out; this is the factor $2^{-\binom{k}{2}}$ in (250).

The second copy comes from the q-binomial side. The interpolation weight at the node -2^k is $w_{n,k} = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$, in which the power of two is prescribed by theorem 10.3.

Their product is the $4^{-\binom{k}{2}}$ of (251). Neither exponent can be adjusted without destroying the identity: one is fixed by the Thue–Morse block, the other by the finite q-binomial theorem.

Finally, (240) lets the centered coordinates be traded for raw ones. For $0 \le r < 2^k$ the substitution $r \mapsto 2^k - 1 - r$ gives $(2^k - 1 - r) - 2^k + Q = -(r + 1 - Q)$, so

$$\sum_{r=0}^{2^k-1} \varepsilon_r (r+Q)^{n+k} = (-1)^k (-1)^{n+k} \sum_{r=0}^{2^k-1} \varepsilon_r \bigl(r - 2^k + 1 - Q\bigr)^{n+k} = (-1)^n \sigma_{k,n+k}(1-Q). \tag{252}$$

The total sign $(-1)^n$ does not depend on k, so it factors out of the whole outer sum, and since $n^2 \equiv n \pmod 2$ it is absorbed by writing $(-2)^{n^2}$ in place of 2^{n^2}. This is the derivation of the raw-coordinate form (209).

10.6 The coefficient-extraction theorem

The two mechanisms now meet. Iterating the formal refinement identity (207) at the arguments $-X, -2X, \ldots, -2^{k-1}X$ telescopes to $\mathscr{A}(-2^k X) = \bigl(\prod_{j<k} E(-2^j X)\bigr)\mathscr{A}(-X)$, and multiplying by e^{-X} produces the bridge

$$\mathscr{E}_k(X)\, \mathscr{A}(-X) = \mathrm{e}^{-X} \mathscr{A}(-2^k X). \tag{253}$$

The left side is the normalized Thue–Morse block of theorem 10.8; the right side is the Fabius moment series evaluated at the geometric node -2^k. Exactly one functional annihilates the node dependence below degree n, and by proposition 10.4 it is the Gaussian row.

Theorem 10.9 (Dyadic Gaussian coefficient extraction)**.** *Let $\mathscr{A}(X) = \sum_{m\geq 0} a_m X^m$ be any formal power series over $\mathbb{Q}$ with $a_0 = 1$, and for each k let $\mathscr{E}_k(X) = \sum_{m\geq 0} \beta_{k,m} X^m$ be a formal power series satisfying* (253)*. Then, with the weights* (231)*,*

$$\sum_{k=0}^{n} w_{n,k}\beta_{k,d} = 0 \qquad (0 \leq d < n), \tag{254}$$

and

$$\sum_{k=0}^{n} w_{n,k}\beta_{k,n} = \mathfrak{m}_n \, a_n. \tag{255}$$

Proof. Extract the coefficient of X^d from (253). On the left, $\mathscr{A}(-X)$ has coefficients $(-1)^{d-i}a_{d-i}$; on the right, e^{-X} has coefficients $(-1)^j/j!$ and $\mathscr{A}(-2^k X)$ has coefficients $a_{d-j}(-2^k)^{d-j}$. Hence, for every k and every d,

$$\sum_{i=0}^{d} \beta_{k,i}(-1)^{d-i}a_{d-i} = \sum_{j=0}^{d} \frac{(-1)^j}{j!} a_{d-j}(-2^k)^{d-j}. \tag{256}$$

Multiply (256) by $w_{n,k}$ and sum over $k = 0, \dots, n$. Note that only the powers $(-2^k)^{d-j} = x_k^{d-j}$ depend on k on the right, and only $\beta_{k,i}$ depends on k on the left.

Suppose first $d < n$, and argue by strong induction on d. Every exponent $d - j$ on the right satisfies $d - j \leq d < n$, so (232) makes every right-hand term vanish; the whole right side is 0. On the left, the terms with $i < d$ carry the factor $\sum_k w_{n,k}\beta_{k,i}$, which vanishes by the induction hypothesis, so only $i = d$ survives. Since $a_0 = 1$, that term is $\sum_k w_{n,k}\beta_{k,d}$. Therefore $\sum_k w_{n,k}\beta_{k,d} = 0$, which is (254). The base case $d = 0$ is the same computation with no induction hypothesis needed.

Now let $d = n$. On the right, every term with $j > 0$ has node exponent $n - j < n$ and vanishes after weighting; the term $j = 0$ is $a_n \sum_k w_{n,k} x_k^n = \mathfrak{m}_n a_n$ by (232). On the left, all terms with $i < n$ vanish by (254), and the term $i = n$ is $\sum_k w_{n,k}\beta_{k,n}$. Equating the two sides gives (255). □

The proof is a double cancellation and nothing else. The finite q-binomial theorem annihilates powers of the geometric nodes below degree n; the coefficient identity (256) recursively transfers that nodal annihilation to the coefficients of the normalized Thue–Morse factors. No summation theorem from the catalogue of basic hypergeometric series is used or needed.

Theorem 10.9 is stated with the geometric nodes -2^k and the half-base weights built in, and its proof reads as though both were needed. Neither was. The only property of the weights the argument used is the annihilation of node powers below degree n, and the only property of the nodes is that they can be substituted into a rescaling. The general statement is the following, and its proof is the proof of theorem 10.9 with the two specializations removed.

Proposition 10.10 (Abstract weighted coefficient isolation)**.** *Let I be a finite index set, let $(x_k)_{k\in I}$ be scalars in a field K of characteristic zero, and let $(w_k)_{k\in I}$ be weights satisfying*

$$\sum_{k\in I} w_k x_k^d = 0 \qquad (0 \leq d < n). \tag{257}$$

Let $\mathscr{A} = \sum_{m\geq 0} a_m X^m$, $\mathscr{B} = \sum_{m\geq 0} b_m X^m$ and $\mathscr{U} = \sum_{m\geq 0} u_m X^m$ be formal power series over K with $a_0 = b_0 = u_0 = 1$, and for each $k \in I$ let $P_k \in K[[X]]$ satisfy

$$P_k(X)\,\mathscr{B}(X) = \mathscr{U}(X)\,\mathscr{A}(x_k X). \tag{258}$$

Then

$$\sum_{k\in I} w_k\,[X^d]P_k = 0 \quad (0 \leq d < n), \qquad \sum_{k\in I} w_k\,[X^n]P_k = \left(\sum_{k\in I} w_k x_k^n\right) a_n. \tag{259}$$

Proof. Extract the coefficient of X^d from (258):

$$\sum_{i=0}^{d}([X^i]P_k)b_{d-i} = \sum_{j=0}^{d} u_j a_{d-j} x_k^{d-j}.$$

Multiply by w_k and sum over $k \in I$. On the right the only k-dependence is x_k^{d-j}.

Let $d < n$ and argue by strong induction on d. Every exponent satisfies $d - j \le d < n$, so the right side vanishes by (257). On the left the terms with $i < d$ carry the factor $\sum_k w_k[X^i]P_k$, which vanishes by the induction hypothesis, and the term $i = d$ is $b_0 \sum_k w_k[X^d]P_k = \sum_k w_k[X^d]P_k$. The case $d = 0$ is the same computation with no hypothesis used.

Let $d = n$. On the right the terms with $j > 0$ have node exponent $n - j < n$ and vanish, and the term $j = 0$ is $u_0 a_n \sum_k w_k x_k^n$. On the left the terms with $i < n$ vanish by what has just been proved, leaving $\sum_k w_k[X^n]P_k$. □

Theorem 10.9 is the case $I = \{0, \dots, n\}$, $x_k = -2^k$, $\mathscr{B}(X) = \mathscr{A}(-X)$, $\mathscr{U}(X) = \mathrm{e}^{-X}$, $P_k = \mathscr{E}_k$, and $w_k = w_{n,k}$, in which (258) is (253) and (257) is the lower half of (232). The dyadic structure of the problem is therefore confined to a single place: verifying (257), which is where the finite q-binomial theorem (202) is used. Any other node set with a matching annihilator would run through the same argument unchanged, and this is what makes the base-b heuristic of section 10.11 more than an analogy.

The same conclusion has a purely interpolatory reading, which explains what the induction is doing. Because $\mathscr{A}(0) = 1$, the series $\mathscr{A}(-X)$ is a unit in $\mathbb{Q}[[X]]$, so one may introduce a polynomial parameter z and set, as in section 9.9,

$$\mathscr{R}(z; X) = \frac{\mathrm{e}^{-X}\mathscr{A}(zX)}{\mathscr{A}(-X)}, \qquad r_n(z) = [X^n]\,\mathscr{R}(z; X). \tag{260}$$

By (253), $\mathscr{R}(-2^k; X) = \mathscr{E}_k(X)$, so $r_n(-2^k) = \beta_{k,n}$.

Proposition 10.11 (The coefficient is a polynomial in the node)**.** *For each n, r_n is a polynomial in z of degree at most n, and its leading coefficient is a_n:*

$$r_n(z) = a_n z^n + (\textit{a polynomial of degree at most } n-1).$$

Proof. Write $\mathrm{e}^{-X}/\mathscr{A}(-X) = \sum_{i\ge 0} u_i X^i$ with $u_0 = 1$. Then $r_n(z) = \sum_{i=0}^{n} u_{n-i} a_i z^i$, a polynomial of degree at most n whose coefficient of z^n is $u_0 a_n = a_n$: the only way to produce $z^n X^n$ is to take $a_n z^n X^n$ from $\mathscr{A}(zX)$ and constant terms from both remaining factors. □

Applying $\mathscr{J}_n$ from (236) to r_n therefore returns a_n, which is (255) again. This packages the formal-series induction of theorem 10.9 into a familiar interpolation statement: the quantity being extracted is the leading coefficient of a polynomial of degree n in the geometric scale node, and the Gaussian row is the divided-difference stencil that extracts it.

The two derivations of (255) given above are not redundant, and the difference between them is not stylistic. The quotient form is the natural analytic statement: $\mathscr{R}(z; X)$ of (260) is a single generating function in which the node z appears as a free parameter, so proposition 10.11 can say in one line what is being extracted, namely the leading coefficient of a polynomial in z, and the divided-difference reading of proposition 10.4 then applies verbatim. That form presupposes the inversion of $\mathscr{A}(-X)$, which is legitimate because $\mathscr{A}(0) = 1$ makes it a unit in $\mathbb{Q}[[X]]$, but it is an extra construction. The product form (253) presupposes nothing: extracting a coefficient from a product is a finite convolution (256), and the induction of theorem 10.9 closes on that convolution alone. A development that works with formal power series — as the formal development of this article does — proves the product form directly and never needs the inverse series, while a reader who wants to see why the stencil is a divided difference wants the quotient form. Both are recorded because each is the shortest route to a different one of the two things being asserted.

We can now assemble the one-block identity. Define the complete numerator

$$\mathcal{N}_{m,n}(Q) = \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!}\,\sigma_{m;k,n+k}(Q), \tag{261}$$

so that (205) says $\mathcal{F}(m/2^n) = \mathcal{N}_{m,n}(Q)/(2^{n^2}(1/2;1/2)_n)$. The single-block case $\mathcal{N}_{1,n}(Q)$ is the quantity later written $\mathcal{N}_n^{(Q)}$ in (294).

Corollary 10.12 (Centered inverse-dyadic evaluation)**.** *For every $n \in \mathbb{N}$,*

$$\mathcal{N}_{1,n}(0) = \mathfrak{m}_n\, a_n, \qquad \frac{\mathcal{N}_{1,n}(0)}{2^{n^2}(1/2;1/2)_n} = 2^{-\binom{n}{2}} a_n = F(2^{-n}). \tag{262}$$

Proof. By (251), the kth summand of $\mathcal{N}_{1,n}(0)$ equals $w_{n,k}\beta_{k,n}$, so $\mathcal{N}_{1,n}(0) = \sum_k w_{n,k}\beta_{k,n} = \mathfrak{m}_n a_n$ by theorem 10.9. For the normalization, insert (204) and use the triangular splitting $n^2 = \binom{n+1}{2} + \binom{n}{2}$:

$$\frac{\mathfrak{m}_n a_n}{2^{n^2}(1/2;1/2)_n} = \frac{\mathfrak{m}_n a_n}{2^{n^2-\binom{n+1}{2}}\mathfrak{m}_n} = 2^{\binom{n+1}{2}-n^2} a_n = 2^{-\binom{n}{2}} a_n,$$

and the last quantity is $F(2^{-n})$ by (136). □

10.7 Arbitrary dyadic numerators

The passage from $m = 1$ to a general numerator costs one extra factor and no new idea. Define the finite Thue–Morse prefix series

$$B_m(X) = \sum_{h=0}^{m-1} \varepsilon_h \mathrm{e}^{(m-1-h)X}, \qquad B_0(X) = 0, \quad B_1(X) = 1, \tag{263}$$

and the long centered block

$$\mathscr{D}_{m,k}(X) = \sum_{r=0}^{m2^k-1} \varepsilon_r \mathrm{e}^{(r-m2^k)X} = \sum_{d\geq 0} \sigma_{m;k,d}(0)\frac{X^d}{d!}. \tag{264}$$

Lemma 10.13 (Long-block factorization)**.** *For all $m, k \in \mathbb{N}$,*

$$\mathscr{D}_{m,k}(X) = B_m(-2^k X)\, C_k(X), \qquad \textit{hence} \qquad \frac{\mathscr{D}_{m,k}(X)}{X^k} = B_m(-2^k X)\mathscr{H}_k(X). \tag{265}$$

Proof. Write $r = h2^k + s$ with $0 \leq h < m$ and $0 \leq s < 2^k$; this is a bijection onto $0 \leq r < m2^k$. By (239), $\varepsilon_r = \varepsilon_h \varepsilon_s$. The exponent check is the only computation. In $B_m(-2^k X)C_k(X)$ the pair (h, s) contributes the exponent

$$-(m-1-h)2^k X + (s-2^k)X = (-m2^k + 2^k + h2^k + s - 2^k)X = (h2^k + s - m2^k)X = (r - m2^k)X,$$

which is the exponent of the same pair in (264). The second identity follows by dividing by X^k and using $\mathscr{H}_k = C_k/X^k$; note in particular that $\mathscr{D}_{m,k}$ still has a zero of order at least k, so $\sigma_{m;k,d}(0) = 0$ for $d < k$. □

Introduce the prefixed parametric series

$$\mathscr{R}_m(z;X) = B_m(zX)\,\mathscr{R}(z;X) = \frac{\mathrm{e}^{-X}B_m(zX)\mathscr{A}(zX)}{\mathscr{A}(-X)}, \qquad r_{m,n}(z) = [X^n]\,\mathscr{R}_m(z;X). \quad (266)$$

Because $B_m(zX)\mathscr{A}(zX) = (B_m\mathscr{A})(zX)$, the argument of proposition 10.11 applies verbatim: $r_{m,n}$ is a polynomial in z of degree at most n, and its leading coefficient is

$$[X^n](B_m(X)\mathscr{A}(X)). \quad (267)$$

Applying $\mathcal{J}_n$ and using $\mathscr{R}_m(-2^k;X) = B_m(-2^kX)\mathscr{E}_k(X)$ gives the general extraction identity

$$\sum_{k=0}^{n} w_{n,k}\,[X^n]\Big(B_m(-2^kX)\mathscr{E}_k(X)\Big) = \mathfrak{m}_n\,[X^n](B_m(X)\mathscr{A}(X)). \quad (268)$$

By (265) and the same weight bookkeeping as in (251), the left side of (268) is exactly $\mathcal{N}_{m,n}(0)$. The right side is the coefficient that the classical rational theory already attaches to a dyadic value.

Lemma 10.14 (The prefix coefficient is the dyadic value)**.** *For all $m,n \in \mathbb{N}$,*

$$\mathcal{F}\left(\frac{m}{2^n}\right) = 2^{-\binom{n}{2}}\,[X^n](B_m(X)\mathscr{A}(X)). \quad (269)$$

Proof. For $m=1$ one has $B_1 = 1$, so the right side is $2^{-\binom{n}{2}}a_n$, and the claim is (136). For general m start from the closed Thue–Morse formula (164) in its signed global form. Put $t_h = 2m-2h-1$, so that the inner polynomial factor is

$$\sum_{k=0}^{\lfloor n/2\rfloor}\binom{n}{2k}c_k t_h^{n-2k} = n!\,[w^n]\left(\mathrm{e}^{t_h w}\sum_{k\ge 0}c_k\frac{w^{2k}}{(2k)!}\right).$$

In the present notation the Fourier relation (118) reads

$$\mathscr{A}(z) = \mathrm{e}^{z/2}\sum_{k\ge 0}c_k\frac{(z/2)^{2k}}{(2k)!},$$

so setting $z=2w$ gives $\sum_k c_k w^{2k}/(2k)! = \mathrm{e}^{-w}\mathscr{A}(2w)$; this is the generating-function form of (125). Since $t_h - 1 = 2(m-1-h)$,

$$\sum_k\binom{n}{2k}c_k t_h^{n-2k} = n!\,[w^n]\Big(\mathrm{e}^{2(m-1-h)w}\mathscr{A}(2w)\Big) = n!\,2^n\,[X^n]\Big(\mathrm{e}^{(m-1-h)X}\mathscr{A}(X)\Big),$$

the last step being the substitution $X = 2w$, under which $[w^n]f(2w) = 2^n[X^n]f(X)$. Multiplying by ε_h and summing over $0 \le h < m$ replaces the exponential by $B_m(X)$. Hence (164) becomes

$$\mathcal{F}\left(\frac{m}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!}\cdot n!\,2^n\,[X^n](B_m(X)\mathscr{A}(X)) = 2^{n-\binom{n+1}{2}}[X^n](B_m(X)\mathscr{A}(X)),$$

and $\binom{n+1}{2} - n = \binom{n}{2}$. □

Combining (268), (269), (204), and $n^2 = \binom{n+1}{2} + \binom{n}{2}$ proves the centered case $Q=0$ of (205); section 10.8 removes the restriction on Q.

Two consequences deserve to be stated separately, because both are easy to misread.

Proposition 10.15 (Representation independence)**.** *For all $m, n \in \mathbb{N}$ and every Q,*

$$\frac{\mathcal{N}_{m,n}(Q)}{2^{n^2}(1/2;1/2)_n} = \frac{\mathcal{N}_{2m,n+1}(Q)}{2^{(n+1)^2}(1/2;1/2)_{n+1}}.$$

Proof. Both quotients equal $\mathcal{F}$ evaluated at the same real number, since $m/2^n = 2m/2^{n+1}$. □

Nothing about the two expressions is termwise comparable. Replacing (m, n) by $(2m, n+1)$ adds one outer term, changes every inner range, changes the exponents $n + k$, and changes the q-Pochhammer denominator. The formula is nevertheless exact for every pair (m, n), with no requirement that $m/2^n$ be in lowest terms. Computationally the reduced representation is almost always preferable: a deliberately unreduced one only aggravates the cancellation analyzed in section 10.13.

The second point concerns the range of m. If $0 \le m \le 2^n$, the argument lies in $[0, 1]$ and (205) computes the bounded value $F(m/2^n)$. If $m > 2^n$, it computes the signed global extension (11), whose sign on the block $[2b, 2b + 2]$ is ε_b by (12). This is not a cosmetic widening of the domain: the endpoint cases are literal finite identities of the same expression. When $m = 0$ every inner range is empty and both sides are zero; when $n = 0$ the formula degenerates to the scale-zero signed block identity, with all conventions — empty products, empty sums, and the natural-power convention $0^0 = 1$ — fixed by finiteness rather than by a limit. For instance, at $m = 3$ and $n = 0$,

$$\frac{\mathcal{N}_{3,0}(0)}{2^0(1/2;1/2)_0} = \varepsilon_0(0-3)^0 + \varepsilon_1(1-3)^0 + \varepsilon_2(2-3)^0 = 1 - 1 - 1 = -1 = \mathcal{F}(3),$$

in agreement with the block rule (12).

10.8 Translation invariance

Theorem 10.16 (Common-translation invariance)**.** *Fix $m, n \in \mathbb{N}$. The map*

$$Q \longmapsto \mathcal{N}_{m,n}(Q) = \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{m2^k-1} \varepsilon_r \left(r - m2^k + Q\right)^{n+k}$$

is a constant polynomial, with the constant value $\mathcal{N}_{m,n}(0) = \mathfrak{m}_n\,[X^n]\big(B_m(X)\mathscr{A}(X)\big)$.

Proof. By (244) and (265), translating every inner power by the same Q multiplies the shifted long-block series by e^{QX}:

$$\sum_{d\ge 0} \sigma_{m;k,d}(Q)\frac{X^d}{d!} = \mathrm{e}^{QX}\mathscr{D}_{m,k}(X), \qquad \frac{\mathrm{e}^{QX}\mathscr{D}_{m,k}(X)}{X^k} = (-1)^k 2^{\binom{k}{2}} \mathrm{e}^{QX} B_m(-2^k X)\mathscr{E}_k(X).$$

Hence, with the bookkeeping of (251),

$$\begin{aligned} \mathcal{N}_{m,n}(Q) &= \sum_{k=0}^{n} w_{n,k}\,[X^n]\Big(\mathrm{e}^{QX} B_m(-2^k X)\mathscr{E}_k(X)\Big) \\ &= \sum_{j=0}^{n} \frac{Q^j}{j!} \sum_{k=0}^{n} w_{n,k}\, r_{m,n-j}(-2^k) = \sum_{j=0}^{n} \frac{Q^j}{j!}\,\mathfrak{m}_n\,\mathcal{J}_n[r_{m,n-j}]. \end{aligned} \tag{270}$$

For $j > 0$ the polynomial $r_{m,n-j}$ has degree at most $n - j < n$, so $\mathcal{J}_n[r_{m,n-j}] = 0$ by (232). Only $j = 0$ survives, and its value is $\mathcal{N}_{m,n}(0)$ by (268). Every coefficient of a positive power of Q therefore vanishes identically. □

The mechanism is the vanishing-moment property and nothing else. Expanding $(r - m2^k + Q)^{n+k}$ introduces contributions of lower effective coefficient degree, and the Gaussian filter has already been engineered to annihilate all of them. Translation invariance is not an extra symmetry of the Fabius function; it is a property of the extraction functional.

Warning 10.17. The qualifier *common* is essential. Theorem 10.16 asserts that one and the same Q may be used in every inner sum. A shift that depends on the scale index k — recentering each block about its own midpoint, for example — destroys the cancellation, because the step from (244) to (270) then produces a different factor $\mathrm{e}^{Q_k X}$ inside each weighted term and the $j > 0$ contributions no longer share the single coefficient $\mathcal{J}_n[r_{m,n-j}]$. Already at $n = 1$ the failure is visible: with $Q_0 = 0$ and $Q_1 = 1$ the numerator becomes $(-1) + \frac{1}{2}((-1)^2 - 0^2) = -\frac{1}{2}$ instead of $\frac{1}{2}$.

Example 10.18 (Symbolic check at $n = 1$)**.** Here $\left[{1 \atop 0}\right]_{1/2} = \left[{1 \atop 1}\right]_{1/2} = 1$ and $4^{\binom{0}{2}} = 4^{\binom{1}{2}} = 1$, so

$$\begin{aligned}\mathcal{N}_{1,1}(Q) &= \frac{\sigma_{0,1}(Q)}{1!} + \frac{\sigma_{1,2}(Q)}{2!}\\ &= (Q-1) + \tfrac{1}{2}\Big((Q-2)^2 - (Q-1)^2\Big)\\ &= (Q-1) + \tfrac{1}{2}(3 - 2Q) = Q - 1 + \tfrac{3}{2} - Q = \tfrac{1}{2},\end{aligned}$$

identically in Q, in agreement with $\mathfrak{m}_1 a_1 = 1 \cdot \frac{1}{2} = \frac{1}{2}$. At larger n the same cancellation occurs coefficient by coefficient, but the direct expansion becomes unwieldy very quickly; the order argument of theorem 10.16 proves all of the cancellations at once.

One further structural point explains why (205) can be asserted for an arbitrary *complex* shift. Theorem 10.16 is not a family of numerical equalities at conveniently chosen shifts; it is the statement that a specific element of the polynomial ring $\mathbb{Q}[Q]$ is constant. Once that is known, evaluation at a scalar is a ring homomorphism from $\mathbb{Q}[Q]$ into any commutative $\mathbb{Q}$-algebra, and it carries the constant polynomial to the same constant. The rational, real, Gaussian-rational, and complex versions of (205) are therefore all instances of one algebraic fact. No density argument, no continuity argument, and no analytic continuation is involved. It is worth separating three logically different assertions that the formula bundles together: the finite sums are algebraic and make sense over any commutative $\mathbb{Q}$-algebra; their interpretation as a value of a real function is analytic and belongs to $\mathbb{R}$; and the complex statement is the image of the rational one under the embedding $\mathbb{Q} \hookrightarrow \mathbb{C}$.

10.9 Worked examples

The smallest cases can be displayed completely, and they already exhibit the double cancellation.

Example 10.19 ($n = 0$)**.** Here $(1/2; 1/2)_0 = 1$, the only outer index is $k = 0$, and the only inner index is $r = 0$. With the centered shift $Q = 0$,

$$\frac{\mathcal{N}_{1,0}(0)}{2^0 \cdot 1} = \frac{(0-1)^0}{0!} = 1 = F(1).$$

The base of the power is -1, so no 0^0 convention is exercised; at $Q = 1$ the base is 0 and the natural-power convention $0^0 = 1$ is used instead, with the same value.

Example 10.20 ($n = 1$)**.** Now $(1/2; 1/2)_1 = 1/2$ and the prefactor is $1/(2^1 \cdot \frac{1}{2}) = 1$. The two outer terms are

$$\begin{aligned} k = 0: &\quad \frac{\left[{1 \atop 0}\right]_{1/2}}{4^0 \cdot 1!}\,\sigma_{0,1}(0) = \frac{(0-1)^1}{1} = -1,\\ k = 1: &\quad \frac{\left[{1 \atop 1}\right]_{1/2}}{4^0 \cdot 2!}\,\sigma_{1,2}(0) = \frac{(0-2)^2 - (1-2)^2}{2} = \frac{4-1}{2} = \frac{3}{2}.\end{aligned}$$

Their sum is $\mathcal{N}_{1,1}(0) = -1 + \frac{3}{2} = \frac{1}{2}$, and $F(1/2) = 1 \cdot \frac{1}{2} = \frac{1}{2}$. The two-stage check reads $\mathfrak{m}_1 a_1 = 1 \cdot \frac{1}{2} = \frac{1}{2}$ and $2^1(1/2;1/2)_1 = 1$.

Example 10.21 ($n = 2$)**.** Now

$$(1/2;1/2)_2 = \left(1-\tfrac{1}{2}\right)\left(1-\tfrac{1}{4}\right) = \tfrac{3}{8}, \qquad \left(\begin{bmatrix}2\\0\end{bmatrix}_{1/2}, \begin{bmatrix}2\\1\end{bmatrix}_{1/2}, \begin{bmatrix}2\\2\end{bmatrix}_{1/2}\right) = \left(1, \tfrac{3}{2}, 1\right),$$

so the global prefactor is $1/(2^4 \cdot \frac{3}{8}) = 1/6$. The inner sums are

$$\begin{aligned}
&\sigma_{0,2}(0) = (0-1)^2 = 1,\\
&\sigma_{1,3}(0) = (0-2)^3 - (1-2)^3 = -8 + 1 = -7,\\
&\sigma_{2,4}(0) = (0-4)^4 - (1-4)^4 - (2-4)^4 + (3-4)^4 = 256 - 81 - 16 + 1 = 160.
\end{aligned}$$

The three outer contributions are

$$k=0: \ \frac{1}{2!} = \frac{1}{2}, \qquad k=1: \ \frac{3/2}{3!}\cdot(-7) = -\frac{7}{4}, \qquad k=2: \ \frac{1}{4\cdot 4!}\cdot 160 = \frac{5}{3},$$

so

$$\mathcal{N}_{1,2}(0) = \frac{1}{2} - \frac{7}{4} + \frac{5}{3} = \frac{6-21+20}{12} = \frac{5}{12}, \qquad F\left(\frac{1}{4}\right) = \frac{1}{6}\cdot\frac{5}{12} = \frac{5}{72}.$$

Both cancellations are visible: the inner sums are already alternating cancellations among integer powers, and the outer combination cancels again before leaving the small rational value. The two-stage check is $\mathfrak{m}_2 a_2 = 3 \cdot \frac{5}{36} = \frac{5}{12}$ and $2^4(1/2;1/2)_2 = 6$.

Example 10.22 ($n = 4$, and why the formula must not be summed in floating point)**.** At $n = 4$ the numerator is $\mathcal{N}_{1,4}(0) = \mathfrak{m}_4 a_4 = 315 \cdot \frac{143}{32400} = \frac{1001}{720}$, the denominator is $2^{16}(1/2;1/2)_4 = 2^6 \cdot 315 = 20160$, and the value is $F(1/16) = \frac{1001}{14515200} = \frac{143}{2073600} \approx 6.90 \times 10^{-5}$. The five outer contributions are shown below.

k	$\begin{bmatrix}4\\k\end{bmatrix}_{1/2}$	$4^{\binom{k}{2}}$	$(4+k)!$	$\sigma_{k,4+k}(0)$	contribution
0	1	1	24	1	1/24
1	15/8	1	120	−31	−31/64
2	35/16	4	720	3304	2891/1152
3	15/8	64	5040	−1057728	−787/128
4	1	4096	40320	903684096	1751/320
				total	1001/720

The single block $k = 4$ contributes $1751/320 = 5.471875$, which is about 3.94 times the entire numerator $1001/720 \approx 1.390278$; the block $k = 3$ contributes $-787/128$, about 4.42 times the numerator in absolute value. The raw inner sum for that last block is $\sigma_{4,8}(0) = 903\,684\,096$, larger than the final answer $F(1/16)$ by a factor of about 1.3×10^{13}. This is the point of the example. The formula is exact, but it is catastrophically cancelling, and it must never be evaluated term by term in floating-point arithmetic. Section 10.13 quantifies the growth and lists the evaluation strategies that work.

10.10 What each factor is doing

Every ingredient of (205) now has exactly one structural job. Table 2 collects them.

The least self-explanatory entry is the normalization 2^{n^2}, and it becomes transparent once the exponent is split as

$$n^2 = \binom{n+1}{2} + \binom{n}{2}. \tag{271}$$

The first triangle is the geometric-grid normalization: it converts the q-Pochhammer product into the Mersenne product through (204), that is, it is the endpoint value of the node polynomial at the first node that is not a root, (218). The second triangle is the inverse-dyadic scaling of the Fabius function itself, the factor in $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$ that comes from (90). So the abrupt-looking square exponent is the sum of a grid normalization and a refinement normalization; it is not a single fitted constant.

The half-base Pochhammer product plays three interchangeable roles: it is the endpoint product of node differences on the dyadic geometric grid; multiplied by $2^{\binom{n+1}{2}}$ it is the odd Mersenne product $\mathfrak{m}_n$; and, up to sign, it is the value of the node polynomial $(z; 1/2)_n$ at $z = 2^n$. It is the geometric analogue of the $n!$ that normalizes an ordinary nth finite difference: on equally spaced nodes $0, 1, \ldots, n$ the endpoint product of differences is $n!$, whereas on the nodes $-1, -2, \ldots, -2^n$ it is a power of two times a Mersenne product, that is, a half-base q-factorial.

Table 2: Structural dictionary for the exact dyadic formula (205).

Ingredient	Structural role
$(1/2; 1/2)_n$	Endpoint value of the dyadic node polynomial at the first non-root; equivalently the normalized Mersenne product $2^{-\binom{n+1}{2}}\mathfrak{m}_n$. The geometric analogue of the $n!$ in an ordinary finite difference.
$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$	Barycentric weight at the node -2^k, by proposition 10.4; equivalently the kth expansion coefficient of $(z; 1/2)_n$. It measures no combinatorial choice inside the Fabius function; it is a closed form for a Lagrange denominator.
ε_r	Multiplicative character of the binary digit sum, whose generating polynomial factors once per dyadic scale, (241). An arbitrary alternating sign pattern would not factor scale by scale.
Exponent $n + k$	Raw moment degree. After the Thue–Morse zero of order k is divided out, what remains is the degree-n coefficient the outer stencil needs.
$4^{\binom{k}{2}}$	Two copies of $2^{-\binom{k}{2}}$: one removes the leading constant of the order-k zero, the other is the weight prescribed by theorem 10.3.
$(n + k)!$	Converts an exponential power moment into a formal-series coefficient.
B_m, the numerator m	Enters only through the finite Thue–Morse prefix (263). The Gaussian row encodes the resolution 2^{-n}, never the numerator.
Shift Q	Multiplication of every block by e^{QX}; invisible once all coefficients below degree n have been annihilated. A gauge freedom of the presentation.
2^{n^2}	The split (271): geometric-grid normalization $2^{\binom{n+1}{2}}$ times inverse-dyadic scaling $2^{\binom{n}{2}}$.

10.11 Two nested notions of finite difference

The formula is the composition of two generalized finite differences of different types. Making that explicit clarifies both the proof and the numerical behavior.

The inner one is additive. For a shift operator $\mathsf{E}^h f(x) = f(x+h)$, the ordinary kth forward difference is $(1 - \mathsf{E})^k$ evaluated at 0, and it annihilates polynomials of degree below k using

$k+1$ nodes. The Thue–Morse block functional

$$\mathcal{U}_k[f] = \sum_{r=0}^{2^k-1} \varepsilon_r f(r) \tag{272}$$

uses 2^k nodes and also annihilates every polynomial of degree below k, and proposition 10.7 says exactly why: its generating polynomial factors into one first difference per dyadic scale, so

$$\mathcal{U}_k[f] = \Big(\prod_{j=0}^{k-1}(1-\mathsf{E}^{2^j})f\Big)\Big|_{x=0}. \tag{273}$$

The inner sum is therefore an iterated *additive* difference, but with dyadically increasing step sizes $1, 2, 4, \dots, 2^{k-1}$ rather than a single step.

The outer one is multiplicative. The functional $\mathcal{J}_n$ of (236) annihilates polynomials of degree below n in the node variable and extracts the leading coefficient in degree n, but its geometry is that of a dilation: moving from one node to the next multiplies by 2. It is the leading divided difference on a geometric grid, which by section 10.1 is what a refinement equation requires.

The exact formula composes the two. First a dyadic additive-difference block is applied to the power function, producing a coefficient of the refinement product; then that coefficient is viewed as a polynomial in the geometric scale node -2^k and the geometric divided difference is applied; what survives is the single leading coefficient a_n, or its prefix-modified counterpart (267). Both operations are triangular with respect to polynomial degree, the inner one starting at degree k and the outer at degree n, and the indices of (205) are arranged so that the two triangles meet at one coefficient:

$$\underbrace{n+k}_{\text{raw moment degree}} - \underbrace{k}_{\text{inner forced zero}} = \underbrace{n}_{\text{outer extracted degree}}. \tag{274}$$

This is the cleanest mnemonic for the shape of the formula.

A base-b heuristic. Suppose a formal refinement equation reads $\mathscr{A}(bX) = E(X)\mathscr{A}(X)$ for an integer $b > 1$, and suppose a signed digit character supplies a block identity producing $\prod_{j<k} E(-b^j X)$. Then a geometric interpolation step on the nodes $-b^k$ must involve

$$q = b^{-1}, \qquad (z; b^{-1})_n = \prod_{j=0}^{n-1}(1 - zb^{-j}), \qquad \begin{bmatrix} n \\ k \end{bmatrix}_{b^{-1}}.$$

The digit character for base b is $\omega^{s_b(r)}$ with $\omega = \mathrm{e}^{2\pi\mathrm{i}/b}$ and s_b the base-b digit sum; its block generating polynomial is $\prod_{j<k}\big(1 + \omega z^{b^j} + \dots + \omega^{b-1} z^{(b-1)b^j}\big)$, and each factor has a simple zero at $z = 1$ because $1 + \omega + \dots + \omega^{b-1} = 0$. The total order of vanishing is again k. The Fabius case is $b = 2$, $\omega = -1$, and it is unusually tight because the Thue–Morse signs, the dyadic blocks, and the factor 2 in (4) are the same 2. The appearance of q-special functions is a general signature of dilation, not a peculiarity of one function.

10.12 Why this is not a basic hypergeometric summation

The outer coefficient of (205) resembles a terminating basic hypergeometric term, and it is tempting to look for a named ${}_r\phi_s$ evaluation [16, 18]. Recall that the defining feature of such a series is that the ratio of consecutive summands is a rational function of q^k. Here the coefficient is

$$\frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!}, \tag{275}$$

and the presence of the *ordinary* factorial $(n+k)!$ already breaks that property: the ratio of consecutive terms acquires the factor $1/(n+k+1)$, which is a rational function of k and not of 2^{-k}. The Thue–Morse power sum $\sigma_{m;k,n+k}(Q)$ breaks it a second time, since it depends on k both through the block length and through the exponent. The formula is therefore not a q-hypergeometric term at all. It is a hybrid of three calculi: finite q-calculus for the geometric scale index k; ordinary exponential generating functions for the moments and derivatives; and automatic-sequence cancellation for the binary digit signs. The interaction among the three is more informative than a forced rewrite as one named function would be.

What the theory does use from q-special functions is exactly one thing: the finite node polynomial $(z;1/2)_n$ and its root set, employed as an annihilator on a geometric grid. The infinite q-binomial theorem, the q-gamma function, and the general ${}_r\phi_s$ hierarchy are context, not ingredients. This also explains a fact recorded later in section 11: the global series is not a new infinite basic hypergeometric identity, because its outer sum is an analytic binary telescope while every q-manipulation inside it stays finite.

Two further framings are worth a sentence each. First, the Jackson q-derivative $(\partial_q f)(x) = (f(x) - f(qx))/((1-q)x)$ samples a multiplicative grid, just as an ordinary derivative samples an additive one. The functional $\mathcal{J}_n$ is not an iterate of $\partial_{1/2}$, because it is normalized as a leading divided difference on the fixed nodes $-1, -2, \ldots, -2^n$ rather than as a derivative at a point; but the two arise from the same principle, that derivatives and ordinary differences are adapted to translations whereas q-derivatives and geometric divided differences are adapted to dilations. Second, there is a spectral reading. Raising the node -2^k to the power d produces the mode 2^{dk} in the scale index k, so the outer sum acts on the finite family of modes indexed by d. The polynomial $(z;1/2)_n$ annihilates the modes with $d < n$ and retains the endpoint mode $d = n$, and its Gaussian coefficients are precisely the taps of that finite spectral filter.

10.13 Numerical practice

Example 10.22 is not an anomaly; it is the generic behavior. There are three independent sources of cancellation in (205), and they act on very different scales.

1. *Inside each Thue–Morse block.* The inner sum $\sigma_{m;k,n+k}(Q)$ has $m2^k$ terms of size roughly $(m2^k)^{n+k}$, but its first k moments vanish identically. This is the dominant effect. For $m = 1$ and $Q = 0$ the largest inner sum in the row of index n exceeds $F(2^{-n})$ by a factor of about 3×10^7 at $n = 3$, 1.3×10^{13} at $n = 4$, 1.4×10^{20} at $n = 5$, 3.8×10^{28} at $n = 6$, and 2.8×10^{49} at $n = 8$. The loss is superexponential in n.

2. *Across the outer interpolation weights.* The weights $w_{n,k}$ alternate in sign, and the outer sum is a divided difference, so the individual contributions again exceed their total. This effect is comparatively mild and appears to stay bounded: the largest single contribution in the row of index n exceeds $|\mathcal{N}_{1,n}(0)|$ by factors 3.0, 4.2, 4.7, 4.4, 4.1, 3.8, 3.7, 3.6 for $n = 1, \ldots, 8$.

3. *Against the global normalizer.* The surviving numerator $\mathcal{N}_{1,n}(0) = \mathfrak{m}_n a_n$ is of moderate size, but it is divided by $2^{n^2}(1/2;1/2)_n = 2^{\binom{n}{2}}\mathfrak{m}_n$, which by (223) grows like 2^{n^2} up to a bounded factor. The final result is thus a small number obtained as a quotient of a moderate number by a very large one.

Consequently the displayed order of summation destroys essentially all significance in fixed precision. Five evaluation strategies are sound.

- *Exact rational arithmetic.* Every quantity in (205) is rational with a denominator that is an explicit power of two times a product of factorials and Mersenne numbers, by (234) and (222). Nothing is lost, and the intermediate integers, though large, are of predictable size.

- *Modular arithmetic with rational reconstruction.* Evaluate the whole double sum modulo several primes that divide neither the factorials, nor $\mathfrak{m}_n$, nor the powers of two involved, then reconstruct the rational value. Because the answer has a known denominator bound, the number of primes needed can be fixed in advance.

- *Interval or ball arithmetic.* If a floating-point route is required, carry rigorous enclosures. The width of the final ball then reports the cancellation honestly instead of silently returning noise; in practice the working precision must exceed the cancellation exponent estimated in item (1) above.

- *Barycentric assembly of the outer weights.* Do not form $2^{-\binom{k}{2}}$ and $\left[{n \atop k}\right]_{1/2}$ separately and multiply. By proposition 10.4 and (238) the weight is a single reciprocal,
$$w_{n,k} = \frac{\mathfrak{m}_n}{(-1)^k 2^{\binom{k}{2}+k(n-k)}\mathfrak{m}_k\mathfrak{m}_{n-k}},$$
so the whole row is built from the two Mersenne prefix products $\mathfrak{m}_0, \dots, \mathfrak{m}_n$ and one exponent per index. Every intermediate is then an integer divided by an explicit power of two, the largest of which is bounded by (234); the raw-power route instead builds a large numerator and a large denominator that cancel afterwards.

- *Direct prefix coefficient for a general numerator.* For $m > 1$ do not expand the double sum of (205), whose inner ranges have total length $m(2^{n+1} - 1)$. Compute instead the single coefficient that (269) identifies with the value. Expanding (263) gives $[X^j]B_m = \left(\sum_{h<m} \varepsilon_h (m-1-h)^j\right)/j!$, hence
$$\mathcal{F}\left(\frac{m}{2^n}\right) = 2^{-\binom{n}{2}} \sum_{i=0}^{n} \frac{a_i}{(n-i)!} \sum_{h=0}^{m-1} \varepsilon_h \left(m-1-h\right)^{n-i},$$
with the a_i taken from (211). This costs $\mathcal{O}(mn)$ rational operations rather than $\mathcal{O}(m2^n)$, and it exhibits none of the cancellation catalogued above, since the only signed sum left is the short Thue–Morse prefix.

The translation freedom of theorem 10.16 is occasionally useful here, since a well-chosen common Q reduces the magnitude of the inner bases without changing the exact value; centering a single block near the mean of its nodes is the natural choice. It reduces the intermediate magnitudes but does not remove the cancellation, whose source is the forced vanishing of the low moments.

Finally, the derivation itself supplies a two-stage debugging check that is far easier to localize than a comparison of the complete nested expression in one step. Verify
$$\mathcal{N}_{1,n}(0) \stackrel{?}{=} \mathfrak{m}_n a_n \qquad \text{and then} \qquad 2^{n^2}(1/2;1/2)_n \stackrel{?}{=} 2^{\binom{n}{2}}\mathfrak{m}_n \tag{276}$$
separately. At $n = 2$ these read $\mathcal{N}_{1,2}(0) = 3 \cdot \frac{5}{36} = \frac{5}{12}$ and $2^4 \cdot \frac{3}{8} = 6$, whose quotient is $\frac{5}{72}$; at $n = 4$ they read $\mathcal{N}_{1,4}(0) = 315 \cdot \frac{143}{32400} = \frac{1001}{720}$ and $2^{16} \cdot \frac{315}{1024} = 20160$, whose quotient is $\frac{143}{2073600}$. A failure in the first equation localizes the error in the Thue–Morse sums, the Gaussian row, or the exponents; a failure in the second localizes it in the normalization alone. The exact data of appendix A give the necessary reference values.

The general lesson is the one already drawn in section 21: the q-binomial formula is a structural identity, not an evaluator. For actual computation of dyadic values, the bit recursion of section 9 and the recurrence (211) are both far better conditioned. What the q-formula supplies is a denominator audit — it records exactly where the powers of two enter, isolates the odd Mersenne part, and reduces the rest to an integer signed-power-sum problem — together with the interpolation picture that makes translation invariance, the raw-coordinate variant (209), and the Toeplitz-row phenomenon of section 12 all instances of one identity.

10.14 How to reconstruct the formula from scratch

The proofs above are arranged for verification: each ingredient is isolated, stated, and checked. A reader who wants to rebuild (205) without them needs a different arrangement, namely the shortest chain of forced steps. That chain is the following, and it fits on one page. Nothing in it is a guess: at every stage the next object is the only one that can appear.

1. *Normalize the half moments.* Define d_n by (115), put $a_n = d_n/n!$ and $\mathscr{A}(X) = \sum_{n\geq 0} a_n X^n$ as in (210), and prove the reciprocal-power theorem $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$, that is (136). From here to the last step the Fabius function is present only through the single rational sequence (a_n).

2. *Package the refinement as a generating-series identity.* The half-moment recurrence (119), in the normalized form (211), is equivalent coefficient by coefficient to the one-line identity $\mathscr{A}(2X) = E(X)\mathscr{A}(X)$ of (207), with $E(X) = (\mathrm{e}^X - 1)/X$. This is an identity in $\mathbb{Q}[[X]]$; no radius of convergence is required for any later step.

3. *Form the Thue–Morse block generating polynomial.* Expanding the binary digits of r gives $\sum_{r<2^k} \varepsilon_r z^r = \prod_{j<k}(1 - z^{2^j})$, which is (206) and proposition 10.7. One factor appears per dyadic scale; an arbitrary alternating sign pattern would not factor this way.

4. *Substitute the exponential, center, and divide out the forced zero.* Put $z = \mathrm{e}^X$ and multiply by $\mathrm{e}^{-2^k X}$, which centers the block at its right endpoint and produces C_k of (243). Each factor is $1 - \mathrm{e}^u = -u\,E(u)$ with $u = 2^j X$, so the product carries an explicit zero of order k with leading constant $(-1)^k 2^{\binom{k}{2}}$; this is (245). Dividing by X^k leaves $\mathscr{E}_k(X) = \mathrm{e}^{-X}\prod_{j<k} E(-2^j X)$ up to that constant, which is (247). Reading the vanishing coefficients off the same display gives Prouhet's cancellation (246) for free.

5. *Identify the quotient series by its product identity with the moment series.* Iterate (207) at $-X, -2X, \ldots, -2^{k-1}X$ and multiply by e^{-X}: the result is the bridge $\mathscr{E}_k(X)\mathscr{A}(-X) = \mathrm{e}^{-X}\mathscr{A}(-2^k X)$ of (253). This is the hinge of the whole derivation. The same product $\prod_{j<k} E(-2^j X)$ has now been produced twice, once by the binary digit character and once by the dilation equation, and the right-hand side depends on k only through the rescaling -2^k.

6. *Introduce the geometric nodes and the half-base weights.* Set $x_k = -2^k$ and $w_{n,k} = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$ as in (231). The base $q = 1/2$ is not chosen: it is the reciprocal of the dilation ratio in (207), so it is forced by step 2.

7. *Kill the low moments with the finite q-binomial theorem.* Expand $(z; 1/2)_n$ by (202), then use that its roots are exactly $1, 2, \ldots, 2^{n-1}$ and that its value at the first non-root $z = 2^n$ is $(-1)^n \mathfrak{m}_n$. In the node variable this is (232): the weights annihilate x_k^d for $d < n$ and return $\mathfrak{m}_n$ at $d = n$.

8. *Isolate the degree-n coefficient.* Two routes are available and give the same identity. The leading-coefficient route observes, as in proposition 10.11, that $[X^n]\mathscr{R}(z; X)$ is a polynomial of degree at most n in the node z with leading coefficient a_n, so step 7 extracts exactly that leading coefficient. The inductive route, theorem 10.9, extracts the coefficient of X^d from (253) and runs a strong induction on d, never dividing by $\mathscr{A}(-X)$. Either yields $\sum_k w_{n,k}\beta_{k,n} = \mathfrak{m}_n\, a_n$, which is (255).

9. *Recognize the weighted summand.* Combining (250) with (231) makes the two signs $(-1)^k$ cancel and the two triangular powers of two combine, so that $w_{n,k}\beta_{k,n}$ is literally the kth summand of (205); this is (251). Hence the numerator (261) at $m = 1$, $Q = 0$ equals $\mathfrak{m}_n\, a_n$.

10. *Normalize.* Convert the half-base Pochhammer product into the odd Mersenne product by (204), and split the exponent as $n^2 = \binom{n+1}{2} + \binom{n}{2}$, which is (271). The first triangle cancels the q-product, the second is the inverse-dyadic scaling already fixed in step 1. The outcome is (262).

11. *Get translation invariance from the order-n vanishing.* A common shift Q multiplies every block series by e^{QX}, by (244). Expanding that exponential contributes only coefficients of degree below n in the node, and step 7 has already annihilated all of them, so every positive power of Q drops out and $\mathcal{N}_{m,n}(Q)$ is a constant polynomial. This is theorem 10.16; the word *common* may not be dropped, by warning 10.17.

12. *Extend to a general numerator by the long-block factorization.* Write $r = h2^k + s$ with $0 \le h < m$ and $0 \le s < 2^k$ and use multiplicativity of the Thue–Morse sign: the long block factors as $\mathscr{D}_{m,k}(X) = B_m(-2^k X)C_k(X)$, which is (265). Steps 6–10 then apply verbatim with $\mathscr{A}$ replaced by $B_m\mathscr{A}$, and the extracted coefficient $[X^n](B_m(X)\mathscr{A}(X))$ is the dyadic value itself by (269). No new cancellation is needed.

13. *Pass to the global series.* Identify each normalized numerator with a Taylor coefficient of $\mathcal{F}$ through (295), and insert those coefficients scale by scale into the absolutely convergent binary-reduction telescope of section 11.

Only the last step involves convergence; steps 1–12 are finite identities in $\mathbb{Q}$ or in $\mathbb{Q}[[X]]$. In particular neither the inverse-dyadic formula (262) nor its independence of the shift Q requires analytic continuation, contour integration, or any infinite q-series summation theorem.

11 The global binary-reduction telescope

The Taylor-block recursion also applies to the signed global extension $\mathcal{F}$. It can therefore be iterated along the binary expansion of an arbitrary nonnegative real number, rather than only along the finite expansion of a dyadic rational. The reductions need not terminate, but their residuals tend geometrically to zero.

Define the finite reduction polynomial

$$\mathcal{T}_m(y) = \sum_{j=0}^{m} 2^{\binom{j+1}{2} - \binom{m-j}{2}} \frac{d_{m-j}}{(m-j)!} \frac{y^j}{j!}. \tag{277}$$

All powers of 2 here are integer powers. In particular, negative exponents are allowed, and $\mathcal{T}_0(y) = 1$. The signed Taylor-block identity is

$$\mathcal{F}(2^{-m} + y) = -\mathcal{F}(y) + \mathcal{T}_m(y), \qquad 0 \le y < 2^{-m}. \tag{278}$$

Fix $x \ge 0$. At scale m, put

$$p_m = \lfloor 2^m x \rfloor, \qquad y_m = x - \frac{p_m}{2^m}, \qquad 0 \le y_m < 2^{-m}. \tag{279}$$

The previous prefix requires a genuine half-scale value at $m = 0$:

$$p^*_{m-1} = \begin{cases} \lfloor x/2 \rfloor, & m = 0, \\ p_{m-1}, & m \ge 1. \end{cases} \tag{280}$$

Uniformly, $p^*_{m-1} = \lfloor p_m/2 \rfloor$. Thus $p_m = 2p^*_{m-1} + \delta_m$ with $\delta_m \in \{0, 1\}$. Define

$$\alpha_m(x) = (-1)^{\mathrm{wt}_2(p_m)}(2p^*_{m-1} - p_m), \qquad R_m(x) = (-1)^{\mathrm{wt}_2(p_m)}\mathcal{F}(y_m). \tag{281}$$

The coefficient α_m is zero or a sign, and it is nonzero precisely when the newly exposed binary digit is 1.

Lemma 11.1 (One-step residual identity)**.** *For $m \geq 1$,*

$$\alpha_m(x)\mathcal{T}_m(y_m) = R_{m-1}(x) - R_m(x). \tag{282}$$

Proof. If $p_m = 2p_{m-1}$, then $y_m = y_{m-1}$, the Thue–Morse signs agree, and $\alpha_m = 0$, so both sides vanish. If $p_m = 2p_{m-1} + 1$, then

$$y_{m-1} = 2^{-m} + y_m, \qquad (-1)^{\mathrm{wt}_2(p_m)} = -(-1)^{\mathrm{wt}_2(p_{m-1})}, \qquad \alpha_m = (-1)^{\mathrm{wt}_2(p_{m-1})}.$$

Apply (278) to y_{m-1} and multiply by the old prefix sign; the two copies of $\mathcal{F}(y_m)$ combine exactly into $R_{m-1} - R_m$. □

The scale-zero identity is separate and retains the parity of the integer block:

$$\mathcal{F}(x) = \alpha_0(x)\mathcal{T}_0(y_0) + R_0(x). \tag{283}$$

Indeed, write $p_0 = 2p^*_{-1} + \delta_0$ and use the two-unit translation law (12). When $\delta_0 = 0$, the polynomial term vanishes. When $\delta_0 = 1$, the identity $\mathcal{F}(1 + y_0) = -\mathcal{F}(y_0) + 1$ supplies it.

Theorem 11.2 (Exact finite binary telescope)**.** *For $x \geq 0$ and $N \in \mathbb{N}$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{N} \alpha_m(x)\mathcal{T}_m(y_m(x)) + R_N(x).} \tag{284}$$

Proof. Start with (283). Summing (282) for $m = 1, \ldots, N$ telescopes all intermediate residuals. The case $N = 0$ is exactly the scale-zero identity. □

For $N \geq 1$, $0 \leq y_N < 2^{-N} \leq 1/2$. On this interval $0 \leq \mathcal{F}(y) = F(y) \leq 2y$, so

$$|R_N(x)| \leq 2^{1-N}. \tag{285}$$

The estimate is independent of x. Therefore, if $B_N(x)$ denotes the finite sum in (284), then

$$\sup_{x \geq 0} |B_N(x) - \mathcal{F}(x)| \leq 2^{1-N}. \tag{286}$$

Lean's natural-floor convention extends every B_N by zero on $x \leq 0$, where $\mathcal{F}$ also vanishes. In that convention (286) holds with the supremum over all of $\mathbb{R}$, so the finite telescopes converge uniformly on the whole line. The one-step identity then gives, for $m \geq 2$,

$$|\alpha_m(x)\mathcal{T}_m(y_m)| \leq |R_{m-1}| + |R_m| \leq 4\, 2^{-(m-1)}. \tag{287}$$

Thus the limiting series is absolutely, not merely conditionally, convergent.

Theorem 11.3 (Absolutely convergent global binary series)**.** *For every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x)\mathcal{T}_m(y_m(x)).} \tag{288}$$

On $0 \leq x \leq 1$, the sum is equivalently $F(x)$.

The index $m = 0$ is indispensable globally. For $0 \leq x < 1$ it vanishes, so a series starting at $m = 1$ happens to be valid there. At $x = 1$, however, the zeroth term is 1 and all later coefficients vanish. More generally it is the zeroth term that distinguishes even and odd integer blocks of the signed extension.

11.1 The corrected parity-power presentation

The same summands have a more literal presentation in terms of prefix parity and inverse dyadic values. Define, using integer exponents,

$$\kappa_j = \frac{((j:\mathbb{Z})+2)((j:\mathbb{Z})-1)}{2} = \binom{j+1}{2} - 1. \tag{289}$$

The integer interpretation matters: $\kappa_0 = -1$, hence $2^{\kappa_0} = 1/2$; natural subtraction would silently give the wrong term. Put

$$\mathcal{I}_m(y) = \sum_{j=0}^{m} 2^{\kappa_j} \mathcal{F}\big(2^{(j:\mathbb{Z})-(m:\mathbb{Z})}\big) \frac{y^j}{j!}. \tag{290}$$

Lemma 11.4. *For every m and y,*

$$\mathcal{I}_m(y) = \frac{1}{2}\mathcal{T}_m(y). \tag{291}$$

Proof. For $j \le m$, set $r = m - j$. The inverse-dyadic identity (136) gives

$$\mathcal{F}(2^{j-m}) = \frac{d_{m-j}}{(m-j)!\, 2^{\binom{m-j}{2}}}.$$

Together with $\kappa_j = \binom{j+1}{2} - 1$, this identifies each summand with one half of the corresponding summand in (277). □

Let $\tau(a) = \mathrm{wt}_2(a) \bmod 2$. Since

$$\frac{(-1)^a - 1}{2} = 2\left\lfloor \frac{a}{2} \right\rfloor - a,$$

the literal summand

$$\begin{aligned} \Lambda_m(x) := {} & \big((-1)^{\lfloor 2^m x\rfloor} - 1\big)(-1)^{\tau(\lfloor 2^m x\rfloor)} \\ & \times \sum_{j=0}^{m} 2^{((j+2)(j-1))/2} \mathcal{F}(2^{j-m}) \frac{(x - \lfloor 2^m x\rfloor/2^m)^j}{j!} \end{aligned} \tag{292}$$

is exactly $\alpha_m(x)\mathcal{T}_m(y_m)$. Both exponents $((j+2)(j-1))/2$ and $j-m$ in this display are integer exponents, not truncated natural subtractions.

Warning 11.5. The same reading discipline applies to the *base* of the inner powers, and there the article's displays give no visual warning at all. In (242) and in (205) the quantity $r - m2^k + Q$ begins life with natural numbers: r is a summation index, m is a numerator, and 2^k is a block length. The subtraction must nevertheless be performed after these have been embedded in a field of characteristic zero. Truncated natural subtraction returns 0 whenever $r < m2^k$, that is for every term of every inner sum, so the whole inner sum would collapse to $\big(\sum_{r<m2^k} \varepsilon_r\big)Q^{n+k}$ and the identity would be destroyed. Already at $m = n = 1$ and $Q = 0$ the truncated reading gives $\sigma_{0,1}(0) = 0$ and $\sigma_{1,2}(0) = 0$, hence numerator 0, whereas the correct value is $\frac{1}{2}$ as in example 10.20. The centered coordinate is negative by design: it is what makes the block of length $m2^k$ end at the evaluation point.

Theorem 11.6 (Corrected parity-power series)**.** *For every $x \ge 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \Lambda_m(x),} \tag{293}$$

and the series converges absolutely. On the half-open unit interval $0 \le x < 1$, the zeroth summand vanishes and one may start at $m = 1$.

Proof. The elementary parity identity and (291) prove $\Lambda_m = \alpha_m \mathcal{T}_m(y_m)$ term by term. The claim follows from (288). The half-open corollary follows from $\lfloor x \rfloor = 0$; the closed endpoint $x = 1$ requires the restored zeroth term. □

11.2 The global complex q-binomial series

The finite constant-polynomial identity from section 9.9 can be inserted at every scale of the binary telescope. For $Q \in \mathbb{C}$, define

$$\mathcal{N}_n^{(Q)} = \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\mathrm{wt}_2(r)} (r - 2^k + Q)^{n+k}. \tag{294}$$

Translation invariance and the inverse-dyadic normalization give

$$\frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}}(1/2;1/2)_n} = \frac{2^n d_n}{n!}. \tag{295}$$

The normalization differs from the 2^{n^2} in (205): here it produces a Taylor coefficient rather than a function value.

For $m \in \mathbb{N}$, put

$$\mathcal{Q}_m^{(Q)}(y) = 2^{-\binom{m+1}{2}} \sum_{n=0}^{m} \frac{(2^{m+1}y)^{m-n}}{(m-n)!} \frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}}(1/2;1/2)_n}. \tag{296}$$

Theorem 11.7 (q-binomial Taylor identity)**.** *For every m, every $y \in \mathbb{R}$, and every $Q \in \mathbb{C}$,*

$$\mathcal{Q}_m^{(Q)}(y) = \mathcal{T}_m(y), \tag{297}$$

where the real polynomial on the right is embedded in $\mathbb{C}$. In particular, each finite summand is pointwise independent of Q before any infinite sum is taken.

Proof. Insert (295) into (296) and set $j = m - n$. Collecting the powers of 2 gives

$$2^{-\binom{m+1}{2}}\, 2^{(m+1)j}\, 2^{m-j} = 2^{\binom{j+1}{2} - \binom{m-j}{2}}.$$

The remaining factorials and moments are exactly those in (277). □

Theorem 11.8 (Global complex q-binomial series)**.** *For fixed $Q \in \mathbb{C}$ and every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x) \mathcal{Q}_m^{(Q)}(y_m(x)),} \tag{298}$$

with the left side embedded in $\mathbb{C}$. The series converges absolutely.

This is not a new infinite basic-hypergeometric identity. The outer sum is the analytic binary telescope, while at each scale a finite q-binomial identity presents the same Taylor polynomial. Expanding one summand makes the architecture explicit:

$$\begin{aligned} &\alpha_m(x)\, 2^{-\binom{m+1}{2}} \sum_{n=0}^{m} \frac{(2^{m+1}x - 2p_m(x))^{m-n}}{(m-n)!} \frac{1}{2^{\binom{n}{2}}(1/2;1/2)_n} \\ &\quad\times \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\mathrm{wt}_2(r)} (r - 2^k + Q)^{n+k}. \end{aligned} \tag{299}$$

Convergence comes entirely from the binary-tail majorant (287); the special-function manipulations are finite and algebraic.

12 A complex q-binomial discrete limit

There is a second way in which the same Gaussian row produces $\mathcal{F}$. It is a genuine limit of finite shifted splines rather than a sequence of partial sums of (298). Reuse the safe midpoint cutoff $N_p(x)$ from (60). For $x \geq 0$ it is equivalent to the inclusive source range

$$0 \leq r \leq \left\lfloor 2^p x - \frac{1}{2} \right\rfloor, \tag{300}$$

including the empty case where the upper endpoint is -1.

For fixed $Q \in \mathbb{C}$ and $x \in \mathbb{R}$, define

$$\begin{aligned}\mathcal{D}_n^{(Q)}(x) = \frac{1}{2^{n^2}(1/2;1/2)_n} &\sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \\ &\times \sum_{r=0}^{N_{n+k}(x)-1} (-1)^{\mathrm{wt}_2(r)} \left(r - 2^{n+k}x + Q\right)^{n+k}.\end{aligned} \tag{301}$$

Theorem 12.1 (Complex discrete-limit formula)**.** *For every fixed $Q \in \mathbb{C}$ and every $x \in \mathbb{R}$,*

$$\boxed{\mathcal{D}_n^{(Q)}(x) \longrightarrow \mathcal{F}(x)} \tag{302}$$

in $\mathbb{C}$. If $0 \leq x \leq 1$, the limit is equivalently $F(x)$.

Proof. If $x \leq 0$, every safe cutoff $N_p(x)$ is zero. Hence every shifted spline and every approximant (301) is zero, as is $\mathcal{F}(x)$. It remains to treat $x \geq 0$.

For a degree p and complex shift Q, introduce the finite shifted spline

$$S_p^{(Q)}(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} (-1)^{\mathrm{wt}_2(r)} (r - 2^p x + Q)^p. \tag{303}$$

At $Q = 1/2$ this is exactly the centered spline (61), so by (68),

$$S_p^{(1/2)}(x) \longrightarrow \mathcal{F}(x).$$

Lemma 12.2 (Fixed-cutoff Taylor expansion of a complex shift)**.** *Read (61) at $p = 0$ as*

$$S_0(x) = \sum_{r=0}^{N_0(x)-1} \varepsilon_r, \tag{304}$$

so that a centered branch S_d is defined for every integer $d \geq 0$. Fix $p \geq 1$ and $x \in \mathbb{R}$, and abbreviate $M = N_p(x)$. For every d with $0 \leq d \leq p$, the degree-d branch carried by that one fixed cutoff satisfies

$$\frac{1}{2^{\binom{d}{2}} d!} \sum_{r=0}^{M-1} \varepsilon_r \left(2^p x - \frac{1}{2} - r\right)^d = S_d\left(2^{p-d}x\right), \tag{305}$$

and in particular $\left|S_d\left(2^{p-d}x\right)\right| \leq 1$. Moreover, for every $Q \in \mathbb{C}$,

$$S_p^{(Q)}(x) - S_p^{(1/2)}(x) = \sum_{d=0}^{p-1} \frac{2^{\binom{d}{2}}}{2^{\binom{p}{2}}(p-d)!} \left(\frac{1}{2} - Q\right)^{p-d} S_d\left(2^{p-d}x\right). \tag{306}$$

Proof. Write the shifted spline in its descending orientation. Since $(-1)^p(r-2^px+Q)^p = (2^px-Q-r)^p$, definition (303) reads

$$S_p^{(Q)}(x) = \frac{1}{2^{\binom{p}{2}}p!}\sum_{r=0}^{M-1}\varepsilon_r\,(2^px-Q-r)^p\,, \qquad M = N_p(x), \tag{307}$$

and at $Q = 1/2$ this is (61). The cutoff M depends on p and x only. It is unaffected by Q, and it is held fixed in what follows while the degree of the summand varies; that is the whole point of (305).

We prove (305). The dilation $x \mapsto 2^{p-d}x$ compensates exactly for the drop in degree, in the sense that

$$N_d\Big(2^{p-d}x\Big) = \max\left(0, \left\lfloor 2^px+\frac{1}{2}\right\rfloor\right) = N_p(x) = M \tag{308}$$

by (60). If $M = 0$, both sides of (305) are empty sums and there is nothing to prove. Assume $M \geq 1$. Every index in the range satisfies

$$r \leq M-1 = \left\lfloor 2^px+\frac{1}{2}\right\rfloor - 1 \leq 2^px-\frac{1}{2},$$

so every base $2^px-\frac{1}{2}-r$ occurring in (305) is nonnegative and the truncation in (62) is inactive. For $d \geq 1$, formula (62) evaluated at the point $2^{p-d}x$, whose own cutoff is M by (308), is therefore literally the left side of (305). For $d = 0$ both sides equal $\sum_{r=0}^{M-1}\varepsilon_r$, by (304) and (308).

For the uniform bound, the case $d \geq 1$ is the global estimate in (68). For $d = 0$ pair the indices: since $\varepsilon_{2r} = \varepsilon_r$ and $\varepsilon_{2r+1} = -\varepsilon_r$, consecutive pairs cancel, so $\sum_{r=0}^{2m-1}\varepsilon_r = 0$ and $\sum_{r=0}^{2m}\varepsilon_r = \varepsilon_m$. Thus S_0 takes only the values 0 and ± 1.

Now expand. Put $w = 2^px-\frac{1}{2}$ for the descending argument at $Q = 1/2$ and $t = \frac{1}{2}-Q$ for its increment, so that $2^px-Q = w+t$. The binomial theorem gives

$$(w+t-r)^p = \sum_{d=0}^{p}\binom{p}{d}\,t^{p-d}(w-r)^d.$$

Multiply by $\varepsilon_r/(2^{\binom{p}{2}}p!)$, sum over $0 \leq r \leq M-1$, and interchange the two finite sums. Using (307) on the left and (305) on each inner sum,

$$\begin{aligned} S_p^{(Q)}(x) &= \sum_{d=0}^{p}\frac{t^{p-d}}{2^{\binom{p}{2}}p!}\binom{p}{d}\sum_{r=0}^{M-1}\varepsilon_r(w-r)^d \\ &= \sum_{d=0}^{p}\frac{2^{\binom{d}{2}}}{2^{\binom{p}{2}}(p-d)!}\,t^{p-d}\,S_d\Big(2^{p-d}x\Big)\,, \end{aligned} \tag{309}$$

because $\binom{p}{d}/p! = 1/(d!\,(p-d)!)$ and the inner sum is $2^{\binom{d}{2}}d!\,S_d(2^{p-d}x)$. The term $d = p$ is $S_p(x) = S_p^{(1/2)}(x)$; subtracting it leaves (306). □

Remark 12.3 (Which convention makes the sign disappear). The expansion (306) is free of alternating signs only because the increment is measured in the *descending* variable 2^px-Q of (307), that is, because the coefficient carries $\left(\frac{1}{2}-Q\right)^{p-d}$ and not $\left(Q-\frac{1}{2}\right)^{p-d}$. If one instead measures the shift by the ascending increment $q = Q-\frac{1}{2}$, which is the natural reading when the branch is written as a function of the ascending argument $r-2^px+Q$ of (303), then the identity acquires exactly the factor $(-1)^{p+d}$:

$$S_p^{(Q)}(x) - S_p^{(1/2)}(x) = \sum_{d=0}^{p-1}(-1)^{p+d}\frac{2^{\binom{d}{2}}}{2^{\binom{p}{2}}(p-d)!}\,q^{p-d}\,S_d\Big(2^{p-d}x\Big)\,, \qquad q = Q-\frac{1}{2}. \tag{310}$$

The two displays are the same statement, since $(-1)^{p+d} = (-1)^{p-d}$ and therefore $(-1)^{p+d}q^{p-d} = \left(\frac{1}{2} - Q\right)^{p-d}$. It is worth being explicit about where the sign does *not* live: it is not a matter of renormalizing the lower branches. The alternating prefactor $(-1)^d$ of (61) is cancelled exactly by the $(-1)^d$ produced when the ascending argument is reversed, so the branch values $S_d(2^{p-d}x)$ appearing in (306) and in (310) are the same numbers. The sign belongs to the increment alone, and the convention that removes it is the descending one.

Remark 12.4 (Where the two constants of the bound come from). Taking absolute values in (306) and using $\left|S_d(2^{p-d}x)\right| \le 1$ gives

$$\left|S_p^{(Q)}(x) - S_p^{(1/2)}(x)\right| \le \sum_{d=0}^{p-1} 2^{\binom{d}{2}-\binom{p}{2}} \frac{\left|Q-\frac{1}{2}\right|^{p-d}}{(p-d)!}. \tag{311}$$

Both constants of (313) are visible in (311). Write $e = p - d$, so that $1 \le e \le p$ and

$$\binom{p}{2} - \binom{d}{2} = \frac{(p-d)(p+d-1)}{2} = \frac{e\,(2p-e-1)}{2}.$$

As a function of e this is a downward parabola, so on $1 \le e \le p$ it attains its minimum at an endpoint; the value at $e = 1$ is $p - 1$ and the value at $e = p$ is $\binom{p}{2} \ge p - 1$. Hence every power of two in (311) is at most $2^{-(p-1)}$, with equality exactly in the top term $d = p - 1$, and

$$\left|S_p^{(Q)}(x) - S_p^{(1/2)}(x)\right| \le 2^{-(p-1)} \sum_{e=1}^{p} \frac{\left|Q-\frac{1}{2}\right|^{e}}{e!} \le 2^{-(p-1)}\left(\exp\left(\left|Q-\tfrac{1}{2}\right|\right) - 1\right). \tag{312}$$

This is (313), marginally sharpened. The factor $2^{-(p-1)}$ is the extremal case $d = p - 1$ of the triangular exponent; the exponential is the truncated exponential series of the increment; and the only input from the normalized Thue–Morse branches is the uniform bound $|S_d| \le 1$, which is why no further constant appears.

Expanding a fixed cutoff branch about $Q = 1/2$ gives an exact finite Taylor combination of lower-degree centered branches. The normalized Thue–Morse bounds then give, for $p \ge 1$,

$$\left|S_p^{(Q)}(x) - S_p^{(1/2)}(x)\right| \le 2^{-(p-1)} \exp\left(|Q - \tfrac{1}{2}|\right). \tag{313}$$

Thus every fixed complex shift has the same spline limit.

Now reindex the outer sum in (301) by $j = n - k$. Direct collection of signs, powers, and factorials gives the exact finite identity

$$\mathcal{D}_n^{(Q)}(x) = \sum_{j=0}^{n} \omega_{n,j} S_{2n-j}^{(Q)}(x), \tag{314}$$

where

$$\omega_{n,j} = \frac{(-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-\binom{j+1}{2}}}{(1/2;1/2)_n}. \tag{315}$$

The finite q-binomial theorem gives the row normalization

$$\sum_{j=0}^{n} \omega_{n,j} = 1. \tag{316}$$

Applying it with the opposite sign also gives the uniform variation bound

$$\sum_{j=0}^{n} |\omega_{n,j}| = \frac{(-1/2;1/2)_n}{(1/2;1/2)_n} \le 16. \tag{317}$$

Finally, every sampled degree satisfies $2n - j \geq n$. A finite-row Toeplitz lemma therefore carries the convergence of $S_p^{(Q)}(x)$ to the weighted rows (314). Explicitly, given $\epsilon > 0$, choose P so that $|S_p^{(Q)}(x) - \mathcal{F}(x)| < \epsilon/16$ for $p \geq P$. If $n \geq P$, then every $2n - j \geq n \geq P$, and the row-sum identity together with (317) gives

$$\left|\sum_{j=0}^{n} \omega_{n,j} S_{2n-j}^{(Q)}(x) - \mathcal{F}(x)\right| \leq \sum_{j=0}^{n} |\omega_{n,j}|\, |S_{2n-j}^{(Q)}(x) - \mathcal{F}(x)| < \epsilon.$$

This proves (302). □

Two distinctions prevent common misreadings of this theorem. First, unlike the complete finite dyadic numerator and each summand of the global q-series, a finite value $\mathcal{D}_n^{(Q)}(x)$ can depend on Q; it is the limit that is independent of a fixed shift. Second, the Toeplitz row (314) is not a finite binary telescope or a partial sum of (298). These finite expressions share a limit but are not equal term by term.

The two properties of the reindexed row (315) used above are both evaluations of the finite q-binomial theorem (202), at $z = 1/2$ and at $z = -1/2$ respectively. Both use the exponent identity $\binom{j}{2} + j = \binom{j+1}{2}$.

For the row sum, evaluate (202) at $z = 1/2$. The left side is $(1/2; 1/2)_n$ by definition, and the right side is

$$(1/2; 1/2)_n = \sum_{j=0}^{n} (-1)^j 2^{-\binom{j}{2}} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} \left(\frac{1}{2}\right)^j = \sum_{j=0}^{n} (-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-\binom{j+1}{2}}. \tag{318}$$

Dividing by $(1/2; 1/2)_n$ gives exactly $\sum_{j=0}^{n} \omega_{n,j} = 1$, which is (316).

For the total variation, evaluate (202) at $z = -1/2$ instead. Each factor $(-1)^j(-1/2)^j = 2^{-j}$ is now positive, so the alternating signs disappear:

$$(-1/2; 1/2)_n = \sum_{j=0}^{n} (-1)^j 2^{-\binom{j}{2}} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} \left(-\frac{1}{2}\right)^j = \sum_{j=0}^{n} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-\binom{j+1}{2}}. \tag{319}$$

Since $\begin{bmatrix} n \\ j \end{bmatrix}_{1/2} > 0$, the summands of (319) are the absolute values of those of (318). Dividing by $(1/2; 1/2)_n$ gives the exact evaluation

$$\sum_{j=0}^{n} |\omega_{n,j}| = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n}, \tag{320}$$

which is the first equality in (317). The uniform bound follows because $(-1/2; 1/2)_n$ increases to $(-1/2; 1/2)_\infty = 2.38423\ldots$ while $(1/2; 1/2)_n$ decreases to $(1/2; 1/2)_\infty = 0.28878\ldots$, so the quotient increases to

$$\frac{(-1/2; 1/2)_\infty}{(1/2; 1/2)_\infty} = 8.25598\ldots < 16. \tag{321}$$

The first values of the quotient are $1, 3, 5, 45/7, 51/7, 1683/217, 12155/1519$ for $n = 0, \ldots, 6$. The constant 16 in (317) is therefore a comfortable rounding of a genuine finite supremum, not an artifact of a lossy estimate.

13 Iterated prefix sums of the signed Thue–Morse sequence

The signed Thue–Morse sequence $\varepsilon_n = (-1)^{\mathrm{wt}_2(n)}$ has already appeared twice in this article: once as the coefficient sequence of the locally finite global expansion (11), and once as the sign pattern that makes the finite dyadic sums of section 9 collapse. There is a third, entirely discrete, way in which the same signs produce the Fabius function. Sum the sequence, sum the result, and iterate. After k summations the sequence has been convolved with a discrete B-spline kernel of order k, and after a dyadic normalization the resulting integer grid reproduces, in the limit, the Rvachev bump and hence F.

This section develops that construction: the convolution law and its explicit binomial kernel, the exact zero runs forced by Prouhet cancellation, the two generating-series identities, the discrete functional equation satisfied by the normalized grid, and the precise sense in which the normalized grid converges. The convergence statement requires one shift of the prefix order and the centered reading of the grid cells; the literal normalization does not converge, and that failure is recorded separately in section 22.7. Two further pieces of foundational material are attached at the end: the lower branch of Lambert's W-function, which the asymptotic sections of this article use but never construct, and an exact binomial–logarithm formula for the Thue–Morse bit.

13.1 Iterated prefix sums and the discrete B-spline law

Definition 13.1 (Iterated prefix sums)**.** Set $\sigma_0(n) = \varepsilon_n$ and, for $k \geq 0$,

$$\sigma_{k+1}(n) = \sum_{j=0}^{n} \sigma_k(j). \tag{322}$$

All sums are *inclusive*: the index n itself is summed.

Every $\sigma_k(n)$ is an integer, and $\sigma_k(0) = 1$ for all k. The inclusive convention has one consequence that must be kept in view throughout, because it is the source of an index shift later:

$$\sigma_{k+1}(n+1) - \sigma_{k+1}(n) = \sigma_k(n+1). \tag{323}$$

The increment of a prefix sum is the lower-order value at the *new* index, not at the old one.

Proposition 13.2 (Odd zeros of the first prefix sum)**.** *For every $m \geq 0$, $\sigma_1(2m+1) = 0$.*

Proof. By (88), $\varepsilon_{2j} + \varepsilon_{2j+1} = 0$ for every j. The range $0 \leq n \leq 2m+1$ is the disjoint union of the pairs $\{2j, 2j+1\}$ for $0 \leq j \leq m$, so the sum of ε_n over that range vanishes. □

The general structure is a discrete convolution. Write

$$\mathcal{B}_k(i) = \binom{i+k-1}{k-1} \qquad (k \geq 1,\ i \geq 0) \tag{324}$$

for the number of ways of writing i as an ordered sum of k nonnegative integers. The family $\mathcal{B}_k$ is the k-fold discrete convolution power of the constant kernel $\mathcal{B}_1 \equiv 1$; it is the discrete B-spline of order k. Because the argument $i = n - m$ is automatically nonnegative in the sums below, no indicator function is needed.

Theorem 13.3 (Discrete B-spline convolution law)**.** *For every $k \geq 1$ and every $n \geq 0$,*

$$\sigma_k(n) = \sum_{m=0}^{n} \mathcal{B}_k(n-m)\,\varepsilon_m = \sum_{m=0}^{n} \binom{n-m+k-1}{k-1} \varepsilon_m. \tag{325}$$

Equivalently, in a form valid for every $k \geq 0$,

$$\sigma_{k+1}(n) = \sum_{m=0}^{n} \binom{n-m+k}{k} \varepsilon_m. \tag{326}$$

Proof. Induct on k in the form (326). For $k = 0$ the binomial coefficient is 1 and the assertion is the definition of σ_1. Assume (326) for k. Then

$$\sigma_{k+2}(n) = \sum_{j=0}^{n} \sigma_{k+1}(j) = \sum_{j=0}^{n}\sum_{m=0}^{j} \binom{j-m+k}{k}\varepsilon_m$$
$$= \sum_{m=0}^{n} \varepsilon_m \sum_{i=0}^{n-m} \binom{i+k}{k} = \sum_{m=0}^{n} \binom{n-m+k+1}{k+1}\varepsilon_m,$$

where the interchange is the triangular reindexing $i = j - m$ and the inner evaluation is the hockey-stick identity $\sum_{i=0}^{N} \binom{i+k}{k} = \binom{N+k+1}{k+1}$. □

Remark 13.4. The kernel in (325) is exactly the coefficient sequence of $(1-z)^{-k}$. Theorem 13.3 is therefore the coefficient form of the series identity proved in theorem 13.10 below, and either statement may be taken as the definition of the other. Both are recorded here because the coefficient form is what one actually evaluates, while the series form is what one manipulates.

13.2 Prouhet annihilation and exact dyadic zero runs

Equation (171) already records that a complete dyadic block of Thue–Morse signs annihilates every polynomial of degree below the block exponent. For the prefix sums we also need the sharp boundary case, which identifies the first power that survives.

Lemma 13.5 (Sharp Prouhet extraction)**.** *Write $S_{r,d} = \sum_{h<2^r} \varepsilon_h h^d$. Then $S_{r,d} = 0$ for $0 \le d < r$, and*

$$S_{r,r} = (-1)^r 2^{\binom{r}{2}} r!. \tag{327}$$

Consequently, for all rational numbers $a_0, \dots, a_r$,

$$\sum_{h<2^r} \varepsilon_h \sum_{d=0}^{r} a_d h^d = a_r\,(-1)^r 2^{\binom{r}{2}} r!, \tag{328}$$

so summation against a complete dyadic block of Thue–Morse signs annihilates every polynomial of degree below r and, on polynomials of degree at most r, extracts the leading coefficient up to the explicit nonzero factor $(-1)^r 2^{\binom{r}{2}} r!$.

Proof. The vanishing statement is (171) with $X = 0$, $Y = 1$. For the boundary case, split the block by parity rather than by halves. Since $\varepsilon_{2j} = \varepsilon_j$ and $\varepsilon_{2j+1} = -\varepsilon_j$ by (88),

$$S_{r+1,d} = \sum_{j<2^r} \varepsilon_j \left[(2j)^d - (2j+1)^d\right] = -\sum_{c=0}^{d-1} \binom{d}{c} 2^c S_{r,c},$$

because $(2j+1)^d - (2j)^d = \sum_{c<d} \binom{d}{c}(2j)^c$. Take $d = r+1$. All terms with $c < r$ vanish by the first assertion, leaving only $c = r$:

$$S_{r+1,r+1} = -\binom{r+1}{r} 2^r S_{r,r} = -(r+1)2^r S_{r,r}.$$

Since $S_{0,0} = \varepsilon_0 = 1$, induction gives $S_{r,r} = (-1)^r r!\, 2^{0+1+\dots+(r-1)} = (-1)^r r!\, 2^{\binom{r}{2}}$, which is (327). Formula (328) follows by exchanging the two sums: every index $d < r$ contributes $a_d S_{r,d} = 0$, and the index $d = r$ contributes $a_r S_{r,r}$. □

Corollary 13.6 (Affine sharp case)**.** *For all $X, Y \in \mathbb{Q}$,*

$$\sum_{h<2^r} \varepsilon_h (X+Yh)^r = (-1)^r Y^r 2^{\binom{r}{2}} r!.$$

In particular the sum does not depend on the translation X.

Proof. Expand $(X+Yh)^r$ by the binomial theorem and apply (328): the coefficient of h^r is Y^r. $\square$

Equivalently, (328) is the statement that the block polynomial of (206) has a zero of exact order r at $z=1$: the mth derivative of $\prod_{j<r}(1-z^{2^j})$ at $z=1$ vanishes for $m<r$, while the rth derivative equals $(-1)^r r! \prod_{j<r} 2^j = (-1)^r r!\, 2^{\binom{r}{2}}$. We use the polynomial form because it applies directly to the binomial kernel (324), which is a polynomial in the summation index.

Theorem 13.7 (Exact dyadic zero runs)**.** *Let $1 \le k \le r$. Then*

$$\sigma_k(n) = 0 \qquad \text{for } 2^r - k \le n \le 2^r - 1, \tag{329}$$

$$\sigma_k(2^r - k - 1) = (-1)^{r-k}, \tag{330}$$

$$\sigma_k(2^r) = -1. \tag{331}$$

Formula (330) also holds for $k=0$, where it reduces to $\varepsilon_{2^r-1} = (-1)^r$, and (331) also holds for $k=0$. Hence the zero run terminating at 2^r-1 has length exactly k: it is bounded on the left by a value of modulus one and on the right by the value -1.

Proof. Endpoint. Put $q = k-1$, so $0 \le q < r$. By (325),

$$\sigma_k(2^r-1) = \sum_{m=0}^{2^r-1} \binom{2^r-1-m+q}{q} \varepsilon_m = \frac{1}{q!} \sum_{m<2^r} \varepsilon_m \prod_{i=1}^{q} (2^r - 1 - m + i).$$

The product is a polynomial in m of degree $q<r$, so the sum vanishes by (328).

Rest of the run. We prove, by induction on d, that $\sigma_k(2^r-1-d) = 0$ whenever $1 \le k \le r$ and $0 \le d < k$. The case $d=0$ is the endpoint. Suppose the claim holds for d and that $d+1<k$, so $k \ge 2$ and $1 \le k-1 \le r$. The induction hypothesis applied at order k gives $\sigma_k(2^r-1-d)=0$, and applied at order $k-1$, which is legitimate because $d<k-1$, gives $\sigma_{k-1}(2^r-1-d)=0$. Now use (323) with $n = 2^r-1-(d+1)$, so that $n+1 = 2^r-1-d$:

$$0 - \sigma_k(n) = \sigma_k(n+1) - \sigma_k(n) = \sigma_{k-1}(n+1) = 0.$$

This proves (329), since the indices $2^r-k, \dots, 2^r-1$ are exactly the 2^r-1-d with $0 \le d < k$.

Left boundary. Induct on k. For $k=0$ the number 2^r-1 has r binary digits, all equal to one, so $\varepsilon_{2^r-1} = (-1)^r$. Assume (330) at order k and let $k+1 \le r$. Put $n = 2^r-(k+1)-1$, so $n+1 = 2^r-k-1$. By the induction hypothesis $\sigma_k(n+1) = (-1)^{r-k}$, and by (329) at order $k+1$ we have $\sigma_{k+1}(n+1) = \sigma_{k+1}(2^r-(k+1)) = 0$. Then (323) gives $-\sigma_{k+1}(n) = (-1)^{r-k}$, that is, $\sigma_{k+1}(n) = (-1)^{r-(k+1)}$.

Right boundary. Induct on k. For $k=0$, $\mathrm{wt}_2(2^r)=1$, so $\varepsilon_{2^r}=-1$. If (331) holds at order k and $k+1 \le r$, then by (323) and (329), $\sigma_{k+1}(2^r) = \sigma_{k+1}(2^r-1) + \sigma_k(2^r) = 0 - 1 = -1$. $\square$

Remark 13.8. The right boundary value is not $+1$. Every iterated prefix sum, at every order $k \le r$, takes the value -1 at the index 2^r. This single sign is what makes the literal normalization of section 13.4 fail at the right endpoint; see section 22.7.

13.3 Two generating-series identities

All series in this subsection are formal power series over $\mathbb{Z}$; no convergence hypothesis on a complex variable is used or needed.

Let

$$\mathcal{K}_r(z) = \sum_{n<2^r} \varepsilon_n z^n, \qquad \mathcal{K}(z) = \sum_{n\geq 0} \varepsilon_n z^n, \qquad \mathcal{S}_k(z) = \sum_{n\geq 0} \sigma_k(n) z^n. \tag{332}$$

Proposition 13.9 (Block product and coefficient stabilization)**.** *For every $r \geq 0$,*

$$\mathcal{K}_r(z) = \prod_{j=0}^{r-1} (1 - z^{2^j}), \tag{333}$$

which is (206). Moreover, for every $n < 2^r$ the coefficient of z^n in the finite product (333) equals ε_n and is therefore independent of r. Consequently the infinite product $\prod_{j\geq 0}(1 - z^{2^j})$ is a well-defined formal power series, and it equals $\mathcal{K}(z)$.

Proof. Induct on r. For $r = 0$ both sides are 1. For the step, split the block of length 2^{r+1} at 2^r and use $\varepsilon_{2^r+n} = -\varepsilon_n$ for $n < 2^r$:

$$\mathcal{K}_{r+1}(z) = \sum_{n<2^r} \varepsilon_n z^n - z^{2^r} \sum_{n<2^r} \varepsilon_n z^n = \mathcal{K}_r(z)(1 - z^{2^r}),$$

and the product formula follows. For the stabilization, $\mathcal{K}_r$ is a polynomial of degree $2^r - 1$ whose coefficient of z^n is ε_n for every $n < 2^r$; adjoining the factor $1 - z^{2^r}$ changes only coefficients of index $\geq 2^r$. Thus for fixed n the coefficient of z^n in $\prod_{j<r}(1 - z^{2^j})$ is constant once $2^r > n$, which is the definition of convergence of the infinite product in the formal topology, and the stable value is ε_n. □

Theorem 13.10 (Prefix generating series)**.** *For every $k \geq 0$,*

$$\mathcal{S}_k(z) = \frac{\mathcal{K}(z)}{(1 - z)^k}, \tag{334}$$

equivalently, in denominator-cleared form,

$$(1 - z)^k \, \mathcal{S}_k(z) = \mathcal{K}(z) = \prod_{j\geq 0} (1 - z^{2^j}). \tag{335}$$

Both identities have a finite verification: for every r and every $n < 2^r$, the coefficient of z^n on either side of (335) may be computed from the polynomial $\prod_{j<r}(1 - z^{2^j})$ alone.

Proof. For $k = 0$ both sides of (334) are $\mathcal{K}(z)$. For $k \geq 1$, expand $(1-z)^{-k} = \sum_{i\geq 0} \binom{i+k-1}{k-1} z^i = \sum_{i\geq 0} \mathcal{B}_k(i) z^i$. The coefficient of z^n in $\mathcal{K}(z)(1 - z)^{-k}$ is then the Cauchy product $\sum_{m=0}^{n} \mathcal{B}_k(n - m)\varepsilon_m$, which is $\sigma_k(n)$ by (325). Since $(1-z)^k$ is a unit multiple of the invertible series $(1-z)^{-k}$, the cleared form (335) is equivalent, and the product expression is proposition 13.9. The finite verification is the stabilization statement of that proposition: no coefficient of index below 2^r sees any factor $1 - z^{2^j}$ with $j \geq r$. □

The two identities of theorem 13.10 connect the prefix sums to the polynomial approximants used in section 13.5. These are the combinatorial step polynomials already introduced in (51), whose degree and total mass are recorded in (53); we restate the product and degree here in the index convention of this section:

$$\mathcal{P}_n(z) = \prod_{i=1}^{n} (1 + z + \cdots + z^{2^i-1}), \qquad g_n = \deg \mathcal{P}_n = \sum_{i=1}^{n} (2^i - 1) = 2^{n+1} - n - 2. \tag{336}$$

Since $1 - z^{2^j} = (1-z)(1+z+\cdots+z^{2^j-1})$ and the factor with $j=0$ is trivial, (333) factors as

$$\mathcal{K}_{n+1}(z) = (1-z)^{n+1}\mathcal{P}_n(z). \tag{337}$$

Comparing (337) with (335) at order $k = n+1$ and using stabilization below 2^{n+1} gives the exact coefficient identification

$$\sigma_{n+1}(m) = [z^m]\mathcal{P}_n(z) \qquad (0 \le m < 2^{n+1}). \tag{338}$$

Note that (338) must fail at $m = 2^{n+1}$, because $g_n < 2^{n+1}$ makes the right side zero while the left side is -1 by (331). That single index is the exceptional right endpoint of theorem 13.16.

13.4 The functional equation of the normalized prefix grid

The natural dyadic normalization of the prefix sums is

$$\mathcal{V}_k(j) = \frac{\sigma_k(j)}{2^{\binom{k}{2}}}, \qquad x_{k,j} = \frac{j}{2^k}, \tag{339}$$

so that $\mathcal{V}_k$ is a rational-valued function on the level-k dyadic grid. The exponent $\binom{k}{2}$ is forced: it is the exponent that appears in the sharp Prouhet value (327), in the centered spline (61), and in the derivative scaling (90).

Theorem 13.11 (Discrete functional equation)**.** *Let $q \ge 0$ and $0 \le j < 2^q$, and put $h = 2^{-(q+1)}$ and $x = x_{q+1,j}$. Then*

$$\frac{\mathcal{V}_{q+1}(j+1) - \mathcal{V}_{q+1}(j)}{h} = 2\,\mathcal{V}_q(j+1), \tag{340}$$

and the abscissa attached to the right-hand value is

$$x_{q,j+1} = 2(x+h) = 2x_{q+1,j+1} \in [0,1]. \tag{341}$$

The same identity holds verbatim for the shifted grid $\widetilde{\mathcal{V}}_k(j) = \sigma_{k+1}(j)/2^{\binom{k}{2}}$ of section 13.5.

Proof. Because $\binom{q+1}{2} = \binom{q}{2} + q$, the forward difference (323) gives

$$\mathcal{V}_{q+1}(j+1) - \mathcal{V}_{q+1}(j) = \frac{\sigma_{q+1}(j+1) - \sigma_{q+1}(j)}{2^{\binom{q}{2}+q}} = \frac{\sigma_q(j+1)}{2^{\binom{q}{2}}} \cdot \frac{1}{2^q} = \frac{\mathcal{V}_q(j+1)}{2^q}.$$

Dividing by $h = 2^{-(q+1)}$ multiplies by 2^{q+1} and gives (340). For (341), $2(x+h) = 2(j+1)/2^{q+1} = (j+1)/2^q = x_{q,j+1}$, and the hypothesis $j < 2^q$ is exactly the condition $j+1 \le 2^q$ that keeps this abscissa in the unit interval. The shifted grid satisfies the same computation with σ replaced by $\sigma_{\bullet+1}$ throughout. □

Equation (340) is the discrete counterpart of the global differential equation (89): a difference quotient at level $q+1$ equals twice a level-q value at the doubled abscissa. Two features of the discrete statement have no continuous analogue and are easy to lose. First, the right-hand side is evaluated at the *forward* node $2(x+h)$ rather than at $2x$; this is the inclusive-prefix convention (323). Second, the identity is an assertion about a level-q value, so it is meaningful as a statement about F or up only when the doubled abscissa lies in the relevant interval, that is, only when $j < 2^q$. Both features are used in section 13.5.

13.5 Corrected normalization and pointwise convergence

We now identify the normalized prefix grid with the histogram approximants of the probabilistic construction, and deduce convergence. Two adjustments to the naive reading are required, and both are forced by identities already proved: the prefix order must be shifted by one, because (338) matches σ_{n+1} with $\mathcal{P}_n$; and the grid index m must be read as labelling the *center* of a cell of width 2^{-n}, not the left endpoint of a cell of width $2^{-(n+1)}$.

Recall the step approximants $\mathcal{W}_n$ from (55) and their pointwise convergence (56). Their bounded-partition interpretation can be expressed through an exact finite discrete law: let $Z_1, \dots, Z_n$ be independent with Z_i uniform on $\{0, 1, \dots, 2^i - 1\}$, so that $Z_1 + \cdots + Z_n$ has generating polynomial $\mathcal{P}_n$ of (336) and total mass $\mathcal{P}_n(1) = 2^{\binom{n+1}{2}}$. Place the mass of the monomial z^m at

$$\xi_{n,m} = \frac{2m - g_n}{2^{n+1}}, \tag{342}$$

and spread it uniformly over the cell $I_{n,m}$ of width 2^{-n} centered at $\xi_{n,m}$. The consecutive centers are spaced 2^{-n} apart, so these finitely many cells tile their support interval

$$\left[-1 + \frac{n+1}{2^{n+1}},\ 1 - \frac{n+1}{2^{n+1}}\right].$$

The resulting density is the step function

$$\mathcal{W}_n(x) = \frac{2^n}{2^{\binom{n+1}{2}}} \sum_{m=0}^{g_n} [z^m]\mathcal{P}_n(z) \cdot \mathbf{1}_{I_{n,m}}(x), \tag{343}$$

where a point shared by two adjacent cells is given the weight $\frac{1}{2}$ in each, so that the two contributions add to the single value the formula is intended to have. This endpoint convention is not cosmetic: with ordinary closed-interval indicators the value at $x = 0$ would be counted twice already for $n = 1$.

Theorem 13.12 (Histogram approximants)**.** *For every $n \geq 0$ the function $\mathcal{W}_n$ satisfies $0 \leq \mathcal{W}_n \leq 1$, is nondecreasing on $(-\infty, 0)$, is nonincreasing on $(0, \infty)$, and satisfies $\mathcal{W}_n(0) = 1$. Moreover $\mathcal{W}_n(x) \to \mathrm{up}(x)$ for every real x.*

Proof. For $n \geq 1$, this is exactly theorem 4.2, since (336) and (51) define the same polynomial, while (342) and (343) give the same cells and normalization as (54) and (55). At $n = 0$ the empty product is $\mathcal{P}_0 = 1$, $g_0 = 0$, and $\mathcal{W}_0$ is the half-weighted indicator of $[-1/2, 1/2]$, so the stated order and normalization properties are immediate. The pointwise convergence of the sequence is already the final assertion of theorem 4.2. □

Theorem 13.13 (Exact identification of the corrected grid)**.** *Put $\widetilde{\mathcal{V}}_n(m) = \sigma_{n+1}(m)/2^{\binom{n}{2}}$. For every $n \geq 0$ and every m with $0 \leq m < 2^{n+1}$,*

$$\widetilde{\mathcal{V}}_n(m) = \mathcal{W}_n(\xi_{n,m}). \tag{344}$$

Proof. If $m \leq g_n$, the point $\xi_{n,m}$ is the center of $I_{n,m}$ and lies in no other cell, so (343) evaluates there to

$$\frac{2^n}{2^{\binom{n+1}{2}}} [z^m]\mathcal{P}_n(z).$$

Since $\binom{n+1}{2} = \binom{n}{2} + n$, this is $[z^m]\mathcal{P}_n/2^{\binom{n}{2}} = \widetilde{\mathcal{V}}_n(m)$ by (338). If $g_n < m < 2^{n+1}$, then $\xi_{n,m}$ lies strictly to the right of the finite support interval above, so $\mathcal{W}_n(\xi_{n,m}) = 0$; meanwhile $[z^m]\mathcal{P}_n = \sigma_{n+1}(m) = 0$ because $\deg \mathcal{P}_n = g_n$. Thus both sides again agree. □

Sampling at the dyadic floor now gives an approximation scheme. For $x \geq 0$ set

$$j_n(x) = \lfloor 2^{n+1}x \rfloor, \qquad \mathcal{J}_n(x) = \widetilde{\mathcal{V}}_n(j_n(x)) = \frac{\sigma_{n+1}(j_n(x))}{2^{\binom{n}{2}}}. \tag{345}$$

Lemma 13.14 (Moving cell centers)**.** *For every $x \in [0,1)$, $\xi_{n,j_n(x)} \to 2x-1$ as $n \to \infty$.*

Proof. By (342), $\xi_{n,j_n(x)} = 2 \cdot j_n(x)2^{-(n+1)} - g_n 2^{-(n+1)}$. The first term tends to $2x$ because $0 \leq 2^{n+1}x - j_n(x) < 1$, and $g_n 2^{-(n+1)} = 1-(n+2)2^{-(n+1)} \to 1$. □

Lemma 13.15 (Stability under moving arguments)**.** *If $u_n \to y$ in $\mathbb{R}$, then $\mathcal{W}_n(u_n) \to \operatorname{up}(y)$.*

Proof. Let $\varepsilon > 0$. Suppose first $y > 0$. By continuity of up choose $d \in (0, y/2]$ with $|\operatorname{up}(y \pm d) - \operatorname{up}(y)| < \varepsilon/2$. For all large n we have $u_n \in [y-d, y+d] \subset (0,\infty)$, so the monotonicity of theorem 13.12 on $(0,\infty)$ gives

$$\mathcal{W}_n(y+d) \leq \mathcal{W}_n(u_n) \leq \mathcal{W}_n(y-d).$$

By pointwise convergence at the two fixed points $y \pm d$, for all large n both outer terms lie within $\varepsilon/2$ of $\operatorname{up}(y \pm d)$, hence within ε of $\operatorname{up}(y)$. The case $y < 0$ is identical using monotonicity on $(-\infty, 0)$. For $y = 0$ the upper bound is $\mathcal{W}_n(u_n) \leq 1 = \operatorname{up}(0)$, and the lower bound is $\mathcal{W}_n(-d)$, $\mathcal{W}_n(0) = 1$, or $\mathcal{W}_n(d)$ according to the sign of u_n; each of the three converges to a value within $\varepsilon/2$ of $\operatorname{up}(0)$. □

Theorem 13.16 (Corrected prefix approximation)**.** *For every $x \in [0,1]$,*

$$\mathcal{J}_n(x) \longrightarrow \operatorname{up}(2x-1), \tag{346}$$

and consequently, for every $x \in [0,1]$,

$$\mathcal{J}_n\left(\frac{x}{2}\right) \longrightarrow F(x). \tag{347}$$

Proof. Let $0 \leq x < 1$. Then $2^{n+1}x < 2^{n+1}$, so $j_n(x) < 2^{n+1}$ and theorem 13.13 applies: $\mathcal{J}_n(x) = \mathcal{W}_n(\xi_{n,j_n(x)})$. Combining lemma 13.14 with lemma 13.15 gives $\mathcal{J}_n(x) \to \operatorname{up}(2x-1)$.

At $x = 1$ the identification fails, because $j_n(1) = 2^{n+1}$ is exactly the first index not covered by (338). Here the value is computed instead from (331) with $k = r = n+1$:

$$\mathcal{J}_n(1) = \frac{\sigma_{n+1}(2^{n+1})}{2^{\binom{n}{2}}} = -\frac{1}{2^{\binom{n}{2}}} \longrightarrow 0 = \operatorname{up}(1),$$

since $\binom{n}{2} \to \infty$ and $\operatorname{up}(1) = F(0) = 0$. This proves (346) on all of $[0,1]$. For (347), apply it at $x/2 \in [0,1]$ and use $2(x/2) - 1 = x - 1 \leq 0$, so that $\operatorname{up}(x-1) = F(x)$ by (7). □

Warning 13.17. Theorem 13.16 is a pointwise statement and nothing more. No uniform rate is asserted, and none is proved. The proof passes through lemma 13.15, whose only quantitative inputs are the shared monotonicity of the $\mathcal{W}_n$ and the modulus of continuity of the limit up at the single point $2x-1$; the sandwich it uses degrades as that modulus degrades. A uniform bound of the kind supplied for the centered splines by (67) is available from section 4.4, but it is a different approximation scheme, and the two should not be conflated.

Remark 13.18. The two corrections are independent of one another and both are necessary. Shifting the prefix order from k to $k+1$ is what makes (338) available; reading the index m as the center (342) of a cell of width 2^{-n} rather than as an abscissa $m2^{-k}$ is what supplies the affine change of variable $x \mapsto 2x-1$ between the unit interval and the support of up. In particular the corrected grid is a sample of the compactly supported bump, not of the distribution function; this is why its right-endpoint limit is 0 and not 1.

13.6 The lower Lambert branch

The asymptotic analysis of section 15 is organized around the coordinate (389), where W_{-1} denotes the lower real branch of Lambert's W-function [11]. That section, and the elementary expansions derived from it in section 17, use three facts about W_{-1} without proving any of them: that the branch exists on $(-\mathrm{e}^{-1}, 0)$, that it is characterized by being the solution below -1, and that it has the two-term expansion recorded below. We supply them here, since they are the foundational input for (391) and for every subsequent estimate in the Lambert variable.

Proposition 13.19 (Existence, defining equation, and uniqueness)**.** *The map $u \mapsto u \log u$ is a strictly decreasing bijection from $(0, \mathrm{e}^{-1})$ onto $(-\mathrm{e}^{-1}, 0)$. For $z \in (-\mathrm{e}^{-1}, 0)$ let $u(z)$ be its unique preimage and set $W_{-1}(z) = \log u(z)$. Then*

$$W_{-1}(z) < -1, \qquad W_{-1}(z)\,\mathrm{e}^{W_{-1}(z)} = z, \tag{348}$$

and $W_{-1}(z)$ is the unique real $w < -1$ with $w\mathrm{e}^{w} = z$.

Proof. On $(0, \mathrm{e}^{-1})$ the derivative of $u \log u$ is $1 + \log u < 0$, so the map is strictly decreasing there. Its limit at $u \downarrow 0$ is 0 and its value at $u = \mathrm{e}^{-1}$ is $-\mathrm{e}^{-1}$; by the intermediate value theorem and strict monotonicity it is a bijection onto $(-\mathrm{e}^{-1}, 0)$. Write $u = u(z)$ and $w = \log u$. Since $0 < u < \mathrm{e}^{-1}$ we get $w < -1$, and $\mathrm{e}^{w} = u$, so $w\mathrm{e}^{w} = u \log u = z$, which is (348). Conversely, if $w < -1$ and $w\mathrm{e}^{w} = z$, put $v = \mathrm{e}^{w}$; then $v \in (0, \mathrm{e}^{-1})$ and $v \log v = w\mathrm{e}^{w} = z$, so $v = u(z)$ by injectivity, hence $w = \log u(z) = W_{-1}(z)$. ☐

Corollary 13.20 (Solution of the saddle equation)**.** *Let $x > 0$ with $Lx < \mathrm{e}^{-1}$, where $L = \log 2$ as in (353), and put $\lambda = -W_{-1}(-Lx)/L$ as in (390). Then $\lambda > 1/L$ and $\lambda 2^{-\lambda} = x$, which is (391).*

Proof. The number $z = -Lx$ lies in $(-\mathrm{e}^{-1}, 0)$, so proposition 13.19 applies. Write $W = W_{-1}(z)$, so $\lambda = -W/L$ and $2^{-\lambda} = \mathrm{e}^{-L\lambda} = \mathrm{e}^{W}$. Hence $\lambda 2^{-\lambda} = (-W/L)\mathrm{e}^{W} = -z/L = x$. Finally $W < -1$ gives $\lambda > 1/L$. ☐

Proposition 13.21 (Two-term expansion)**.** *As $\epsilon \downarrow 0$,*

$$W_{-1}(-\epsilon) - \big(\log \epsilon - \log|\log \epsilon|\big) \longrightarrow 0. \tag{349}$$

Proof. For $0 < \epsilon < \mathrm{e}^{-1}$ put $y = y(\epsilon) = -W_{-1}(-\epsilon) > 1$. Negating (348) gives the defining equation in the convenient form

$$y\,\mathrm{e}^{-y} = \epsilon.$$

The map $t \mapsto t\mathrm{e}^{-t}$ is strictly decreasing on $[1, \infty)$ with limit 0 at infinity, so $y(\epsilon) \to \infty$ as $\epsilon \downarrow 0$. Taking logarithms in the displayed equation gives $\log \epsilon = \log y - y$, hence

$$\frac{-\log \epsilon}{y} = 1 - \frac{\log y}{y} \longrightarrow 1,$$

because $\log y/y \to 0$. Finally, $\epsilon < 1$ makes $\log \epsilon < 0$, so $|\log \epsilon| = -\log \epsilon$, and

$$\begin{aligned} W_{-1}(-\epsilon) - \big(\log \epsilon - \log|\log \epsilon|\big) &= -y - (\log y - y) + \log(-\log \epsilon) \\ &= \log \frac{-\log \epsilon}{y} \longrightarrow \log 1 = 0. \end{aligned}$$ ☐

Remark 13.22. Proposition 13.21 is an $o(1)$ statement, not a statement with a rate, and it is the first two terms only. Neither property is a defect of the proof; both are reasons why the elementary displays of section 17 cannot simply substitute it for the exact phase $\lambda(x)$ in (393). The main exponent there contains $-L\lambda^2/2$, so a phase error of size η moves the answer by about $L\lambda\eta$; the phase must therefore be resolved several orders beyond (349), and the periodic term must still be evaluated at the exact phase.

13.7 A binomial–logarithm formula for the Thue–Morse bit

The zero-one Thue–Morse bit is $\tau(n) = \mathrm{wt}_2(n) \bmod 2$, so that $\varepsilon_n = 1 - 2\tau(n)$; the primitive-recursive evaluation (541) computes it one binary digit at a time. It also admits a closed formula in which the entire binary structure is hidden inside a single row of Pascal's triangle. Set

$$\mathcal{X}(n) = \sum_{k=0}^{n} (-1)^{\binom{n}{k}} \in \mathbb{Z}. \tag{350}$$

Lemma 13.23 (Glaisher count)**.** *The number of odd entries in the nth row of Pascal's triangle is $2^{\mathrm{wt}_2(n)}$.*

Proof. Let $\nu(n)$ denote that number, so that $\nu(n)$ is the number of nonzero coefficients of $(1+z)^n$ in $\mathbb{F}_2[z]$. There $(1+z)^2 = 1 + z^2$, hence

$$(1+z)^{2n} = (1+z^2)^n, \qquad (1+z)^{2n+1} = (1+z^2)^n(1+z).$$

The substitution $z \mapsto z^2$ does not change the number of nonzero coefficients, so $\nu(2n) = \nu(n)$; and multiplying a polynomial in z^2 by $1+z$ produces two summands with disjoint supports, one even and one odd, so $\nu(2n+1) = 2\nu(n)$. Since $\mathrm{wt}_2(2n) = \mathrm{wt}_2(n)$ and $\mathrm{wt}_2(2n+1) = \mathrm{wt}_2(n)+1$ by (87), and $\nu(0) = 1 = 2^{\mathrm{wt}_2(0)}$, induction on the binary representation gives $\nu(n) = 2^{\mathrm{wt}_2(n)}$. □

Proposition 13.24 (The logarithm is exact)**.** *For every $n \geq 0$,*

$$n + 1 - \mathcal{X}(n) = 2^{\mathrm{wt}_2(n)+1}. \tag{351}$$

In particular $n+1-\mathcal{X}(n)$ is a positive power of two, and $\log_2(n+1-\mathcal{X}(n)) = \mathrm{wt}_2(n)+1$ is an integer.

Proof. Each summand of (350) is -1 at an odd entry and $+1$ at an even entry, and the row has $n+1$ entries. Hence $\mathcal{X}(n) = (n+1) - 2\nu(n)$ with ν as in lemma 13.23, so $n+1-\mathcal{X}(n) = 2\nu(n) = 2^{\mathrm{wt}_2(n)+1}$. □

Theorem 13.25 (Binomial–logarithm formula)**.** *For every $n \geq 0$,*

$$\tau(n) = \frac{1 + (-1)^{\log_2\left(n+1-\sum_{k=0}^{n}(-1)^{\binom{n}{k}}\right)}}{2}, \qquad \varepsilon_n = 1 - 2\tau(n). \tag{352}$$

Proof. By proposition 13.24 the exponent equals $\mathrm{wt}_2(n)+1$. If $\mathrm{wt}_2(n)$ is even then $\tau(n) = 0$ and $(-1)^{\mathrm{wt}_2(n)+1} = -1$, so the right side is 0; if $\mathrm{wt}_2(n)$ is odd then $\tau(n) = 1$ and $(-1)^{\mathrm{wt}_2(n)+1} = 1$, so the right side is 1. The second identity is $\varepsilon_n = (-1)^{\mathrm{wt}_2(n)} = 1 - 2(\mathrm{wt}_2(n) \bmod 2)$. □

Remark 13.26. The order of the two steps matters. As printed, (352) raises -1 to a logarithm, an operation that is meaningful only because the argument of that logarithm has first been proved to be exactly a power of two. The formal development therefore proves (351) before it interprets the outer expression at all, evaluates the logarithm on the natural numbers, and reads the final division in $\mathbb{Q}$. A floating-point evaluation of the same display, or an interpretation of the logarithm as a real-valued function whose output is then rounded, is an approximation to a formula rather than the formula.

14 Endpoint asymptotics: exact logarithmic decomposition

We now turn to the behavior of $F(x)$ as $x \downarrow 0$. This is the technically deepest part of the theory and the place where a very small, but mathematically unavoidable, log-periodic fluctuation appears.

Throughout the asymptotic sections, put

$$L = \log 2. \tag{353}$$

We use the standard Stieltjes convention

$$\zeta(1+s) = \frac{1}{s} + \gamma - \gamma_1 s + \mathcal{O}(s^2), \tag{354}$$

where γ is Euler's constant and γ_1 is the first Stieltjes constant.

14.1 An exact delay identity and the coarse quadratic scale

Before introducing the saddle point, it is useful to isolate what follows directly from the differential equation. Put

$$\phi(t) = F(2^{-t}), \qquad \Gamma(t) = -\log \phi(t). \tag{355}$$

For $t \geq 1$, the argument 2^{-t} lies in the left half of the unit interval, and the chain rule with $F'(x) = 2F(2x)$ gives

$$\phi'(t) = -L\, 2^{1-t}\phi(t-1), \qquad \Gamma'(t) = L\, 2^{1-t}\frac{\phi(t-1)}{\phi(t)} > 0. \tag{356}$$

Taking logarithms of the second identity yields the exact delay relation

$$\boxed{\Gamma(t) - \Gamma(t-1) = \log \Gamma'(t) - \log L - (1-t)L, \qquad t \geq 1.} \tag{357}$$

The restriction $t \geq 1$ is essential: it is precisely what licenses the left-half bounded differential law.

The recurrence and monotonicity already prove, independently of the sharper periodic saddle decomposition below,

$$\frac{\Gamma(t)}{t^2} \longrightarrow \frac{L}{2}, \tag{358}$$

and, for all sufficiently large t,

$$\left|\Gamma(t) - \frac{L}{2}t^2\right| \leq 16t\log t. \tag{359}$$

At integral scales the elementary estimate is sharper:

$$\left|\Gamma(n) - \frac{L}{2}n^2\right| \leq 3n\log(n+1) \qquad (n \geq 1). \tag{360}$$

Monotonicity of Γ and $\lfloor t \rfloor \leq t < \lfloor t \rfloor + 1$ transfer the integer estimate to real t. These facts are formalized separately in the log-scale modules: they establish the quadratic law before the analysis below resolves the linear and logarithmic drift, the exact phase, and its unavoidable periodic fluctuation.

14.2 The negative Laplace product

The function G from (116) is the moment generating function of the random variable X in (40):

$$G(z) = \mathbb{E}e^{zX}. \tag{361}$$

Indeed, this also follows from (118) and $Y = 2X - 1$. Therefore, for $s > 0$, $G(-s)$ is the ordinary Laplace transform of the probability law.

Iterating the functional equation (117) in the negative direction gives an exact product.

Proposition 14.1 (Negative Laplace product)**.** *For $s > 0$,*

$$G(-s) = \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j}. \tag{362}$$

Consequently

$$\Lambda(s) := \log G(-s) = \sum_{j=1}^{\infty} \kappa(s/2^j), \qquad \kappa(u) = \log \frac{1 - e^{-u}}{u}. \tag{363}$$

The series converges absolutely and locally uniformly on $(0, \infty)$.

Proof. Putting $z = -s/2$ in (117) gives

$$G(-s) = \frac{1 - e^{-s/2}}{s/2} G(-s/2).$$

After N iterations, the remaining factor is $G(-s/2^N)$, which tends to $G(0) = 1$. For small u,

$$\kappa(u) = -\frac{u}{2} + \mathcal{O}(u^2),$$

so the logarithmic series is dominated on compact s-intervals by a geometric series. □

The product has the exact dilation law

$$\Lambda(2s) = \kappa(s) + \Lambda(s). \tag{364}$$

This equation resembles a renewal equation on the logarithmic scale. Its homogeneous solutions are periodic functions of $\log_2 s$, which is the conceptual origin of the fluctuation in the final asymptotic.

14.3 Removing the quadratic drift

Define the forward tail

$$T(s) = \sum_{m=0}^{\infty} \log(1 - e^{-s2^m}) < 0 \tag{365}$$

and the positive tail error

$$E(s) = -T(s) \geq 0. \tag{366}$$

The series is exponentially small as $s \to \infty$.

Lemma 14.2 (Forward-tail bound)**.** *For $s > 0$,*

$$0 \leq E(s) \leq \frac{e^{-s}}{(1 - e^{-s})^2}. \tag{367}$$

In particular $E(s) \leq 4e^{-s}$ for $s \geq \log 2$.

Proof. For $0 < q < 1$, $-\log(1-q) \le q/(1-q)$. Hence

$$E(s) \le \frac{1}{1-\mathrm{e}^{-s}} \sum_{m\ge 0} \mathrm{e}^{-s2^m}.$$

Since $2^m \ge m+1$, the last sum is at most $\sum_{m\ge 0} \mathrm{e}^{-s(m+1)} = \mathrm{e}^{-s}/(1-\mathrm{e}^{-s})$. □

For $t \in \mathbb{R}$, set

$$P(t) = \Lambda(2^t) + \frac{L}{2}(t^2 - t) + T(2^t). \tag{368}$$

Theorem 14.3 (Exact periodic decomposition)**.** *The function P is 1-periodic and smooth. For every $s > 0$, with $t = \log_2 s$, one has the exact identity*

$$\boxed{\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{\log s}{2} + P(\log_2 s) + E(s).} \tag{369}$$

Proof. Write $s = 2^t$. Using (364),

$$\Lambda(2s) = \Lambda(s) + \log(1-\mathrm{e}^{-s}) - \log s.$$

Also

$$T(2s) = T(s) - \log(1-\mathrm{e}^{-s}).$$

The change in the quadratic correction is

$$\frac{L}{2}((t+1)^2 - (t+1) - (t^2 - t)) = Lt = \log s.$$

The three changes cancel, proving $P(t+1) = P(t)$. Rearranging (368) and using $E = -T$ yields (369). Smoothness follows by differentiating locally uniformly convergent series; periodicity then makes every derivative bounded. □

Let

$$\overline{P} = \int_0^1 P(t)\,\mathrm{d}t, \qquad \Psi(t) = P(t) - \overline{P}. \tag{370}$$

Then Ψ is smooth, real-valued, 1-periodic, and has mean zero.

14.4 Fourier coefficients of the periodic correction

The periodic term is not an unspecified bounded nuisance: all of its Fourier coefficients are explicit. Put

$$\chi_k = \frac{2\pi \mathrm{i} k}{L} \qquad (k \in \mathbb{Z}), \tag{371}$$

and use the convention

$$\widehat{\Psi}(k) = \int_0^1 \Psi(t)\mathrm{e}^{-2\pi \mathrm{i} k t}\,\mathrm{d}t. \tag{372}$$

Theorem 14.4 (Gamma–zeta Fourier coefficients)**.** *One has $\widehat{\Psi}(0) = 0$, and for $k \ne 0$,*

$$\boxed{\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L}.} \tag{373}$$

Every nonzero Fourier coefficient is nonzero. In particular, Ψ is genuinely nonconstant.

Proof. The zero coefficient is zero by definition. For $k \neq 0$, change variables $s = 2^t$ in (372). Dyadic periodization turns the integral over one logarithmic period into a Mellin transform; this is the standard harmonic-sum framework for dyadic periodicities [12]. The basic Mellin identity is

$$\int_0^\infty s^{z-1} \log(1 - \mathrm{e}^{-s})\, \mathrm{d}s = -\Gamma(z)\zeta(1+z), \qquad \Re z > 0. \tag{374}$$

It follows by expanding $\log(1-\mathrm{e}^{-s}) = -\sum_{m\geq 1} \mathrm{e}^{-ms}/m$ and integrating term by term. Near $s = 0$ the kernel contains logarithmic and linear divergences; the quadratic correction in (368) is exactly the finite-part subtraction that removes them. Analytic continuation of the regularized Mellin transform to the imaginary axis gives (373) at $z = -\chi_k$.

The Gamma function has no zeros. Moreover $1 - \chi_k$ lies on the line $\Re z = 1$ away from the pole at 1, and the zeta function has no zeros there. Hence the product in (373) is nonzero. □

Conjugate symmetry $\widehat{\Psi}(-k) = \overline{\widehat{\Psi}(k)}$ reconstructs a real-valued Fourier series. Stirling's estimate for Γ on vertical lines implies exponential decay of the coefficients, so the Fourier series and all of its differentiated series converge rapidly.

The first coefficient is tiny:

$$\widehat{\Psi}(1) \approx (7.5586475410 - 7.4701035590\,\mathrm{i}) \times 10^{-7}, \tag{375}$$

with magnitude approximately $1.0627116252 \times 10^{-6}$. The second is smaller by six orders of magnitude,

$$\widehat{\Psi}(2) \approx (3.2481234772 + 5.8517589871\,\mathrm{i}) \times 10^{-13}, \tag{376}$$

of magnitude $6.6927863679 \times 10^{-13}$, so the first harmonic already describes Ψ to better than one part in a million of its own size.

Warning 14.5 (A numerical hazard specific to these points). Evaluating (373) numerically is delicate for a reason that has nothing to do with rounding. Many implementations of the zeta function compute it from the alternating Dirichlet series through

$$\zeta(s) = \frac{1}{1 - 2^{1-s}} \sum_{m\geq 1} \frac{(-1)^{m-1}}{m^s}. \tag{377}$$

At the points that (373) needs, $s = 1 - \chi_k$, one has $2^{1-s} = 2^{\chi_k} = \mathrm{e}^{\chi_k L} = \mathrm{e}^{2\pi \mathrm{i} k} = 1$ exactly, so the prefactor in (377) has a pole precisely there. The singularity is removable in ζ but not in that formula, and a routine built on it returns an overflow or, worse, a plausible wrong number. This is not a rounding story; it is a defect of the algorithm at exactly the arguments this problem cares about, and it explains how two independently published decimal values for $\widehat{\Psi}(1)$ can agree with each other to eight figures and both be wrong.

The symptom that identifies the defect is that raising the working precision does not shrink the error. It moves it:

working digits	value returned for $10^7\widehat{\Psi}(1)$	
30	$7.55865005 - 7.47011001\,\mathrm{i}$	
50	$7.55864674 - 7.47009636\,\mathrm{i}$	
80	$7.55865060 - 7.47008702\,\mathrm{i}$	
any	$7.55864754 - 7.47010356\,\mathrm{i}$	(Euler–Maclaurin)

The first three rows come from a widely used arbitrary-precision library whose complex zeta routes through (377); the last is a hand-written Euler–Maclaurin evaluation, which is stable at all three precisions and agrees with (375).[1] Digits that wander as the precision rises are the signature of a removable singularity being divided by, and they are exactly what would persuade a careful reader to trust a wrong value. Use an Euler–Maclaurin or Hurwitz evaluation instead, and check the answer at two working precisions.

Thus the visible oscillation is on the scale of a few parts in a million in the logarithm. Tiny is not the same as zero: its nonvanishing changes the logical status of an asymptotic equivalent.

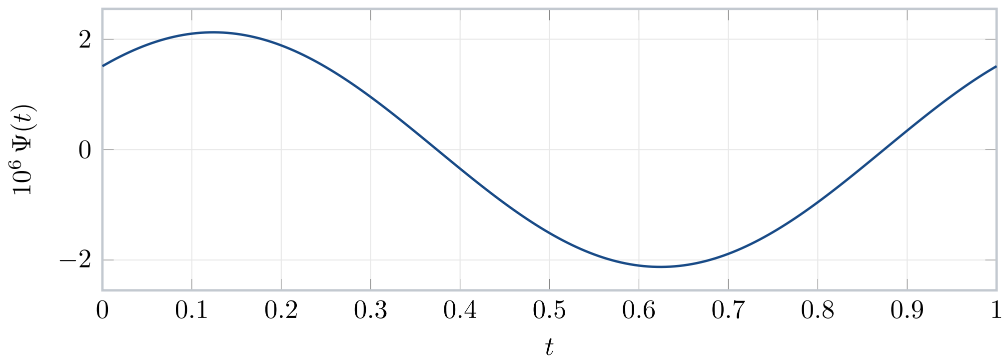

Figure 3: The periodic correction Ψ, magnified by 10^6. The curve is indistinguishable from its first Fourier harmonic at this resolution, and the half-period antisymmetry $\Psi(t+\frac{1}{2}) \approx -\Psi(t)$ visible here is the statement that the even harmonics are negligible. The vertical scale is the point of the picture: the oscillation is a few parts in a million in the logarithm, but it does not tend to zero.

14.5 The finite part at the Mellin origin

The constant that enters the saddle asymptotic comes from the finite part of $-\Gamma(z)\zeta(1+z)$ at $z=0$. Using

$$\begin{aligned}\Gamma(z) &= \frac{1}{z} - \gamma + \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12}\right) z + \mathcal{O}(z^2),\\ \zeta(1+z) &= \frac{1}{z} + \gamma - \gamma_1 z + \mathcal{O}(z^2),\end{aligned}$$

one finds

$$-\Gamma(z)\zeta(1+z) + \frac{1}{z^2} = A_0 + \mathcal{O}(z), \tag{378}$$

where

$$A_0 = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12}. \tag{379}$$

The final constant is

$$C_F = \frac{A_0}{L} - \frac{7L}{12} - \frac{1}{2}\log\pi \approx -2.02798389863885942397. \tag{380}$$

The terms $-7L/12$ and $-(1/2)\log\pi$ arise from the dyadic Euler–Maclaurin finite part and the leading Gaussian saddle mass, respectively.

[1] The precision table is due to an independent recomputation carried out alongside this article; the Euler–Maclaurin values were checked here at two truncations, 60 terms with 12 Bernoulli corrections and 100 with 16, which agree with each other to 37 digits.

14.5.1 The Mellin transform in the relevant strip

The identity (374) is stated for the kernel $\log(1-\mathrm{e}^{-s})$ in the half-plane $\Re z > 0$. The kernel that actually occurs in (363) is not that one but

$$\kappa(u) = \log\frac{1-\mathrm{e}^{-u}}{u} = \log(1-\mathrm{e}^{-u}) - \log u,$$

and its Mellin transform lives in a different strip. Both facts are needed, and it is worth saying at once that they do not compete.

Lemma 14.6 (Mellin transform of the dyadic kernel)**.** *The integral*

$$\widetilde{\kappa}(z) = \int_0^\infty \kappa(u)u^{z-1}\,\mathrm{d}u \tag{381}$$

converges absolutely precisely in the strip $-1 < \Re z < 0$, and there

$$\widetilde{\kappa}(z) = -\Gamma(z)\zeta(1+z). \tag{382}$$

Proof. As $u \downarrow 0$ one has $\kappa(u) = -u/2 + \mathcal{O}(u^2)$, so $\kappa(u)u^{z-1} = \mathcal{O}(u^{\Re z})$ and the integral converges at the origin exactly when $\Re z > -1$. As $u \to \infty$ one has $\kappa(u) = -\log u + \mathcal{O}(\mathrm{e}^{-u})$, so the integrand is of size $u^{\Re z-1}\log u$ and the integral converges at infinity exactly when $\Re z < 0$.

Fix z in the strip. Differentiating,

$$\kappa'(u) = \frac{1}{\mathrm{e}^u - 1} - \frac{1}{u}. \tag{383}$$

Integration by parts is legitimate because the boundary term $\kappa(u)u^z/z$ vanishes at both ends: near 0 it is $\mathcal{O}(u^{1+\Re z})$ and near ∞ it is $\mathcal{O}(u^{\Re z}\log u)$, and $-1 < \Re z < 0$. Hence

$$\widetilde{\kappa}(z) = -\frac{1}{z}\int_0^\infty \kappa'(u)u^z\,\mathrm{d}u = -\frac{1}{z}\int_0^\infty \left(\frac{1}{\mathrm{e}^u-1} - \frac{1}{u}\right)u^z\,\mathrm{d}u.$$

The subtracted Bose integral is the classical analytic continuation of $\int_0^\infty u^{w-1}(\mathrm{e}^u-1)^{-1}\,\mathrm{d}u = \Gamma(w)\zeta(w)$ from $\Re w > 1$ to the critical strip: for $0 < \Re w < 1$,

$$\int_0^\infty u^{w-1}\left(\frac{1}{\mathrm{e}^u-1} - \frac{1}{u}\right)\mathrm{d}u = \Gamma(w)\zeta(w). \tag{384}$$

Applying (384) with $w = z+1$, which lies in $0 < \Re w < 1$ exactly when $-1 < \Re z < 0$, and using $\Gamma(z+1) = z\Gamma(z)$ gives

$$\widetilde{\kappa}(z) = -\frac{\Gamma(z+1)\zeta(z+1)}{z} = -\Gamma(z)\zeta(1+z).$$

□

Remark 14.7 (The two Mellin identities continue each other). A reader who has just met (374) will notice that (382) has the same right-hand side in a disjoint region, and may suspect a clash. There is none. The kernel $\log(1-\mathrm{e}^{-s})$ decays exponentially at infinity but is logarithmically singular at the origin, so its transform converges exactly in $\Re z > 0$. Subtracting $\log s$ replaces that logarithmic singularity by a zero of order one, which moves the left edge of the strip from 0 to -1; the price is that the subtracted term destroys convergence at infinity, which caps the strip at $\Re z < 0$. The two strips therefore abut along the line $\Re z = 0$, and the two integrals are the restrictions to the two sides of that line of the single meromorphic function $-\Gamma(z)\zeta(1+z)$, which has a double pole at $z = 0$. Neither identity is more fundamental; (382) is simply the one adapted to the harmonic sum (363).

Because Λ is the dyadic harmonic sum of κ, its Mellin transform acquires a geometric factor. For $-1 < \Re z < 0$ the substitution $s \mapsto 2^j s$ gives $\int_0^\infty \kappa(s/2^j)s^{z-1}\,\mathrm{d}s = 2^{jz}\widetilde{\kappa}(z)$, and $\sum_{j\ge 1} 2^{jz}$ converges in that strip, so

$$\widetilde{\Lambda}(z) = \int_0^\infty \Lambda(s)s^{z-1}\,\mathrm{d}s = \frac{2^z}{1-2^z}\big(-\Gamma(z)\zeta(1+z)\big), \qquad -1 < \Re z < 0. \tag{385}$$

The factor $2^z/(1-2^z)$ is where the discrete dyadic scale enters, and it is the source of both the periodicity and the constant $\overline{P}$. Writing $w = Lz$ and using $(\mathrm{e}^w-1)^{-1} = w^{-1} - \frac{1}{2} + w/12 + \mathcal{O}(w^3)$,

$$\frac{2^z}{1-2^z} = -1 - \frac{1}{\mathrm{e}^w - 1} = -\frac{1}{Lz} - \frac{1}{2} - \frac{Lz}{12} + \mathcal{O}(z^3). \tag{386}$$

Its poles are simple and lie exactly at the points $z = \chi_k = 2\pi\mathrm{i}k/L$ of (371), since $1-2^z$ vanishes there with derivative $-L$.

Proposition 14.8 (Derivation of the mean of the periodic function)**.** *The mean $\overline{P}$ of (370) is*

$$\overline{P} = \frac{A_0}{L} - \frac{L}{12}, \tag{387}$$

with A_0 as in (379).

Proof. Combining (386) with the finite-part expansion (378), that is $-\Gamma(z)\zeta(1+z) = -z^{-2} + A_0 + \mathcal{O}(z)$, the product (385) has a triple pole at the origin with Laurent expansion

$$\widetilde{\Lambda}(z) = \frac{1}{Lz^3} + \frac{1}{2z^2} + \frac{1}{z}\left(\frac{L}{12} - \frac{A_0}{L}\right) + \mathcal{O}(1). \tag{388}$$

Mellin inversion writes $\Lambda(s)$ as an integral of $\widetilde{\Lambda}(z)s^{-z}$ along a vertical line inside the strip; displacing that line to the right past $\Re z = 0$ contributes minus the residues at the poles crossed. Expanding $s^{-z} = 1 - z\log s + \frac{1}{2}z^2\log^2 s - \cdots$ and reading off the coefficient of z^{-1} in $\widetilde{\Lambda}(z)s^{-z}$ gives

$$\operatorname*{Res}_{z=0}\widetilde{\Lambda}(z)s^{-z} = \frac{\log^2 s}{2L} - \frac{\log s}{2} + \frac{L}{12} - \frac{A_0}{L},$$

so the contribution of the triple pole is

$$-\frac{(\log s)^2}{2L} + \frac{\log s}{2} + \frac{A_0}{L} - \frac{L}{12}.$$

Comparison with the exact decomposition (369), which is proved independently and elementarily, identifies the first two terms with the explicit quadratic drift and the constant with the mean of P; the remaining displaced integral is smaller than every power of $1/\log s$ and is the forward tail $E(s)$. This proves (387). □

The same residue computation reproduces (373) and thereby confirms the normalizations. Near $z = \chi_k$ with $k \neq 0$ one has $1 - 2^z \sim -L(z-\chi_k)$ and $2^z \to 1$, so

$$\operatorname*{Res}_{z=\chi_k}\widetilde{\Lambda}(z)s^{-z} = \frac{\Gamma(\chi_k)\zeta(1+\chi_k)}{L}\,s^{-\chi_k}.$$

With $s = 2^t$ one has $s^{-\chi_k} = \mathrm{e}^{-2\pi\mathrm{i}kt}$. After replacing k by $-k$, minus the sum of these residues is

$$\sum_{k\neq 0} -L^{-1}\Gamma(-\chi_k)\zeta(1-\chi_k)\mathrm{e}^{2\pi\mathrm{i}kt},$$

which is precisely the Fourier series of Ψ with the coefficients (373). Thus the mean and every oscillating mode come from the single factorization (385): the triple pole at the origin produces the nonperiodic part together with $\overline{P}$, and the simple poles on the imaginary axis produce Ψ.

15 The corrected sharp asymptotic

15.1 The lower Lambert coordinate

For sufficiently small $x > 0$, define

$$W = W_{-1}(-Lx), \tag{389}$$

where W_{-1} is the lower real branch of Lambert's W-function [11], and put

$$\lambda = \lambda(x) = -\frac{W}{L}. \tag{390}$$

Then $\lambda \to \infty$ as $x \downarrow 0$ and

$$\lambda 2^{-\lambda} = x. \tag{391}$$

Equivalently,

$$r = 2^{\lambda} = \frac{\lambda}{x}. \tag{392}$$

The real number r is the explicit Lambert saddle radius used in the contour analysis.

15.2 The main term

Define

$$\mathcal{M}(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda), \tag{393}$$

where C_F is given by (380) and $\lambda = \lambda(x)$. The same expression can be written compactly in the Lambert variable W:

$$\mathcal{M}(x) = -\frac{7L}{12} - \frac{1}{2}\log(\pi x) + \frac{A_0 - W - \frac{1}{2}W^2}{L} + \Psi\left(-\frac{W}{L}\right). \tag{394}$$

The equivalence follows from $W = -L\lambda$ and $\log x = \log\lambda - L\lambda$.

Theorem 15.1 (Corrected sharp endpoint asymptotic)**.** *As $x \downarrow 0$,*

$$\boxed{\log F(x) = \mathcal{M}(x) + \mathcal{O}\left(\frac{1}{\lambda(x)}\right) = \mathcal{M}(x) + \mathcal{O}\left(\frac{1}{-\log x}\right).} \tag{395}$$

Equivalently,

$$F(x) \sim \exp\mathcal{M}(x). \tag{396}$$

The periodic term is evaluated at the *exact* phase $\lambda(x)$, not at a truncated log–log approximation to it. This matters at the stated error scale.

15.3 Bromwich inversion

The contour representation used below rests on one elementary identity, which is worth recording separately because it is what converts a statement about the probability law into a statement about a Laplace transform with an explicit pole at the origin.

Proposition 15.2 (Laplace transform of the distribution function)**.** *For every $s > 0$,*

$$\int_0^{\infty} \mathrm{e}^{-sx} F(x)\,\mathrm{d}x = \frac{G(-s)}{s}, \tag{397}$$

where G is the moment generating function (361).

Proof. By (44) the function F is the distribution function of the random variable X of (40), so $F(0) = 0$, $F(x) = 1$ for $x \geq 1$, and F is continuous and nondecreasing. Split the integral at $x = 1$.

On $[0, 1]$, Stieltjes integration by parts gives

$$G(-s) = \int_0^1 \mathrm{e}^{-sx}\,\mathrm{d}F(x) = \left[\mathrm{e}^{-sx}F(x)\right]_0^1 + s\int_0^1 \mathrm{e}^{-sx}F(x)\,\mathrm{d}x = \mathrm{e}^{-s} + s\int_0^1 \mathrm{e}^{-sx}F(x)\,\mathrm{d}x, \quad (398)$$

because $F(1) = 1$ and $F(0) = 0$. Hence

$$\int_0^1 \mathrm{e}^{-sx}F(x)\,\mathrm{d}x = \frac{G(-s)}{s} - \frac{\mathrm{e}^{-s}}{s}.$$

On $[1, \infty)$ the integrand is the constant tail e^{-sx}, and

$$\int_1^\infty \mathrm{e}^{-sx}\,\mathrm{d}x = \frac{\mathrm{e}^{-s}}{s}.$$

The boundary term e^{-s}/s produced by the integration by parts and the contribution of the constant tail are therefore equal and opposite, and adding the two pieces gives (397). □

Two features of (397) decide the geometry of the inversion. First, since X is bounded, $G(-z) = \mathbb{E}\mathrm{e}^{-zX}$ is entire, so the only singularity of the transform $G(-z)/z$ is the simple pole at $z = 0$ coming from the constant tail of F; the integral in (397) converges absolutely for every z with $\Re z > 0$ and diverges at $z = 0$. The Bromwich representation

$$F(x) = \frac{1}{2\pi\mathrm{i}}\int_{r-\mathrm{i}\infty}^{r+\mathrm{i}\infty} \frac{\mathrm{e}^{zx}G(-z)}{z}\,\mathrm{d}z$$

of (399) is consequently valid on *every* vertical line $\Re z = r$ with $r > 0$, and on no vertical line with $r \leq 0$. The freedom in r is exactly what the saddle choice $r = 2^\lambda$ of (392) exploits. Second, absolute convergence of the contour integral on each such line follows from the rapid decay of $\tau \mapsto G(-r - \mathrm{i}\tau)$, which is the Fourier transform of a smooth compactly supported density up to an exponential tilt and therefore decays faster than every power (section 5). No contour shifting hypothesis is needed: the line $\Re z = r$ is admissible for all $r > 0$ simultaneously.

Because $G(-z) = \mathbb{E}\mathrm{e}^{-zX}$ is entire and F is continuous, Laplace–Stieltjes inversion gives, for every $r > 0$ and $0 < x < 1$,

$$F(x) = \frac{1}{2\pi\mathrm{i}}\int_{r-\mathrm{i}\infty}^{r+\mathrm{i}\infty} \frac{\mathrm{e}^{zx}G(-z)}{z}\,\mathrm{d}z. \quad (399)$$

Set $z = r(1 + \mathrm{i}\theta)$ and choose $r = 2^\lambda = \lambda/x$. Then

$$F(x) = \frac{\mathrm{e}^{\lambda+\Lambda(r)}}{2\pi}\int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}\lambda\theta}\frac{G(-r(1+\mathrm{i}\theta))}{G(-r)}\frac{\mathrm{d}\theta}{1+\mathrm{i}\theta}. \quad (400)$$

The exact real decomposition (369) at $r = 2^\lambda$ gives

$$\lambda + \Lambda(r) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda + P(\lambda) + E(r). \quad (401)$$

Since

$$\overline{P} = \frac{A_0}{L} - \frac{L}{12}, \qquad \overline{P} - \frac{1}{2}\log(2\pi) = C_F, \quad (402)$$

the missing factors in (393) are exactly the leading Gaussian mass $1/\sqrt{2\pi\lambda}$ and the centered fluctuation $\Psi = P - \overline{P}$.

15.4 The normalized saddle mass

It is convenient to give the entire remaining error a name. Everything in (400) except the scaled oscillatory integral is explicit, and that integral is a positive real number divided by the Gaussian mass it is supposed to reproduce.

Definition 15.3. For small $x > 0$, with $\lambda = \lambda(x)$ from (390) and $r = 2^\lambda$ from (392), put

$$\mathcal{G}(x) = \frac{F(x)\sqrt{2\pi\lambda}}{\exp(rx + \Lambda(r))}, \tag{403}$$

where $\Lambda = \log G(-\,\cdot\,)$ is the log-transform of (363).

The symbol $\mathcal{G}$ is used nowhere else in this article. Because $F(x) > 0$ for $x > 0$, the quantity $\mathcal{G}(x)$ is a positive real number, and by (400) together with $rx = \lambda$ it has the exact integral representation

$$\mathcal{G}(x) = \sqrt{\frac{\lambda}{2\pi}} \int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}\lambda\theta} \frac{G(-r(1+\mathrm{i}\theta))}{G(-r)} \frac{\mathrm{d}\theta}{1+\mathrm{i}\theta} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\Phi_\lambda\!\left(\frac{v}{\sqrt{\lambda}}\right) \mathrm{d}v, \tag{404}$$

the second form arising from the substitution (415) and the local phase (412) of the next subsection. Taking logarithms in (403) and using $rx = \lambda$ gives the exact identity

$$\boxed{\log F(x) = rx + \Lambda(r) - \frac{1}{2}\log(2\pi\lambda) + \log\mathcal{G}(x).} \tag{405}$$

No approximation whatever has been made in (405): it is a rearrangement of the definition of $\mathcal{G}$, valid for every x small enough that $\lambda(x)$ is defined. All of the analysis still to be done is the analysis of the single positive number $\mathcal{G}(x)$.

Proposition 15.4 (Exact separation of the error)**.** *For small $x > 0$, with $r = 2^{\lambda(x)}$,*

$$rx + \Lambda(r) - \frac{1}{2}\log(2\pi\lambda) = \mathcal{M}(x) + E(r), \tag{406}$$

where $\mathcal{M}$ is the main term (393) and E is the forward-tail error (366). Consequently

$$\boxed{\log F(x) - \mathcal{M}(x) = \log\mathcal{G}(x) + E(r).} \tag{407}$$

Proof. By (401) and $rx = \lambda$,

$$rx + \Lambda(r) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda + P(\lambda) + E(r).$$

Subtract $\frac{1}{2}\log(2\pi\lambda) = \frac{1}{2}\log\lambda + \frac{1}{2}\log(2\pi)$ and split $P = \overline{P} + \Psi$ as in (370). By (402) one has $\overline{P} - \frac{1}{2}\log(2\pi) = C_F$, so the right-hand side becomes

$$-\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda) + E(r),$$

which is $\mathcal{M}(x) + E(r)$ by (393). Subtracting (406) from (405) gives (407). □

Equation (407) splits the total error into two parts of completely different nature. The term $E(r)$ is beyond all orders: by (367),

$$0 \le E(2^\lambda) \le 4\exp(-2^\lambda), \tag{408}$$

which is smaller than every power of $1/\lambda$, and indeed smaller than $\exp(-\lambda^K)$ for every K. It can never be the obstruction to any inverse-power expansion, and it may be carried along inertly through every order. Everything else is $\log \mathcal{G}(x)$, an algebraic quantity computed from a single explicit integral.

This bookkeeping is preferable to working with the phase integral directly. The phase integral in (400) is a complex oscillatory object whose normalization changes at every stage of the argument, so error statements about it must always be qualified by which prefactors have already been extracted. By contrast, $\mathcal{G}$ is dimensionless and pinned: it is a single positive real number whose distance from 1 *is* the entire error, up to the beyond-all-orders term. The leading theorem (395) is exactly the statement

$$\mathcal{G}(x) = 1 + \mathcal{O}\left(\frac{1}{\lambda(x)}\right), \tag{409}$$

the first correction (428) is the statement that $\log \mathcal{G}(x) = A_1(\lambda)/\lambda + \mathcal{O}(\lambda^{-2})$, and the all-orders theorem (429) is the statement that

$$\log \mathcal{G}(x) = \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}\left(\lambda^{-N}\right) \tag{410}$$

for every N. In each case the periodic coefficients are those of the relative saddle mass (430), and no further normalization has to be tracked.

15.5 The local phase

Let

$$u = \log(1 + \mathrm{i}\theta), \tag{411}$$

with the principal logarithm near the saddle. Subtracting the value at $\theta = 0$ from the complexified version of (369), and including the factor $(1+\mathrm{i}\theta)^{-1}$ in (400), gives the local phase

$$\Phi_\lambda(\theta) = \lambda(\mathrm{i}\theta - u) - \frac{u}{2} - \frac{u^2}{2L} + \Psi\left(\lambda + \frac{u}{L}\right) - \Psi(\lambda) + \mathcal{E}_r(\theta), \tag{412}$$

where $\mathcal{E}_r(0) = 0$ and, on a fixed neighborhood of zero, every derivative of $\mathcal{E}_r$ is exponentially small in r. This last term comes from the forward tail E and is far smaller than every power of $1/\lambda$.

The elementary expansion

$$\log(1 + \mathrm{i}\theta) = \mathrm{i}\theta + \frac{\theta^2}{2} - \frac{\mathrm{i}\theta^3}{3} - \frac{\theta^4}{4} + \mathcal{O}(\theta^5) \tag{413}$$

shows that

$$\lambda(\mathrm{i}\theta - u) = -\frac{\lambda\theta^2}{2} + \frac{\mathrm{i}\lambda\theta^3}{3} + \frac{\lambda\theta^4}{4} + \mathcal{O}(\lambda|\theta|^5). \tag{414}$$

Thus the natural central scale is

$$\theta = \frac{v}{\sqrt{\lambda}}. \tag{415}$$

Uniformly for $|v|$ growing sufficiently slowly with λ,

$$\Phi_\lambda(v/\sqrt{\lambda}) = -\frac{v^2}{2} + \lambda^{-1/2} P_1(v;\lambda) + \lambda^{-1} P_2(v;\lambda) + \mathcal{O}\left(\lambda^{-3/2}(1 + |v|^9)\right), \tag{416}$$

where

$$A(u) = \frac{\Psi'(u)}{L} - \frac{1}{2}, \tag{417}$$

$$B(u) = -\frac{1}{4} + \frac{1}{2L} + \frac{\Psi'(u)}{2L} - \frac{\Psi''(u)}{2L^2}, \tag{418}$$

$$P_1(v; u) = \mathrm{i}\left(A(u)v + \frac{v^3}{3}\right), \tag{419}$$

$$P_2(v; u) = \frac{v^4}{4} + B(u)v^2. \tag{420}$$

In (416), the second argument is $u = \lambda$.

15.6 Central and tail estimates

For completeness, the quantitative saddle proof is organized by the following estimates. They are stated here in the form needed for the argument; the formal development proves them with explicit domains and constants.

Lemma 15.5 (Central expansion)**.** *For every fixed truncation order N, there is $\delta > 0$ such that on $|\theta| \le \delta$ the exponent in (400) admits a Taylor expansion through order N after the scaling (415). Its coefficients are smooth 1-periodic functions of λ, and the remainder is dominated by a polynomial in v times the next power of $\lambda^{-1/2}$.*

Proof. The logarithm in (411) is analytic for $|\theta| < 1$. The periodic function Ψ is smooth with bounded derivatives of every order. Taylor's theorem applied to each term of (412), together with the exponentially small bounds on $\mathcal{E}_r$, gives a uniform expansion. The dependence on λ occurs only through $\Psi^{(j)}(\lambda)$, hence is periodic. □

Lemma 15.6 (Gaussian domination)**.** *There are positive constants c, δ and λ_0 such that, for $\lambda \ge \lambda_0$ and $|\theta| \le \delta$,*

$$\Re\Phi_\lambda(\theta) \le -c\lambda\theta^2. \tag{421}$$

Proof architecture. The exact function $\Re(\mathrm{i}\theta - \log(1 + \mathrm{i}\theta)) = -(1/2)\log(1 + \theta^2)$ supplies the negative quadratic term. All remaining terms in (412) have bounded first and second derivatives on a fixed neighborhood. For large λ the $-\lambda\theta^2/2$ contribution dominates them. □

Lemma 15.7 (Off-saddle decay)**.** *For each N one may choose the central window so that the contribution to (400) outside that window is $O(\lambda^{-N})$ relative to the leading saddle mass.*

Proof architecture. Split the negative-Laplace product at the dyadic index nearest λ. In the intermediate range, a positive proportion of the factors have arguments of order one and their moduli are strictly smaller off the real axis, yielding exponential decay in $\lambda\theta^2$. In the far range, repeated use of the sinc/Laplace product bounds gives arbitrarily high polynomial decay. The two bounds overlap and cover the whole complement of the central window. The factor $(1 + \mathrm{i}\theta)^{-1}$ is harmless. □

These lemmas justify extending the scaled central integral to the entire real v-axis and integrating its expansion term by term against $\mathrm{e}^{-v^2/2}$.

15.7 Completion of the leading proof

Keeping only the Gaussian term in (416) gives

$$\int_{\mathbb{R}} \mathrm{e}^{\Phi_\lambda(\theta)}\,\mathrm{d}\theta = \frac{1}{\sqrt{\lambda}}\int_{\mathbb{R}} \mathrm{e}^{-v^2/2}\,\mathrm{d}v\,\Big(1+\mathcal{O}(\lambda^{-1})\Big) = \sqrt{\frac{2\pi}{\lambda}}\,\Big(1+\mathcal{O}(\lambda^{-1})\Big). \tag{422}$$

The term of order $\lambda^{-1/2}$ integrates to zero because P_1 is odd in v and purely imaginary. Combining (401), (402), and (422) yields

$$\log F(x) = -\frac{L}{2}\lambda^2 + \left(1+\frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda) + \mathcal{O}(\lambda^{-1}),$$

which is (395). Finally, $\lambda \asymp -\log x$ follows from (391), so the two displayed error scales are equivalent.

16 First correction and the all-orders saddle expansion

The leading theorem leaves an error of order $1/\lambda$. The same contour analysis can be continued to arbitrary order. The coefficient functions are periodic because every local coefficient is built from derivatives of Ψ evaluated at the exact phase λ.

16.1 The first correction

Expand the exponential in (416):

$$\mathrm{e}^{\Phi_\lambda(v/\sqrt{\lambda})} = \mathrm{e}^{-v^2/2}\Big[1+\lambda^{-1/2}P_1(v;\lambda) \tag{423}$$

$$+\lambda^{-1}\big(P_2(v;\lambda)+\tfrac{1}{2}P_1(v;\lambda)^2\big)+\mathcal{O}(\lambda^{-3/2}(1+|v|^{12}))\Big]. \tag{424}$$

Let Z be a standard normal random variable. Since

$$\mathbb{E}Z^2 = 1, \qquad \mathbb{E}Z^4 = 3, \qquad \mathbb{E}Z^6 = 15, \tag{425}$$

the normalized coefficient of λ^{-1} in the saddle mass is

$$\begin{aligned} b_1(u) &= \mathbb{E}\left[P_2(Z;u)+\tfrac{1}{2}P_1(Z;u)^2\right] \\ &= \frac{3}{4}+B(u)-\frac{1}{2}\left(A(u)^2+2A(u)+\frac{5}{3}\right). \end{aligned} \tag{426}$$

Substituting (417) and (418) simplifies the expression considerably.

Theorem 16.1 (First saddle correction)**.** *Define*

$$A_1(u) = \frac{1}{24}+\frac{1}{2L}-\frac{\Psi''(u)+\Psi'(u)^2}{2L^2}. \tag{427}$$

Then

$$\boxed{\log F(x) = \mathcal{M}(x)+\frac{A_1(\lambda(x))}{\lambda(x)}+\mathcal{O}\Big(\lambda(x)^{-2}\Big).} \tag{428}$$

Proof. The order-$\lambda^{-1/2}$ term in (424) is odd and integrates to zero. Equation (426) gives the first nonzero relative correction to the integral. A direct simplification yields (427). If the normalized saddle mass is $1+b_1/\lambda+O(\lambda^{-2})$, then its logarithm is $b_1/\lambda+O(\lambda^{-2})$, so the same coefficient appears in the logarithmic expansion. The off-saddle contribution is smaller than the stated remainder by the tail estimates of the preceding section. □

The constant part $1/24+1/(2L)$ is the usual type of Gaussian/curvature correction. The terms involving Ψ' and Ψ'' record the fact that the log-periodic environment is not constant across the saddle width.

16.2 All orders

At order N, Taylor expansion of (412) produces finitely many polynomials in v whose coefficients are differential polynomials in

$$\Psi'(u), \Psi''(u), \dots, \Psi^{(2N)}(u).$$

After exponentiation, all odd powers of $\lambda^{-1/2}$ integrate to zero by parity. The remaining Gaussian moments are integers, so one obtains integer powers of $1/\lambda$.

Theorem 16.2 (Full logarithmic expansion)**.** *There are smooth* 1*-periodic functions* $A_j : \mathbb{R} \to \mathbb{R}$*,* $j \geq 0$*, with* $A_0 = 0$ *and* A_1 *given by* (427)*, such that for every* $N \geq 1$*,*

$$\boxed{\log F(x) = \mathcal{M}(x) + \sum_{j=1}^{N-1} \frac{A_j(\lambda(x))}{\lambda(x)^j} + \mathcal{O}\Big(\lambda(x)^{-N}\Big)} \tag{429}$$

as $x \downarrow 0$*.*

Proof architecture. Fix N. Expand the analytic local phase through sufficiently high order in $\lambda^{-1/2}$, with a Gaussian-dominated remainder. Exponentiate using complete Bell polynomials, integrate term by term, and discard odd terms. The quantitative tail theorem makes the complement of the central region $O(\lambda^{-N})$. This gives a relative saddle mass expansion

$$1 + \sum_{j=1}^{N-1} \frac{b_j(\lambda)}{\lambda^j} + O(\lambda^{-N}), \tag{430}$$

where every b_j is smooth and periodic. Taking the formal logarithm defines the A_j. If

$$\log\left(1 + \sum_{j\geq 1} b_j z^j\right) = \sum_{j\geq 1} A_j z^j,$$

then the coefficients may be generated recursively by

$$A_n = b_n - \frac{1}{n}\sum_{k=1}^{n-1} k A_k b_{n-k}. \tag{431}$$

All operations preserve smooth periodic dependence on the phase. □

This form of the expansion is preferable to a very long series in nested logarithms. The exact Lambert coordinate absorbs the dominant implicit inversion, while the exact periodic phase prevents phase errors from being magnified into false remainder claims.

16.3 The saddle jets in closed form

The description just given defines the A_j only as the output of a chain: a differential recursion for the periodic quantities that enter the local phase, then exponentiation, then Gaussian contraction, then a formal logarithm. Every step after the first is algebraic. The first is not, and it can be solved.

Write $\mathcal{H}_n = \sum_{i\leq n} 1/i$ for the harmonic numbers, and let $\mathcal{Z}_n$ be the periodic *jets* generated by

$$\mathcal{Z}_0 = \frac{1}{2} + \frac{\Psi'}{L}, \qquad \mathcal{Z}_{n+1} = \left(\frac{\mathrm{D}}{L} - (n+1)\right)\mathcal{Z}_n + \frac{(-1)^{n+1} n!}{L}, \tag{432}$$

where D is differentiation in the phase variable. The operators $\mathrm{D}/L - k$ commute, since $(\mathrm{D}/L - j)(\mathrm{D}/L - k)$ is symmetric in j and k, so the homogeneous part of the solution is the falling-factorial operator $\prod_{k=1}^{n}(\mathrm{D}/L - k)$ applied to $\mathcal{Z}_0$, and each inhomogeneous constant is an eigenvector of the remaining factors. Carrying this out gives a closed form.

Theorem 16.3 (Closed form of the jets)**.** *Let $c(n,m)$ be defined by*

$$\prod_{k=1}^{n}(z-k)=\sum_{m=0}^{n}c(n,m)z^m. \tag{433}$$

Then for every $n\geq 0$,

$$\mathcal{Z}_n(u)=(-1)^n n!\left(\frac{1}{2}+\frac{\mathcal{H}_n}{L}\right)+\sum_{m=0}^{n}c(n,m)\,\frac{\Psi^{(m+1)}(u)}{L^{m+1}}, \tag{434}$$

$$\widetilde{\mathcal{Z}}_n(u)=(-1)^n n!\left(\frac{\mathcal{H}_n}{L}-\frac{1}{2}\right)+\sum_{m=0}^{n}c(n,m)\,\frac{\Psi^{(m+1)}(u)}{L^{m+1}}, \tag{435}$$

where $\widetilde{\mathcal{Z}}_n=\mathcal{Z}_n+(-1)^{n+1}n!$ are the bounded exponent jets. The polynomial in (433) is monic of degree exactly n.

In terms of these jets the exponent polynomials collapse: the mth one is

$$\Xi_m(u,v)=\mathrm{i}^m\left(\frac{\widetilde{\mathcal{Z}}_{m-1}(u)\,v^m}{m!}+\frac{(-1)^{m+1}v^{m+2}}{m+2}\right), \tag{436}$$

with no factorial surviving in the second term. Substituting (435) therefore exhibits every A_j as a finite polynomial in $\Psi',\ldots,\Psi^{(2j)}$ with coefficients in $\mathbb{Q}[1/L]$, and hence, through (373), as an explicit Gamma–zeta Fourier object. This is what makes Theorem 16.13 below a statement about a formula rather than about a recursion.

Warning 16.4 (Which Stirling numbers). The weights (433) are signed Stirling numbers of the first kind *shifted by one in each index.* With the standard convention $x(x-1)\cdots(x-n+1)=\sum_k s(n,k)x^k$, multiplying (433) by z gives $\prod_{k=0}^{n}(z-k)$, whence

$$c(n,m)=s(n+1,m+1)=(-1)^{n-m}\mathfrak{s}(n+1,m+1), \tag{437}$$

with $\mathfrak{s}$ the unsigned numbers already used in (448). The shift is not cosmetic. One has $s(n,0)=0$ for every $n\geq 1$, whereas $c(n,0)=\prod_{k=1}^{n}(-k)=(-1)^n n!$, and that constant column is precisely what carries the harmonic-number term in (434). Reading $c(n,m)$ as $s(n,m)$ therefore deletes the entire non-oscillatory part of the jet. Explicitly,

$$c(2,\cdot)=(2,-3,1),\quad s(2,\cdot)=(0,-1,1);\qquad c(3,\cdot)=(-6,11,-6,1),\quad s(3,\cdot)=(0,2,-3,1).$$

Remark 16.5. Theorem 16.3 and (436), together with the monicity and degree statement, are machine-checked in the formal development. The identification (437) is elementary and is verified here; note that it was the one item in this circle of results that neither a compiler nor a numerical check could have caught, since every number in sight is computed from the product (433) and never from the name.

Written in the jets, the first two coefficients are far shorter than in the derivatives of Ψ:

$$A_1=-\frac{\widetilde{\mathcal{Z}}_0^2}{2}-\widetilde{\mathcal{Z}}_0-\frac{\widetilde{\mathcal{Z}}_1}{2}-\frac{1}{12}, \tag{438}$$

$$\begin{aligned}A_2=\frac{1}{24}\Big(&8\widetilde{\mathcal{Z}}_0^3+12\widetilde{\mathcal{Z}}_0^2\widetilde{\mathcal{Z}}_1+12\widetilde{\mathcal{Z}}_0^2+48\widetilde{\mathcal{Z}}_0\widetilde{\mathcal{Z}}_1+12\widetilde{\mathcal{Z}}_0\widetilde{\mathcal{Z}}_2\\&+6\widetilde{\mathcal{Z}}_1^2+24\widetilde{\mathcal{Z}}_1+20\widetilde{\mathcal{Z}}_2+3\widetilde{\mathcal{Z}}_3\Big).\end{aligned} \tag{439}$$

Both are machine-checked. Expanding (438) through (435) reproduces (427) exactly: the terms in Ψ' cancel, and the constants collapse to $1/24+1/(2L)$. That is a useful check in its own right, because (427) predates the jet parametrization and so is an independent anchor for it.

These forms also make Theorem 16.13 below legible: the top jet enters (438) with coefficient $-\frac{1}{2}$ and (439) with coefficient $\frac{3}{24}=\frac{1}{8}$, which are $(-1)^j/(2^j j!)$ at $j=1$ and $j=2$.

16.4 Every coefficient in closed form

With the jets solved, the remaining steps are pure algebra and can be carried out once and for all, replacing the recursion (431) by a formula. Write $(m_1, \dots, m_r) \vDash N$ for a composition of N into r positive parts, as in Section 8.1.

Theorem 16.6 (Explicit saddle mass coefficients)**.** *Let $1 + \sum_{j \geq 1} a_j(u)\lambda^{-j}$ be the normalized saddle mass. Then for $j \geq 1$,*

$$a_j(u) = (-1)^j \sum_{r=1}^{2j} \frac{1}{r!} \sum_{(m_1,\dots,m_r)\vDash 2j} \sum_{S\subseteq\{1,\dots,r\}} (2j + 2|S| - 1)!! \\ \times \prod_{i\in S} \frac{(-1)^{m_i+1}}{m_i + 2} \prod_{i\notin S} \frac{\widetilde{\mathcal{Z}}_{m_i-1}(u)}{m_i!}. \tag{440}$$

The set S records which of the two monomials of (436) is taken from each factor: $i \in S$ selects the universal tail $v^{m_i+2}/(m_i + 2)$ and $i \notin S$ selects the jet term. The total degree is then $2j + 2|S|$, always even, and the double factorial is its Gaussian moment. The prefactor is i^{2j}.

Theorem 16.7 (Explicit logarithmic coefficients)**.** *With $a_0 = 1$, for every $j \geq 1$,*

$$A_j(u) = \sum_{r=1}^{j} \frac{(-1)^{r-1}}{r} \sum_{(q_1,\dots,q_r)\vDash j} a_{q_1}(u) \cdots a_{q_r}(u). \tag{441}$$

Substituting (435) into these two leaves no recursion anywhere: every A_j is a finite expression with rational coefficients in $1/L$ and $\Psi'(u), \dots, \Psi^{(2j)}(u)$. Three structural facts follow immediately. No derivative beyond $\Psi^{(2j)}$ occurs, because the highest jet reached at order $2j$ is $\widetilde{\mathcal{Z}}_{2j-1}$. Every A_j is smooth, one-periodic and bounded, being a polynomial in such functions. And the mean of A_j is its Ψ-free part plus the means of its terms of degree two and higher, which is the content of Warning 16.17.

Remark 16.8. Unlike Theorem 16.3, these two are not machine-checked. They were verified symbolically against the recursions they replace for $j \leq 3$, with identically vanishing residuals, and (440) is checked here at $j = 1$: it returns $-\widetilde{\mathcal{Z}}_0^2/2 - \widetilde{\mathcal{Z}}_0 - \widetilde{\mathcal{Z}}_1/2 - 1/12$, which is (438), and $A_1 = a_1$.

For reference, the Ψ-free parts of the first four coefficients — the values obtained by setting $\Psi \equiv 0$, which by Warning 16.17 are close to but not equal to the means — are

j	Ψ-free part of A_j	value
1	$\frac{1}{24} + \frac{1}{2L}$	0.7630141871
2	$\frac{1}{4L^2}$	0.5203422453
3	$-\frac{7}{2880} - \frac{1}{24L} - \frac{1}{4L^2} + \frac{1}{6L^3}$	−0.0824216430
4	$\frac{-L^3 - 72L^2 - 816L + 144}{1152L^4}$	−1.7167954639

The order-two entry is unexpectedly clean: it is exactly $1/(4\log^2 2)$, with no rational term and no term in $1/L$.

16.5 The phase to all orders

The proof of the corrected elementary expansion (461) appeals to the lower-branch Lambert expansion without displaying it, even though the precision of that expansion is what controls the residual. We display it here, in the general variable, and state the all-orders form.

Throughout, $S = -\log x$ and $a = \log L$ as in (459), and we write

$$z = \log S - \log L = m - a, \tag{442}$$

so that z is the second logarithm of the primary scale measured in units of L. Taking logarithms in (391) gives the equation that defines the phase,

$$L\lambda - \log\lambda = S, \qquad \text{equivalently} \qquad L\lambda - \log(L\lambda) = S - a. \tag{443}$$

Theorem 16.9 (Lambert phase expansion)**.** *As $x \downarrow 0$,*

$$L\lambda(x) = S + z + \frac{z}{S} + \frac{z - \frac{1}{2}z^2}{S^2} + \frac{z - \frac{3}{2}z^2 + \frac{1}{3}z^3}{S^3} + \mathcal{O}\left(\frac{(1+|z|)^4}{S^4}\right). \tag{444}$$

More generally, there are unique polynomials p_j, with $\deg p_j = j$ and $p_j(0) = 0$, such that for every $N \geq 1$

$$\boxed{L\lambda(x) = S + z + \sum_{j=1}^{N-1} \frac{p_j(z)}{S^j} + \mathcal{O}\left(\frac{(1+|z|)^N}{S^N}\right).} \tag{445}$$

The first three are

$$p_1(z) = z, \qquad p_2(z) = z - \frac{z^2}{2}, \qquad p_3(z) = z - \frac{3z^2}{2} + \frac{z^3}{3}. \tag{446}$$

Proof. Write $w = L\lambda$. By (443), $w = S + \log w - a$. Substituting $w = S + z + \delta$ and using $\log(S + z + \delta) = \log S + \log(1 + (z+\delta)/S)$ together with $z = \log S - a$ turns the equation into the fixed-point form

$$\delta = \log\left(1 + \frac{z+\delta}{S}\right). \tag{447}$$

Since $\lambda \to \infty$ and $z = o(S)$, one has $\delta \to 0$, and (447) may be solved by successive substitution. Expanding the logarithm,

$$\delta = \frac{z+\delta}{S} - \frac{(z+\delta)^2}{2S^2} + \frac{(z+\delta)^3}{3S^3} - \cdots,$$

and inserting $\delta = p_1(z)S^{-1} + p_2(z)S^{-2} + \cdots$ determines the p_j recursively: comparing coefficients of S^{-1} gives $p_1 = z$; of S^{-2} gives $p_2 = p_1 - z^2/2 = z - z^2/2$; of S^{-3} gives

$$p_3 = p_2 - z^2 + \frac{z^3}{3} = z - \frac{3z^2}{2} + \frac{z^3}{3}.$$

At each stage the coefficient of S^{-j} on the right involves only $p_1, \ldots, p_{j-1}$, so the recursion is triangular and the p_j are unique; an induction on j shows $\deg p_j = j$, with leading coefficient $(-1)^{j+1}/j$ and $p_j(0) = 0$. The remainder bound follows by tracking the same recursion one step further, all terms at level N being polynomials of degree N in z. Truncating after $j = 3$ gives (444). □

Remark 16.10 (Closed form of the coefficients)*.* The recursion has the classical solution in unsigned Stirling numbers of the first kind $\mathfrak{s}(j,k)$ familiar from the asymptotic series of Lambert W [11]:

$$p_j(z) = \sum_{m=1}^{j} \frac{(-1)^{m+1}}{m!}\, \mathfrak{s}(j,\, j-m+1)\, z^m. \tag{448}$$

For $j=4$ this gives $p_4(z) = z - 3z^2 + \frac{11}{6}z^3 - \frac{1}{4}z^4$, and for $j=5$, $p_5(z) = z - 5z^2 + \frac{35}{6}z^3 - \frac{25}{12}z^4 + \frac{1}{5}z^5$.

Remark 16.11 (Substituting a truncated phase is a separate step)*.* It is tempting to insert (445) into the all-orders formula (429) and read off a purely elementary logarithmic expansion of $\log F(x)$ to any order. That step needs its own argument, for two reasons that reinforce each other.

First, the main exponent contains $-L\lambda^2/2$, so an error $\delta\lambda$ in the phase propagates to an error of size about $L\lambda\,\delta\lambda$ in the answer; the phase must therefore be known one order beyond the naive count, which is precisely why the fourth term of (444) is required to reach the residual $\mathcal{O}(1/S)$ claimed in (461).

Second, the truncation error in (445) is $\mathcal{O}\big((1+|z|)^N S^{-N}\big)$, not $\mathcal{O}(S^{-N})$; the factor $(1+|z|)^N$ is a power of $\log\log(1/x)$ and is not absorbed by the stated order. When the truncated phase is substituted into Ψ and into the coefficients A_j, these logarithmic factors are inherited and then multiplied by the amplification just described. The periodic function is smooth with bounded derivatives, so it is Lipschitz in the abstract; but the error one is allowed to make in its argument is exactly of the same order as the terms being claimed, so Lipschitz control by the truncation error alone does not close the estimate. A genuine all-orders elementary statement requires composing the two expansions with matched precision and re-collecting the resulting cross terms, and that composition is not assumed silently here. Keeping the exact phase $\lambda(x)$ inside Ψ and inside every A_j, as in (429), avoids the entire issue.

Finally, (444) specializes correctly to the dyadic scale. Put $x = 2^{-t}$, so that $S = Lt$ and, by (442), $z = \log(Lt) - \log L = \log t$. Dividing (444) by L gives

$$\lambda_t = t + \frac{\log t}{L} + \frac{\log t}{L^2 t} + \frac{\log t - \frac{1}{2}\log^2 t}{L^3 t^2} + \frac{\log t - \frac{3}{2}\log^2 t + \frac{1}{3}\log^3 t}{L^4 t^3} + \mathcal{O}\!\left(\frac{\log^4 t}{t^4}\right), \tag{449}$$

whose first four terms are exactly (465), and whose fourth term is the explicit form of the error term displayed there.

16.6 A numerical check at reciprocal powers

The values $F(2^{-n})$ of (136) are exact rationals, obtained from the half-moment recurrence (119) without any floating-point step, so they give an unusually clean test of (428). For example

$$F(2^{-10}) = \frac{134926369}{1246394851358539387238350848000}, \tag{450}$$

whose logarithm is $-50.5775682823\ldots$, is computed in exact arithmetic and then converted to a high-precision float once, at the end.

Table 3 reports, for $n = 10, 20, 40$: the exact Lambert coordinate λ_n solving (464); the residual $\log F(2^{-n}) - \mathcal{M}(2^{-n})$ against the main term (393); that residual multiplied by λ_n; and the predicted first coefficient $A_1(\lambda_n)$ of (427). All entries were recomputed from the definitions; the periodic function and its derivatives were evaluated from the Gamma–zeta Fourier series (373), and independently from the defining series (368), the two agreeing to fifteen digits.

Three things are visible in the table. First, the unscaled residual does tend to zero, which is (395). Second, the scaled residual does not tend to zero and does not grow: it stays near

n	λ_n	$\log F(2^{-n}) - \mathcal{M}$	$\lambda_n(\log F(2^{-n}) - \mathcal{M})$	$A_1(\lambda_n)$
10	13.78503056	0.05801333439	0.7997155875	0.7629678468
20	24.62186833	0.03182756966	0.7836542295	0.7629268731
40	45.50804986	0.01701388954	0.7742689337	0.7629490545

Table 3: Exact-rational test of the first saddle correction at $x = 2^{-n}$. The third column tends to zero; the fourth, which is the third scaled by the exact Lambert coordinate, tends to the periodic coefficient A_1 of the fifth column rather than to any other constant.

0.78, which is the content of (428), and it converges to the value of the periodic coefficient A_1 evaluated at the exact phase, not to a fitted constant of its own. The differences between the fourth and fifth columns are 0.03675, 0.02073, 0.01132; scaled by λ_n they are 0.507, 0.510, 0.515, confirming the remainder $\mathcal{O}(\lambda^{-2})$ of (428) and showing that the next coefficient A_2 is of size about one half.

Warning 16.12. That last observation is a trap, and it is worth spelling out because the trap is generic to this whole expansion. Extending the same computation along $n = 10, 20, 40, 80, 160, 320$ gives scaled second differences 0.5066, 0.5103, 0.5151, 0.5179, 0.5193, 0.5199, which look like a sequence converging to a constant near 0.520. They are not. The theorem of Section 16 says every coefficient A_j is one-periodic in λ, and A_2 is no exception. Doubling n increases $\log_2 \lambda$ by roughly one, so $\lambda_n \bmod 1$ returns to nearly the same place; a geometric sample of n therefore holds the phase almost fixed and displays only the residual drift in λ. Sampling n densely instead separates the two effects. Over $300 \le n \le 400$, where λ moves by only a third, the quantity $\lambda^2(\log F - \mathcal{M} - A_1(\lambda)/\lambda)$ sweeps from 0.52105 down to 0.51810 and back up, tracking $\lambda \bmod 1$, a spread of 2.95×10^{-3}. Held at one phase instead and varied in magnitude, it moves by only 3.7×10^{-4} from $n = 173$ to $n = 352$, consistent with an $\mathcal{O}(\lambda^{-1})$ correction. So A_2 is a genuinely nonconstant periodic function.

Neither of those sampled numbers should be mistaken for a property of A_2 itself, and the distinction is worth drawing because it is the same trap one level down. The sample covers $0.268 \le \lambda \bmod 1 \le 0.675$, which is 41% of a period and misses the maximum; and it carries an A_3/λ displacement of about -3×10^{-4}. Computed from the coefficient rather than from the sample, A_2 has

$$\max A_2 = 0.522282117, \quad \min A_2 = 0.518402361, \quad \langle A_2 \rangle = 0.5203422413, \tag{451}$$

attained at $\lambda \bmod 1 = 0.101128$ and 0.601131, so the true peak-to-peak swing is 3.880×10^{-3} rather than the 2.95×10^{-3} visible in the window. The measured minimum at 0.602 does agree with the true one.

The amplitude is not an accident, and its growth with j is governed by an exact law, which is the subject of Section 16.7 below. Any numerical extraction of a later coefficient must fix the phase first.

16.7 The highest derivative in each coefficient

The coefficients A_j of (429) are polynomials in L^{-1} and in the derivatives of Ψ. Two facts about them are worth isolating, because together they explain both the size of the oscillation seen above and the ultimate divergence of the series.

Theorem 16.13 (Leading-derivative law)**.** *For every $j \ge 1$ the coefficient A_j involves no derivative of Ψ beyond the $2j$th, that derivative occurs linearly, and its coefficient is exactly*

$$\frac{(-1)^j}{2^j\, j!\, L^{2j}}\, \Psi^{(2j)}(u). \tag{452}$$

Proof architecture. The $(2j)$th derivative reaches A_j through exactly one route. It enters the exponent expansion only in the top exponent polynomial, and there linearly, with coefficient $\mathrm{i}^{2j}/(2j)! = (-1)^j/(2j)!$. The Gaussian contraction replaces v^{2j} by the moment $(2j-1)!!$, and the passage from the saddle mass to its logarithm leaves a linear term untouched. The stated constant is then the identity

$$\frac{(2j-1)!!}{(2j)!} = \frac{1}{2^j\, j!},$$

which is $(2j)! = 2^j j!\,(2j-1)!!$. □

The law is visible in the coefficients that have been computed. For $j=1$ it gives $-\Psi''/(2L^2)$, which is the Ψ'' term of (427); for $j=2$ it gives $+\Psi^{(4)}/(8L^4)$; for $j=3$, $-\Psi^{(6)}/(48L^6)$; for $j=4$, $+\Psi^{(8)}/(384L^8)$.

Remark 16.14 (What is formalized here, and what is not). The substance of Theorem 16.13 is machine-checked for general j, and not merely verified in the four cases just listed. What the formal development proves is the statement about the Gaussian *mass* coefficient: for an arbitrary jet sequence and every j, the order-$(j+1)$ mass coefficient depends on the top jet $\widetilde{\mathcal{Z}}_{2j+1}$ exactly linearly, with coefficient $(-1)^{j+1}/(2^{j+1}(j+1)!)$. The engine of that proof is the observation that the exponential recurrence is affine in its top exponent coefficient, which is what makes the statement provable at once for all orders rather than checkable one order at a time.

The passage from the mass coefficient to the logarithmic coefficient A_j is not formalized. It is immediate — taking the formal logarithm subtracts from the order-j mass coefficient only products of strictly lower mass coefficients, and no such product reaches $\widetilde{\mathcal{Z}}_{2j-1}$, so the linear term survives unchanged — but it is an elementary observation and the article does not claim otherwise.

Corollary 16.15 (Growth of the oscillation)**.** *Differentiating multiplies the first Fourier mode by $(2\pi)^{2j}$, so that mode of $\Psi^{(2j)}$ has amplitude $2|\widehat{\Psi}(1)|(2\pi)^{2j}$, and the leading term* (452) *therefore contributes an oscillation of amplitude*

$$2|\widehat{\Psi}(1)|\,\frac{1}{j!}\left(\frac{2\pi^2}{L^2}\right)^j, \qquad \frac{2\pi^2}{L^2} = 41.0845769104. \tag{453}$$

Numerically this predicts 8.73×10^{-5}, 1.79×10^{-3}, 2.46×10^{-2} and 2.52×10^{-1} for $j=1,2,3,4$.

Warning 16.16. Equation (453) is a growth rate, not the amplitude. It uses only the top derivative; the lower ones contribute in phase and inflate the true swing by a factor that itself grows. Measuring amplitudes — half the peak-to-peak swing, on the same convention as (453) itself — the true values are 8.73×10^{-5}, 1.94×10^{-3}, about 3×10^{-2} and about 0.41 for $j=1,2,3,4$, against the estimates 8.73×10^{-5}, 1.79×10^{-3}, 2.46×10^{-2}, 2.52×10^{-1}. The ratio of truth to estimate drifts 1.00, 1.08, about 1.2, about 1.6. Use (453) to see how fast the oscillation grows, never to size a particular coefficient.

Warning 16.17 (Products of derivatives are not negligible either). It is tempting to argue that since Ψ oscillates by only about 2×10^{-6}, products of its derivatives are of order 4×10^{-12} and may be dropped. That reasoning contradicts the one above and is false. A mean of a product is

$$\langle \Psi^{(a)}\Psi^{(b)}\rangle = \sum_{k\neq0}(2\pi\mathrm{i}k)^a(-2\pi\mathrm{i}k)^b\big|\widehat{\Psi}(k)\big|^2, \tag{454}$$

which carries the same amplification $(2\pi k)^{a+b}$ as everything else, and it vanishes unless $a+b$ is even. Numerically $\langle(\Psi')^2\rangle = 8.917\times10^{-11}$ but $\langle(\Psi''')^2\rangle = 1.390\times10^{-7}$, four decades larger.

The practical consequence is that the Ψ-free part of A_j is *not* its mean. The means themselves are available in closed form. Every mean of a derivative vanishes, and integration by parts over a period kills or pairs the rest:

$$\langle \Psi'\Psi'' \rangle = 0, \qquad \langle (\Psi')^2\Psi'' \rangle = 0, \qquad \langle \Psi'\Psi''' \rangle = -\langle (\Psi'')^2 \rangle, \tag{455}$$

the first two because each integrand is the derivative of a periodic function. Applying these to (427) and to the corresponding expression one order further leaves

$$\langle A_1 \rangle = \frac{1}{24} + \frac{1}{2L} - \frac{\langle (\Psi')^2 \rangle}{2L^2}, \tag{456}$$

$$\langle A_2 \rangle = \frac{1}{4L^2} - \frac{\langle (\Psi')^2 \rangle}{2L^3} - \frac{\langle (\Psi'')^2 \rangle}{4L^4} - \frac{\langle (\Psi')^3 \rangle}{6L^3}. \tag{457}$$

Both are exact. With $\langle (\Psi')^2 \rangle = 8.917038 \times 10^{-11}$, $\langle (\Psi'')^2 \rangle = 3.520305 \times 10^{-9}$ and $\langle (\Psi')^3 \rangle = 1.115006 \times 10^{-21}$ they give $\langle A_1 \rangle = 0.7630141870184$ and $\langle A_2 \rangle = 0.5203422413049$.

The cubic term in (457) is -5.58×10^{-22}, twelve orders below the smallest term beside it, and it is the only one with no partner to cancel against. It is kept because (457) is an identity and not an estimate; dropping it would change the equals sign into an approximation, which is a different claim.

The gap between the mean and the Ψ-free part grows with j like everything else here: it is -9.3×10^{-11}, -3.9×10^{-9} and -1.14×10^{-7} at $j = 1, 2, 3$, so it is invisible at $j = 1$ and reaches the seventh digit by $j = 3$. The Ψ-free part remains the right practical value for the mean, but the reason is a ratio and not an absolute bound: at each order it sits four to five decades below the oscillation of A_j itself.

The same two facts explain why the expansion (429) is asymptotic and not convergent, and the mechanism is worth naming because it is not the usual one. Stirling's estimate for Γ on vertical lines gives

$$|\widehat{\Psi}(k)| \asymp \mathrm{e}^{-\pi^2 k/L}, \qquad \frac{\pi^2}{L} = 14.2385595, \tag{458}$$

so the Fourier modes of Ψ die extremely fast; already $|\widehat{\Psi}(2)|/|\widehat{\Psi}(1)| \approx 6.3 \times 10^{-7}$. But mode k enters A_j amplified by $(2\pi k)^{2j}/(2^j j! L^{2j})$, so its contribution behaves like $\exp\bigl(2j \log k - \pi^2 k/L\bigr)$, which is maximized at $k \approx 0.14j$. For small j the first mode dominates and the coefficients are minute; past roughly $j = 8$ the higher modes take over and $\sup |A_j|$ grows faster than geometrically. The divergence of the expansion is therefore caused by the *high-frequency* content of the periodic correction, not by the factorial growth one expects from a saddle-point expansion. This paragraph is a size estimate, not a theorem.

Third, and for honesty about what the table can and cannot show at these sizes: the periodic part of A_1 is minute. The constant part $1/24 + 1/(2L) = 0.7630141871$ of (427) is modulated by $-(\Psi'' + (\Psi')^2)/(2L^2)$, whose amplitude is about 8.7×10^{-5} (the second derivative multiplies the first Fourier mode by $(2\pi)^2$), while $\Psi(\lambda_n)$ itself is -1.13×10^{-6}, -2.13×10^{-6} and -1.59×10^{-6} at $n = 10, 20, 40$. These are far below the $\mathcal{O}(\lambda^{-2})$ remainder at $n \le 40$. The table therefore confirms the size and value of the periodic coefficient A_1; it does not, by itself, isolate the oscillation. The proof that the oscillation cannot be dropped is the subsequential-limit argument of (470), not a table. The forward tail plays no role at all here: at $n = 10$ the saddle radius is $r = \lambda_{10} 2^{10} \approx 1.41 \times 10^4$, so by (408) the term $E(r)$ is smaller than 10^{-6100}.

17 Elementary log–log forms and the missing periodic term

Lambert W gives the cleanest formula, but an expansion in elementary logarithms is useful for comparison with published displays. It must, however, retain Ψ at the exact Lambert phase.

17.1 The literal small-x expansion

Put

$$S = -\log x, \qquad m = \log S, \qquad a = \log L. \tag{459}$$

Thus $S \to \infty$ as $x \downarrow 0$. Define the nonperiodic elementary expression

$$\begin{aligned}\mathcal{E}_0(x) = & -\frac{S^2}{2L} - \frac{Sm}{L} + \left(\frac{1}{2} + \frac{1+a}{L}\right) S \\ & - \frac{m^2}{2L} + \frac{am}{L} + \frac{6\gamma^2 + 12\gamma_1 - \pi^2 - 6a^2}{12L} - \frac{7L}{12} - \frac{1}{2}\log\pi \\ & - \frac{m^2}{2LS} + \frac{am}{LS}.\end{aligned} \tag{460}$$

Theorem 17.1 (Corrected elementary expansion)**.** *As $x \downarrow 0$,*

$$\boxed{\log F(x) = \mathcal{E}_0(x) + \Psi(\lambda(x)) + \mathcal{O}\left(\frac{1}{-\log x}\right).} \tag{461}$$

Proof architecture. The standard lower-branch Lambert expansion solves $\lambda L - \log\lambda = S$ by successive substitution. Substituting that expansion into (393), and retaining enough phase terms to control the amplified quadratic contribution, gives (460). A coarse two-term approximation to λ is insufficient: because the main exponent contains $-L\lambda^2/2$, an error $\delta\lambda$ contributes about $-L\lambda\delta\lambda$, so the phase must be known one order more accurately than a naive count suggests. The fourth Lambert phase term in the formal proof is exactly what reduces the residual to $O(1/S)$. □

In the original variable $\ell = \log x < 0$, the same expression is

$$\begin{aligned}\mathcal{E}_0(x) = & -\frac{\ell^2}{2L} + \frac{\ell\log(-\ell)}{L} - \left(\frac{1}{2} + \frac{1+\log L}{L}\right)\ell \\ & - \frac{\log^2(-\ell)}{2L} + \frac{\log L\,\log(-\ell)}{L} \\ & + \frac{6\gamma^2 + 12\gamma_1 - \pi^2 - 6\log^2 L}{12L} - \frac{7L}{12} - \frac{1}{2}\log\pi \\ & + \frac{\log^2(-\ell)}{2L\ell} - \frac{\log L\,\log(-\ell)}{L\ell}.\end{aligned} \tag{462}$$

This is the literal elementary display formalized in the repository, with the periodic term added separately.

17.2 The dyadic logarithmic scale

Set $x = 2^{-t}$ and let

$$\lambda_t = -\frac{1}{L} W_{-1}(-L2^{-t}). \tag{463}$$

Then λ_t is the unique large solution of

$$\lambda_t - \log_2\lambda_t = t. \tag{464}$$

Its first terms are

$$\lambda_t = t + \frac{\log t}{L} + \frac{\log t}{L^2 t} + \frac{\log t - \frac{1}{2}\log^2 t}{L^3 t^2} + \mathcal{O}\left(\frac{\log^3 t}{t^3}\right). \tag{465}$$

Define

$$\mathcal{D}_0(t) = -\frac{L}{2}t^2 - t\log t + \left(1 + \frac{L}{2}\right)t - \frac{\log^2 t}{2L} + C_F - \frac{\log^2 t}{2L^2 t}. \tag{466}$$

Theorem 17.2 (Corrected dyadic-scale expansion)**.** *As $t \to \infty$,*

$$\boxed{\log F(2^{-t}) = \mathcal{D}_0(t) + \Psi(\lambda_t) + \mathcal{O}(t^{-1}).} \tag{467}$$

The literal substitution of $x = 2^{-t}$ into (460) differs from $\mathcal{D}_0(t)$ by the explicit term

$$\frac{\log^2 L}{2L^2 t}, \tag{468}$$

which is itself $O(1/t)$. Thus both nonperiodic displays are valid at the stated precision, but they are not algebraically identical.

17.3 Why the uncorrected formula is false at the claimed scale

Let $\mathcal{M}_0(x) = \mathcal{M}(x) - \Psi(\lambda(x))$ be the nonperiodic Lambert main. From (395),

$$\log F(x) - \mathcal{M}_0(x) = \Psi(\lambda(x)) + O(1/\lambda(x)). \tag{469}$$

Since Ψ is nonconstant, the right side does not tend to zero.

Proposition 17.3 (Failure without the periodic correction)**.** *The relation*

$$\log F(x) = \mathcal{M}_0(x) + O(1/(-\log x)) \tag{470}$$

is false. In fact $F(x)/\exp(\mathcal{M}_0(x))$ has distinct subsequential limits.

Proof. Choose $u, v \in [0, 1)$ with $\Psi(u) \neq \Psi(v)$. Set

$$\lambda_n = n + u, \qquad x_n = \lambda_n 2^{-\lambda_n}.$$

Then $x_n \downarrow 0$ and the exact Lambert phase of x_n is λ_n. Hence

$$\log F(x_n) - \mathcal{M}_0(x_n) \longrightarrow \Psi(u).$$

The analogous sequence with phase $n + v$ tends to $\Psi(v)$. Since $1/(-\log x) \to 0$, the bound (470) is impossible. Exponentiating gives distinct ratio limits $e^{\Psi(u)}$ and $e^{\Psi(v)}$. $\square$

Numerically the discrepancy is minute because the first Fourier mode has size about 10^{-6}, but asymptotic equivalence is a limit statement, not a graphical tolerance. A nonzero bounded oscillation cannot be hidden inside an error term that tends to zero.

17.4 The periodic correction is unique

The previous subsection shows that the periodic term cannot be deleted. A little more is true and is worth recording, because it removes the last escape route: the periodic term cannot be replaced either. There is exactly one 1-periodic function that makes the sharp formula correct, and it is Ψ.

Theorem 17.4 (Uniqueness of the periodic correction)**.** *Let $\mathcal{M}_0(x) = \mathcal{M}(x) - \Psi(\lambda(x))$ be the nonperiodic Lambert main term, and let $\widetilde{\Psi} : \mathbb{R} \to \mathbb{R}$ be continuous, 1-periodic and of mean zero. Suppose that*

$$\log F(x) = \mathcal{M}_0(x) + \widetilde{\Psi}(\lambda(x)) + o(1) \qquad (x \downarrow 0). \tag{471}$$

Then $\widetilde{\Psi} = \Psi$ identically.

Proof. By (395) one has $\log F(x) = \mathcal{M}_0(x) + \Psi(\lambda(x)) + \mathcal{O}(1/\lambda(x))$, and $\lambda(x) \to \infty$ as $x \downarrow 0$. Subtracting (471) gives

$$(\Psi - \widetilde{\Psi})(\lambda(x)) \longrightarrow 0 \qquad (x \downarrow 0). \tag{472}$$

Fix $t \in \mathbb{R}$ and, for integers n with $n + t > 1/L$, set

$$x_n = (n+t)\, 2^{-(n+t)}. \tag{473}$$

Then $x_n \downarrow 0$, and by (391) the number $n + t$ is a solution of $\lambda 2^{-\lambda} = x_n$ on the branch where $\lambda > 1/L$; since that solution is unique and equals $\lambda(x_n)$, the sequence has exact Lambert coordinate

$$\lambda(x_n) = n + t.$$

This is the point of the construction: the sequence is chosen so that the Lambert coordinate, not the naive scale $\log_2(1/x_n)$, lands on a prescribed residue. By 1-periodicity,

$$(\Psi - \widetilde{\Psi})(\lambda(x_n)) = (\Psi - \widetilde{\Psi})(n + t) = (\Psi - \widetilde{\Psi})(t)$$

for every n, so the left-hand side of (472) is a constant sequence along (x_n). A constant sequence that tends to zero is zero, whence $\Psi(t) = \widetilde{\Psi}(t)$. As $t \in \mathbb{R}$ was arbitrary, the two functions agree everywhere. □

Remark 17.5. The proof uses only 1-periodicity: continuity is never invoked, since the argument produces equality at each real t separately rather than on a dense set. Uniqueness therefore holds in the class of all 1-periodic functions. The mean-zero normalization is likewise not used here, because (471) fixes the nonperiodic main term, including the constant C_F; its role is to remove the trivial ambiguity that arises if one allows the additive constant in $\mathcal{M}_0$ to float, in which case the pair (C_F, Ψ) is determined only up to moving a constant between them, and centering pins it. In particular C_F is itself uniquely determined by (395).

This matters because it changes the logical status of the correction. One might hope that Ψ is an artifact of the Mellin route, and that some other bounded oscillation, or some cleverer nonperiodic reshuffling, would make the elementary formula true. Theorem 17.4 closes that possibility: the order-one periodic term is pinned by the asymptotic statement itself, independently of how it was derived. Any correct sharp formula with this main term must contain exactly the Gamma–zeta function Ψ of (373), evaluated at exactly the Lambert phase. Together with the nonvanishing of every Fourier coefficient, this makes the fluctuation an intrinsic feature of the Fabius law at the endpoint rather than a feature of one proof.

17.5 Symmetry at the right endpoint

The symmetry $F(1-x) = 1 - F(x)$ immediately transfers the left-endpoint expansion to the right:

$$1 - F(1-x) = F(x). \tag{474}$$

Thus the same Lambert phase, periodic fluctuation, and all-orders coefficients describe the survival probability near 1.

18 Consequences of the endpoint expansion

The full sharp formula immediately settles several qualitative questions that are otherwise awkward to read from the differential equation.

Theorem 18.1 (Quadratic logarithmic law)**.** *As $x \downarrow 0$,*

$$\frac{\log F(x)}{(\log x)^2} \longrightarrow -\frac{1}{2\log 2}. \tag{475}$$

Equivalently, on the dyadic scale,

$$\log F(2^{-t}) \sim -\frac{\log 2}{2} t^2. \tag{476}$$

Proof. In (393), $\lambda \sim S/L$ with $S = -\log x$. The quadratic term $-L\lambda^2/2$ dominates the linear term, the logarithmic term, and the bounded periodic correction. Substitution gives (475). □

Corollary 18.2 (Between powers and pure endpoint exponentials)**.** *For every $N > 0$ and every $c > 0$,*

$$F(x) = o(x^N), \tag{477}$$

and

$$e^{-c/x} = o(F(x)) \tag{478}$$

as $x \downarrow 0$.

Proof. The logarithm of x^N is $-NS$, whereas $\log F(x) \sim -S^2/(2L)$, so $F(x)/x^N \to 0$. On the other hand, $\log e^{-c/x} = -ce^S$, which is much more negative than $-S^2/(2L)$; therefore $e^{-c/x}/F(x) \to 0$. □

This places the endpoint decay in an intermediate regime. The function is flatter than any finite algebraic order, as every compactly supported smooth function must be at its support boundary, but it is vastly larger than the familiar model $e^{-1/x}$.

Corollary 18.3 (Asymptotics of reciprocal-power values)**.** *For integers $n \to \infty$,*

$$\begin{aligned}\log F(2^{-n}) = &-\frac{L}{2}n^2 - n\log n + \left(1 + \frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F \\ &- \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + O(n^{-1}),\end{aligned} \tag{479}$$

where λ_n is defined by (463). Combining this with (136) gives an equally sharp expansion for the half moments d_n.

The formula shows why a fixed multiplicative constant cannot complete a recurrence-based ansatz for $F(2^{-n})$: the fractional parts of λ_n drift, and the factor $\exp \Psi(\lambda_n)$ is nonconstant.

The corollary above was left at the level of a remark; the resulting expansion of the half moments is worth displaying, because the cancellation is exact and total.

Corollary 18.4 (Sharp asymptotic for the half moments)**.** *Let λ_n be the solution of $\lambda_n - (\log \lambda_n)/L = n$ given by (463). Then, as $n \to \infty$,*

$$\log d_n = \frac{1}{2}\log(2\pi n) + C_F + \Psi(\lambda_n) - \frac{(\log n)^2}{2L} - \frac{(\log n)^2}{2L^2 n} + \mathcal{O}\left(\frac{1}{n}\right), \tag{480}$$

and in particular

$$\boxed{d_n \sim \sqrt{2\pi n}\, \exp\left(C_F + \Psi(\lambda_n) - \frac{(\log n)^2}{2L}\right).} \tag{481}$$

Proof. By the bridge (137),

$$\log d_n = \log n! + \binom{n}{2} L + \log F(2^{-n}). \tag{482}$$

Insert Stirling's formula $\log n! = n\log n - n + \frac{1}{2}\log(2\pi n) + \mathcal{O}(n^{-1})$, the identity $\binom{n}{2}L = \frac{L}{2}n^2 - \frac{L}{2}n$, and the expansion (479). The three leading blocks cancel identically:

$$\begin{aligned} n^2 \text{ terms:} \quad & \frac{L}{2}n^2 - \frac{L}{2}n^2 = 0, \\ n\log n \text{ terms:} \quad & n\log n - n\log n = 0, \\ n \text{ terms:} \quad & -n - \frac{L}{2}n + \left(1 + \frac{L}{2}\right)n = 0. \end{aligned}$$

What survives is

$$\log d_n = \frac{1}{2}\log(2\pi n) + C_F + \Psi(\lambda_n) - \frac{(\log n)^2}{2L} - \frac{(\log n)^2}{2L^2 n} + \mathcal{O}(n^{-1}),$$

which is (480). The constant is exactly C_F of (380); no separate constant is generated by the cancellation. Since $(\log n)^2/n \to 0$, exponentiating and absorbing the last two terms into a factor tending to 1 gives (481). □

The three cancellations are the reason the half moments look deceptively simple: the superexponentially small factor $F(2^{-n})$ and the superexponentially large factor $n!\,2^{\binom{n}{2}}$ agree through the quadratic, the $n\log n$, and the linear order, and their quotient retains only a Gaussian-type prefactor $\sqrt{2\pi n}$, the endpoint constant, the log-square drift, and the periodic fluctuation. Note that Ψ survives the cancellation untouched. Consequently the normalized sequence $d_n \exp((\log n)^2/(2L))/\sqrt{2\pi n}$ does not converge. Indeed (464) gives $\lambda_n - n = \log_2 \lambda_n$, which tends to infinity with increments tending to zero, so the fractional parts of λ_n are dense in $[0,1)$; since Ψ is continuous and nonconstant, the set of limit points of the normalized sequence is the full interval

$$\left[\mathrm{e}^{C_F + \min\Psi},\ \mathrm{e}^{C_F + \max\Psi}\right].$$

Any conjecture that this sequence tends to a limit is false, even though its numerical variation is only a few parts in a million.

The comparison (478) is far from sharp. The true decay is governed by $(\log x)^2$, and every negative power of x in the exponent beats that, however small the power.

Proposition 18.5 (Strengthened decay comparison)**.** *For every $c > 0$ and every $\alpha > 0$,*

$$\exp\left(-\frac{c}{x^\alpha}\right) = o\big(F(x)\big) \qquad (x \downarrow 0). \tag{483}$$

Proof. Put $S = -\log x \to \infty$. By (475), $\log F(x) = -(1 + o(1))S^2/(2L)$, while

$$\log\exp\left(-\frac{c}{x^\alpha}\right) = -c\,\mathrm{e}^{\alpha S}.$$

Hence

$$\log\frac{\exp(-cx^{-\alpha})}{F(x)} = -c\,\mathrm{e}^{\alpha S} + (1 + o(1))\frac{S^2}{2L} = -c\,\mathrm{e}^{\alpha S}\left(1 - (1 + o(1))\frac{S^2}{2cL\mathrm{e}^{\alpha S}}\right).$$

For every fixed $\alpha > 0$ one has $S^2\mathrm{e}^{-\alpha S} \to 0$, so the bracket tends to 1 and the whole expression tends to $-\infty$. Exponentiating gives (483). □

Taking $\alpha = 1$ recovers (478). The gain is that no exponent is too small: even $\exp(-x^{-1/100})$, which decays extremely slowly by the standards of endpoint models, is still negligible compared with F. Combined with (477), the Fabius endpoint is squeezed between the two scales from both sides, and the squeeze is now tight enough to be decisive for the candidate approximations discussed in section 22. In particular, the quotient of exponentials examined in the subsection on a visually good quotient with the wrong boundary scale has a numerator of the form $\exp(-|x|^{-\alpha})$ with $\alpha = \frac{3}{4}\sqrt{2\pi}$; by Proposition 18.5 that factor alone already forces the candidate to be $o(F)$ at the endpoint, without any appeal to the sharper bound $\exp(-1/(2y))$ used there. The failure is therefore not an artifact of the particular constant in that bound: no expression whose logarithm decays like a negative power of the distance to the endpoint can match a function whose logarithm decays like the square of the logarithm of that distance.

19 The Fourier–Legendre expansion of up

A second exact spectral representation uses the ordinary Legendre polynomials P_n on $[-1, 1]$. Unlike the Fourier transform, this is a polynomial expansion on the natural support interval, and its coefficients can be expressed by the reciprocal-power dyadic values already computed.

19.1 Orthogonality and coefficients

Normalize the Legendre polynomials by $P_n(1) = 1$. They satisfy

$$-\frac{\mathrm{d}}{\mathrm{d}x}\Big((1-x^2)P_n'(x)\Big) = n(n+1)P_n(x), \tag{484}$$

with orthogonality

$$\int_{-1}^{1} P_m(x)P_n(x)\,\mathrm{d}x = \frac{2}{2n+1}\,\delta_{mn}. \tag{485}$$

For a continuous function f, define

$$\alpha_n(f) = \frac{2n+1}{2}\int_{-1}^{1} f(x)P_n(x)\,\mathrm{d}x. \tag{486}$$

Because up is even and $P_n(-x) = (-1)^n P_n(x)$,

$$\alpha_{2n+1}(\mathrm{up}) = 0. \tag{487}$$

Put

$$u_n = \alpha_{2n}(\mathrm{up}) = \frac{4n+1}{2}\int_{-1}^{1} \mathrm{up}(x)P_{2n}(x)\,\mathrm{d}x. \tag{488}$$

19.2 Exact coefficient formula

The even Legendre polynomial has the finite power expansion

$$P_{2n}(x) = 4^{-n}\sum_{k=0}^{n}(-1)^{n+k}\binom{2n}{n+k}\binom{2n+2k}{2n}x^{2k}. \tag{489}$$

The moment-to-dyadic bridge is

$$c_k = 2(2k)!\,2^{\binom{2k+1}{2}}F(2^{-2k-1}). \tag{490}$$

This is (142) with an even exponent, together with symmetry.

Theorem 19.1 (Exact Legendre coefficients)**.** *For every $n \ge 0$,*

$$u_n = 4^{-n}(4n+1)\sum_{k=0}^{n}(-1)^{n+k}\binom{2n}{n+k}\binom{2n+2k}{2n} \times (2k)!\,2^{\binom{2k+1}{2}}F(2^{-2k-1}). \tag{491}$$

In particular every u_n is rational.

Proof. Insert (489) into (488) and interchange the finite sum with the integral. The integral of $x^{2k}\,\mathrm{up}(x)$ is c_k. Substitution of (490) cancels the factor $1/2$ in (488) and gives (491). □

Thus the entire polynomial spectral expansion is anchored in the same exact arithmetic table $F(2^{-m})$ that drives the bit-recursive evaluator.

Example 19.2 (The first two Legendre coefficients)**.** Take $n=0$ and $n=1$ in (488). Since $P_0 = 1$ and $P_2(x) = (3x^2-1)/2$,

$$u_0 = \frac{1}{2}\int_{-1}^{1}\mathrm{up}(x)\,\mathrm{d}x = \frac{c_0}{2} = \frac{1}{2},$$
$$u_1 = \frac{5}{2}\int_{-1}^{1}\mathrm{up}(x)\frac{3x^2-1}{2}\,\mathrm{d}x = \frac{5}{4}(3c_1-c_0) = \frac{5}{4}\left(\frac{1}{3}-1\right) = -\frac{5}{6}.$$

The exact formula (491) returns the same two numbers without reference to the moments. For $n=0$ the sum has the single term $\binom{0}{0}\binom{0}{0}\,0!\,2^{\binom{1}{2}}F(2^{-1}) = \frac{1}{2}$, so $u_0 = \frac{1}{2}$. For $n=1$,

$$u_1 = \frac{5}{4}\left[-\binom{2}{1}\binom{2}{2}\,0!\,2^{\binom{1}{2}}F(2^{-1}) + \binom{2}{2}\binom{4}{2}\,2!\,2^{\binom{3}{2}}F(2^{-3})\right] = \frac{5}{4}\left[-1+\frac{96}{288}\right] = -\frac{5}{6},$$

using $\binom{1}{2}=0$, $\binom{3}{2}=3$, $F(2^{-1}) = \frac{1}{2}$ and $F(2^{-3}) = \frac{1}{288}$ from (143). Hence the Fourier–Legendre expansion (495) begins

$$\mathrm{up}(t) = \frac{1}{2} - \frac{5}{6}P_2(t) + \cdots. \tag{492}$$

Both values are stated in the normalization (488) used here, in which $u_n = \alpha_{2n}(\mathrm{up})$ already carries the factor $(4n+1)/2$ of (486); the bare integrals are $\int_{-1}^{1}\mathrm{up}\,P_0 = 1$ and $\int_{-1}^{1}\mathrm{up}\,P_2 = -\frac{1}{3}$. The negative sign of u_1 is genuine and reflects the fact that up is more sharply peaked at the origin than the constant $\frac{1}{2}$.

19.3 Absolute and uniform convergence

The Legendre Sturm–Liouville operator

$$\mathcal{L}f = -\frac{\mathrm{d}}{\mathrm{d}x}\Big((1-x^2)f'(x)\Big) \tag{493}$$

is formally self-adjoint on $[-1,1]$. Boundary terms vanish because of the factor $1-x^2$. Repeated integration by parts therefore gives, for $n \ge 1$,

$$\alpha_n(f) = \frac{2n+1}{2[n(n+1)]^m}\int_{-1}^{1}(\mathcal{L}^m f)(x)P_n(x)\,\mathrm{d}x. \tag{494}$$

Since $|P_n(x)| \le 1$ on $[-1,1]$, any C^∞ function has coefficients that decay faster than every power of n.

Theorem 19.3 (Uniform Legendre series)**.** *On the entire closed interval* $[-1,1]$,

$$\boxed{\mathrm{up}(x) = \sum_{n=0}^{\infty} u_n P_{2n}(x).} \tag{495}$$

The series converges absolutely and uniformly, including at both endpoints.

Proof. Apply (494) to $f = \mathrm{up}$. Choosing $m \geq 2$ gives $|\alpha_n(\mathrm{up})| \leq Cn^{-3}$, say. Since $|P_n| \leq 1$, the Weierstrass test gives absolute and uniform convergence. Let the uniform sum be g. Orthogonality shows that g and up have the same Legendre coefficients. Completeness of the Legendre polynomials in $L^2[-1,1]$ implies $g = \mathrm{up}$ almost everywhere, and continuity upgrades equality to every point. Odd coefficients vanish by (487), leaving (495). □

The endpoint statement is worth emphasizing. Generic Fourier–Legendre convergence theorems are often quoted only on the open interval, or with an endpoint average. Here smoothness and absolute uniform convergence give the literal values $\mathrm{up}(\pm 1) = 0$.

19.4 A practical polynomial approximation

Truncating (495) at $n = N$ gives a polynomial of degree $2N$ with exact rational coefficients after expanding the P_{2n}. The error is bounded by

$$\left\| \mathrm{up} - \sum_{n=0}^{N} u_n P_{2n} \right\|_{\infty} \leq \sum_{n>N} |u_n|. \tag{496}$$

Repeated use of (494) supplies arbitrarily high algebraic tail bounds. For moderate precision this method is attractive: all coefficients are exact, evaluation is stable on $[-1,1]$, and the approximation respects evenness automatically.

19.4.1 The translated series on $[0,2]$

For computation it is usually the signed extension on $[0,2]$, not the bump on $[-1,1]$, that one wants to evaluate, and there the same coefficients serve with a shift of the argument.

Proposition 19.4 (Translated Legendre series)**.** *For* $0 \leq x \leq 2$,

$$\mathcal{F}(x) = \sum_{n=0}^{\infty} u_n \, P_{2n}(x-1), \tag{497}$$

with the u_n *of* (488)*. The series converges absolutely and uniformly on* $[0,2]$*, including at both endpoints.*

Proof. The single-block formula (12) with $b = 0$ gives $\mathcal{F}(x) = \varepsilon_0 \, \mathrm{up}(x-1) = \mathrm{up}(x-1)$ for $0 \leq x \leq 2$, since $\varepsilon_0 = 1$. Substituting $x - 1$ for the variable in (495) gives (497), and the convergence statement transfers unchanged. □

Lemma 19.5 (Power expansion of the translated Legendre polynomials)**.** *For every* $n \geq 0$ *and every* x,

$$P_{2n}(x-1) = \sum_{j=0}^{2n} (-1)^j 2^{-j} \binom{2n}{j} \binom{2n+j}{j} x^j. \tag{498}$$

Proof. Let $N \geq 0$ and put $t = x/2$, so that $x - 1 = 2t - 1$. Rodrigues' formula $P_N(y) = (2^N N!)^{-1}(\mathrm{d}/\mathrm{d}y)^N(y^2-1)^N$ together with $\mathrm{d}/\mathrm{d}y = \frac{1}{2}\,\mathrm{d}/\mathrm{d}t$ and $(2t-1)^2 - 1 = 4(t^2 - t)$ gives

$$P_N(2t-1) = \frac{1}{2^N N!}\cdot\frac{1}{2^N}\frac{\mathrm{d}^N}{\mathrm{d}t^N}\left[4^N(t^2-t)^N\right] = \frac{1}{N!}\frac{\mathrm{d}^N}{\mathrm{d}t^N}(t^2-t)^N.$$

Expanding $(t^2-t)^N = \sum_{j=0}^{N}\binom{N}{j}(-1)^{N-j}t^{N+j}$ and differentiating N times turns t^{N+j} into $\frac{(N+j)!}{j!}t^j$, so

$$P_N(2t-1) = \sum_{j=0}^{N}(-1)^{N-j}\binom{N}{j}\binom{N+j}{j}t^j.$$

Now set $N = 2n$, so that $(-1)^{N-j} = (-1)^j$, and substitute $t = x/2$. □

The identity is easy to misquote, so two instances are printed in full. For $n = 1$,

$$P_2(x-1) = \frac{3(x-1)^2 - 1}{2} = \frac{3}{2}x^2 - 3x + 1,$$

while (498) gives the coefficients $\binom{2}{0}\binom{2}{0} = 1$, $-\frac{1}{2}\binom{2}{1}\binom{3}{1} = -3$, and $\frac{1}{4}\binom{2}{2}\binom{4}{2} = \frac{3}{2}$. For $n = 2$, expanding $P_4(y) = (35y^4 - 30y^2 + 3)/8$ at $y = x - 1$ gives

$$P_4(x-1) = \frac{35}{8}x^4 - \frac{35}{2}x^3 + \frac{45}{2}x^2 - 10x + 1,$$

while (498) gives, for $j = 0, 1, 2, 3, 4$ in turn,

$$1, \qquad -\tfrac{1}{2}\binom{4}{1}\binom{5}{1} = -10, \qquad \tfrac{1}{4}\binom{4}{2}\binom{6}{2} = \tfrac{45}{2},$$
$$-\tfrac{1}{8}\binom{4}{3}\binom{7}{3} = -\tfrac{35}{2}, \qquad \tfrac{1}{16}\binom{4}{4}\binom{8}{4} = \tfrac{35}{8}.$$

The two computations agree in every coefficient.

Combining (497) with (498) converts the truncation (500) into an ordinary power polynomial:

$$S_N(x-1) = \sum_{j=0}^{2N}\left[\sum_{n=\lceil j/2\rceil}^{N}(-1)^j 2^{-j}u_n\binom{2n}{j}\binom{2n+j}{j}\right]x^j, \tag{499}$$

and the error bound (496) applies verbatim, since the substitution $x \mapsto x - 1$ is a bijection of $[0, 2]$ onto $[-1, 1]$.

This is the form to use in practice. Every u_n is rational by (491), and every inner bracket in (499) is a finite rational combination of them, so the whole approximant is a single polynomial with exact rational coefficients that can be computed once and stored. Evaluating it costs one Horner sweep of length $2N$ over the rationals or over any floating-point type, with no Legendre recurrence and no evaluation of P_{2n} at run time; and the natural variable is the one the caller already has, since $\mathcal{F}$ and F agree on $[0, 1]$. The price is the usual one for the power basis: the coefficients in (499) alternate in sign and grow, so the conversion should be performed in exact arithmetic and only the final evaluation rounded.

19.5 Least-squares optimality of the partial sums

The Legendre truncation is not only a convenient polynomial approximation; it is the exact orthogonal projection in $L^2[-1, 1]$. Define

$$S_N(x) = \sum_{n=0}^{N}u_n P_{2n}(x). \tag{500}$$

Theorem 19.6 (Pythagorean error identity)**.** *For every real polynomial p of degree at most $2N+1$,*

$$\int_{-1}^{1} \big(\mathrm{up}(x) - p(x)\big)^2 \,\mathrm{d}x = \int_{-1}^{1} \big(\mathrm{up}(x) - S_N(x)\big)^2 \,\mathrm{d}x + \int_{-1}^{1} \big(p(x) - S_N(x)\big)^2 \,\mathrm{d}x. \tag{501}$$

Consequently S_N is the unique least-squares approximant among all such polynomials.

Proof. Every polynomial of degree at most $2N+1$ is a linear combination of $P_0, \dots, P_{2N+1}$. By construction, $\mathrm{up} - S_N$ is orthogonal to each even basis element through P_{2N}; by evenness it is also orthogonal to every odd basis element. Hence

$$\int_{-1}^{1} (\mathrm{up} - S_N)(p - S_N) = 0.$$

Expanding $\mathrm{up} - p = (\mathrm{up} - S_N) - (p - S_N)$ gives (501). Equality with the minimum forces the final nonnegative integral to vanish. A polynomial vanishing almost everywhere vanishes identically, so $p = S_N$. □

The one-degree slack is genuine: the odd coefficient of P_{2N+1} is zero. Thus a partial sum visibly of degree $2N$ is already optimal in the larger space of degree at most $2N+1$.

20 Additional summation and spectral identities

20.1 The cosine series on the support

Periodize up with period 2. Since neighboring support intervals meet only at zeros, the periodization agrees with up on $[-1, 1]$. Its Fourier coefficients are $\widehat{\mathrm{up}}(n/2)/2$, so rapid Fourier decay gives the uniformly convergent cosine series

$$\mathrm{up}(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \widehat{\mathrm{up}}(n/2) \cos(\pi n x), \qquad -1 \le x \le 1. \tag{502}$$

Using (70), every coefficient is an explicit infinite product. Many coefficients vanish because one of the sinc factors is evaluated at a nonzero integer.

The vanishing can be made exact. Precisely half of the coefficients in (502) are zero, and the surviving ones are never zero.

Proposition 20.1 (Odd cosine series)**.** *In (502) the coefficient $\widehat{\mathrm{up}}(n/2)$ vanishes for every even $n \ge 2$ and for no odd n. Hence, for $-1 \le x \le 1$,*

$$\mathrm{up}(x) = \frac{1}{2} + \sum_{m=0}^{\infty} \widehat{\mathrm{up}}\left(\frac{2m+1}{2}\right) \cos((2m+1)\pi x), \tag{503}$$

with

$$\widehat{\mathrm{up}}\left(\frac{2m+1}{2}\right) = \frac{2(-1)^m}{\pi(2m+1)} \prod_{n=1}^{\infty} \mathrm{sinc}\left(\frac{\pi(2m+1)}{2^{n+1}}\right) \neq 0. \tag{504}$$

The series converges absolutely and uniformly on $[-1, 1]$.

Proof. Write $n = 2m$ with $m \ge 1$. Then $\widehat{\mathrm{up}}(n/2) = \widehat{\mathrm{up}}(m) = 0$ by theorem 5.4, or directly because the factor $\mathrm{sinc}(\pi m)$ of (70) vanishes. Deleting these terms from (502) leaves (503).

For an odd cosine index $2m+1$, the factor indexed by $j \ge 0$ in (70), evaluated at $z = (2m+1)/2$, is $\mathrm{sinc}\big(\pi(2m+1)/2^{j+1}\big)$; it could vanish only if 2^{j+1} divided the odd integer $2m+1$. Thus every factor is nonzero. Moreover the tail differs from 1 by a summable sequence (quadratically in 2^{-j}), so the convergent product is nonzero and $\widehat{\mathrm{up}}((2m+1)/2) \neq 0$. The displayed value of the first factor is

$$\operatorname{sinc}\left(\frac{\pi(2m+1)}{2}\right) = \frac{\sin(\pi(2m+1)/2)}{\pi(2m+1)/2} = \frac{2(-1)^m}{\pi(2m+1)},$$

which gives (504). Absolute and uniform convergence follow from (80). $\square$

The sign pattern in (504) is worth noting: the leading factor alternates and decays like $1/(2m+1)$, while the remaining product has magnitude at most 1. The rapid decay beyond this explicit reciprocal factor is supplied by that remaining product, as is also forced by the Schwartz estimate (80).

20.2 A sampling identity

Poisson summation at the origin converts a single value of the bump into a rapidly convergent sum of transform samples.

Proposition 20.2 (Sampling identity)**.** *For every $a > 0$,*

$$\sum_{m\in\mathbb{Z}} \mathrm{up}(am) = \frac{1}{a}\sum_{m\in\mathbb{Z}} \widehat{\mathrm{up}}\left(\frac{m}{a}\right). \tag{505}$$

If moreover $a \geq \frac{1}{2}$, the left side has at most three nonzero terms and

$$1 + 2\,\mathrm{up}(a) = \frac{1}{a}\sum_{m\in\mathbb{Z}} \widehat{\mathrm{up}}\left(\frac{m}{a}\right). \tag{506}$$

Both series on the right converge absolutely, and faster than every power of $|m|$.

Proof. Take $t = 0$ and $u = a$ in (82); this is (505). For the second statement, $\mathrm{up}(am) \neq 0$ forces $|am| < 1$, that is $|m| < 1/a \leq 2$, so only $m \in \{-1, 0, 1\}$ can contribute. Evenness of up and $\mathrm{up}(0) = 1$ then give $\sum_m \mathrm{up}(am) = 1 + 2\,\mathrm{up}(a)$. Absolute convergence and the rate are (80). $\square$

The identity is a genuine evaluation device: the left side is a single value of a function with no elementary closed form, and the right side is a sum whose terms are explicit infinite products (70) decaying faster than any power. The values $a = 1$ and $a = \frac{1}{2}$ give two useful consistency checks. At $a = 1$ every term with $m \neq 0$ is $\widehat{\mathrm{up}}(m) = 0$, so (506) reads $1 + 2\,\mathrm{up}(1) = 1$, which is the vanishing of up at the edge of its support. At $a = \frac{1}{2}$ only the even samples $\widehat{\mathrm{up}}(2m)$ appear, all of which vanish except $m = 0$, so the right side is $2\widehat{\mathrm{up}}(0) = 2$ and (506) gives $\mathrm{up}(\frac{1}{2}) = \frac{1}{2}$, matching $F(\frac{1}{2}) = \frac{1}{2}$ through (8).

The source restricts this specialization to $\frac{1}{2} \leq a \leq 1$, but the proof uses no upper bound. Lean therefore retains source-compatible wrappers and also exposes the stronger ray statements `rvachev_poisson_support_specialization_unscaled_of_one_half_le` and `rvachev_poisson_support_specialization_of_one_half_le` for every $a \geq \frac{1}{2}$.

20.3 Scaled partitions

Equation (83) may be differentiated term by term. For $j \geq 1$,

$$\sum_{k\in\mathbb{Z}} \mathrm{up}^{(j)}\left(t + \frac{k}{m}\right) = 0. \tag{507}$$

The sums are locally finite. Combining this with (91) turns the identity into exact finite cancellations among signed translates of $\mathcal{F}$ at dyadic scales.

The case $j = 0$ of (83) has a two-term form on the unit interval that deserves to be recorded separately, because it is the cleanest visible expression of the self-similarity.

Proposition 20.3 (Derivative partition)**.** *For* $0 \le x \le 1$*,*

$$F'(x) = 2\ \mathrm{up}(2x - 1). \tag{508}$$

Consequently, for $0 \le x \le \frac{1}{2}$*,*

$$F'(x) + F'\Big(\tfrac{1}{2} - x\Big) = 2. \tag{509}$$

Proof. The first identity is the restriction to $[0, 1]$ of the global unified derivative formula (14) proved earlier.

Take $m = 1$ in (83), that is (84), and evaluate at $t \in [0, 1]$. Since up vanishes outside $[-1, 1]$, only the indices $k = 0$ and $k = -1$ can contribute, so

$$\mathrm{up}(t) + \mathrm{up}(t - 1) = 1 \qquad (0 \le t \le 1). \tag{510}$$

Put $t = 2x$ with $0 \le x \le \frac{1}{2}$, so that $\mathrm{up}(2x) + \mathrm{up}(2x - 1) = 1$. By (508) and evenness of up,

$$F'\Big(\tfrac{1}{2} - x\Big) = 2\ \mathrm{up}(2(\tfrac{1}{2} - x) - 1) = 2\ \mathrm{up}(-2x) = 2\ \mathrm{up}(2x),$$

and adding $F'(x) = 2\,\mathrm{up}(2x - 1)$ gives (509). □

Equation (509) says that the graph of F' on $[0, \frac{1}{2}]$ is antisymmetric about the point $(\frac{1}{4}, 1)$. Every scaling identity in this article is ultimately a statement about a single dilation by two; this one displays it without any transform, any series, or any auxiliary function, and it is the identity to check first against numerical output.

20.4 Mean and variance of the Fabius law

From the random-series representation,

$$\mathbb{E}X = \sum_{n \ge 0} \frac{1/2}{2^{n+1}} = \frac{1}{2}. \tag{511}$$

Since $\mathrm{Var}(U_n) = 1/12$ and the coordinates are independent,

$$\mathrm{Var}(X) = \frac{1}{12} \sum_{n \ge 0} 4^{-n-1} = \frac{1}{36}. \tag{512}$$

For $Y = 2X - 1$, this gives $\mathbb{E}Y = 0$ and $\mathrm{Var}(Y) = 1/9$, agreeing with $c_1 = 1/9$. Higher even moments are exactly the c_n of (110).

20.5 An exact self-convolution law

Splitting the random series into its even and odd coordinates gives two independent scaled copies with different weights. At the transform level this is the factorization

$$\widehat{\mathrm{up}}(z) = \widehat{\mathrm{up}}(z/2)\,\mathrm{sinc}(\pi z). \tag{513}$$

In physical space,

$$\mathrm{up}(x) = 2 \int_{-1/2}^{1/2} \mathrm{up}(2x - 2t)\,\mathrm{d}t = \int_{2x-1}^{2x+1} \mathrm{up}(y)\,\mathrm{d}y, \tag{514}$$

with the normalization understood as convolution with the uniform density on $[-1/2, 1/2]$. Differentiating this integral refinement recovers (28). Conversely, integrating the differential equation and using compact support recovers the convolution law.

21 Computation in exact and numerical regimes

The identities above do not lead to a single universally best numerical method. They lead to a collection of algorithms adapted to different questions. This section makes the distinctions explicit, since confusing an exact arithmetic formula with an endpoint asymptotic is as unhelpful as using a dense dyadic table to evaluate one value at 2^{-10^6}.

21.1 Which representation should be used?

Representation	Principal strength	Principal limitation
Contraction fixed point	A direct constructive proof; after m iterations the uniform error is geometrically small.	Produces approximating functions rather than exact individual values.
Finite convolution / probability	Positive approximants, natural sampling, and a transparent explanation of smoothness.	High resolution requires many convolution levels.
Centered finite spline	An explicit finite formula with certified uniform error at most 2^{-p}.	Direct summation has exponentially many terms unless its recursion is compiled.
Fourier sinc product	Entire transform, rapid frequency decay, and efficient evaluation away from extreme endpoint cancellation.	Recovering a tiny value of $F(x)$ by oscillatory inversion may lose many digits.
Exact dyadic formula	A closed rational answer and an algebraic proof of representation invariance.	The literal Thue–Morse sum may contain exponentially many terms.
Bit-recursive Taylor evaluator	Exact rational arithmetic with at most one recursive call per set bit of the numerator.	Rational numerators and denominators themselves become very large.
Legendre series	Uniform polynomial approximation on the full closed support, with exact coefficients.	Not specially adapted to resolving the boundary layer at a tiny x.
Lambert saddle expansion	Stable evaluation of $\log F(x)$ at extraordinarily small positive x.	Asymptotic rather than exact; the periodic correction must be retained.

For the fixed-point construction, let T be the transform of (24). If f_0 is any admissible continuous function, then

$$\|T^m f_0 - F\|_\infty \le 2^{-m}\|f_0 - F\|_\infty \le 2^{-m}. \tag{515}$$

Thus the contraction proof itself is a certified numerical scheme. It is excellent for coarse graphs and for constructing a first approximation without any prior arithmetic tables. Exact dyadic computation, however, should use the rational recurrences.

21.2 Precomputing the reciprocal-power table

Put

$$V_n = F(2^{-n}). \tag{516}$$

Eliminating d_n from (119) and (136) gives an especially convenient recurrence involving only earlier V_k:

$$V_0 = 1, \qquad V_{n+1} = \frac{1}{2^{n+1}-1} \sum_{k=0}^{n} \frac{V_k}{2^{\binom{n+1}{2}-\binom{k}{2}}(n+2-k)!}. \tag{517}$$

Every operation is rational. A practical implementation stores the values in an array, reduces each rational after every addition, and reuses factorials and powers of two. This table is precisely the data required by the Taylor blocks in (178).

There are three useful normalization choices.

1. Store V_n directly. This makes the public API simple and is ideal when values are eventually returned as rational numbers.
2. Store $d_n = n!2^{\binom{n}{2}}V_n$. The recurrence has smaller explicit powers of two, and the parity theorem provides a strong integrity check.
3. Store the division-free natural numerators D_n of the half moments, defined by (120). This avoids fraction reduction inside the recurrence but requires restoring a known structured denominator at the end.

The third choice is usually best for bulk generation; the first is usually best for a small library interface.

21.3 Operation count of the bit recursion

Suppose $0 < a < 2^n$, and write the set-bit positions of a in decreasing order as

$$b_1 > b_2 > \cdots > b_h \geq 0, \qquad a = \sum_{j=1}^{h} 2^{b_j}. \tag{518}$$

The evaluator of algorithm 9.12 makes exactly $h = \operatorname{popcount}(a)$ nontrivial recursive calls. At the jth call, the Taylor order is $n - b_j$. Hence the number of rational multiply-add steps in straightforward Horner evaluation is

$$\sum_{j=1}^{h}(n-b_j) \leq nh \leq n^2. \tag{519}$$

This is an arithmetic-operation count, not a bit-complexity estimate. The bit lengths of the exact numerator and denominator grow quadratically in n already at $a = 1$, as the factor $2^{\binom{n}{2}}$ in (136) makes unavoidable.

By contrast, the literal formula (164) sums over every $0 \leq h < a$. When a is comparable with 2^n, it therefore has exponentially many outer terms. Its value is conceptual rather than algorithmic: it proves rationality, exposes Thue–Morse cancellation, and connects the answer to q-binomial identities. The Taylor recursion is the compiled form of that cancellation.

Unreduced inputs cause no mathematical difficulty. The exact formula and the executable evaluator satisfy

$$F\left(\frac{2a}{2^{n+1}}\right) = F\left(\frac{a}{2^n}\right). \tag{520}$$

Reducing the pair (a, n) first merely avoids unnecessary work. Signed inputs should be handled at the outermost layer: clamp to 0 for $a \leq 0$, clamp to 1 for $a \geq 2^n$, and call the unit-interval recursion only in the interior.

21.4 Stable endpoint evaluation

For a very small non-dyadic x, direct evaluation of $F(x)$ is the wrong numerical variable. One should compute $\log F(x)$. Set

$$S = -\log x, \qquad L = \log 2. \tag{521}$$

The exact saddle coordinate is the positive solution of

$$L\lambda - \log\lambda = S. \tag{522}$$

Although a library implementation may use the W_{-1} branch directly, Newton iteration is simple and stable:

$$\lambda_{j+1} = \lambda_j - \frac{L\lambda_j - \log\lambda_j - S}{L - \lambda_j^{-1}}. \tag{523}$$

For large S, the starting value

$$\lambda_0 = \frac{S + \log S - \log L}{L} \tag{524}$$

already has small relative error. Once λ is known, evaluate (393) and, when needed, add $A_1(\lambda)/\lambda$ from (427).

The periodic function Ψ should be evaluated by its Fourier series (373). The Gamma factor in (373) decays exponentially with the frequency, while the zeta factor has only moderate growth on the relevant vertical line. Consequently a handful of modes usually suffices for precision far beyond that of ordinary floating-point arithmetic. Since Ψ is real,

$$\Psi(u) = \widehat{\Psi}(0) + 2\Re\sum_{k=1}^{K}\widehat{\Psi}(k)\mathrm{e}^{2\pi iku} + \text{exponentially small Fourier tail}. \tag{525}$$

Reducing u modulo 1 before evaluating the exponentials avoids loss of phase accuracy.

The saddle expansion and the exact dyadic evaluator overlap over a substantial range. Their agreement is a powerful independent test. For $x = 2^{-n}$, an implementation can compare the exact rational logarithm (computed at high precision) against (479). The difference after the first correction should be $O(n^{-2})$, with the phase evaluated at the exact λ_n, not at n.

21.5 A verification suite suggested by the mathematics

An exact implementation should test identities, not merely decimal snapshots. Particularly effective checks are

$$F(a/2^n) + F((2^n - a)/2^n) = 1, \tag{526}$$

$$F(2a/2^{n+1}) = F(a/2^n), \tag{527}$$

$$v_2(F(2^{-n})) = -\binom{n}{2} - 1 - v_2(n!), \tag{528}$$

$$F(5/16) = \frac{305857}{2073600}, \tag{529}$$

$$R_n \equiv 1 \pmod 2. \tag{530}$$

The Taylor-block identity can be tested for every a on a modest dyadic grid; the q-binomial expression can be evaluated at several unrelated translations q to test its proved translation independence. For floating-point code, the partition of unity (83), Fourier refinement (71), and agreement between the Legendre and Fourier reconstructions provide complementary global tests.

21.5.1 Two tests for the global extension and the periodic term

The checks listed above all take arguments in the unit interval, where the bounded function and the signed extension agree, and all evaluate the periodic correction at no phase at all. Two further tests close those gaps.

The first is a global value. It costs nothing and it fails loudly.

Proposition 21.1 (Integer values of the signed extension)**.** *For every integer $b \geq 0$,*

$$\mathcal{F}(2b) = 0, \qquad \mathcal{F}(2b+1) = \varepsilon_b = (-1)^{\mathrm{wt}_2(b)}. \tag{531}$$

In particular $\mathcal{F}(1) = 1$, $\mathcal{F}(3) = -1$, $\mathcal{F}(5) = -1$, $\mathcal{F}(7) = 1$, and $\mathcal{F}$ vanishes at every even integer.

Proof. On the block $2b \leq x \leq 2b+2$ the single-block formula (12) gives $\mathcal{F}(x) = \varepsilon_b \operatorname{up}(x-2b-1)$. At $x = 2b+1$ the argument of up is 0, and $\operatorname{up}(0) = 1$; this proves the second identity. At $x = 2b+2$ the argument is 1 and $\operatorname{up}(1) = 0$, so $\mathcal{F}(2b+2) = 0$; the block $[2b+2, 2b+4]$ gives the same value from the argument -1. For $b = 0$ the left endpoint gives $\mathcal{F}(0) = 0$ as well, in agreement with the vanishing of $\mathcal{F}$ on $(-\infty, 0]$. Finally $\mathrm{wt}_2(1) = 1$ and $\mathrm{wt}_2(2) = 1$, so $\varepsilon_1 = \varepsilon_2 = -1$, while $\mathrm{wt}_2(0) = 0$ and $\mathrm{wt}_2(3) = 2$ give $\varepsilon_0 = \varepsilon_3 = 1$. □

The single most informative regression test in the suite is therefore

$$\mathcal{F}(3) = -1. \tag{532}$$

Its value lies outside $[0,1]$ and has the opposite sign to the value 1 that the clamped distribution function returns at the same argument, so no tolerance and no rounding convention can hide a substitution of one function for the other. That substitution is the single most common error in this subject, and it is the reason for the three-symbol discipline of Section 2 and for the crosswalk of Appendix D: a library whose global entry point silently clamps passes every test in the existing suite, because every one of them evaluates inside the unit interval. The companion identity $\mathcal{F}(2b) = 0$ tests the flat gluing between consecutive blocks and is a useful second line.

The second test locks the phase of the periodic correction. Fix $u \in [0,1)$ and put

$$x_m = (m+u)2^{-(m+u)}, \qquad m \in \mathbb{N}. \tag{533}$$

By (391) the exact Lambert coordinate of x_m is $\lambda(x_m) = m + u$, so the fractional phase is u at every point of the sequence. Write $\mathcal{M}_0$ for the nonperiodic part of the main term, that is, for (393) with $\Psi(\lambda)$ deleted. Then (395) and the one-periodicity of Ψ give

$$\log F(x_m) - \mathcal{M}_0(x_m) \longrightarrow \Psi(u) \qquad (m \to \infty). \tag{534}$$

The limit depends on u and is not a universal constant; choosing u and v with $\Psi(u) \neq \Psi(v)$ produces two sequences with different limits, which is the quantitative content of (470).

In an implementation the test is run in three parts. The Lambert solver applied to x_m must return $m + u$ to full working precision, which tests (390) and the Newton iteration used for it. The periodic factor must then be evaluated at that phase and must reproduce $\Psi(u)$, computed independently from the Fourier series (373). Finally the same computation must be repeated for a second value of u and must give a different answer. Code that has dropped Ψ, or frozen it at its mean, returns 0 for every u and fails the third part while passing the first two.

A suite that samples only $x = 2^{-n}$ cannot perform this test. If the periodic argument is taken to be the dyadic scale variable itself rather than the exact phase, then it is the integer n

at every sample, Ψ contributes the single number $\Psi(0)$, and the corrected formula becomes indistinguishable from the uncorrected one with C_F of (380) replaced by $C_F + \Psi(0)$. Even at the exact phase the situation is no better in practice: by (465) the phase λ_n of (463) exceeds n by $\log n/L$ plus lower-order terms, so its fractional part advances by only about $1/(Ln)$ from one sample to the next. Over any range of n that can actually be computed the phase is nearly frozen, and the residual looks like a constant to within the $\mathcal{O}(n^{-1})$ error of (467). Varying u in (533) sweeps a full period at will and is the only inexpensive way to see the fluctuation at all.

One caution belongs with this test. By (375) the oscillation of Ψ has amplitude only about 2×10^{-6}, whereas the proved remainder in (395) is $\mathcal{O}(1/\lambda)$. A direct numerical confrontation of $\log F$ with $\mathcal{M}_0$ therefore cannot exhibit (534) at any feasible m: the remainder dominates until λ is of order 10^6. The test is a test of the asymptotic evaluator, not a measurement of the remainder, and its purpose is to verify that the implementation carries the phase and the fluctuation rather than a constant.

21.6 Effective uniform continuity and sequential computability

Computability has a precise recursion-theoretic meaning here, not merely the informal meaning that a numerical routine can be written. We use the classical Grzegorczyk formulation [10], expressed through fast dyadic names.

Definition 21.2 (Fast signed-dyadic name)**.** A signed-natural numerator is a pair $c = (c^+, c^-) \in \mathbb{N}^2$. At precision p it represents

$$\langle c \rangle_p = \frac{c^+ - c^-}{2^p}. \tag{535}$$

A real sequence $(x_i)_{i \geq 0}$ is *computable* if one recursive algorithm, uniform in both i and p, produces pairs $a(i,p)$ such that

$$|x_i - \langle a(i,p) \rangle_p| \leq 2^{-p}. \tag{536}$$

A function $f \colon \mathbb{R} \to \mathbb{R}$ is *sequentially computable* if it sends every computable real sequence to a computable real sequence.

The two-natural representation is deliberately redundant. It is extensionally the usual signed dyadic representation, but its arithmetic can be certified using primitive recursion on natural numbers without hiding an integer or rational coding convention.

Definition 21.3 (Effective uniform continuity)**.** A function $f \colon \mathbb{R} \to \mathbb{R}$ is effectively uniformly continuous if there is a recursive $d \colon \mathbb{N} \to \mathbb{N}$ such that, for every positive n,

$$d(n) > 0, \qquad |x - y| < \frac{1}{d(n)} \implies |f(x) - f(y)| < \frac{1}{n}. \tag{537}$$

It is a computable real function in this sense if it is both sequentially computable and effectively uniformly continuous.

Restricting (537) to $n > 0$ and requiring $d(n) > 0$ are substantive, not cosmetic. At $n = 0$ the reciprocal accuracy request is meaningless; in a totalized division formalism, allowing $d(n) = 0$ would make the implication vacuous.

Theorem 21.4 (Computability of the bounded Fabius function)**.** *The constant-tail extension of the bounded Fabius distribution function is sequentially computable and effectively uniformly continuous. A primitive-recursive modulus is*

$$d(n) = 2n \qquad (n > 0). \tag{538}$$

Consequently $F \colon \mathbb{R} \to [0,1]$ is a computable real function in the preceding sense.

Proof. The effective-continuity clause follows immediately from the global 2-Lipschitz estimate: if $|x-y| < 1/(2n)$, then

$$|F(x) - F(y)| \le 2|x-y| < \frac{1}{n}.$$

Multiplication by the constant 2 is primitive recursive, so (538) is an effective modulus.

It remains to give one recursive transformer of fast names. Suppose an output of precision p is requested, and put $s = p+3$. Read the input numerator $c = (c^+, c^-)$ at precision s, and clamp it to the unit interval by the natural-number operation

$$a = \begin{cases} 0, & c^+ \le c^-, \\ 2^s, & 2^s + c^- \le c^+, \\ c^+ - c^-, & \text{otherwise.} \end{cases} \tag{539}$$

Because F has constant tails, this clamp does not change its value. Evaluate the centered spline at the matched grid point $a/2^s$. Formula (61) simplifies to the exact rational number

$$S_s\left(\frac{a}{2^s}\right) = \frac{\sum_{r=0}^{a-1} \varepsilon_r (2a - 2r - 1)^s}{2^{\binom{s+1}{2}} s!}. \tag{540}$$

On this clamped grid the numerator is nonnegative. It can nevertheless be computed without signed arithmetic: keep separate positive and negative accumulators according to the Thue–Morse bit.

Every part of this evaluator is primitive recursive. More explicitly, compute the Thue–Morse bit from

$$b(0) = 0, \qquad b(m) = (b(\lfloor m/2 \rfloor) + (m \bmod 2)) \bmod 2 \quad (m > 0), \tag{541}$$

fold the finite range $0 \le r < a$ using addition, multiplication, subtraction, and powers on natural numbers, and compute the denominator from

$$\delta_0 = 1, \qquad \delta_{s+1} = \delta_s \, 2^{s+1}(s+1), \qquad \delta_s = 2^{\binom{s+1}{2}} s!. \tag{542}$$

Finally round the rational in (540) to the nearest multiple of 2^{-p}, breaking ties upward. If N and D are its nonnegative numerator and positive denominator, the returned numerator is

$$Q = \left\lfloor \frac{2(2^p N) + D}{2D} \right\rfloor. \tag{543}$$

Bounded range folds, natural division, and the displayed recursion are all closed under primitive recursion; thus $(c, p) \mapsto (Q, 0)$ is one certified recursive algorithm.

It remains to verify its precision. The centered-spline error (67) at level $s = p+3$ costs one unit of 2^{-s}. Nearest-grid rounding costs

$$\frac{1}{2}\, 2^{-p} = 4 \cdot 2^{-(p+3)}.$$

Hence the evaluator itself has error at most

$$5 \cdot 2^{-(p+3)}. \tag{544}$$

When the input is the sth approximation to an arbitrary real x, propagation through the 2-Lipschitz function costs another

$$2 \cdot 2^{-(p+3)}.$$

The total is therefore

$$(2+5)2^{-(p+3)} = 7 \cdot 2^{-(p+3)} \le 8 \cdot 2^{-(p+3)} = 2^{-p}.$$

Applying the same algorithm uniformly to the two indices (i,p) produces a fast name for $F(x_i)$ from any fast name for (x_i). This proves sequential computability and completes the theorem. □

Remark 21.5 (Scope of the formal computability theorem)*.* The signed global extension $\mathcal{F}$ is also globally 2-Lipschitz, so the same modulus $d(n) = 2n$ proves its effective uniform continuity. The formal development packages the primitive-recursive spline evaluator and the combined sequential-computability theorem for the bounded constant-tail function F. It does not claim a packaged global sequential computability theorem for $\mathcal{F}$.

22 Corrections, conventions, and proof audit

A formal development is most useful when it records not only what is true but also exactly where an attractive informal shortcut ceases to be valid. The following points are the ones most likely to create a silent error in a human treatment.

22.1 Source-level defects corrected in the statements

The corrections collected here are of a different kind from those discussed below. They do not concern a mathematical strategy but the literal content of printed statements: an index that should be finite, an endpoint weight, a missing variable in an exponential, an omitted hypothesis, a removable singularity, a power of two, a factor of two. Each has been incorporated silently into the statements of this article, and each is recorded here so that a reader comparing a formula with its source can tell which differences are deliberate.

22.1.1 The approximating convolution is finite and depends on the stage

The construction of up by discrete convolution introduces, for each stage m, a probability measure μ_m on $[-1,1]$ whose Fourier transform is the finite product

$$\widehat{\mu_m}(z) = \prod_{k=1}^{m} \left(\cos\frac{\pi z}{2^k}\right)^k. \tag{545}$$

The printed convolution representation of μ_m carries the upper index ∞: it convolves the symmetric two-point law with masses $1/2$ at $\pm 2^{-k-1}$, taken with multiplicity k, over all $k \ge 1$.

The upper index cannot be infinite. The right side of an all-k product is a single measure with no reference to m, so the display would define the same object for every stage and would erase precisely the dependence on which the approximation argument turns; it also contradicts (545), which is the definition of μ_m. The convolution must stop at $k = m$, the factors with $k > m$ being absent. Since $\operatorname{sinc}(\pi z) = \prod_{k\ge1} \cos(\pi z/2^k)$, letting $m \to \infty$ in (545) recovers exactly the sinc product (70), which is the correct statement that μ_m converges weakly to the measure with density up. This is precisely the finite atomic construction (46)–(48), whose stage dependence is explicit in the notation.

22.1.2 Half weights at shared dyadic endpoints

The step approximants of the same source are built from a generating polynomial by replacing each monomial z^j with a multiple of the indicator of a *closed* dyadic cell of width 2^{-m-1}, the cells for consecutive j being adjacent. Two adjacent closed cells share an endpoint, and there the printed approximant returns the *sum* of the two neighboring cell values instead of their average, hence asymptotically twice the intended value.

Every integral, every moment, and the weak limit are unaffected, because the grid is a null set. What fails is exactly the pointwise statement: at a shared endpoint the approximants do not converge to up. The correction is to give each shared endpoint the weight $1/2$, equivalently to use half-open cells and to read the endpoint value as the average of the two adjacent constants. With that convention the approximants converge pointwise everywhere, and the value at every grid point is the one the normalization intends.

22.1.3 Poisson summation: the translation parameter cannot be dropped

The printed Poisson formula reads

$$\sum_{k\in\mathbb{Z}} \mathrm{up}(t+uk) \stackrel{?}{=} \frac{1}{u}\sum_{k\in\mathbb{Z}} \widehat{\mathrm{up}}(k/u)\,\mathrm{e}^{2\pi \mathrm{i}k/u}, \tag{546}$$

with the spatial variable t absent from the exponential. The right side of (546) is then a constant, while the left side is not: at $u=2$ the left side equals $\mathrm{up}(t)$ for $-1\le t\le 1$, since no other translate is supported there. The identity as printed would therefore make up constant. The correct exponential is $\mathrm{e}^{2\pi \mathrm{i}kt/u}$, as in (82); at $u=2$ it produces the cosine series (502) rather than a number.

The identity derived from it by subdividing the period likewise needs its parameter to be a *strictly positive* integer. The specialization is stated for an integer subdivision m, and at $m=0$ the translates k/m are undefined and the asserted value m is 0; the correct hypothesis is $m\ge 1$, as in (83). Finally, the sampling identity obtained by setting $t=0$ is printed with inconsistent scaling: its left side has been multiplied by the sampling step while its right side retains the reciprocal step. For $1/2\le a\le 1$ only three translates are supported, so the identity in the convention of (69) is

$$1+2\,\mathrm{up}(a)=\frac{1}{a}\sum_{k\in\mathbb{Z}}\widehat{\mathrm{up}}\left(\frac{k}{a}\right), \qquad \text{equivalently} \qquad a+2a\,\mathrm{up}(a)=\sum_{k\in\mathbb{Z}}\widehat{\mathrm{up}}\left(\frac{k}{a}\right). \tag{547}$$

22.1.4 The translated-derivative lemma needs a nonnegative combined order

The arithmetic source proves a remainder-translation lemma in the following form: for real $x>0$, for the integer k with $2^k\le x<2^{k+1}$, for $y=x-2^k$, and for every integer $n\ge 0$,

$$\int_{2^k}^{x}(x-t)^n\mathcal{F}^{(n+1)}(t)\,\mathrm{d}t \stackrel{?}{=} -\int_0^y (y-t)^n\mathcal{F}^{(n+1)}(t)\,\mathrm{d}t. \tag{548}$$

No relation between k and n is imposed.

The proof substitutes $t=2^k(1+u)$ and then rewrites $\mathcal{F}^{(n+1)}(2^k(1+u))$ by means of the iterated dilation identity (90), which converts it into $\mathcal{F}^{(k+n+1)}(u+1)$. That step presupposes $k+n\ge 0$; for a negative scale k and a small order n it calls for a derivative of negative order and is unavailable. The statement itself is then false, not merely unproved. Take $k=-1$ and $n=0$, so that the shell is $1/2\le x<1$. At $x=3/4$, hence $y=1/4$, the left side of (548) is $F(3/4)-F(1/2)=1/2-F(1/4)$ by (3), while the right side is $-F(1/4)$; the two differ by $1/2$. The corrected statement carries the hypothesis

$$0\le k+n, \tag{549}$$

which is exactly the condition under which the substitution in the proof is legitimate.

22.1.5 A removable singularity must be filled before coefficients are compared

The functional equation for the moment generating function is printed with the factor $(\mathrm{e}^z-1)/z$ written as a bare quotient, and the half-moment recurrence is then obtained by comparing power-series coefficients at the origin. The quotient is undefined at $z=0$, which is precisely the point at which the comparison is performed, so the argument as printed divides by zero at its only interesting point.

The repair is not cosmetic bookkeeping but a change of object: the quotient must first be completed to the entire function

$$z \longmapsto \sum_{j\geq 0} \frac{z^j}{(j+1)!}, \tag{550}$$

whose value at the origin is 1. Only then is $G(2z) = ((\mathrm{e}^z-1)/z)G(z)$ an identity between entire functions, as it is stated in (117), and only then does coefficient comparison yield (119). The same completion is required wherever the factor $(\mathrm{e}^z-1)/z$ or its reciprocal is expanded, in particular in the Bernoulli-number recurrences of Section 7.

22.1.6 The Taylor-remainder reduction must stay inside its valid range

The dyadic reduction that removes the leading bit is derived by applying Taylor's theorem at 2^{-r} with a free order N and then identifying the integral remainder through (548). As printed, the order and the shell index are independent, and the shell index is allowed to be negative.

The identification of the remainder is legitimate only for pairs of scale and order that satisfy (549), since that is the range in which the iterated dilation identity (90) may be applied inside the remainder integral. The corrected statement ties the two together: at the shell $2^{-r} \leq x < 2^{-r+1}$ the Taylor order is taken to be r, which is the form of (178), and its global counterpart (278) makes the same choice for $\mathcal{F}$. When the shell index is negative the point 2^{-r} is an even integer, all derivatives there vanish by (12) and the flatness of up at ± 1, and any nonnegative order is admissible; that degenerate case must be stated separately rather than absorbed into the generic one.

22.1.7 Normalization in the change of variables for the closed dyadic formula

The closed formula (164) is obtained from the Taylor integral (165) by inserting the derivative identity (91) and then changing variables blockwise, first by $s = 2^{n+1}(t+1)$ and then by $u = s - 2h - 1$.

Both the polynomial factor and the differential are rescaled by that substitution, and both must be tracked. The differential contributes $2^{-(n+1)}$, while the factor $(x-1-t)^n$ contributes a further $2^{-n(n+1)}$ when it is rewritten in the variable u. Keeping only the differential leaves the prefactor $2^{\binom{n+2}{2}-(n+1)}$, whereas the correct prefactor is

$$2^{\binom{n+2}{2}-(n+1)-n(n+1)} = 2^{-\binom{n+1}{2}}, \tag{551}$$

which is the one displayed in (164). The normalization is not a matter of taste: it is the only one compatible with the refinement invariance (520) and with the values in (143), and it is what makes the closed formula agree with the bit-recursive evaluator of Algorithm 9.12.

22.1.8 The naturality identity in the parity argument carries a factor of two

The proof of the parity theorem for the half moments introduces the odd integer

$$(2n-1)!!\, M_{\lfloor n/2\rfloor}, \qquad M_m = \prod_{k=1}^{m} (2^{2k}-1), \tag{552}$$

observes that these multipliers are nested, and then asserts that $d_k(2n-1)!!\,M_{\lfloor n/2\rfloor}$ is an integer for every $1 \le k \le n$.

The assertion is false as printed, already at the smallest index. What the integrality of the Reshetnikov numbers supplies is $R_k = 2d_k(2k-1)!!\,M_{\lfloor k/2\rfloor}$, and the half-moment normalization puts the factor 2 on the left. For $k = n = 1$ the multiplier (552) equals 1 while $d_1 = 1/2$, so the printed product is 1/2. The corrected assertion is that

$$2d_k(2n-1)!!\,M_{\lfloor n/2\rfloor} \in \mathbb{Z} \qquad (1 \le k \le n), \tag{553}$$

which follows from (158) together with the nestedness of the multipliers, and which is what the remainder of the parity argument actually uses. With the factor restored, the argument gives $v_2(2d_n) = 0$ for $n \ge 1$ and hence the exact valuation (156).

22.2 Bounded versus signed-global Fabius functions

The bounded function F takes values in $[0,1]$ and is constant to the right of 1. The signed extension $\mathcal{F}$ is not bounded in that way and instead obeys

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \qquad (x \in \mathbb{R}).$$

They agree for every $x \le 1$, hence in particular on $[0,1]$, but they are globally distinct. Thus a statement such as "$F'(x) = 2F(2x)$ for all real x" is correct only when F denotes the signed extension. The explicit Thue–Morse series (11) makes the distinction unavoidable and proves that the global sum is locally finite rather than merely conditionally convergent.

Likewise, the support statement for up concerns the topological support: $\operatorname{supp}\mathrm{up} = [-1,1]$, even though $\mathrm{up}(\pm 1) = 0$. Endpoint formulas must distinguish closed support from the open set on which the function is positive.

22.3 The Reshetnikov normalization

Equation (158) contains the factor $2^{+\binom{n-1}{2}}$. The negative exponent printed in the abstract of one version of the arithmetic paper is incompatible with its own equation (27) and with the integrality theorem. The positive exponent is the one proved by the exact recurrence. The resulting first values are

$$R_1, R_2, \ldots, R_7 = 1,\ 5,\ 15,\ 1001,\ 5985,\ 2853675,\ 26261235, \tag{554}$$

all odd as predicted.

The strongest common-denominator formula in the arithmetic paper depends on a separate divisibility conjecture. The formal development keeps the unconditional multiplier bound and the conditional exact denominator statement logically separate. A computational match for the first many levels is evidence for the conjecture, not a proof.

22.4 The local asymptotic derivation

Write the logarithmic endpoint profile in a dyadic variable as

$$g(t) = \log F(2^{-t}). \tag{555}$$

A proposed elementary derivation substitutes a main term $G(t)$ into the exact difference-differential relation. Two steps in that derivation cannot be made for an unspecified remainder $R = g - G$:

1. a coefficient comparison drops the actual difference $R(t) - R(t-1)$;

2. a later estimate uses a bound on $R'(t)$ that has not been assumed or proved.

Even the explicit main term has a residual of order

$$\left(\frac{\log t}{t}\right)^2, \tag{556}$$

with a nonzero leading coefficient; it is not $O(t^{-2})$. Therefore the printed iteration does not establish the claimed bounded periodic remainder. The rigorous route is the exact negative-Laplace decomposition of section 14, followed by Bromwich inversion and a quantitative saddle analysis.

There is also a small but conceptually useful conversion issue. Substituting $x = 2^{-t}$ in the literal elementary small-x expression does not give the separately displayed dyadic expression term for term. Their difference is

$$\frac{(\log\log 2)^2}{2(\log 2)^2\, t}, \tag{557}$$

which is $O(t^{-1})$. It is harmless for a coarser theorem but must be retained when the display itself claims accuracy through the inverse-logarithmic term.

22.5 Why the periodic term cannot be averaged away

Replacing $\Psi(\lambda)$ by its mean changes the logarithm by a bounded quantity, so it preserves only very coarse statements such as (475). It does not preserve a sharp multiplicative asymptotic. Indeed, for every nonzero integer k,

$$\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{\log 2} \neq 0.$$

The nonvanishing follows from the absence of zeros of Γ, the fact that $\zeta(s) \neq 0$ on $\Re s = 1$ away from its pole, and $\chi_k \neq 0$. Hence Ψ is nonconstant. Choosing two fractional saddle phases where it takes different values produces two sequences along which the ratio to an uncorrected main term has different limits. This is a structural obstruction, not a matter of choosing a better constant.

22.6 A visually good quotient of exponentials has the wrong boundary scale

A proposed elementary approximation to the displaced bump has the form

$$Q(x) = \frac{1+\exp(-|x|^{-\alpha})}{1+\exp((1-2|x|)/(x^2-|x|))}, \qquad \alpha = \frac{3}{4}\sqrt{2\pi}. \tag{558}$$

It can look strikingly accurate on a plot. Put $y = x+1$, the distance from the left support endpoint. For $0 < y \le 1/4$ one has the rigorous bound

$$Q(-1+y) \le 2\exp\left(-\frac{1}{2y}\right). \tag{559}$$

But corollary 18.2 shows

$$\exp(-c/y) = o(F(y)) \qquad (y \downarrow 0) \tag{560}$$

for every $c > 0$. Therefore

$$Q(-1+y) = o(F(y)), \tag{561}$$

so the quotient is not asymptotically equivalent at the endpoint. This is a useful warning: a graph on a linear scale is almost blind to the distinction between $\exp[-\Theta((\log(1/y))^2)]$ and $\exp[-\Theta(1/y)]$ once both values are tiny.

22.7 The literal Thue–Morse prefix normalization does not converge

The construction of section 13 has a shorter and more attractive reading, in which the normalized grid $\mathcal{V}_k$ of (339) is plotted directly against the abscissae $x_{k,j} = j/2^k$ and the resulting broken lines are asserted to converge, uniformly on $[0,1]$, to the Fabius function. That assertion is false, and it fails in the strongest possible way: not merely without a rate, and not merely nonuniformly, but pointwise at the right endpoint.

Theorem 22.1 (Endpoint obstruction)**.** *For every $k \geq 0$,*

$$\mathcal{V}_k(2^k) = -\frac{1}{2^{\binom{k}{2}}} < 0, \qquad |\mathcal{V}_k(2^k) - 1| = 1 + \frac{1}{2^{\binom{k}{2}}} > 1. \tag{562}$$

In particular the sequence $k \mapsto \mathcal{V}_k(2^k)$ of values at $x = 1$ does not converge to $F(1) = 1$; every term is negative, so any limit is at most 0.

Proof. The index 2^k is the right boundary index of theorem 13.7 with $r = k$, so $\sigma_k(2^k) = -1$ by (331). Divide by $2^{\binom{k}{2}}$ to obtain the first identity, and note $-2^{-\binom{k}{2}} < 0 < 1$ for the second. A sequence of nonpositive numbers cannot converge to 1. □

The obstruction is not an artifact of reading the displayed floor function literally, which produces a step function rather than a curve. Let Π_k denote the polygon obtained by joining the consecutive points $(x_{k,j}, \mathcal{V}_k(j))$ by line segments, which is the object the construction intends. By construction Π_k interpolates the grid: $\Pi_k(x_{k,j}) = \mathcal{V}_k(j)$ for every j, and in particular

$$\Pi_k(1) = \mathcal{V}_k(2^k) = -\frac{1}{2^{\binom{k}{2}}}. \tag{563}$$

So $\Pi_k(1) \not\to 1$ either, whether the polygon is formed over $\mathbb{Q}$ or over $\mathbb{R}$. Passing from the step reading to the polygonal reading changes nothing at the right endpoint, because the two agree there.

The failure is localized precisely. It is not a failure of the iterated prefix sums, whose values (329)–(331) are exactly right; it is a failure of the pair consisting of the prefix order and the abscissa convention. Shifting the order from k to $k+1$ and reading the index as a cell center rather than a left endpoint yields genuine pointwise convergence, by theorem 13.16. In that corrected reading the right endpoint is still exceptional, but for a reason that is now visible and harmless: the limit there is $\mathrm{up}(1) = 0$, and the exact value $-2^{-\binom{n}{2}}$ of (331) converges to it. The corrected statement is a separate theorem about a different scheme, not a reinterpretation that rescues the literal one.

22.8 A proposed maximum proxy has no maximum

A second construction attached to the same prefix sums proposes to approximate F near the origin by the largest term of an explicit family. Write

$$\mathcal{Y}_n(x) = \frac{(2^n x)^n}{n!\, 2^{\binom{n}{2}}} \qquad (n \geq 1), \tag{564}$$

and consider the claim that for small $x > 0$ the quantity $\max_{n \geq 1} \mathcal{Y}_n(x)$ exists and serves as a proxy for $F(x)$. The maximum does not exist for any $x > 0$.

Theorem 22.2 (Unboundedness of the proxy family)**.** *Fix $x > 0$. Then*

$$\frac{\mathcal{Y}_{n+1}(x)}{\mathcal{Y}_n(x)} = \frac{x\, 2^{n+1}}{n+1} \longrightarrow \infty \qquad (n \to \infty), \tag{565}$$

the family $\{\mathcal{Y}_n(x) : n \geq 1\}$ is unbounded above, and there is no index $n \geq 1$ with $\mathcal{Y}_m(x) \leq \mathcal{Y}_n(x)$ for all $m \geq 1$.

Proof. Since $\binom{n+1}{2} - \binom{n}{2} = n$,

$$\frac{\mathcal{Y}_{n+1}(x)}{\mathcal{Y}_n(x)} = \frac{(2^{n+1}x)^{n+1}}{(2^n x)^n} \cdot \frac{1}{(n+1)\,2^n} = \frac{2^{2n+1}x}{(n+1)2^n} = \frac{x\,2^{n+1}}{n+1},$$

which tends to infinity because $2^{n+1}/(n+1) \to \infty$ and $x > 0$ is fixed. Choose N with $x2^{n+1}/(n+1) \geq 2$ for all $n \geq N$. Then $\mathcal{Y}_{N+m}(x) \geq 2^m \mathcal{Y}_N(x)$ for every $m \geq 0$, and $\mathcal{Y}_N(x) > 0$, so the family is unbounded above. Unboundedness contradicts the existence of a largest term. □

Since $0 \leq F \leq 1$, an unbounded family cannot serve as a pointwise proxy for F, and the displayed maximum cannot be evaluated at any positive argument. The monotone structure is in fact the reverse of the one a maximum requires. The ratio (565) is nondecreasing in n, because $2^{n+2}/(n+2) \geq 2^{n+1}/(n+1)$ for every $n \geq 0$, so it crosses the value 1 at most once. Hence for $0 < x < 1/2$ the terms first decrease to a single interior minimum and then increase without bound, and for $x \geq 1/2$ they increase from the start. The distinguished index is therefore a minimum of the family, not a maximum, and the analytically controlled way to select a scale is the saddle-point argument of section 15, where the contour radius is $r = 2^\lambda$ with λ given by corollary 13.20.

22.9 A stationary substitution that leaves a linear term

Even after the index in (564) is treated as a real variable, so that a stationary point exists, a separate algebraic defect remains. The Stirling proxy attached to (564) is

$$\mathcal{H}(x,n) = \frac{L}{2}n^2 + n\log x - n\log n + n, \qquad L = \log 2, \tag{566}$$

and the claim under discussion is that substituting the stationary equation into (566) produces $-\frac{L}{2}n^2$ up to an error $\mathcal{O}(\log n)$ inherited from Stirling's formula.

The Stirling input itself is correct and we record it in the sharp form actually used.

Lemma 22.3 (Stirling with logarithmic error)**.** *As $n \to \infty$,*

$$\log(n!) = n\log n - n + \mathcal{O}(\log n). \tag{567}$$

More precisely, for every $n \geq 1$,

$$0 \leq \log(n!) - \left(n\log n - n + \frac{1}{2}\log n + \frac{1}{2}\log(2\pi)\right) \leq \frac{1}{12n}.$$

Proof. The two-sided bound is Robbins' form of Stirling's formula, obtained from the Stirling sequence $s_n = n!\,e^n n^{-n}(2n)^{-1/2}$: the increments $\log s_n - \log s_{n+1}$ are positive and bounded by $\frac{1}{12n} - \frac{1}{12(n+1)}$, so telescoping and passing to the limit $\log s_n \to \frac{1}{2}\log\pi$ gives $0 \leq \log s_n - \frac{1}{2}\log\pi \leq \frac{1}{12n}$, which is the display after unwinding the definition of s_n. Since $\frac{1}{2}\log n + \frac{1}{2}\log(2\pi) + \frac{1}{12n} = \mathcal{O}(\log n)$, the display implies (567). □

Theorem 22.4 (The residual linear term)**.** *Let $x > 0$ and $n > 0$ be real with $n\log 2 + \log x = \log n$, equivalently $n2^{-n} = x$. Then*

$$\mathcal{H}(x,n) = -\frac{L}{2}n^2 + n. \tag{568}$$

The residual term n is not $\mathcal{O}(\log n)$ as $n \to \infty$; consequently the substitution does not yield $-\frac{L}{2}n^2 + \mathcal{O}(\log n)$.

Proof. Differentiating (566) in n gives $\partial_n \mathcal{H}(x,n) = nL + \log x - \log n$, so the stationary equation is $nL + \log x = \log n$. For $x, n > 0$ it is equivalent to $n2^{-n} = x$, since both sides of the latter are positive and taking logarithms is injective there: $\log(n2^{-n}) = \log n - nL$. Substituting $\log x - \log n = -nL$ into (566),

$$\mathcal{H}(x,n) = \frac{L}{2}n^2 + n(\log x - \log n) + n = \frac{L}{2}n^2 - Ln^2 + n = -\frac{L}{2}n^2 + n.$$

Suppose $n = \mathcal{O}(\log n)$ as $n \to \infty$. Composing with $\log n = o(n)$ would give $n = o(n)$, which is impossible for a quantity that is eventually nonzero. Hence the term n in (568) cannot be absorbed into a logarithmic remainder. □

The defect sits inside the main term rather than in the error term, and the amount by which it does is computable. The stationary equation of theorem 22.4 is $n2^{-n} = x$, which is exactly (391); so the stationary index is the Lambert phase, $n = \lambda(x)$. Discarding the term n in (568) therefore replaces the coefficient of λ in the answer by 0, whereas (393) has the coefficient $1 + \frac{L}{2}$.

Remark 22.5. The proxy (566) loses a second linear term even before the substitution. Since $\binom{n}{2} = \frac{1}{2}n^2 - \frac{1}{2}n$, the exact logarithm of (564) is

$$\log \mathcal{Y}_n(x) = \frac{L}{2}n^2 + \frac{L}{2}n + n\log x - \log(n!),$$

so lemma 22.3 gives $\log \mathcal{Y}_n(x) = \mathcal{H}(x,n) + \frac{L}{2}n - \frac{1}{2}\log n - \frac{1}{2}\log(2\pi) + \mathcal{O}(n^{-1})$; the proxy (566) omits the term $\frac{L}{2}n$. Reading the index as a real variable and substituting $\log x - \log n = -nL$ at $n = \lambda(x)$ turns this into

$$-\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda - \frac{1}{2}\log(2\pi) + \mathcal{O}\left(\frac{1}{\lambda}\right),$$

which reproduces the quadratic, linear, and logarithmic coefficients of the corrected main term (393). What it does not reproduce is the constant C_F and, more importantly, the nonconstant periodic function $\Psi(\lambda)$. Both omitted linear terms, n and $\frac{L}{2}n$, must be restored to reach even that much; and restoring them still leaves the periodic obstruction described earlier in this section.

22.10 The uniform local slope bound fails at the second level

The last claim attached to the literal grid is a quantitative one: that the polygon Π_k of section 22.7 has, on each cell of width $h = 2^{-k}$, a slope within $4h$ of the derivative of F there, and hence a uniform error controlled by the same quantity. The bound fails at the first index at which it can be tested.

Theorem 22.6 (A counterexample at $k = 2$). *Let $k = 2$, $j = 1$, so that $h = 2^{-2}$ and $x_{2,1} = 1/4$. The forward slope of Π_2 on $[1/4, 1/2]$ equals -2, whereas $F'(1/4) = 1$. The discrepancy is 3, while $4h = 1$.*

Proof. The relevant values are collected in the following table, computed from (322) and $\binom{2}{2} = 1$.

j	0	1	2	3
ε_j	1	-1	-1	1
$\sigma_1(j)$	1	0	-1	0
$\sigma_2(j)$	1	1	0	0
$\mathcal{V}_2(j)$	$\frac{1}{2}$	$\frac{1}{2}$	0	0

Hence
$$\frac{\mathcal{V}_2(2) - \mathcal{V}_2(1)}{h} = 2^2 \left(0 - \frac{1}{2}\right) = -2. \tag{569}$$
On the other side, (4) and the symmetry value $F(1/2) = 1/2$ give $F'(1/4) = 2F(1/2) = 1$. Therefore $|-2-1| = 3$, while $4h = 4 \cdot 2^{-2} = 1$. □

The sign is the informative part. The bounded Fabius function is strictly increasing on $(0, 1)$, so any approximation whose slope is negative in the interior is not approximating F at all on that cell. This is the same phenomenon as theorem 22.1, seen at the level of first differences rather than at the level of values: the literal grid samples a compactly supported bump on a centered window, and a bump has decreasing stretches. The corrected reading of section 13.5 makes that identification explicit, and (329) shows that the zero runs responsible for the flat stretches sit exactly at the top of each dyadic block.

22.11 What the formalization proves, and what it deliberately does not claim

The public aggregate contains exact proofs of the two source papers' named results, together with the additional dyadic, q-binomial, Legendre, and sharp-asymptotic theorems described here. It does not promote a source conjecture to a theorem, does not infer derivative bounds for an unspecified asymptotic remainder, and does not identify two global functions merely because they coincide on $[0, 1]$. These negative design decisions are part of the mathematics: they preserve the hypotheses needed by every later theorem.

23 Conclusion

The Fabius function is unusual not because it has one exotic property, but because several mathematical structures coincide in one object.

- A contraction on symmetric distribution functions constructs it and proves uniqueness.
- An infinite sum of independent uniform variables explains positivity, symmetry, and smoothing.
- Rvachev's fold is a compactly supported C^∞ density satisfying an exact functional-differential equation.
- Its Fourier transform is an entire sinc product with superalgebraic real-axis decay.
- Thue–Morse splitting creates a signed global solution and exact formulas for every derivative.
- Rational moment recurrences evaluate every dyadic argument exactly.
- A Taylor recursion turns the closed Thue–Morse formula into an efficient bit-sensitive algorithm.
- Binary-prefix reduction gives a finite telescope with an explicit residual, an absolutely convergent global series, and the corrected parity-power presentation.
- Half-base Gaussian finite differences explain the translated complex dyadic formulas, the pointwise global q-series, and the Toeplitz discrete limit.
- A negative-Laplace product and a lower-Lambert saddle expose the true endpoint scale, its Gamma–zeta periodic fluctuation, and its complete asymptotic expansion.

- A uniformly convergent Legendre series converts the same exact dyadic data into globally optimal polynomial approximants on the support.

The dyadic and asymptotic theories are not separate curiosities. The exact values $F(2^{-n})$ determine the moments and the Legendre coefficients; their logarithmic growth is governed by the same saddle geometry that controls arbitrary small arguments. The periodic term is the analytic shadow of dyadic self-similarity, while the Thue–Morse signs encode the same binary splitting algebraically. Seen together, these descriptions turn a famous “pathological example” into a remarkably coherent piece of analysis.

23.1 What each viewpoint makes transparent

The table in Section 21 sorts these same representations by algorithmic strength and limitation and answers the question which one to compute with; the table below sorts them by explanatory power and answers the question which one to think with.

Viewpoint	What it makes transparent that the others do not
Contraction fixed point	That the function exists, is unique, and is determined by a self-map: any admissible starting function converges to it at a certified rate (515), so the existence proof and the first numerical scheme are the same object.
Probability law	That F is a distribution function at all, together with positivity, strict monotonicity on $[0, 1]$, the symmetry (3), and the exact mean and variance. These are invisible in every analytic presentation.
Finite atomic approximation	Why weak convergence can be proved through characteristic functions: the finite product (47) converges to the transform of the probability measure with density up, yielding (48).
Sinc product	That the transform is entire of exponential type with superalgebraic decay, and that its zeros at the nonzero integers are the exact mechanism behind the partitions of unity (83).
Thue–Morse signed extension	That differentiation is a rescaling rather than an approximation: (90) gives every derivative in closed form, and (12) localizes the global function to one translate per block.
Moment recurrences	That the entire arithmetic layer is rational and triangular: each level is a finite rational combination of earlier levels (110), (119), and the endpoint values are moments in disguise (136).
Bit recursion	Where the computational work actually is: one Taylor block per set bit of the numerator, with both the operation count and the growth of the exact rational made explicit.
q-binomial form	Why visibly different finite sums return the same rational number: the half-base Gaussian coefficients make the value independent of the translation parameter Q (205).
Laplace product and saddle representation	The true endpoint scale, and the origin of the log-periodic fluctuation: the exact dilation of the negative Laplace product forces the periodic decomposition (369), whose Fourier coefficients (373) are explicit and nonzero.
Legendre expansion	Optimality: the partial sums are the best mean-square polynomial approximations on the whole closed support, and their coefficients are finite combinations of the same exact dyadic data (486).

The two tables disagree in a useful way. The exact dyadic formula (164) is a poor algorithm and an excellent explanation; the bit recursion is the reverse. A treatment that keeps only one of them keeps either a proof without a program or a program without a reason.

A Exact data

The following tables collect enough exact values to reproduce the examples in the text and to seed independent implementations.

A.1 Moment and half-moment sequences

n	c_n	d_n
0	1	1
1	$\frac{1}{9}$	$\frac{1}{2}$
2	$\frac{19}{675}$	$\frac{5}{18}$
3	$\frac{583}{59535}$	$\frac{1}{6}$
4	$\frac{132809}{32531625}$	$\frac{143}{1350}$
5	$\frac{46840699}{24405225075}$	$\frac{19}{270}$
6	$\frac{4068990560161}{4133856862760625}$	$\frac{1153}{23814}$
7	$\frac{1204567303451311}{2232691548877164375}$	$\frac{583}{17010}$
8	—	$\frac{1616353}{65063250}$

The c_n column satisfies (110); the d_n column satisfies (119). Substitution in (136) gives the next table.

A.2 Constants of the endpoint expansion

The sharp asymptotic (393) needs three real constants that are otherwise reachable only through a table of Stieltjes constants. To twenty significant figures,

$$\begin{aligned} L = \log 2 &= 0.69314718055994530942, \\ \gamma_1 &= -0.072815845483676724861, \\ A_0 = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12} &= -0.72869391700393060594, \\ \overline{P} = \frac{A_0}{L} - \frac{L}{12} &= -1.1090453654341866821 9, \\ C_F = \overline{P} - \tfrac{1}{2}\log(2\pi) &= -2.0279838986388594239 7. \end{aligned} \tag{570}$$

Here γ_1 is the first Stieltjes constant, appearing through the Laurent expansion of ζ at 1. The last line is (380) rewritten through $\overline{P}$; the two forms agree, which is a useful check on an implementation, since one routes through $-7L/12 - \frac{1}{2}\log\pi$ and the other through $-L/12 - \frac{1}{2}\log(2\pi)$.

A.3 Reciprocal powers and normalized odd integers

n	$F(2^{-n})$	$v_2(F(2^{-n}))$	R_n
0	1	0	—
1	$\frac{1}{2}$	-1	1
2	$\frac{5}{72}$	-3	5
3	$\frac{1}{288}$	-5	15
4	$\frac{143}{2073600}$	-10	1001
5	$\frac{19}{33177600}$	-14	5985
6	$\frac{1153}{561842749440}$	-20	2853675
7	$\frac{583}{179789679820800}$	-26	26261235

The valuation column is computed from (156); it is not obtained by factoring the displayed decimal-sized denominators. This is exactly the kind of redundancy that makes the table useful as a test vector.

A.4 Half-base q-products and the q-binomial layer

The next table collects the exact quantities entering the q-binomial dyadic formula (205), in the normalization of section 10. Here $(1/2;1/2)_n$ is the half-base q-Pochhammer product (201), $\mathfrak{m}_n = \prod_{\ell=1}^{n}(2^\ell - 1)$ is the odd Mersenne product, and $a_n = d_n/n!$ is the normalized half moment. The two products are linked by (204), and the last column is $2^{-\binom{n}{2}}a_n$ by (136).

n	$(1/2;1/2)_n$	$\mathfrak{m}_n$	a_n	$F(2^{-n})$
0	1	1	1	1
1	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$
2	$\frac{3}{8}$	3	$\frac{5}{36}$	$\frac{5}{72}$
3	$\frac{21}{64}$	21	$\frac{1}{36}$	$\frac{1}{288}$
4	$\frac{315}{1024}$	315	$\frac{143}{32400}$	$\frac{143}{2073600}$
5	$\frac{9765}{32768}$	9765	$\frac{19}{32400}$	$\frac{19}{33177600}$
6	$\frac{615195}{2097152}$	615195	$\frac{1153}{17146080}$	$\frac{1153}{561842749440}$

The corresponding global normalizers $2^{n^2}(1/2;1/2)_n = 2^{\binom{n}{2}}\mathfrak{m}_n$ of (205) are

$$1, \quad 1, \quad 6, \quad 168, \quad 20160, \quad 9999360, \quad 20158709760 \qquad (n = 0, \dots, 6),$$

and the complete centered numerators $\mathcal{N}_{1,n}(0) = \mathfrak{m}_n a_n$ are

$$1, \quad \frac{1}{2}, \quad \frac{5}{12}, \quad \frac{7}{12}, \quad \frac{1001}{720}, \quad \frac{4123}{720}, \quad \frac{35743}{864} \qquad (n = 0, \dots, 6).$$

The first half-base Gaussian rows $\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$ are

$$\begin{aligned}
n=0:&\quad 1,\\
n=1:&\quad 1,\ 1,\\
n=2:&\quad 1,\ \frac{3}{2},\ 1,\\
n=3:&\quad 1,\ \frac{7}{4},\ \frac{7}{4},\ 1,\\
n=4:&\quad 1,\ \frac{15}{8},\ \frac{35}{16},\ \frac{15}{8},\ 1,\\
n=5:&\quad 1,\ \frac{31}{16},\ \frac{155}{64},\ \frac{155}{64},\ \frac{31}{16},\ 1,\\
n=6:&\quad 1,\ \frac{63}{32},\ \frac{651}{256},\ \frac{1395}{512},\ \frac{651}{256},\ \frac{63}{32},\ 1.
\end{aligned}$$

Each row is symmetric under $k \leftrightarrow n-k$. Multiplying the entry at k by $2^{k(n-k)}$ produces the integer Gaussian coefficient $\begin{bmatrix} n \\ k \end{bmatrix}_2$, which counts the k-dimensional subspaces of $\mathbb{F}_2^n$ and is odd; equivalently $v_2\big(\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}\big) = -k(n-k)$. These rows and the valuation give a compact independent check on any implementation of the q-binomial layer.

B Proof dependency and Lean module guide

The Lean development is intentionally split into small files, while this article groups results by mathematical theme. The tables below cover every module of the development, not merely its principal entry points. File names are relative to

`Analysis/FabiusFunction/Lean/FabiusFunction/`,

the sole exception being the root aggregate `FabiusFunction.lean`, which sits one directory higher and is the module a user actually imports.

At the time of writing the development consists of 175 Lean files: 174 modules in the directory named above, together with that root aggregate. They comprise roughly 62,400 lines, 2967 top-level `theorem` or `lemma` declarations — of which 537 are `private`, leaving 2430 in the public interface — and 536 top-level `def` or `abbrev` declarations.

Dependency order runs broadly downward through the clusters. The exact-arithmetic module `Arithmetic.lean` imports nothing from the project at all, and `Basic.lean` imports only it; between them they sit below essentially everything else. The moment, dyadic, Fourier, and Legendre layers depend on the foundations and on each other but not on any asymptotic module. Within the endpoint asymptotics the order is genuinely less linear: the dyadic-cumulant branch and the vertical-Bromwich branch are developed in parallel from a shared negative-Laplace root and first meet in `FabiusDyadicSharpAsymptotic.lean`, then again in `FabiusSharpAsymptotic.lean`, which imports both.

A handful of modules are pure infrastructure with no counterpart anywhere in the article: the generic Gaussian-contraction and Gaussian-tail estimates, the formal exponential and logarithm coefficient algebra of the `SaddleExpansion` / `SaddleLog` files, the abstract Toeplitz row lemma, and the generic weighted-path-sum solver are all stated for arbitrary inputs and are used only to keep the named Fabius theorems short.

B.1 Foundations: definitions, existence, and the original characterization

These modules underlie everything else and correspond to Section 1, Section 2, and Section 3.

Module	Role in the proof
`Arithmetic.lean`	Keeps all exact quantities in $\mathbb{Q}$ and $\mathbb{N}$: binary weight, Thue–Morse signs, c_n and d_n, division-free numerators, reciprocal values, closed dyadic formulas, and executable rational evaluators. Imports nothing from the rest of the project.
`Basic.lean`	Separates the three functions that share the letter F: the bounded CDF-valued `BoundedFabius` subtype with its characterization `IsFabius`, the signed extension on all of $\mathbb{R}$, and the compactly supported up. Defines the Fourier products and moment integrals used later.
`Existence.lean`	Constructs F as the fixed point of a strict contraction on symmetric $[0, 1]$-valued continuous functions, extracts the integral fixed-point equation, and proves uniqueness from agreement on the dense set of dyadic rationals.
`Differential.lean`	Derives the Rvachev differential equation, gluing the two one-sided derivatives at the fold point $x = 0$, and bootstraps the equation to smoothness of every finite order.
`OriginalCharacterization .lean`	Formalizes Theorem 1 of the compact-support paper starting from an unspecified constant k, derives positivity of k from the other hypotheses, and then obtains the exact support, nonnegativity, translate identity, normalization, and the forced value $k = 2$. It also proves that every bounded Fabius solution folds to an original compact-support solution.
`OriginalUniqueness.lean`	Proves the uniqueness half of that theorem by the Fourier refinement argument: every solution has transform value one at the origin and satisfies the same finite and infinite dyadic sinc-product formulas. The final theorems identify the exact kernel of folding, give the sharp fixed-candidate iff after restoring the strict right tail, and package original solutions as exactly the scale-two folds of unique bounded Fabius solutions.
`OriginalPaperSupplement. lean`	Packages Theorems 4, 6, and 7 of the compact-support paper, the finite Taylor expansion at dyadic points, the nowhere-analyticity corollary, and the exact even and odd integer values of the signed extension.

B.2 Regularity, shape, and the global extension

The sharp shape statements collected here are used throughout Section 2 and supply the exact derivative identities of Section 6.

Module	Role in the proof
`Regularity.lean`	Records the global 2-Lipschitz estimate for both functions, proves that the constant is attained at the midpoint, and isolates the linear majorants $F(x) \le 2x$ and $\mathrm{up}(x) \le 2(1 - \lvert x\rvert)$ used as quantitative input elsewhere.
`Monotonicity.lean`	Collects the order-theoretic consequences of the derivative formula: strict monotonicity of F on $[0, 1]$ and the resulting bijection of $[0, 1]$ onto itself, the exact nonzero set (ordinary `Function.support`) $(-1, 1)$ and topological support $[-1, 1]$ of up, and its strict unimodality.
`Convexity.lean`	Turns the monotonicity of up into convexity of F on $(-\infty, 1/2]$ and concavity on $[1/2, \infty)$, strictly inside the unit interval, so that the midpoint is exactly the inflection point.

Module	Role in the proof
`GlobalExtension.lean`	Proves that the Thue–Morse translate series defining the signed extension is locally finite, then derives the global differential equation and the closed formulas for every iterated derivative of both $\mathcal{F}$ and up.
`GlobalBounds.lean`	Observes that only the b-th translate can be nonzero on $[2b, 2b+2]$ (and that it vanishes at the endpoints), hence $\lvert\mathcal{F}\rvert \le 1$ everywhere, and converts the derivative formula into the sharp attained bound $\lvert\mathcal{F}^{(k)}\rvert \le 2^{\binom{k+1}{2}}$.
`BoundedDerivatives.lean`	Transfers those sharp derivative bounds from the signed extension to the bounded CDF by a germ argument below one, a local-constancy argument above one, and continuity at the remaining point $x = 1$.
`NowhereAnalytic.lean`	Determines the exact real-analytic locus: F is analytic at x if and only if $x \notin [0, 1]$. The right endpoint needs a separate argument, since the reflection identity is not an identity of germs there.
`FabiusFlatness.lean`	The two-sided qualitative flatness statements at both endpoints: in particular $\mathcal{F}(x) = o(\lvert x\rvert^n)$ and $F(x) = o(\lvert x\rvert^n)$ at 0, and $1 - F(x) = o(\lvert 1 - x\rvert^n)$ at 1, for every n.
`EffectiveFlatness.lean`	An elementary effective form $F(x) \le 2^{\binom{n+1}{2}} x^n$ on $2^n x \le 1$, proved by iterating the mean value theorem, kept because it avoids importing any integral machinery.
`SharpFlatness.lean`	Recovers the factorial that the mean value theorem discards by integrating instead, giving $F(x) \le 2^{\binom{n+1}{2}} x^n/n!$; the remaining overshoot at $x = 2^{-n}$ is exactly $1/d_n$.

B.3 Probability, finite splines, and computability

This cluster realizes the probabilistic model of Section 4 and supplies the evaluators and error bounds of Section 21 and Section 21.6.

Module	Role in the proof
`ProbabilityRepresentation.lean`	Builds the infinite product probability space, proves independence and the uniform law of each coordinate, identifies the CDF of the random weighted series with F globally on $\mathbb{R}$, and obtains up as the corresponding density.
`FabiusComplexMGF.lean`	Identifies the entire generating function with Mathlib's complex moment-generating function of that law, so that all complex derivatives are tilted moments and vertical-line norms are dominated by real tilted moments.
`EarlyApproximants.lean`	Formalizes the generating polynomials, their product and recursion laws, exact degree and coefficient normalization, the associated atomic measures and finite Bernoulli convolutions, and the corrected half-endpoint step representatives.
`EarlyMeasureBridge.lean`	Proves that the atomic measure encoded by the generating polynomial really is the finite Bernoulli convolution appearing in the source.
`WeakConvergence.lean`	Proves weak-$*$ convergence of the finite convolution measures to up $\,d\lambda$, with explicit test-function formulations over an arbitrary `RCLike` field.
`StepMeasureBridge.lean`	Identifies half-endpoint cells with ordinary interval indicators almost everywhere, computes their masses exactly, and sandwiches each histogram integral between shrunken and enlarged interval masses.

Module	Role in the proof
`StepApproximationLimit.lean`	Supplies coefficient unimodality, monotonicity of the step functions, and the interval average squeeze that upgrades convergence of interval integrals to pointwise convergence against a continuous limit.
`FabiusUniformSpline.lean`	Identifies the centered Thue–Morse power sums with distribution functions of finite weighted sums of independent uniform coordinates, giving support bounds, monotonicity, block-translation identities, and pointwise signed-global convergence.
`FabiusComputability.lean`	Formalizes both clauses of computability in the sense of computable analysis: effective uniform continuity from the global derivative equation, and preservation of computable sequences by rounded dyadic spline evaluation. It also proves the all-real error $\lvert S_p - \mathcal{F}\rvert \le 2^{-p}$ and packages global uniform convergence.
`FabiusComputableSpline.lean`	Certifies a primitive-recursive natural-number evaluator: clamp a signed-natural numerator, evaluate a finite centered spline, round half-up, and inherit the explicit error bound $5 \cdot 2^{-(p+3)}$.

B.4 Fourier transform, Poisson summation, and the Laplace bridge

These are the analytic identities of Section 5, together with the spectral consequences of Section 20 and the transform normalization that the asymptotic analysis later requires.

Module	Role in the proof
`FourierProduct.lean`	Proves that the removable sinc factors form a genuine convergent infinite product, that the transform satisfies the finite refinement identity, and that the limit identifies the two functions.
`FourierAnalytic.lean`	Shows that the transform of up is entire, establishes Fourier inversion, and relates its values at imaginary arguments to the half-moment generating function.
`PoissonSummation.lean`	Formalizes Poisson summation and the partition-of-unity identities, correcting two typographical errors in the source: a missing dependence on t in the exponent and a spurious factor $1/a$ after multiplication.
`LaplaceTransform.lean`	Proves that the negative generating function divided by the spectral variable is the one-sided Laplace transform of F, in both complex and real form, with the integrability needed to avoid detouring through complex integrals.

B.5 Moments, generating functions, and exact arithmetic

Everything here stays in $\mathbb{Q}$ or $\mathbb{N}$, so no denominator or 2-adic valuation is ever taken of a real number. The cluster carries Section 7 and Section 8.

Module	Role in the proof
`MomentPowerSeries.lean`	Proves in exact rational arithmetic that the half moments are the even-binomial transform of the moments, by packaging the defining recurrence as a formal power series identity.
`NormalizedMoments.lean`	Shows that the executable rational recurrence for d_n agrees with the division-free recurrence for its natural numerator, independently of any analytic construction.

Module	Role in the proof
`NormalizedEvenMoments.lean`	The same normalization statement for the even moments c_n, identifying the division-free natural recurrence with the numerator of the rational one.
`AnalyticMoments.lean`	Supplies the integral interpretations of both exact sequences, their inverse-power specializations, and the generating-function dilation identity.
`BernoulliRecurrences.lean`	Connects the executable moment recurrences to Mathlib's formal generating series for the Bernoulli numbers.
`LaplaceMoments.lean`	Differentiates the negative Laplace transform under its compact integral, producing the tilted moment family with $M_k' = -M_{k+1}$ and $M_k(0) = d_k$, the exact moment input for the endpoint comparison.
`Parity.lean`	Proves the binary-coefficient count behind the oddness of every integral moment numerator, using Lucas's theorem at the prime 2, together with the truncated odd-coefficient count needed two-adically.
`TwoAdic.lean`	Establishes the exact numerator and denominator parity of the half moments and converts it into the dyadic valuation $v_2(F(2^{-n}))$, using Legendre's formula to trade the factorial valuation for binary weight.
`HalfMomentDenominator.lean`	Proves that the reduced denominator of $2d_n$ is odd for every n, inductively at even indices and through $d_{2m+1} = (2m+1)c_m/2$ at odd ones.
`DenominatorBound.lean`	Puts the moments in the closed dyadic formula over a common denominator, shows the scaled value is an integer after the factorial and power-of-two cancellations, and derives the lcm denominator bound.
`ExactInversePower.lean`	Arithmetic proofs of the explicit inverse-power identities relating $F(2^{-n})$ to the half moments and to the normalized odd integers R_n.
`FabiusRecurrenceSequence.lean`	Records the factorially normalized half-moment sequence $a_n = d_n/n!$, its elementary recurrence, its exact relation to $F(2^{-n})$, and the associated entire series and dyadic product.
`FabiusInverseDyadicClosedForm.lean`	Solves the triangular recurrence for $F(2^{-n})$ nonrecursively as a weighted sum over ordered compositions of n, via a generic path-weight layer with last-edge and edge-count decompositions; the empty composition covers $n = 0$.

B.6 Dyadic closed forms and finite q-binomial identities

This is the exact evaluation machinery of Section 9, whose algorithmic side is discussed in Section 21.

Module	Role in the proof
`ScaleTranslation.lean`	Proves that the Thue–Morse series changes sign under translation by a power of two over the corresponding initial block, and combines this with the derivative formula to obtain the translated Taylor-remainder identity.
`TaylorReduction.lean`	Turns that cancellation into the finite recursive reduction formula, treating nonnegative indices by Taylor expansion and negative ones directly by the sign-translation law.

Module	Role in the proof
`DyadicCorrectness.lean`	Discharges the structural half of the evaluator proof: strong induction upgrades the single-step dyadic recurrence to agreement of the terminating bit recursion with the closed formula at every unit dyadic grid point.
`DyadicClosedForm.lean`	Proves the affine Thue–Morse cancellation and the Taylor-block identity, identifies the inverse-power lookup table, and concludes that the fast evaluator agrees with the closed formula on $[0, 1]$.
`DyadicAnalytic.lean`	Closes the loop analytically: endpoint compatibility of the finite Taylor recurrence determines the inverse-power values, so the rational evaluator really computes the analytic function.
`GlobalDyadic.lean`	Extends exact evaluation beyond the unit interval by reducing modulo a length-two block, folding the residue, and applying the Thue–Morse sign; local finiteness identifies the one contributing translate.
`HalfQBinomial.lean`	Develops the finite q-Pochhammer symbol and the Gaussian binomial coefficient at $q = 1/2$ over $\mathbb{Q}$: vanishing criteria, Mersenne evaluation, symmetry, both Pascal recurrences, and the finite q-binomial theorem.
`ThueMorseExponential.lean`	Packages the centered Thue–Morse power sums into a formal exponential generating series, shows Prouhet cancellation forces divisibility by X^k, and computes the effect of translating the inner power by a constant.
`FabiusQBinomialFormula.lean`	Proves the source-literal centered q-binomial formula, first at inverse powers and then for arbitrary numerators, together with the stronger statement that a common rational translation leaves the numerator unchanged.
`FabiusQBinomialTaylor.lean`	Identifies that finite q-binomial polynomial with the Taylor reduction sum by viewing the translation as an indeterminate over $\mathbb{Q}$; constancy then permits evaluation in any characteristic-zero field.
`FabiusQBinomialScalarFormula.lean`	Restores the normalizing prefactor at scalar level, exposing the centered identity for arbitrary real and complex translations.
`FabiusRawQBinomialFormula.lean`	Proves the raw-coordinate variant with inner power $(r + q)^{n+k}$ by reflecting each dyadic Thue–Morse block, the combined sign being cancelled exactly by the normalization $(-2)^{n^2}$.
`FabiusRawQBinomialScalar.lean`	Lifts the raw identity from $\mathbb{Q}$ to any $\mathbb{Q}$-algebra by the same constant-polynomial argument, with real and complex wrappers and fully expanded finite-sum statements.
`FabiusDyadicQBinomialScalar.lean`	Does the same for the arbitrary-numerator dyadic formula, so that a single statement covers $\mathbb{R}$, $\mathbb{C}$, and every other characteristic-zero scalar field.

B.7 Global binary reduction and the discrete limit

These modules prove the telescope of Section 11 and the convergence statement of Section 12.

Module	Role in the proof
`FabiusBinaryReductionSeries.lean`	Iterates the finite Taylor reduction along the binary expansion, keeping the outer index at $m = 0$ so that the even and odd unit blocks are distinguished, and upgrades the signed residual identity to absolute summability and all-real uniform convergence with error 2^{1-N} through inclusive scale $N \geq 1$.

Module	Role in the proof
`FabiusParityPowerSeries.lean`	Formalizes the corrected scale-zero parity-power series, valid for every nonnegative real input; the series indexed from $m = 1$ remains valid only on $0 \le x < 1$.
`FabiusGlobalQBinomialSeries.lean`	Combines the constant-polynomial identity with the binary telescope to give the absolutely convergent pointwise q-binomial expansion, over $\mathbb{R}$ and $\mathbb{C}$ alike, for every real or complex translation.
`FabiusComplexShiftSpline.lean`	Identifies the complex-shifted finite spline with a fixed polynomial branch, expands it around the centered shift $q = 1/2$, and proves an exponentially small translation bound, so every fixed shift has the same limit.
`FabiusDiscreteLimitComplexShift.lean`	Isolates the complex-analytic ingredient: translating a normalized finite Thue–Morse power sum has an exact finite Taylor expansion in the lower-degree branches.
`FabiusDiscreteLimitToeplitz.lean`	Proves that each row of finite q-binomial weights has mass one, gives an exact formula and a uniform bound for its total variation, and supplies a general finite-row Toeplitz convergence theorem.
`FabiusDiscreteLimitIntegration.lean`	Performs the outer q-binomial reindexing, applies Toeplitz convergence to the complex-shift spline limit, and extends the result from the nonnegative axis to all of $\mathbb{R}$ by exact empty-prefix vanishing.

B.8 The Fourier–Legendre expansion

The whole cluster corresponds to Section 19.

Module	Role in the proof
`LegendrePolynomial.lean`	Supplies the ordinary Rodrigues-normalized Legendre polynomials on $[-1, 1]$, which Mathlib does not carry, with endpoint, parity, norm, and differential-equation facts and the affine rule relating them to the shifted basis.
`LegendreSeriesConvergence.lean`	Provides the analytic convergence layer from the self-adjoint Sturm–Liouville operator $f \mapsto -((1 - x^2)f')'$, whose weight makes both boundary terms vanish, and iterates the Green identity to move eigenvalue powers onto the expanded function.
`FabiusLegendreCoefficients.lean`	Evaluates the coefficient u_n of the even function up in closed form in terms of the inverse-power values $F(2^{-k})$.
`FabiusLegendreSeries.lean`	Combines coefficients and convergence: absolute and uniform convergence on $[-1, 1]$, including both endpoints, plus the full orthogonality relation and absolute summability of the coefficient sequence.
`FabiusTranslatedLegendreSeries.lean`	Translates the even expansion to $[0, 2]$, where $\mathcal{F}(x) = \mathrm{up}(x - 1)$, and expands the shifted Legendre polynomials into monomials to give the exact double finite inner sum.
`FabiusLegendreLeastSquares.lean`	Proves a generic Pythagorean identity for the unweighted $L^2[-1, 1]$ error and deduces that S_N is the unique least-squares minimizer through degree $2N + 1$, one degree better than its visible class.

B.9 The negative-Laplace product and its periodic correction

This is the exact decomposition of Section 14 together with the Mellin evaluation of the constant appearing in Section 15.

Module	Role in the proof
`NegativeLaplace.lean`	Constructs the logarithm of the dyadic product as an absolutely convergent series and splits its dilation equation into a quadratic solution, an exact one-periodic correction, and a forward tail bounded by $4\mathrm{e}^{-s}$, using only elementary bounds.
`PeriodicCorrection.lean`	Proves continuity of that exact correction by treating the two infinite sums locally on compact positive intervals with geometric majorants, establishing the regularity interface used downstream.
`PeriodicRegularity.lean`	Differentiates the forward tail termwise through order four with superexponential majorants, proving that Ψ is C^4 and that its first two derivatives are one-periodic and globally bounded.
`PeriodicSmooth.lean`	Upgrades that to C^∞ by encoding the $(k+1)$st derivative of each summand as a recursively generated polynomial in e^{-s2^n}, then propagates smoothness through the saddle-jet recurrence.
`PeriodicMean.lean`	Integrates the negative-Laplace logarithm and its forward tail over a dyadic interval, evaluates the mean of the exact correction, and pins down the unit-interval integral of the centered normalization Ψ.
`PeriodicFourier.lean`	Computes every Fourier coefficient of Ψ through the regularized Bose–Mellin kernel, proves the first coefficient nonzero — hence Ψ nonconstant — and gives the summable Gamma–zeta Fourier series.
`StieltjesConstant.lean`	Fixes the sign convention for the first Stieltjes constant and records the Laurent expansion of ζ at its pole.
`MellinBose.lean`	Proves the real Mellin identity for $\log(1-\mathrm{e}^{-x})$ with value $-\Gamma(a)\zeta(a+1)$ by expanding the logarithm and justifying termwise integration.
`MellinFinitePart.lean`	Regularizes $\Gamma(s)\zeta(1+s)$ at the double pole $s=0$ and identifies the finite part with the Euler–Stieltjes combination.
`BoseFinitePartIntegral.lean`	Realizes that finite part as a convergent split integral — singular logarithm removed near zero, kernel divided by x near infinity — and supplies the two kernels reused by the mean and Fourier computations.
`GammaSecondOrder.lean`	Supplies the second-order Taylor expansion of Γ at 1 with an explicit remainder, obtained from analyticity and the reflection formula.
`FabiusSharpConstant.lean`	Combines the mean of the periodic correction with the Gaussian saddle prefactor and records the resulting constant in both compact Gamma–zeta and expanded Euler–Stieltjes form.

B.10 Tilted moments, endpoint cumulants, and the sharp dyadic formula

This branch turns Section 14 into the unconditional dyadic statement quoted in Section 15.

Module	Role in the proof
`NegativeLaplaceDerivatives.lean`	Expresses the first four logarithmic derivatives in terms of normalized tilted moments $R_k = M_k/M_0$ satisfying $R_k' = -R_{k+1} + R_k R_1$, so that they become the usual cumulant polynomials.
`NegativeLaplaceDerivativeBounds.lean`	Differentiates the dilation equation, bounds the elementary dilation kernel with polynomial weights, and iterates to prove $q^{(j)}(n) = \mathcal{O}(\log n/n^j)$ for $1 \le j \le 4$.

Module	Role in the proof
`NegativeLaplaceAllOrderJets.lean`	Packages every ordinary derivative into one recursive sequence agreeing with `iteratedDeriv`, splitting each dyadically rescaled derivative into a linear drift, a smooth periodic jet, and a forward-tail remainder.
`NegativeLaplaceScaledTailFlat.lean`	Bounds every positive-order forward-tail summand by a polynomial times an exponential on $[1, \infty)$ and shows the scaled dyadic sum is o of every real power of the radius.
`LaplaceMoments.lean`	(See Appendix B.5.) Its tilted moment family is the entry point of this branch.
`LaplaceMomentBounds.lean`	Makes the endpoint comparison unconditional through $R_k(s)^2 \leq s((4/s)^k k!)^2$ for $s \geq 2$, proved by Hölder log-convexity at the intermediate tilt $3s/4$ with no periodic regularity input at all.
`LaplaceCumulantAsymptotics.lean`	Inverts the first four cumulant polynomials and packages the bookkeeping that turns the logarithmic derivative bounds into an $\mathcal{O}(1/n)$ endpoint error.
`LaplacePeriodicSecondOrder.lean`	Differentiates the quadratic-plus-periodic decomposition, controls the forward-tail derivative, and shows the linear logarithmic terms cancel in $n(q'' + (q')^2)/2$, leaving the explicit periodic-derivative expansion.
`EndpointLaplaceComparison.lean`	Compares $(1 - x)^n$ with $\mathrm{e}^{-nx}(1 - nx^2/2)$ pointwise, bridges to survival-function moments by Fubini, and transfers d_n to the negative Laplace transform at scale n with an explicit logarithmic correction.
`DyadicSharpDecomposition.lean`	Performs the exact algebraic assembly before any estimate: elementary main term, exact periodic correction, exponentially small product tail, endpoint-versus-Laplace error, and sharp Stirling remainder, with no hidden cancellation.
`DyadicSharpConditional.lean`	Quantifies every contribution except the two isolated moment estimates, reducing the complete dyadic formula to exactly those.
`FabiusDyadicSharpCumulant.lean`	Combines the exact Euler–Stieltjes constant with the unconditional endpoint comparison, reaching the final $\mathcal{O}(1/n)$ precision while keeping the second-order cumulant exact.
`FabiusDyadicSharpAsymptotic.lean`	Recognizes that cumulant as the first-order displacement of the periodic correction from $\log_2 n$ to the exact Lambert phase, and performs the Taylor transfer with a uniform quadratic remainder.

B.11 Vertical estimates, Bromwich inversion, and the central saddle

The contour-free analytic input to Section 15. There is no contour-shifting argument anywhere: every hypothesis of Fourier inversion is discharged explicitly.

Module	Role in the proof
`NegativeLaplaceVertical.lean`	Proves the canonical complex dyadic product and extracts finitely many factors to obtain the minor-arc bound $\|P(-rw)\|/P(-r) \leq C(r, N)\|w\|/\|w\|^N$; already $N = 2$ gives integrability of the vertical kernel.
`NegativeLaplaceMinorArc.lean`	Isolates the elementary criterion bounding the finite coth product by $\exp(4N\mathrm{e}^{-b})$, separating that estimate from any later choice of the number of extracted factors.
`NegativeLaplaceVerticalLog.lean`	Constructs a branch-safe logarithm on each vertical line by integrating the logarithmic derivative rather than selecting a branch of log, and proves that exponentiating it recovers the transform ratio.

Module	Role in the proof
`NegativeLaplaceVerticalSmooth.lean`	Promotes that normalized logarithm to C^∞, since the underlying transform curve is smooth and nonvanishing and the fundamental theorem of calculus transfers the regularity.
`NegativeLaplaceVerticalTaylor.lean`	Gives the branch-safe cubic Taylor expansion with explicit integral remainder, together with continuity and C^4 smoothness of the normalized vertical moments.
`NegativeLaplaceVerticalFourthBound.lean`	Supplies the uniform off-axis fourth derivative bound used by the first-order central estimate.
`NegativeLaplaceVerticalAllOrderBound.lean`	Continues the one-factor derivative holomorphically and applies Cauchy's estimate on disks of radius 1/2 to give the scale-independent increment bound $m!\,5/(1/2)^m$, then iterates to $\mathcal{O}(b)$ at $r = 2^b$ for every order.
`NegativeLaplaceVerticalOrdinaryJets.lean`	Defines the holomorphic logarithmic derivative on the open right half-plane and matches it on the real axis with the ordinary derivatives of the negative-Laplace logarithm.
`BromwichSaddle.lean`	Derives the real Bromwich formula directly from Mathlib's Fourier inversion theorem and performs the exact change of variables putting a transform $P(z)/z$ into dimensionless saddle coordinates.
`FabiusBromwichInput.lean`	Discharges the hypotheses of that framework for F: exponential tilting gives an integrable continuous input whose transform is the negative generating function over the vertical coordinate.
`QuantitativeSaddle.lean`	The measure-theoretic end of the argument. Bounds on a central set and its complement give a relative estimate $1 + \mathcal{O}(1/b)$, with the vanishing of the signed odd correction made explicit.
`FabiusSaddleReduction.lean`	Separates the exact Bromwich formula into a real prefactor and a normalized kernel integral, so that no positivity or logarithm manipulation survives into the central-arc estimates.
`FabiusSaddleCentral.lean`	Bounds the central L^1 error against a standard Gaussian plus the retained odd linear-plus-cubic correction, at a general radius and then at the standardized one.
`FabiusSaddleCentralLambert.lean`	Instantiates that estimate on the dyadic Lambert orbit, extracting the concrete linear, cubic, and curvature coefficients and proving the normalized kernel mass is $1 + \mathcal{O}(1/\lambda)$.

B.12 The lower-Lambert phase and the corrected sharp asymptotic

The exact saddle coordinate and the source-facing statements of Section 15.

Module	Role in the proof
`LowerLambertW.lean`	Supplies the minimal real W_{-1} infrastructure: a totalized branch with specification and uniqueness on $(-\mathrm{e}^{-1}, 0)$ and the standard first two terms of its expansion.
`FabiusLambertPhase.lean`	Defines the coordinate λ solving $\lambda 2^{-\lambda} = 2^{-t}$, proves its exact fixed-point equation, and records its first two asymptotic terms; keeping it exact is what avoids losing a logarithmic factor.
`FabiusLambertSaddle.lean`	Packages λ with the positive Laplace radius $r = 2^\lambda$, turning the defining equation into the identity $rx = \lambda$ without committing to an implicitly defined saddle point.
`FabiusLambertRates.lean`	Compares the reciprocals of the Lambert and logarithmic coordinates by $\mathcal{O}$ and converts the rate into the literal $1/(-\log x)$ scale.
`FabiusLambertHigherExpansion.lean`	Retains the next phase term, since substitution into a quadratic action magnifies phase errors by one power of t; without it the final error would be $\log t/t$ rather than $1/t$.

Module	Role in the proof
`FabiusLambertDerivativeBounds.lean`	Bounds the first three logarithmic derivative residuals along the real orbit at a Lambert radius and expresses the second and third derivatives in periodic form.
`FabiusLambertMinorArc.lean`	Chooses the factor count $m(t) = \lceil \lambda(t)/4 \rceil$, so that $2m(t) = \lambda(t)/2 + \mathcal{O}(1)$ and the finite coth product is uniformly bounded by e^4.
`FabiusLambertTailFlat.lean`	Records once and for all that the forward Laplace tail is smaller than every inverse power of λ, so it can be discarded at every finite order.
`FabiusSmallArgumentScale.lean`	Packages the exact inverse relationship between $t \mapsto 2^{-t}$ and $x \mapsto -\log_2 x$ and transfers $\mathcal{O}$ estimates between the two forms.
`FabiusSharpExactReduction.lean`	Isolates the exact identity behind the transfer: on the lower-Lambert branch, $\log F$ minus the corrected main term is exactly the logarithm of the real saddle ratio plus the exponentially small forward tail.
`FabiusSharpLambertMain.lean`	Isolates the purely algebraic part of the corrected small-argument asymptotic, leaving the normalized Bromwich kernel estimate as the sole analytic input.
`FabiusSharpLambertTransfer.lean`	Converts a relative kernel error of order $1/\lambda$ into the corrected sharp logarithmic asymptotic, absorbing the forward tail unconditionally via $\lambda \le 2^\lambda$.
`FabiusSharpAsymptoticTransfer.lean`	Packages the last source-facing reduction, producing the corrected formula with its literal $\mathcal{O}(1/(-\log x))$ error as $x \downarrow 0$.
`FabiusExplicitSharpTransfer.lean`	Combines the compact Lambert transfer with the expansion into logarithms and Euler–Stieltjes constants, discharging every phase and formula-conversion error.
`FabiusFirstSaddleCorrection.lean`	Expands the normalized kernel one order further and records the resulting periodic coefficient of $1/\lambda$, contracted against the Gaussian moments $(1, 3, 15)$, independently of the analytic remainder estimate.
`FabiusSharpAsymptotic.lean`	The headline module: it closes the quantitative saddle argument, shows the elementary printed expression is missing a genuine nonconstant periodic term, and imports the quotient-approximation obstruction.

B.13 The all-orders expansion

The machinery behind Section 16. The first eight modules are generic: they know nothing about the Fabius function and would serve any saddle-point problem with bounded periodic coefficients.

Module	Role in the proof
`SaddleExpansionAlgebra.lean`	Turns a formal exponent $E(\varepsilon) = \sum_{j \ge 1} E_j \varepsilon^j$ into the correction coefficients of $\exp E$, in any commutative $\mathbb{Q}$-algebra, and defines the bounded-coefficient Poincaré expansion API.
`SaddleExpansionFiniteRemainder.lean`	Makes the coefficient identity a finite polynomial one: the discrepancy is an exact multiple of X^L, with canonical quotient defined by iterating `divX` so that parameter dependence stays algebraic.
`SaddleExpansionFlat.lean`	Bridges from estimates at every algebraic order to the parameter-dependent expansion API.
`SaddleExpansionSmallArgument.lean`	Transports a full expansion from the dyadic logarithmic scale to $x \downarrow 0$.

Module	Role in the proof
`SaddleLogExpansionAlgebra.lean`	Builds the coefficient sequence of the formal logarithm of a unit-constant series from the recurrence $B'A = A'$.
`SaddleLogExpansionPowerSeries.lean`	Identifies that recurrence with Mathlib's `PowerSeries.logOf`, connecting the finite recursion to the standard substitution calculus.
`SaddleLogAsymptoticTransfer.lean`	Turns a full real expansion with constant coefficient one into the expansion of its logarithm, combining the formal divisibility statement with the analytic Taylor remainder and a stability estimate.
`SaddleAllOrders.lean`	Supplies the two-source exponential remainder bound, the Gaussian-weighted reference, and the criteria under which a normalized integral matches its even expansion.
`GaussianPolynomialContraction.lean`	Identifies normalized Gaussian integration with a linear functional on complex polynomials and records the integrability and coefficient estimates reused throughout.
`GaussianPolynomialTail.lean`	Bounds the complementary-interval integral of a Gaussian times a finite polynomial coefficient by coefficient, making any fixed reference smaller than any requested inverse power.
`GaussianPolynomialTailAllOrders.lean`	Upgrades that estimate to the order-dependent radius, so a phase-dependent polynomial family contributes a tail of order $b^{-(N+1)}$ whenever its coefficient weight stays bounded.
`GaussianPolynomialWholeIntegral.lean`	Controls the full L^1 norm of the same references by the same factorial coefficient weight.
`FabiusSaddleCoefficientRecurrence.lean`	Records the concrete periodic-jet algebra: $r^{n+1}q^{(n+1)}(r) = s_n t + p_n(t)$ with $s_n = (-1)^{n+1}n!$, together with the parity $E_m(t,-v) = (-1)^m E_m(t,v)$ that kills all odd half-powers.
`FabiusSaddlePolynomialCoefficients.lean`	Packages that exponent coefficient as an element of $\mathbb{C}[X]$ and propagates coefficientwise conjugation, so every even exponential coefficient is fixed.
`FabiusSaddleExpansionCoefficients.lean`	Exponentiates and contracts against Gaussian moments, proves every mass coefficient real, lifts the recurrence to continuous maps for continuity and periodic boundedness, and applies the logarithmic recurrence.
`FabiusSaddleExponentAllOrders.lean`	Splits the exact centered exponent jet of each order into its bounded part, its linear-in-t slope part, and the flat forward-product correction.
`FabiusSaddleCentralAllOrders.lean`	Identifies the finite polynomial reference on the dyadic Lambert orbit, contracts its odd terms, and decomposes the exact centered exponent into a periodic truncation plus explicit remainders.
`FabiusSaddleCentralRadiusAsymptotics.lean`	Shows the logarithmic central radius grows more slowly than $\sqrt{\lambda}$ for every fixed degree, controlling all fixed-order Taylor polynomials uniformly on the expanding interval.
`FabiusSaddleReferenceWeight.lean`	Proves the Gaussian tail weight of the finite reference polynomial uniformly bounded, in the periodic phase and in every substitution parameter of norm at most one.
`FabiusSaddleReferenceTail.lean`	Supplies the explicit Gaussian moment tail estimates on the complement of $[-A, A]$ used by that weight argument.
`FabiusSaddleTail.lean`	Integrates the polynomial kernel bound $C(r, 2m)/(1 + v^2/b)^m$ by splitting into an intermediate Gaussian region and a far geometric region, giving an $\mathcal{O}(1/b)$ tail at the standardized radius.
`FabiusSaddleTailAllOrders.lean`	Repeats that at the order-dependent radius $\sqrt{32(N+1)\log b}$, so the complementary tail is below every requested inverse power.
`FabiusSaddleMassAllOrders.lean`	Combines vertical Taylor expansion, finite exponential substitution, Gaussian contraction, and the minor-arc estimates into the full Poincaré expansion of the normalized kernel mass.

Module	Role in the proof
`FabiusFullAsymptoticExpansion.lean`	Transfers that expansion through the exact real saddle identity and the logarithm, adds the flat forward tail, and moves the result to $x \downarrow 0$, keeping every coefficient evaluated at the exact Lambert phase.
`FabiusLambertAllOrderAlgebra.lean`	Defines the displacement coefficient polynomials $a_n(\ell)$ by their exact recursive algebra and connects the first nontrivial truncation to the quantitative expansion already proved.
`FabiusLambertFormalLog.lean`	Identifies those coefficients with the logarithm of the formal unit series $1 + uA(u)$, supplying the cancellations needed to make the formal series analytic.
`FabiusLambertAllOrderRemainder.lean`	Upgrades the formal recurrence to a rigorous expansion of every finite order, bounding the varying-coefficient Taylor remainder and using stability of $u - \log u/\log 2$ on the large positive branch.
`FabiusLambertAllOrderSmallArgument.lean`	Transports the all-order phase expansion to $x \downarrow 0$ and writes it purely in terms of $-\log x$ and $\log(-\log x)$.

B.14 Elementary log–log forms and endpoint consequences

These modules carry Section 17 and Section 18, and several of the negative results of Section 22.

Module	Role in the proof
`FabiusLogScale.lean`	Introduces the profiles $F(2^{-t})$ and $-\log F(2^{-t})$ and proves their exact differential-delay identity, carrying the explicit hypothesis $t \geq 1$ that the source omits and without which the statement is false.
`FabiusDyadicLogBounds.lean`	Combines the exact identity $d_n = n!\,2^{\binom{n}{2}} F(2^{-n})$ with elementary integral bounds to prove $\lvert \log F(2^{-n}) + (\log 2/2)n^2 \rvert \leq 3n\log(n+1)$, with no periodic ansatz.
`FabiusLogSquaredAsymptotic.lean`	Promotes that natural-index estimate to the real logarithmic scale by squeezing t between consecutive integers, giving the rigorous leading asymptotic $\log 2/2$.
`FabiusLogMainDefect.lean`	Substitutes the draft's proposed main term into the exact delay equation and determines its residual: a nonzero multiple of $(\log t/t)^2$, which is exactly the term the source later discards.
`FabiusWikipediaMain.lean`	Defines the compact lower-Lambert form of the published expression, including the centered periodic correction, and proves it equals the Lambert-coordinate saddle main term.
`FabiusWikipediaExpansion.lean`	Expands that compact form into the elementary printed logarithms, retaining the periodic correction at its exact phase; the fourth phase term is what keeps the error at $1/t$.
`FabiusWikipediaObstruction.lean`	Shows the periodic correction is nonconstant and its phase visits every far integer translate, so deleting the term makes an $\mathcal{O}(1/(-\log x))$ error impossible.
`FabiusDecayComparison.lean`	Proves the second qualitative comparison: although F is flatter than every power, it decays strictly more slowly than $\exp(-c/x)$ for every $c > 0$.
`FabiusQuotientExponentialMismatch.lean`	Formalizes the endpoint obstruction for the visually accurate quotient of exponentials, whose ordinary $\exp(-c/y)$ decay is strictly faster than the true bump.
`StirlingAsymptotics.lean`	Records the precise $\mathcal{O}(\log n)$ form of Stirling's formula invoked without proof in the draft.

B.15 Iterated Thue–Morse prefixes and draft refutations

The machine-checked half of Section 22: what the local drafts get right, stated correctly, and what they get wrong, refuted.

Module	Role in the proof
`ThueMorsePrefix.lean`	Defines the iterated inclusive prefix sums, proves the convolution kernel formula, the Prouhet vanishing of low-degree power sums, the affine power-sum identity, and the exact zero runs at dyadic indices.
`ThueMorseGenerating.lean`	Builds the signed Thue–Morse block polynomial and its finite product form, then shows the draft's literal grid normalization has endpoint value bounded away from one, refuting its convergence claim.
`ThueMorseApproximation.lean`	Identifies the corrected iterated prefixes with the polynomial and histogram approximants, exactly at dyadic cell centers and asymptotically along moving cells, including the exceptional right endpoint and the rescaling that recovers F.
`ThueMorseBinomialLog.lean`	Proves the binomial-parity formula for the Thue–Morse sequence, interpreting the finite sum in $\mathbb{Z}$ and the outer division in $\mathbb{Q}$, and showing the logarithm's argument is exactly $2^{\mathrm{wt}_2(n)+1}$.
`DraftCounterexamples.lean`	Records three explicit obstructions while preserving the draft's definitions literally: the nonexistent maximum, the uncancelled linear term in the printed proxy, and a local error of 3 where the draft claims $4h = 1$.

B.16 Public aggregates

The root module `FabiusFunction.lean` is what a user imports. It pulls in five paper aggregates — `Paper05442.lean`, `Paper06487.lean`, `PaperStatements.lean`, `PaperFabiusAsymptotic.lean`, and `PaperKFoldThueMorse.lean` — the last two of which cover the two local drafts audited in Section 22, together with the regularity, flatness, Legendre, negative-Laplace, and computability layers. Only `PaperStatements.lean` is reached indirectly, through `Paper06487.lean`.

Module	Role in the proof
`FabiusFunction.lean`	The root aggregate, one directory above all the others. Its module docstring is the authoritative summary of the public surface: the two formalized papers, the two audited drafts, exact arithmetic, the Legendre expansion, the sharp asymptotic, the regularity layer, and Grzegorczyk computability.
`Paper05442.lean`	Public import for the compact-support paper, naming a proved Lean counterpart for every theorem, lemma, corollary, and prose proposition, and flagging where the development proves strictly more than the source-facing domain.
`Paper06487.lean`	Public import for the arithmetic paper, exposing the analytic, integrality, divisibility, and two-adic statements, both orders of summation in the dyadic formula, and the denominator consequences of the conjecture.
`PaperStatements.lean`	The numbered index for the arithmetic paper: every proposition, theorem, lemma, question, definition, and conjecture, with the two deliberate corrections — the bounded versus signed codomain split, and the missing order hypothesis in Lemma 1.

Module	Role in the proof
`Paper06487Supplement.lean`	Exposes the prose-level and proof-internal assertions of the same paper that are not numbered theorem environments, several with weaker hypotheses or wider boundary cases than the prose needs.
`PaperFabiusAsymptotic.lean`	Claim-level audit of the local asymptotics draft, which contains no theorem environments at all: it tracks numbered equations and prose claims, proves the valid ones, disproves the $\mathcal{O}(t^{-2})$ residual claim, and points to the correct saddle analysis.
`PaperKFoldThueMorse.lean`	Claim-level audit of the local K-fold Thue–Morse draft: the identities, zero runs, and corrected approximation theorem are proved, while the printed normalization, the maximum, and the proxy's missing linear term are refuted.

C Concordance with the source papers

The preceding guide maps this article onto the formal development. The following table maps it onto the printed literature: it lists the numbered results of Arias de Reyna's two papers, [7] and [8], and records where each of them is proved here. Two warnings apply throughout. The source papers use F for what is written $\mathcal{F}$ here, and φ or up for the bump; and the arithmetic paper writes F_n and G_n for the integer numerators of c_n and d_n, which are C_n and D_n here, while its D_n is the least common denominator, which is Δ_n here. Rows in which this article strengthens or corrects a source statement say so.

Source result	Statement	Location in this article
1982 paper, Theorem 1	Existence and uniqueness of a C^∞ function supported on $[-1,1]$, positive inside, equal to 1 at the origin, and satisfying a two-translate refinement equation with an undetermined constant; the constant is forced to be 2.	Section 3; the refinement equation is (28). Uniqueness is obtained here from a contraction on symmetric distribution functions and transferred to up by the fold.
1982 paper, Lemma 1	The stage-m discrete convolution measures converge weak-$*$ to the measure with density up.	Section 4, via the finite atomic laws (46)–(48). Corrected: the printed convolution index is finite, see Section 22.1.
1982 paper, Theorem 2	up is the limit of the explicit step functions attached to the generating polynomials.	Section 4. Corrected: shared dyadic endpoints carry weight 1/2, without which the pointwise claim fails on the grid.
1982 paper, Theorem 3	Probability representation of $\mathrm{up}(x)$ for $-1 \le x \le 0$ as the measure of a set of sequences.	Section 4, from the random series (40). Strengthened: the identity is proved here for every real threshold, not only on $[-1,0]$.
1982 paper, Theorem 4	The signed translate sum is C^∞, obeys a pure dilation equation, and expresses every derivative of up.	Section 6: (11), (13), (90), and (91).
1982 paper, corollary after Theorem 4	up is nowhere real analytic.	Section 6, as a consequence of the terminating dyadic Taylor series (99).
1982 paper, Theorem 5	Poisson summation for the translates of up, and the partition-of-unity specializations.	Section 5: (82) and (83). Corrected: the printed exponential omits the spatial variable, and the subdivision parameter must be a positive integer.

Source result	Statement	Location in this article
1982 paper, Theorem 6	The moment integral over $[0,1]$ equals a normalized endpoint value, and the even moment integral is half of c_n.	Section 8: (142) and (136).
1982 paper, Theorem 7	up takes rational values at every dyadic point, with an explicit finite formula.	Section 9, (164). The powers of two are tracked through both the polynomial factor and the differential, see Section 22.1.
Arithmetic paper, Proposition 1	Recurrence for the even moments c_n and their integer normalization.	Section 7: (110), (112), and the division-free form (114).
Arithmetic paper, Proposition 2	Functional equation of the moment generating function with the factor $(\mathrm{e}^z-1)/z$.	(117). Corrected: the factor carries its removable value 1 at the origin before any coefficient is compared.
Arithmetic paper, Proposition 3	The half moments are a binomial transform of the even moments.	(125).
Arithmetic paper, Proposition 4	Existence of natural numerators for the half moments after a structured denominator is cleared.	(120); the numerators are written D_n here.
Arithmetic paper, Question 5	Reshetnikov's question whether the normalized numbers R_n are integers.	(158), with the first values (554). Corrected: the exponent of two is positive, see Section 22.
Arithmetic paper, Proposition 6	R_n expressed through d_n, a double factorial, and the even Mersenne product.	Section 8; the same expression appears in (553).
Arithmetic paper, Theorem 7	R_{2n+1} is an integer multiple of the moment numerator.	Section 8, using (112).
Arithmetic paper, Proposition 8	Explicit formula for R_{2n}, whose denominator can only be a power of two.	Section 8.
Arithmetic paper, Theorem 9	Every R_n is a natural number.	Section 8; the article also proves the stronger statement that each R_n is odd.
Arithmetic paper, Lemma 1	Translation of a Taylor remainder across one dyadic shell, with a sign reversal.	Section 9, inside the proof of (178). Corrected: the combined scale-plus-order must be nonnegative, see Section 22.1.
Arithmetic paper, Proposition 10	The Taylor-block reduction that removes the leading bit.	(178) and its half-moment form (181); Algorithm 9.12 is the executable version.
Arithmetic paper, Definition 12	The least common denominator of the dyadic values at level n.	Section 9, written Δ_n here to avoid the collision with the half-moment numerators.
Arithmetic paper, Theorem 13	An explicit universal multiplier that makes every level-n dyadic value an integer.	(190), with the divisibility form (191).
Arithmetic paper, Proposition 15	The level denominator is a multiple of the denominator of the reciprocal-power value.	Section 9, combined with (136).
Arithmetic paper, Conjecture 16	A conjectural exact form for the level denominators.	Kept conditional; see Section 22, where the unconditional bound and the conditional exact formula are deliberately separated.
Arithmetic paper, Theorem 17	Lucas's congruence for binomial coefficients modulo a prime.	Quoted from [13] and used in Section 7 and Section 8.

Source result	Statement	Location in this article
Arithmetic paper, Proposition 18	The number of odd binomial coefficients in the relevant rows is a power of two determined by the binary weight.	Section 7, in the oddness induction after (114).
Arithmetic paper, Proposition 19	Every moment numerator is odd.	Section 7, immediately after (114).
Arithmetic paper, Theorem 20	The rational number $2d_n$ has odd numerator and odd denominator for $n \geq 1$.	Section 7, the normalized parity theorem. Corrected: the proof-internal naturality assertion needs the factor two, (553).
Arithmetic paper, Theorem 21	R_n is odd, equivalently the exact two-adic valuation of $F(2^{-n})$.	(156), made explicit in binary by (157).
Arithmetic paper, Proposition 22	Bernoulli-number recurrences for both moment sequences.	Section 7, in the subsection on Bernoulli-number recurrences; the article also proves the equivalence of the Bernoulli and binomial forms.

Results of this article that have no counterpart in either paper – the Fourier–Legendre expansion, the q-binomial and discrete-limit formulas, the global binary reduction, the sharp endpoint asymptotic with its periodic term, and the computability theorem – are placed by Section 1.1 instead.

D Notation crosswalk

Different sources use F, φ, θ, and up for overlapping objects. The following crosswalk records the convention used here.

This article	Definition	Frequent source notation
$F(x)$	Bounded CDF on all of $\mathbb{R}$, constant 0 to the left and 1 to the right.	Fabius function on $[0,1]$; sometimes the same letter is reused for the global extension.
$\operatorname{up}(x)$	Even compactly supported density $F(1-\lvert x\rvert)$ on $[-1,1]$.	$\operatorname{up}(x)$ or $\varphi(x)$.
$\mathcal{F}(x)$	Signed Thue–Morse extension, zero on $(-\infty,0]$ and satisfying $\mathcal{F}' = 2\mathcal{F}\circ(2\,\cdot)$.	$F(x)$ or $\theta(x)$ in papers that work globally.
c_n	Even moment $\int x^{2n}\operatorname{up}(x)\,\mathrm{d}x$.	c_n in the arithmetic paper.
d_n	Half moment $n\int_0^1 x^{n-1}\operatorname{up}(x)\,\mathrm{d}x$, with $d_0 = 1$.	d_n in the arithmetic paper.
Ψ	Centered or explicitly mean-separated one-periodic correction in the negative-Laplace finite part.	Often absent from elementary accounts.
λ	Exact positive solution of $\lambda 2^{-\lambda} = x$.	Lower-Lambert saddle coordinate $-W_{-1}(-x\log 2)/\log 2$.

One collision of letters deserves an explicit warning, because it is invisible in a formula read out of context. In the q-special-function objects $(a;q)_n$ of (201) and $\left[{n \atop k}\right]_q$ of (224), the symbol q is the *base*, and throughout this article it is fixed at $q = 1/2$ by the dilation ratio of (207); it is never a free parameter and never a shift. The arbitrary common translation in (205) is written Q here for exactly that reason.

The 2019 source formula [19] that this article cites for the shape of (205) uses the same lowercase letter for the shift, so that its $(r - 2^k + q)^{n+k}$ has q meaning the translation and not the base, while the Gaussian coefficient standing beside it is at base $1/2$. A reader comparing the two presentations term by term should substitute $q \mapsto Q$ in the power and leave the base alone; no other change is needed, and in particular the numerical value $1/2$ appears in that source in both roles at once, since the shift chosen there is also $1/2$.

When comparing formulas, one should first identify which global convention is in force and only then compare signs, supports, and endpoint values.

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